

Egyptian Baccalaureate Mathematics

Student Edition • Open Workbook / Guided Practice • All_Chapters

Eng Eslam Ahmed | 01120009622

Assistant Lecturer • Faculty of Engineering • Cairo University

Concept 1 | Binomial and Multinomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Binomial

$$\binom{n}{r} a^r b^{n-r}$$

Multinomial

$$\binom{n}{r} \binom{n-r}{s} a^r b^s c^{n-r-s}$$

Exponent check

$$r + (n - r) = n$$

Classified Practice

Binomial and Multinomial Theorems

Q001 Challenge

For $(2x - 3)^5$, find the coefficient of x^3 and the sum of all coefficients without writing the full expansion:

Choose the correct option.

- A Coefficient = 720; sum = -1
- B Coefficient = 720; sum = 1
- C Coefficient = 540; sum = -1
- D Coefficient = 240; sum = 32

Classified Practice

Binomial and Multinomial Theorems

Q002 Challenge

Find the coefficient of x^2y^3z in $(x + y + z)^6$, then find the coefficient of x^2yz^3 in $(x - 2y + z)^6$:

Choose the correct option.

- A First = 60; second = -120
- B First = 60; second = 120
- C First = 120; second = -60
- D First = -60; second = -120

Classified Practice

Binomial and Multinomial Theorems

Q003 Challenge

For $(x^2 + x + 1)^4$, find the greatest coefficient and verify it by identifying the coefficient of x^4 :

Choose the correct option.

- A Greatest coefficient = 19
- B Greatest coefficient = 16
- C Greatest coefficient = 20
- D Greatest coefficient = 24

Classified Practice

Binomial and Multinomial Theorems

Q004 **Written****Expand $(x + 2)^5$ using the Binomial Theorem.**

Q005 MCQ**Expand $(3x - 2)^4$ using the Binomial Theorem:****Choose the correct option.**

A $81x^4 - 216x^3 - 96x + 16$

B $81x^4 - 216x^3 + 432x^2 - 96x + 16$

C $-(3x - 2)^4$

D $81x^4 - 216x^3 + 216x^2 - 96x + 16$

Classified Practice

Binomial and Multinomial Theorems

Q006 Written

Expand $(x + 2)^6$ using the Binomial Theorem.

Q007 MCQ**Expand $(x + 1)^4$ using the Binomial Theorem:****Choose the correct option.**

A $x^4 + 4x^3 + 4x + 1$

B $x^4 + 4x^3 + 6x^2 + 4x + 1$

C $x^4 + 4x^3 + 12x^2 + 4x + 1$

D $-(x + 1)^4$

Classified Practice

Binomial and Multinomial Theorems

Q008 MCQ

Expand $(a + b)^4$ using the Binomial Theorem:**Choose the correct option.**

A $a^4 + 4a^3b + 4ab^3 + b^4$

B $a^4 + 4a^3b + 12a^2b^2 + 4ab^3 + b^4$

C $a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

D $-(a + b)^4$

Q009 Written**Expand $(a - 2b)^4$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q010 MCQ

Expand $(x - 2)^5$ using the Binomial Theorem:**Choose the correct option.**

A $x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$

B $x^5 - 10x^4 + 40x^3 - 80x^2 - 32$

C $x^5 - 10x^4 + 40x^3 - 80x^2 + 160x - 32$

D $-(x - 2)^5$

Q011 Written**Expand $(3x + 2y)^5$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q012 Written

Expand $(2a^2 - 1)^5$ using the Binomial Theorem.

Q013 Written**Expand $(x - 3)^6$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q014 Written

Expand $(2x - 3y)^6$ using the Binomial Theorem.

Q015 Written**Expand $\left(x + \frac{1}{3}\right)^6$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q016 MCQ

Expand $(x + 3)^3$ using the Binomial Theorem:**Choose the correct option.**

A $x^3 + 9x^2 + 27$

B $x^3 + 9x^2 + 27x + 27$

C $x^3 + 9x^2 + 54x + 27$

D $-(x + 3)^3$

Classified Practice

Binomial and Multinomial Theorems

Q017 MCQ

Expand $(a + 2b)^5$ using the Binomial Theorem:**Choose the correct option.**

A $a^5 + 10a^4b + 40a^3b^2 + 80ab^4 + 32b^5$

B $a^5 + 10a^4b + 40a^3b^2 + 160a^2b^3 + 80ab^4 + 32b^5$

C $-(a + 2b)^5$

D $a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$

Classified Practice

Binomial and Multinomial Theorems

Q018 MCQ

Expand $(x + 2)^2$ using the Binomial Theorem:**Choose the correct option.**

A $x^2 + 4$

B $x^2 + 8x + 4$

C $-(x + 2)^2$

D $x^2 + 4x + 4$

Q019 MCQ

Expand $(a - b)^3$ using the Binomial Theorem:**Choose the correct option.**

A $a^3 - 3a^2b + 3ab^2 - b^3$

B $a^3 - 3a^2b - b^3$

C $a^3 - 3a^2b + 6ab^2 - b^3$

D $-(a - b)^3$

Classified Practice

Binomial and Multinomial Theorems

Q020 MCQ

Expand $(2x + 1)^5$ using the Binomial Theorem:**Choose the correct option.**

A $32x^5 + 80x^3 + 40x^2 + 10x + 1$

B $32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$

C $32x^5 + 160x^4 + 80x^3 + 40x^2 + 10x + 1$

D $-(2x + 1)^5$

Q021 Written**Expand $(3a - 2b)^5$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q022 MCQ

Expand $(2x - 1)^4$ using the Binomial Theorem:**Choose the correct option.**

A $16x^4 - 32x^3 - 8x + 1$

B $16x^4 - 32x^3 + 48x^2 - 8x + 1$

C $16x^4 - 32x^3 + 24x^2 - 8x + 1$

D $-(2x - 1)^4$

Q023 MCQ**Expand $(x - 3)^4$ using the Binomial Theorem:****Choose the correct option.**

A $x^4 - 12x^3 - 108x + 81$

B $x^4 - 12x^3 + 54x^2 - 108x + 81$

C $x^4 - 12x^3 + 108x^2 - 108x + 81$

D $-(x - 3)^4$

Classified Practice

Binomial and Multinomial Theorems

Q024 MCQ

Expand $(2a + b)^3$ using the Binomial Theorem:**Choose the correct option.**

A $8a^3 + 6ab^2 + b^3$

B $8a^3 + 24a^2b + 6ab^2 + b^3$

C $8a^3 + 12a^2b + 6ab^2 + b^3$

D $-(2a + b)^3$

Q025 MCQ

Expand $(2a^2 + 3)^4$ using the Binomial Theorem:**Choose the correct option.**

A $16a^8 + 96a^6 + 216a^2 + 81$

B $16a^8 + 96a^6 + 432a^4 + 216a^2 + 81$

C $16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$

D $-(2a^2 + 3)^4$

Classified Practice

Binomial and Multinomial Theorems

Q026 MCQ

Expand $(3a^2 - 2)^3$ using the Binomial Theorem:**Choose the correct option.**

A $27a^6 - 54a^4 - 8$

B $27a^6 - 54a^4 + 36a^2 - 8$

C $27a^6 - 54a^4 + 72a^2 - 8$

D $-(3a^2 - 2)^3$

Q027 Written**Find the coefficient of a^3b^2 in the expansion of $(a + b)^5$.**

Classified Practice

Binomial and Multinomial Theorems

Q028 Written

Find the coefficient of x^4y^2 in the expansion of $(x - 2y)^6$.

Q029 Written**Find the coefficient of x^4 in the expansion of $(2x^2 - 3)^7$.**

Classified Practice

Binomial and Multinomial Theorems

Q030 Written

Find the coefficient of x^6y in the expansion of $(2x - 3y)^7$.

Q031 MCQ**Find the coefficient of x^2 in the expansion of $(3x - 2)^5$:****Choose the correct option.**

- A 720
 - B -720
 - C -1440
 - D -719
-
-
-

Classified Practice

Binomial and Multinomial Theorems

Q032 Written

Find the coefficient of a^6 in the expansion of $(3a^2 - 2)^8$.

Q033 MCQ**Find the coefficient of x^4y^4 in the expansion of $(2x - y)^8$:****Choose the correct option.**

- A 1120
 - B -1120
 - C 2240
 - D 1121
-
-
-

Classified Practice

Binomial and Multinomial Theorems

Q034 MCQ

Find the coefficient of ab^3 in the expansion of $(3a + b)^4$:

Choose the correct option.

A -12

B 12

C 24

D 13

Q035 MCQ**Find the coefficient of x^3y^2 in the expansion of $(x - 4y)^5$:****Choose the correct option.**A -160 B 320 C 160 D 161

Classified Practice

Binomial and Multinomial Theorems

Q036 MCQ

Find the coefficient of x^4 in the expansion of $(3x + 1)^5$:

Choose the correct option.

A -405

B 810

C 406

D 405

Q037 Written**Find the coefficient of a in the expansion of $(2a - 3)^4$.**

Classified Practice

Binomial and Multinomial Theorems

Q038 Written

Find the coefficient of x^3 in the expansion of $(3x + 2)^5$.

Q039 Written**Find the coefficient of x^6 in the expansion of $(2x^2 - 1)^6$.**

Classified Practice

Binomial and Multinomial Theorems

Q040 MCQ

Find the coefficient of x^4 in the expansion of $(3x^2 - 1)^5$:

Choose the correct option.

- A 90
- B -180
- C -89
- D -90

Q041 Written**Find the coefficient of x^6 in the expansion of $(3x^2 + 2)^6$.**

Classified Practice

Binomial and Multinomial Theorems

Q042 Written

If $(x + 1)^5 = a_0 + a_x + \dots$, find the sum of its coefficients.

Q043 MCQ**Find the coefficient of xyz in the expansion of $(x + y + z)^3$ ∴****Choose the correct option.**

- A 6
- B -6
- C 3
- D 12

Classified Practice

Binomial and Multinomial Theorems

Q044 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a + b + c)^8$:

Choose the correct option.

A -280

B 560

C 280

D 281

Q045 MCQ**Find the coefficient of a^3bc^2 in the expansion of $(a + b + c)^6$:****Choose the correct option.**

A 60

B -60

C 120

D 61

Classified Practice

Binomial and Multinomial Theorems

Q046 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + b + c)^6$:

Choose the correct option.

A -90

B 90

C 180

D 91

Q047 MCQ**Find the coefficient of xy^5z in the expansion of $(x + y + z)^7$:****Choose the correct option.****A 42****B -42****C 84****D 43**

Classified Practice

Binomial and Multinomial Theorems

Q048 Written

Find the coefficient of x^4y^2z in the expansion of $(x + y + z)^7$.

Q049 Written**Find the coefficient of $x^2y^3z^2$ in the expansion of $(x + y + z)^7$.**

Classified Practice

Binomial and Multinomial Theorems

Q050 Written

Find the coefficient of x^6yz^2 in the expansion of $(x + y + z)^9$.

Q051 Written

Find the coefficient of $x^3y^3z^3$ in the expansion of $(x + y + z)^9$.

Classified Practice

Binomial and Multinomial Theorems

Q052 Written

Find the coefficient of $x^2y^4z^3$ in the expansion of $(x + y + z)^9$.

Q053 MCQ**Find the coefficient of a^4b^3c in the expansion of $(a - 2b + 3c)^8$:****Choose the correct option.**

- A 6720
 - B -13440
 - C -6719
 - D -6720
-
-
-

Classified Practice

Binomial and Multinomial Theorems

Q054 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + 3b - 2c)^6$:

Choose the correct option.

A -3240

B 6480

C 3240

D 3241

Q055 MCQ**Find the coefficient of ab^2c^3 in the expansion of $(a + 3b - 2c)^6$:****Choose the correct option.**

- A 4320
 - B -8640
 - C -4319
 - D -4320
-
-
-

Classified Practice

Binomial and Multinomial Theorems

Q056 Written

Find the coefficient of $a^2b^3c^2$ in the expansion of $(a + 2b - 3c)^7$.

Q057 Written**Find the coefficient of abc^5 in the expansion of $(a + 2b - 3c)^7$.**

Classified Practice

Binomial and Multinomial Theorems

Q058 Written

Find the coefficient of a^2b^4c in the expansion of $(a + 2b - 3c)^7$.

Q059 Written**Find the coefficient of $a^2b^2c^4$ in the expansion of $(2a - b + 3c)^8$.**

Classified Practice

Binomial and Multinomial Theorems

Q060 Written

Find the coefficient of a^4bc^3 in the expansion of $(2a - b + 3c)^8$.

Q061 Written

Find the coefficient of $a^3b^3c^2$ in the expansion of $(2a - b + 3c)^8$.

Classified Practice

Binomial and Multinomial Theorems

Q062 Written

Find the greatest coefficient in the expansion of $(x^2 + x + 1)^4$.

Quiz MCQ 1 Concept Quiz · MCQ

The coefficient of x^4 in $(2x - 3)^5$ is:

Choose the correct option.

- A -240
- B 240
- C -120
- D 120

Classified Practice

Binomial and Multinomial Theorems

Quiz MCQ 2 **Concept Quiz · MCQ**

The coefficient of $x^2y^2z^2$ in $(x + y + z)^6$ is:

Choose the correct option.

A 30

B 60

C 90

D 120

Quiz MCQ 3 Concept Quiz · MCQ

The coefficient of x^6 in $(3x^2 - 2)^6$ is:

Choose the correct option.

- A -4320
- B -2160
- C 2160
- D 4320

Classified Practice

Binomial and Multinomial Theorems

Quiz MCQ 4 **Concept Quiz · MCQ**

The coefficient of x^5 in $(2x^2 + 1)^4$ is:

Choose the correct option.

A 0

B 8

C 16

D 32

Quiz WRITTEN 5 Concept Quiz · Written**Expand $(2x - 1)^4$ fully.**

Classified Practice

Binomial and Multinomial Theorems

Quiz WRITTEN 6 Concept Quiz · Written

Find the coefficient of x^4 in $(3x^2 + 2)^5$. Show the exponent check.

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the coefficient of x^2y^3z in $(x - 2y + z)^6$.

Classified Practice

Binomial and Multinomial Theorems

Quiz WRITTEN 8 Concept Quiz · Written

Find the greatest coefficient in $(x^2 + x + 1)^4$. Show enough work to justify the maximum.

Answer key

Use the question number shown on each page.

Q001 A	Q018 D	Q036 D	Q054 C
Q002 A	Q019 A	Q037 -216	Q055 D
Q003 A	Q020 B	Q038 1080	Q056 15120
Q004 $x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$	Q021 $243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$	Q039 -160	Q057 -20412
Q005 D	Q022 C	Q040 D	Q058 -5040
Q006 $x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$	Q023 B	Q041 4320	Q059 136080
Q007 B	Q024 C	Q042 32	Q060 -120960
Q008 C	Q025 C	Q043 A	Q061 -40320
Q009 $a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$	Q026 B	Q044 C	Q062 19
Q010 A	Q027 10	Q045 A	Quiz MCQ 1 A
Q011 $243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$	Q028 60	Q046 B	Quiz MCQ 2 C
Q012 $32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$	Q029 -20412	Q047 A	Quiz MCQ 3 A
Q013 $x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$	Q030 -1344	Q048 105	Quiz MCQ 4 A
Q014 $64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$	Q031 B	Q049 210	Quiz WRITTEN 5 $16x^4 - 32x^3 + 24x^2 - 8x + 1$
Q015 $x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$	Q032 -48384	Q050 252	Quiz WRITTEN 6 720
Q016 B	Q033 A	Q051 1680	Quiz WRITTEN 7 -480
Q017 D	Q034 B	Q052 1260	Quiz WRITTEN 8 19
	Q035 C	Q053 D	

Concept 2 | Polynomial Division

English Student Edition • Focused formulas and guided practice

Formula opener

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder

$$R = P(a) \quad \text{for divisor } x - a$$

Degree check

$$\deg R < \deg B$$

Classified Practice

Polynomial Division

Q001 Challenge

Find the quotient and the remainder when $2x^3 + x^2 - 7x + 3$ is divided by $x^2 + x - 2$. Then verify the result using $A = B \times Q + R$:

Choose the correct option.

A $Q = 2x - 2, \quad R = -2x + 2$

B $Q = 2x - 1, \quad R = -2x + 1$

C $Q = 2x - 1, \quad R = 2x - 1$

D $Q = 2x, \quad R = -2x + 1$

Classified Practice

Polynomial Division

Q002 Challenge

Find the polynomial A when it is divided by $x^2 - 2x + 3$, the quotient is $3x - 1$, and the remainder is $x + 2$. Then divide your answer back to check it:

Choose the correct option.

A $3x^3 - 7x^2 + 12x - 1$

B $(3x - 1) \times (x^2 - 2x + 3)$

C $3x^3 - 7x^2 + 10x - 5$

D $x^2 + 2x + 4$

Classified Practice

Polynomial Division

Q003 Challenge

When $x^3 + 5x^2 + 7x - 7$ is divided by B, the quotient is $x + 3$ and the remainder is $2x - 4$. Find B. Then verify the degree condition:

Choose the correct option.

A $\frac{x^3+5x^2+9x-11}{x+3}$

B $\frac{-x^2-2x+1}{x^3+5x^2+5x-3}$

C $\frac{x^3+5x^2+5x-3}{x+4}$

D $x^2 + 2x - 1$

Classified Practice

Polynomial Division

Q004 Challenge

A cuboid has volume $2x^3 + 9x^2 + 4x - 15$ and base area $x^2 + 2x - 3$. Write its height in terms of x . State the mathematical check Which of the following is correct?:

Choose the correct option.

A $2x^3 + 10x^2 + 6x - 18$

B $2x + 6$

C $2x + 5$

D $\frac{1}{2x+5}$

Classified Practice

Polynomial Division

Q005 Challenge

Find the quotient and the remainder when $3x^4 - x^3 + 2x^2 - 4x + 1$ is divided by $x^2 - 2x + 2$. Decide whether the result is exact and justify from R:

Choose the correct option.

A $Q = 3x^2 + 5x + 5, \quad R = -2x - 10$

B $Q = 3x^2 + 5x + 6, \quad R = -2x - 11$

C $Q = 3x^2 + 5x + 6, \quad R = 2x + 11$

D $Q = 3x^2 + 5x + 7, \quad R = -2x - 11$

Classified Practice

Polynomial Division

Q006 Challenge

Find the quotient and the remainder when $x^4 - 3x^2 + 2x + 6$ is divided by $x + 2$. Explain why a zero placeholder is needed:

Choose the correct option.

A $Q = x \times (x^2 - 2x + 1), \quad R = 6$

B $Q = x \times (x^2 - 2x + 1), \quad R = -6$

C $Q = x^3 - 2x^2 + x + 1, \quad R = 6$

D $Q = x^3 - 2x^2 + x - 1, \quad R = 7$

Classified Practice

Polynomial Division

Q007 Written

Find the quotient and the remainder when $x^2 - 5x + 1$ is divided by $x - 2$.

Q008 Written**Find the quotient and the remainder when $x^3 - 4x + 1$ is divided by $x - 3$.**

Classified Practice

Polynomial Division

Q009 MCQ

Find the quotient and the remainder when $3x^3 + 2x^2 - 7x + 5$ is divided by $x - 2$:

Choose the correct option.

A $Q = 3x^2 + 8x + 9, \quad R = -23$

B $Q = 3x^2 + 8x + 10, \quad R = 23$

C $Q = 3x^2 + 8x + 8, \quad R = 24$

D $Q = 3x^2 + 8x + 9, \quad R = 23$

Classified Practice

Polynomial Division

Q010 MCQ

Find the quotient and the remainder when $x^3 + 6x^2 - 4$ is divided by $2x - 3$:

Choose the correct option.

A $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{37}{8}, R = \frac{111}{8}$

B $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, R = \frac{103}{8}$

C $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, R = -\frac{103}{8}$

D $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{53}{8}, R = \frac{103}{8}$

Classified Practice

Polynomial Division

Q011 MCQ

Find the quotient and the remainder when $3x^3 - 4x^2 + 2x + 7$ is divided by $x - 1$:

Choose the correct option.

A $Q = x \times (3x - 1), \quad R = 9$

B $Q = 3x^2 - x + 1, \quad R = 8$

C $Q = 3x^2 - x + 1, \quad R = -8$

D $Q = 3x^2 - x + 2, \quad R = 8$

Q012 MCQ

Find the quotient and the remainder when $2x^4 + x^3 - 5x + 4$ is divided by $x + 2$:

Choose the correct option.

A $Q = 2x^3 - 3x^2 + 6x - 17, \quad R = 38$

B $Q = 2x^3 - 3x^2 + 6x - 17, \quad R = -38$

C $Q = 2x^3 - 3x^2 + 6x - 16, \quad R = 38$

D $Q = 2x^3 - 3x^2 + 6x - 18, \quad R = 39$

Classified Practice

Polynomial Division

Q013 MCQ

Find the quotient and the remainder when $4x^4 - 3x^3 + x - 6$ is divided by $x - 2$:

Choose the correct option.

A $Q = 4x^3 + 5x^2 + 10x + 21, \quad R = -36$

B $Q = 4x^3 + 5x^2 + 10x + 22, \quad R = 36$

C $Q = 4x^3 + 5x^2 + 10x + 20, \quad R = 37$

D $Q = 4x^3 + 5x^2 + 10x + 21, \quad R = 36$

Classified Practice

Polynomial Division

Q014 MCQ

Find the quotient and the remainder when $2x^3 - 9x + 1$ is divided by $3x - 2$:

Choose the correct option.

A $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{100}{27}, \quad R = -\frac{92}{27}$

B $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = -\frac{119}{27}$

C $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = \frac{119}{27}$

D $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{46}{27}, \quad R = -\frac{119}{27}$

Classified Practice

Polynomial Division

Q015 Written

Find the quotient and the remainder when $2x^3 - 5x^2 - 2$ is divided by $2x^2 - x - 1$.

Q016 Written**Find the quotient and the remainder when $2x^3 - 13x + 1$ is divided by $x^2 + 2x - 3$.**

Classified Practice

Polynomial Division

Q017 Written

Find the quotient and the remainder when $2x^3 - 3x^2 + 3x + 4$ is divided by $x^2 - 2x + 3$.

Q018 Written**Find the quotient and the remainder when $x^3 - 7x + 6$ is divided by $x^2 + 2x - 3$.**

Classified Practice

Polynomial Division

Q019 MCQ

Find the quotient and the remainder when $x^3 + 2x^2 - 5x + 4$ is divided by $x^2 + x - 2$:

Choose the correct option.

A $Q = x + 1, \quad R = 4x - 6$

B $Q = x + 2, \quad R = -4x + 6$

C $Q = x, \quad R = -4x + 7$

D $Q = x + 1, \quad R = -4x + 6$

Q020 MCQ

Find the quotient and the remainder when $x^4 - 3x^3 + 2x - 5$ is divided by $x^2 + 2x - 1$:

Choose the correct option.

A $Q = x^2 - 5x + 10, \quad R = -25x + 7$

B $Q = x^2 - 5x + 11, \quad R = -25x + 6$

C $Q = x^2 - 5x + 11, \quad R = 25x - 6$

D $Q = x^2 - 5x + 12, \quad R = -25x + 6$

Classified Practice

Polynomial Division

Q021 MCQ

Find the quotient and the remainder when $2x^3 + x^2 - 4x + 5$ is divided by $x^2 - 2x + 2$:

Choose the correct option.

A $Q = 2x + 5, \quad R = -2x + 5$

B $Q = 2x + 6, \quad R = 2x - 5$

C $Q = 2x + 4, \quad R = 2x - 4$

D $Q = 2x + 5, \quad R = 2x - 5$

Q022 MCQ

Find the quotient and the remainder when $x^4 - 2x^3 + 5x^2 - x + 3$ is divided by $x^2 + 3x - 1$:

Choose the correct option.

A $Q = x^2 - 5x + 22, \quad R = -69x + 24$

B $Q = x^2 - 5x + 20, \quad R = -69x + 25$

C $Q = x^2 - 5x + 21, \quad R = -69x + 24$

D $Q = x^2 - 5x + 21, \quad R = 69x - 24$

Classified Practice

Polynomial Division

Q023 MCQ

Find the quotient and the remainder when $3x^3 - 5x + 2$ is divided by $x^2 - x - 1$:

Choose the correct option.

A $Q = 3x + 2, \quad R = x + 6$

B $Q = 3x + 3, \quad R = x + 5$

C $Q = 3x + 3, \quad R = -x - 5$

D $Q = 3x + 4, \quad R = x + 5$

Q024 MCQ

Find the quotient and the remainder when $2x^4 - x^3 + 4x^2 - 7$ is divided by $x^2 + x + 3$:

Choose the correct option.

A $Q = 2x^2 - 3x + 1, \quad R = 8x - 10$

B $Q = 2x^2 - 3x + 1, \quad R = -8x + 10$

C $Q = 2x^2 - 3x + 2, \quad R = 8x - 10$

D $Q = x \times (2x - 3), \quad R = 8x - 9$

Classified Practice

Polynomial Division

Q025 Written

Find the quotient and the remainder when $6x^3 - 8x^2 + 5$ is divided by $2x^2 - 2x - 5$.

Q026 Written**Find the quotient and the remainder when $3x^4 - 5x^3 + x - 1$ is divided by $-x^2 + 3x + 2$.**

Classified Practice

Polynomial Division

Q027 MCQ

Find the quotient and the remainder when $4x^3 - x^2 + 8x - 3$ is divided by $2x^2 + x - 4$:

Choose the correct option.

A $Q = 2x - \frac{3}{2}, R = -\frac{35x}{2} + 9$

B $Q = 2x - \frac{1}{2}, R = \frac{35x}{2} - 9$

C $Q = 2x - \frac{5}{2}, R = \frac{35x}{2} - 8$

D $Q = 2x - \frac{3}{2}, R = \frac{35x}{2} - 9$

Q028 MCQ

Find the quotient and the remainder when $x^3 - 8x + 2$ is divided by $x^2 - 4$:

Choose the correct option.

A $Q = x - 1, \quad R = -4x + 3$

B $Q = x, \quad R = -4x + 2$

C $Q = x, \quad R = 4x - 2$

D $Q = x + 1, \quad R = -4x + 2$

Classified Practice

Polynomial Division

Q029 MCQ

Find the quotient and the remainder when $5x^3 - 4x^2 + x - 6$ is divided by $2x^2 - x + 1$:

Choose the correct option.

- A $Q = \frac{5x}{2} - \frac{3}{4}, R = \frac{9x}{4} + \frac{21}{4}$
- B $Q = \frac{5x}{2} + \frac{1}{4}, R = -\frac{9x}{4} - \frac{21}{4}$
- C $Q = \frac{5x}{2} - \frac{7}{4}, R = -\frac{9x}{4} - \frac{17}{4}$
- D $Q = \frac{5x}{2} - \frac{3}{4}, R = -\frac{9x}{4} - \frac{21}{4}$

Classified Practice

Polynomial Division

Q030 MCQ

Find the quotient and the remainder when $3x^4 + 2x^3 - x^2 + 5$ is divided by $4x^2 + x - 2$:

Choose the correct option.

- A $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{67}{64}, R = \frac{37x}{64} + \frac{163}{32}$
- B $Q = \frac{3x^2}{4} + \frac{5x}{16} - \frac{61}{64}, R = \frac{37x}{64} + \frac{195}{32}$
- C $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, R = \frac{37x}{64} + \frac{163}{32}$
- D $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, R = -\frac{37x}{64} - \frac{163}{32}$

Classified Practice

Polynomial Division

Q031 MCQ

Find the quotient and the remainder when $3x^3 + 4x^2 - x + 9$ is divided by $2x^2 - x + 3$:

Choose the correct option.

- A $Q = \frac{3x}{2} + \frac{11}{4}, R = \frac{11x}{4} - \frac{3}{4}$
- B $Q = \frac{3x}{2} + \frac{15}{4}, R = -\frac{11x}{4} + \frac{3}{4}$
- C $Q = \frac{3x}{2} + \frac{7}{4}, R = -\frac{11x}{4} + \frac{7}{4}$
- D $Q = \frac{3x}{2} + \frac{11}{4}, R = -\frac{11x}{4} + \frac{3}{4}$

Classified Practice

Polynomial Division

Q032 MCQ

Find the quotient and the remainder when $6x^3 - 5x + 1$ is divided by $2x^2 + 3x - 2$:

Choose the correct option.

A $Q = 3x - \frac{11}{2}, R = \frac{29x}{2} - 7$

B $Q = 3x - \frac{9}{2}, R = \frac{29x}{2} - 8$

C $Q = 3x - \frac{9}{2}, R = -\frac{29x}{2} + 8$

D $Q = 3x - \frac{7}{2}, R = \frac{29x}{2} - 8$

Q033 Written

Find the quotient and the remainder when $-9a^2 - 9ax + 4x^2$ is divided by $-3a + x$. Treat it as a polynomial in x ; other letters are constants.

Classified Practice

Polynomial Division

Q034 Written

Find the quotient and the remainder when $-5a^3 + 5a^2x - 3ax^2 + x^3$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Q035 Written

Find the quotient and the remainder when $-6a^2 - 5ax + 4x^2$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Classified Practice

Polynomial Division

Q036 Written

Find the quotient and the remainder when $-a^3 + a^2x + ax^2 + x^3$ is divided by $2a + x$. Treat it as a polynomial in x ; other letters are constants.

Q037 Written

Find the quotient and the remainder when $3a^3 - 2ax^2 + x^3$ is divided by $a + x$. Treat it as a polynomial in x ; other letters are constants.

Classified Practice

Polynomial Division

Q038 MCQ

Find the quotient and the remainder when $-2a^3 + 7a^2x - 4ax^2 + x^3$ is divided by $-a + x$. Treat it as a polynomial in x ; other letters are constants:

Choose the correct option.

A $Q = 4a^2 - 3ax + x^2$, $R = -2a^3$

B $Q = 4a^2 - 3ax + x^2 + 1$, $R = 2a^3$

C $Q = 4a^2 - 3ax + x^2 - 1$, $R = 2a^3 + 1$

D $Q = 4a^2 - 3ax + x^2$, $R = 2a^3$

Q039 MCQ

Find the quotient and the remainder when $x^2 + 3xy - 4y^2$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants:

Choose the correct option.

A $Q = x + 2y - 1, \quad R = -6y^2 + 1$

B $Q = x + 2y, \quad R = -6y^2$

C $Q = x + 2y, \quad R = 6y^2$

D $Q = x + 2y + 1, \quad R = -6y^2$

Classified Practice

Polynomial Division

Q040 Written

Find the quotient and the remainder when $x^3 + 3x^2y - 2y^3$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants.

Q041 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $x + 3$, and the remainder is $-2x + 5$.

Classified Practice

Polynomial Division

Q042 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $2x - 3$, and the remainder is $-2x$.

Q043 Written

Find the polynomial A when it is divided by $x^2 - 3x + 2$, the quotient is $2x + 6$, and the remainder is $2x - 3$.

Classified Practice

Polynomial Division

Q044 Written

Find the polynomial A when it is divided by $2x^2 - x + 2$, the quotient is $2x - 1$, and the remainder is 2.

Q045 Written

Find the polynomial A when it is divided by $2x^2 + x + 1$, the quotient is $3x + 2$, and the remainder is $-2x + 6$.

Classified Practice

Polynomial Division

Q046 **Written**

When $4x^3 + 8x - 9$ is divided by B , the quotient is $2x - 3$ and the remainder is $x + 15$. Find B .

Q047 Written

When $x^3 - x^2 + 3x + 1$ is divided by B , the quotient is $x^2 - 2x + 5$ and the remainder is -4 . Find B .

Classified Practice

Polynomial Division

Q048 **Written**

When $2x^3 + x^2 - 3x + 5$ is divided by B, the quotient is $x + 1$ and the remainder is $-5x + 2$. Find B.

Q049 Written

When $6x^3 + 11x^2 - 2$ is divided by B , the quotient is $2x^2 + 3x - 1$ and the remainder is -1 . Find B .

Classified Practice

Polynomial Division

Q050 Written

When $2x^3 + 4x^2 + 7$ is divided by B , the quotient is $x + 2$ and the remainder is $3x + 13$. Find B .

Q051 Written

When $x^4 + x^2 + 3x - 1$ is divided by B , the quotient is $x^2 + x + 1$ and the remainder is $3x - 2$. Find B .

Classified Practice

Polynomial Division

Q052 Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Write its height in terms of x .

Quiz MCQ 1 Concept Quiz · MCQ

For $x^2 - 5x + 1$ divided by $x - 2$, what is R?:

Choose the correct option.

A -5

B -1

C 5

D 1

Classified Practice

Polynomial Division

Quiz MCQ 2 **Concept Quiz · MCQ**

In a polynomial in x , a is treated as... Which of the following is correct?:

Choose the correct option.

A *another variable to divide by*

B a constant coefficient

C *zero*

D the divisor

Quiz MCQ 3 Concept Quiz · MCQ

To recover **B** from $A = B \times Q + R$, use... Which of the following is correct?:

Choose the correct option.

A $B = A \times Q + R$

B $B = \frac{A+R}{Q}$

C $B = \frac{A-R}{Q}$

D $B = \frac{Q}{A-R}$

Classified Practice

Polynomial Division

Quiz MCQ 4 **Concept Quiz · MCQ**

If B has degree 2, a valid remainder can have degree...:

Choose the correct option.

A 3

B 2

C 1

D 4

Quiz WRITTEN 5 Concept Quiz · Written

Divide $3x^3 - 5x^2 + 2x - 7$ by $x - 2$. State Q , R , and verify $A = B \times Q + R$.

Classified Practice

Polynomial Division

Quiz WRITTEN 6 Concept Quiz · Written**Recover A when $B=x^2 - x - 2$, $Q=2x + 3$, $R=-x + 1$.**

Quiz WRITTEN 7 Concept Quiz · Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Find its height.

Classified Practice

Polynomial Division

Quiz WRITTEN 8 Concept Quiz · Written

Divide $x^4 - 2x^3 + 3x - 1$ by $x^2 - x + 1$ and explain why the remainder is valid.

Answer key

Use the question number shown on each page.

Q001 B	Q016 $Q = 2x - 4$, $R = x - 11$	Q031 D	Q046 $2x^2 + 3x + 8$
Q002 A	Q017 $Q = 2x + 1$, $R = -x + 1$	Q032 B	Q047 $x + 1$
Q003 D	Q018 $Q = x - 2$, $R = 0$	Q033 $Q = 3a + 4x$, $R = 0$	Q048 $2x^2 - x + 3$
Q004 C	Q019 D	Q034 $Q = 3a^2 - ax + x^2$, $R = a^3$	Q049 $3x + 1$
Q005 B	Q020 B	Q035 $Q = 3a + 4x$, $R = 0$	Q050 $2x^2 - 3$
Q006 A	Q021 D	Q036 $Q = 3a^2 - ax + x^2$, $R = -7a^3$	Q051 $x^2 - x + 1$
Q007 $Q = x - 3$, $R = -5$	Q022 C	Q037 $Q = 3a^2 - 3ax + x^2$, $R = 0$	Q052 $2x + 3$
Q008 $Q = x^2 + 3x + 5$, $R = 16$	Q023 B	Q038 D	Quiz MCQ 1 A
Q009 D	Q024 A	Q039 B	Quiz MCQ 2 B
Q010 B	Q025 $Q = 3x - 1$, $R = 13x$	Q040 $Q = x^2 + 2xy - 2y^2$, $R = 0$	Quiz MCQ 3 C
Q011 B	Q026 $Q = -3x^2 - 4x - 18$, $R = 63x + 35$	Q041 $x^3 + x^2 - 9x + 2$	Quiz MCQ 4 C
Q012 A	Q027 D	Q042 $2x^3 - 7x^2 + 2x + 3$	Quiz WRITTEN 5 $Q = 3x^2 + x + 4$, $R = 1$
Q013 D	Q028 B	Q043 $2x^3 - 12x + 9$	Quiz WRITTEN 6 $2x^3 + x^2 - 8x - 5$
Q014 B	Q029 D	Q044 $4x^3 - 4x^2 + 5x$	Quiz WRITTEN 7 $2x + 3$
Q015 $Q = x - 2$, $R = -x - 4$	Q030 C	Q045 $6x^3 + 7x^2 + 3x + 8$	Quiz WRITTEN 8 $Q = x^2 - x - 2$, $R = 2x + 1$

Concept 3 | Rational Expressions

English Student Edition • Focused formulas and guided practice

Formula opener

Restriction

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Classified Practice

Rational Expressions

Q001 Challenge

After simplifying this transfer chain before substitution, which result is correct for 2():

Choose the correct option.

A $\frac{2x(x+3)}{(x-3)(x-1)}$

B $\frac{-2x^2-6x}{x^2-4x+3}$

C $\frac{4x^2+12x}{x^2-4x+3}$

D $\frac{x^2+3x}{x^2-4x+3}$

Classified Practice

Rational Expressions

Q002 Challenge

After simplifying this transfer chain before substitution, which result is correct for 3():

Choose the correct option.

A $\frac{-3x^2+3x+18}{x^2+3x}$

B $\frac{3(x-3)(x+2)}{x(x+3)}$

C $\frac{6x^2-6x-36}{x^2+3x}$

D $\frac{3x^2-3x-18}{2x^2+6x}$

Classified Practice

Rational Expressions

Q003 Challenge

After simplifying this transfer chain before substitution, which result is correct for 4():

Choose the correct option.

A $\frac{-8x^2+24x+32}{x^2-5x+6}$

B $\frac{16x^2-48x-64}{x^2-5x+6}$

C $\frac{8(x-4)(x+1)}{(x-3)(x-2)}$

D $\frac{4x^2-12x-16}{x^2-5x+6}$

Classified Practice

Rational Expressions

Q004 Challenge

After simplifying this transfer chain before substitution, which result is correct for 8():

Choose the correct option.

A $-\frac{8}{x}$

B $\frac{16}{x}$

C $-\frac{x}{4}$

D $-\frac{8}{x}$

Classified Practice

Rational Expressions

Q005 Challenge

After simplifying this transfer chain before substitution, which result is correct for 9():

Choose the correct option.

A $\frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$

B $-\frac{9x^5}{2} + \frac{9}{2}$

C $9x^5 - 9$

D $\frac{9x^5}{4} - \frac{9}{4}$

Classified Practice

Rational Expressions

Q006 Challenge

After simplifying this transfer chain before substitution, which result is correct for 10 ()?:

Choose the correct option.

A $\frac{-10x^3+10x^2+160x-160}{x^3-5x^2+6x}$

B $\frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$

C $\frac{20x^3-20x^2-320x+320}{x^3-5x^2+6x}$

D $\frac{5x^3-5x^2-80x+80}{x^3-5x^2+6x}$

Classified Practice

Rational Expressions

Q007 Written

Simplify $\frac{12ab^5}{9a^3b^2}$.

Q008 Written**Simplify** $\frac{6x^2y^3}{20x^3y}$.

Classified Practice

Rational Expressions

Q009 Written

Simplify $\frac{6a^2bc^2}{8ab^4c^2}$.

Q010 Written**Simplify** $\frac{2a^2b}{8a^3b^3}$.

Classified Practice

Rational Expressions

Q011 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{12ab^5}{9a^3b^2}\right)$?:

Choose the correct option.

A $-\frac{4b^3}{a^2}$

B $\frac{4b^3}{a^2}$

C $\frac{8b^3}{a^2}$

D $\frac{2b^3}{a^2}$

Q012 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6x^2y^3}{20x^3y}\right)$?:

Choose the correct option.

- A $\frac{9y^2}{10x}$
B $-\frac{9y^2}{10x}$
C $\frac{9y^2}{5x}$
D $\frac{5x}{9y^2}$

Classified Practice

Rational Expressions

Q013 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{6a^2bc^2}{8ab^4c^2}\right)$?:

Choose the correct option.

- A $\frac{3a}{b^3}$
B $-\frac{3a}{b^3}$
C $\frac{6a}{b^3}$
D $\frac{3a}{2b^3}$

Q014 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{2a^2b}{8a^3b^3}\right)$?:

Choose the correct option.

A $-\frac{1}{2ab^2}$

B $\frac{1}{2ab^2}$

C $\frac{1}{ab^2}$

D $\frac{1}{4ab^2}$

Classified Practice

Rational Expressions

Q015 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{18a^3b^2}{24ab^5}\right)$?:

Choose the correct option.

- A $\frac{3a^2}{2b^3}$
B $-\frac{3a^2}{2b^3}$
C $\frac{3a^2}{b^3}$
D $\frac{3a^2}{4b^3}$

Q016 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{-15x^4y^3}{20x^2y}\right)$?:

Choose the correct option.

A $\frac{9x^2y^2}{4}$

B $-\frac{4}{9x^2y^2}$

C $-\frac{4}{9x^2y^2}$

D $-\frac{2}{8x^2y^2}$

Classified Practice

Rational Expressions

Q017 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{45a^2b^4c}{60a^5bc^2}\right)$?:

Choose the correct option.

A $-\frac{3b^3}{a^3c}$

B $\frac{6b^3}{a^3c}$

C $\frac{3b^3}{a^3c}$

D $\frac{3b^3}{2a^3c}$

Q018 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{14x^5y^2}{21x^2y^4}\right)$?:

Choose the correct option.

A $-\frac{4x^3}{3y^2}$

B $\frac{8x^3}{3y^2}$

C $\frac{2x^3}{3y^2}$

D $\frac{4x^3}{3y^2}$

Classified Practice

Rational Expressions

Q019 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{32a^4}{48ab^2}\right)$?:

Choose the correct option.

A $\frac{2a^3}{b^2}$

B $-\frac{2a^3}{b^2}$

C $\frac{4a^3}{b^2}$

D $\frac{a^3}{b^2}$

Q020 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{27x^3y^5}{36xy^2}\right)$?:

Choose the correct option.

A $-3x^2y^3$

B $3x^2y^3$

C $6x^2y^3$

D $\frac{3x^2y^3}{2}$

Classified Practice

Rational Expressions

Q021 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-42a^2c^3}{56a^5c}\right)$?:

Choose the correct option.

A $\frac{3c^2}{2a^3}$

B $-\frac{3c^2}{a^3}$

C $-\frac{3c^2}{2a^3}$

D $-\frac{3c^2}{4a^3}$

Q022 Written**Simplify** $\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}$.

Classified Practice

Rational Expressions

Q023 Written

Simplify $\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}$.

Q024 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}\right)$?:

Choose the correct option.

A $-\frac{12c^2}{a}$

B $\frac{24c^2}{a}$

C $\frac{12c^2}{a}$

D $\frac{6c^2}{a}$

Classified Practice

Rational Expressions

Q025 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}\right)$?:

Choose the correct option.

A $-\frac{8a^2c^3}{b^2}$

B $\frac{16a^2c^3}{b^2}$

C $\frac{4a^2c^3}{b^2}$

D $\frac{8a^2c^3}{b^2}$

Q026 MCQ

After reducing coefficients and powers, which result is correct for $3 \left(\frac{6a^2b}{5c} \times \frac{15c^2}{4ab^3} \right)$?:

Choose the correct option.

A $-\frac{27ac}{2b^2}$

B $\frac{27ac}{b^2}$

C $\frac{27ac}{4b^2}$

D $\frac{4b^2}{27ac}$

Classified Practice

Rational Expressions

Q027 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{9x^3y}{14a^2} \times \frac{21a^3}{6xy^2}\right)$?:

Choose the correct option.

A $\frac{9ax^2}{y}$

B $-\frac{9ax^2}{y}$

C $\frac{18ax^2}{y}$

D $\frac{9ax^2}{2y}$

Q028 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-8ab^2}{3c^2} \times \frac{12c^3}{-5a^2b}\right)$?:

Choose the correct option.

A $-\frac{64bc}{5a}$

B $\frac{64bc}{5a}$

C $\frac{5a}{128bc}$

D $\frac{5a}{32bc}$

Classified Practice

Rational Expressions

Q029 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{10x^2y^3}{9ab} \times \frac{27a^2b^2}{25xy}\right)$?:

Choose the correct option.

A $-\frac{18abxy^2}{5}$

B $\frac{36abxy^2}{5}$

C $\frac{18abxy^2}{5}$

D $\frac{9abxy^2}{5}$

Q030 Written**Simplify** $\frac{x^2-4}{3x^2+7x+2}$.

Classified Practice

Rational Expressions

Q031 Written

Simplify $\frac{2x^2-3x-5}{x^2+3x+2}$.

Q032 Written**Simplify** $\frac{x^2+3x}{x^2-9}$.

Classified Practice

Rational Expressions

Q033 Written

Simplify $\frac{x^2+x-2}{x^2-5x+4}$.

Q034 Written**Simplify** $\frac{2x^2-4xy+2y^2}{x^3+3x^2y-4xy^2}$.

Classified Practice

Rational Expressions

Q035 Written

Simplify $\frac{x^3-1}{x^2+2x-3}$.

Q036 Written

Simplify $\frac{x^2-11x+24}{x^2-6x-16} \times \frac{x^2+2x}{x^2-6x+9}$.

Classified Practice

Rational Expressions

Q037 Written

Simplify $\frac{x+1}{x^2-4} \times \frac{x^2-2x-8}{x^2+x}$.

Q038 Written

Simplify $\frac{x^2-9}{x^2-2x} \times \frac{x^2-7x+10}{x^2+x-12}$.

Classified Practice

Rational Expressions

Q039 Written

Simplify $\frac{x^2+x-6}{2x^2+x-6} \times \frac{x^2+6x+8}{3x^2-5x-2}$.

Q040 Written

Simplify $\frac{x^2-4x-5}{2x^2+5x+2} \div \frac{x^2-x-20}{2x^2-5x-3}$.

Classified Practice

Rational Expressions

Q041 Written

Simplify $\frac{x^2-9}{x^2-x-2} \div \frac{2x^2+5x-3}{x+1}$.

Q042 Written

Simplify $\frac{2xy}{3ab} \div \frac{8x^3y}{9a^2b^3}$.

Classified Practice

Rational Expressions

Q043 Written

Simplify $\frac{2x^2-x-10}{x^2-2x-3} \div \frac{x^2+x-2}{x^2-6x+9}$.

Q044 Written

Simplify $\frac{x^2-3x-10}{2x^2+3x-2} \div \frac{x^2-2x-15}{6x-3}$.

Classified Practice

Rational Expressions

Q045 Written

Simplify $\frac{x^2-y^2}{x^2-2xy+y^2} \div \frac{x^2+xy}{x-y}$.

Q046 Written

Simplify $\frac{x^3+8}{x^2-x-6} \times \frac{x^2-6x+9}{x^2-2x+4} \div \frac{2x-6}{x^5-1}$ **for** $x = \sqrt{123}$.

Classified Practice

Rational Expressions

Q047 **Challenge****Simplify**
$$\left(\frac{x+1}{x-2} - \frac{2}{x+3} + \frac{3}{x+3} - \frac{1}{x-2}\right); \text{ state every excluded value.}$$

Classified Practice

Rational Expressions

Q048 Written

Simplify

$$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}.$$

Classified Practice

Rational Expressions

Q049 MCQ

Simplify

$$\frac{x^2 + 2x + 3}{x + 1} + \frac{2x - 1}{x + 1}:$$

Choose the correct option.

A $\frac{x^2 + 5x + 3}{x + 1}$

B $\frac{x^2 + 3x + 1}{x + 1}$

C $\frac{x^2 + 4x + 2}{x + 1}$

D $-\frac{x^2 + 4x + 2}{x + 1}$

Classified Practice

Rational Expressions

Q050 MCQ

Simplify

$$\frac{2x^2 - 5}{x^2 - 4} + \frac{x + 6}{x^2 - 4}:$$

Choose the correct option.

A $\frac{3x^2 + x - 3}{(x - 2)(x + 2)}$

B $\frac{x^2 + x + 5}{(x - 2)(x + 2)}$

C $-\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$

D $\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$

Classified Practice

Rational Expressions

Q051 MCQ

Simplify

$$\frac{3x^2 + x - 2}{(x - 2)(x + 1)} + \frac{4x - 3}{(x - 2)(x + 1)}:$$

Choose the correct option.

A $\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$

B $\frac{4x^2 + 4x - 7}{(x - 2)(x + 1)}$

C $\frac{2x^2 + 6x - 3}{(x - 2)(x + 1)}$

D $-\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$

Classified Practice

Rational Expressions

Q052 MCQ

Simplify

$$\frac{x^3 - 1}{x + 2} + \frac{2x^2 + x + 4}{x + 2}:$$

Choose the correct option.

A $\frac{x^3 + 2x^2 + 2x + 5}{x + 2}$

B $\frac{x^3 + 2x^2 + x + 3}{x + 2}$

C $\frac{x^3 + 2x^2 + 1}{x + 2}$

D $-\frac{x^3 + 2x^2 + x + 3}{x + 2}$

Classified Practice

Rational Expressions

Q053 Written

Simplify

$$\frac{3}{x^2 + x} - \frac{-x + 3}{x^2 + x}.$$

Classified Practice

Rational Expressions

Q054 MCQ

Simplify

$$\frac{x^2 + 4x + 4}{x + 2} - \frac{x + 4}{x + 2}:$$

Choose the correct option.

A $\frac{x(x + 3)}{x + 2}$

B $\frac{x^2 + 4x + 2}{x + 2}$

C $\frac{x^2 + 2x - 2}{x + 2}$

D $-\frac{x(x + 3)}{x + 2}$

Classified Practice

Rational Expressions

Q055 MCQ

Simplify

$$\frac{3x^2 - 2x + 1}{x^2 - 1} - \frac{2x - 5}{x^2 - 1}:$$

Choose the correct option.

A $\frac{4x^2 - 4x + 5}{(x - 1)(x + 1)}$

B $\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$

C $\frac{2x^2 - 4x + 7}{(x - 1)(x + 1)}$

D $-\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$

Classified Practice

Rational Expressions

Q056 MCQ

Simplify

$$\frac{x^2 - 6x + 8}{(x - 1)(x + 2)} - \frac{x^2 - 3x + 2}{(x - 1)(x + 2)}:$$

Choose the correct option.

A $\frac{x^2 - 2x + 4}{(x - 1)(x + 2)}$

B $-\frac{x^2 + 4x - 8}{(x - 1)(x + 2)}$

C $-\frac{3(x - 2)}{(x - 1)(x + 2)}$

D $\frac{3(x - 2)}{(x - 1)(x + 2)}$

Classified Practice

Rational Expressions

Q057 MCQ

Simplify

$$\frac{2x^3 + x}{x - 3} - \frac{x^2 + 7}{x - 3}:$$

Choose the correct option.

A $\frac{2x^3 - x^2 + 2x - 10}{x - 3}$

B $\frac{2x^3 - x^2 - 4}{x - 3}$

C $\frac{2x^3 - x^2 + x - 7}{x - 3}$

D $-\frac{2x^3 - x^2 + x - 7}{x - 3}$

Classified Practice

Rational Expressions

Q058 Written

Simplify

$$\frac{x+4}{x^2-2x} - \frac{3}{x^2-3x+2}$$

Q059 **Written****Simplify**

$$\frac{x-1}{x^2+4x+3} + \frac{2}{x^2+5x+6}.$$

Classified Practice

Rational Expressions

Q060 MCQ

Simplify

$$\frac{x-1}{(x+1)(x+3)} + \frac{2x+1}{(x+3)(x+5)}:$$

Choose the correct option.

A $\frac{x^3 + 12x^2 + 30x + 11}{(x+1)(x+3)(x+5)}$

B $\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$

C $-\frac{x^3 + 6x^2 + 16x + 19}{(x+1)(x+3)(x+5)}$

D $-\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$

Classified Practice

Rational Expressions

Q061 MCQ

Simplify

$$\frac{x+3}{(x+2)(x+4)} + \frac{x-2}{(x+4)(x+6)}:$$

Choose the correct option.

- A $\frac{x^3 + 14x^2 + 53x + 62}{(x+2)(x+4)(x+6)}$
- B $-\frac{x^3 + 10x^2 + 35x + 34}{(x+2)(x+4)(x+6)}$
- C $\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$
- D $-\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$

Classified Practice

Rational Expressions

Q062 MCQ

Simplify

$$\frac{3x + 2}{(x + 1)(x + 4)} + \frac{x + 5}{(x + 4)(x + 7)}:$$

Choose the correct option.

A $\frac{x^3 + 16x^2 + 68x + 47}{(x + 1)(x + 4)(x + 7)}$

B $-\frac{x^3 + 8x^2 + 10x + 9}{(x + 1)(x + 4)(x + 7)}$

C $-\frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$

D $\frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$

Classified Practice

Rational Expressions

Q063 MCQ

Simplify

$$\frac{x+1}{(x+2)(x+5)} + \frac{4x-3}{(x+5)(x+8)}:$$

Choose the correct option.

- A $\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$
- B $\frac{x^3 + 20x^2 + 80x + 82}{(x+2)(x+5)(x+8)}$
- C $-\frac{x^3 + 10x^2 + 52x + 78}{(x+2)(x+5)(x+8)}$
- D $-\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$

Classified Practice

Rational Expressions

Q064 MCQ

Simplify

$$\frac{2x + 1}{(x + 3)(x + 6)} + \frac{x + 4}{(x + 6)(x + 9)}:$$

Choose the correct option.

A $\frac{x^3 + 21x^2 + 125x + 183}{(x + 3)(x + 6)(x + 9)}$

B $\frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$

C $-\frac{x^3 + 15x^2 + 73x + 141}{(x + 3)(x + 6)(x + 9)}$

D $-\frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$

Classified Practice

Rational Expressions

Q065 MCQ

Simplify

$$\frac{x-2}{(x+1)(x+5)} + \frac{3x+1}{(x+5)(x+8)}:$$

Choose the correct option.

- A $\frac{x^3 + 18x^2 + 63x + 25}{(x+1)(x+5)(x+8)}$
- B $-\frac{x^3 + 10x^2 + 43x + 55}{(x+1)(x+5)(x+8)}$
- C $\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$
- D $-\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$

Classified Practice

Rational Expressions

Q066 Written

Simplify

$$\frac{x}{x-1} - \frac{2}{x^2-1}.$$

Q067 Written**Simplify**

$$\frac{x+2}{x} - \frac{x-2}{x-1}.$$

Classified Practice

Rational Expressions

Q068 Written

Simplify

$$\frac{2}{x-1} + \frac{3}{x^2+5x-6}.$$

Classified Practice

Rational Expressions

Q069 MCQ

Simplify

$$\frac{x+2}{(x+1)(x+3)} - \frac{2}{x+1};$$

Choose the correct option.

A $\frac{x^2 + 3x - 1}{(x+1)(x+3)}$

B $-\frac{x^2 + 5x + 7}{(x+1)(x+3)}$

C $-\frac{x+4}{(x+1)(x+3)}$

D $\frac{x+4}{(x+1)(x+3)}$

Classified Practice

Rational Expressions

Q070 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2}.$$

Q071 Written

Simplify

$$\frac{x+4}{x^2+2x} - \frac{x+5}{x^2+x-2}$$

Classified Practice

Rational Expressions

Q072 Written

Simplify

$$\frac{x+8}{x^2+x-2} - \frac{x+5}{x^2-1}.$$

Q073 Written**Simplify**

$$\frac{x-3}{x^2+x} + \frac{x}{x^2-1}.$$

Classified Practice

Rational Expressions

Q074 Written

Simplify

$$\frac{4x}{x^2 - 1} - \frac{x - 1}{x^2 + x}.$$

Q075 Written

Simplify

$$\frac{1}{x^2 + 5x + 6} + \frac{1}{x^2 + 7x + 12}.$$

Classified Practice

Rational Expressions

Q076 Written

Simplify

$$\frac{x-1}{x^2+3x+2} - \frac{x-3}{x^2+4x+3}$$

Classified Practice

Rational Expressions

Q077 MCQ

A learner combines

$$\frac{2x-1}{(x-1)(x+2)} - \frac{x+4}{x+2}$$

Correct the numerator line, then simplify the expression:

Choose the correct option.

A
$$\frac{1}{(x-1)(x+2)}$$

B
$$-\frac{x^2+x-3}{(x-1)(x+2)}$$

C
$$-\frac{2x^2+2x-5}{(x-1)(x+2)}$$

D
$$\frac{x^2+x-3}{(x-1)(x+2)}$$

Classified Practice

Rational Expressions

Q078 Written

Simplify

$$\frac{x^3 + 8x}{x^2 - x - 2} - \frac{8}{x - 2}.$$

Classified Practice

Rational Expressions

Q079 **Challenge**

Simplify the single-inner fraction together with the reciprocal-pair quotient; state the full domain.

Classified Practice

Rational Expressions

Q080 Written

Simplify

$$\frac{1}{1 - \frac{1}{x-1}}.$$

Q081 Written**Simplify**

$$\frac{1}{1 - \frac{x}{x+1}}.$$

Classified Practice

Rational Expressions

Q082 MCQ

Simplify

$$\frac{1}{1 - \frac{x+2}{x+1}}:$$

Choose the correct option.

A $-x$ B $-x - 2$ C $x + 1$ D $-x - 1$

Q083 MCQ

Simplify

$$\frac{1}{1 - \frac{x+3}{x+2}}:$$

Choose the correct option.

A $-x - 2$

B $-x - 1$

C $-x - 3$

D $x + 2$

Classified Practice

Rational Expressions

Q084 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+1}{x+1}}:$$

Choose the correct option.

A $-\frac{1}{x}$

B $-\frac{x}{x+1}$

C $-\frac{x}{2x+1}$

D $\frac{x+1}{x}$

Q085 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+3}{x+1}}:$$

Choose the correct option.

- A $\frac{1}{x+2}$
B $-\frac{2x+3}{x+2}$
C $-\frac{x+1}{x+2}$
D $\frac{x+1}{x+2}$

Classified Practice

Rational Expressions

Q086 MCQ

Simplify

$$\frac{1}{1 - \frac{3x+1}{x+2}}:$$

Choose the correct option.

- A $\frac{x-3}{2x-1}$
- B $-\frac{3x+1}{2x-1}$
- C $\frac{x+2}{2x-1}$
- D $-\frac{x+2}{2x-1}$

Q087 MCQ

Simplify

$$\frac{1}{1 - \frac{x+4}{x+3}}:$$

Choose the correct option.

A $-x - 3$

B $-x - 2$

C $-x - 4$

D $x + 3$

Classified Practice

Rational Expressions

Q088 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+5}{x+1}}:$$

Choose the correct option.

- A $\frac{3}{x+4}$
- B $-\frac{x+1}{x+4}$
- C $-\frac{2x+5}{x+4}$
- D $\frac{x+1}{x+4}$

Q089 Written**Simplify**

$$\frac{1}{1-x} + \frac{1}{1+x} \div \left(\frac{1}{1-x} - \frac{1}{1+x} \right).$$

Classified Practice

Rational Expressions

Q090 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+4} \div \left(\frac{1}{x+1} - \frac{1}{x+4} \right):$$

Choose the correct option.

- A $\frac{2x+5}{3}$
B $\frac{2(x+4)}{3}$
C $\frac{2(x+1)}{3}$
D $-\frac{2x+5}{3}$

Q091 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+5} \div \left(\frac{1}{x+1} - \frac{1}{x+5} \right):$$

Choose the correct option.

A $\frac{x+3}{2}$

B $\frac{x+5}{2}$

C $\frac{x+1}{2}$

D $-\frac{x+3}{2}$

Classified Practice

Rational Expressions

Q092 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2} \div \left(\frac{1}{x+2} - \frac{1}{x-2} \right).$$

Q093 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+3} \div \left(\frac{1}{x+1} + \frac{1}{x+3} \right):$$

Choose the correct option.

A 2

B 0

C 1

D -1

Classified Practice

Rational Expressions

Q094 MCQ

Simplify

$$\frac{1}{x+2} + \frac{1}{x+5} \div \left(\frac{1}{x+2} + \frac{1}{x+5} \right):$$

Choose the correct option.

A 2

B 0

C -1

D 1

Q095 MCQ

Simplify

$$\frac{1}{x+3} + \frac{1}{x+6} \div \left(\frac{1}{x+3} + \frac{1}{x+6} \right):$$

Choose the correct option.

A 2

B 1

C 0

D -1

Classified Practice

Rational Expressions

Q096 Written

Simplify

$$\frac{1}{x+y} + \frac{1}{x-y} \div \left(\frac{1}{x+y} - \frac{1}{x-y} \right).$$

Q097 Written

Simplify

$$\frac{1}{1 - \frac{1}{1 - \frac{1}{x}}}.$$

Classified Practice

Rational Expressions

Q098 Written

Simplify

$$\frac{\frac{1}{1 - \frac{1}{1 + \frac{1}{x}}}}$$

Q099 Written**Simplify**

$$\left(1 + \frac{x-y}{x+y}\right) \div \left(1 - \frac{x-y}{x+y}\right).$$

Classified Practice

Rational Expressions

Q100 Written

Simplify

$$\left(x - \frac{2}{x-1}\right) \div \left(\frac{x}{x-1} - 2\right).$$

Q101 Written**Simplify**

$$\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right) \div \left(\frac{x+1}{x-1} + \frac{x-1}{x+1}\right).$$

Classified Practice

Rational Expressions

Q102 Written

Simplify

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + x}}}.$$

Quiz MCQ 1 Concept Quiz · MCQ

Simplify $\frac{x^2-4}{3x^2+7x+2}$:

Choose the correct option.

- A $\frac{x-2}{3x+1}$
B $\frac{3x+1}{x+2}$
C $\frac{3x+1}{x-2}$
D $\frac{3x+2}{x^2-4}$

Classified Practice

Rational Expressions

Quiz MCQ 2 Concept Quiz · MCQ

Simplify

$$\frac{x+1}{x-2} + \frac{2}{x-2}:$$

Choose the correct option.

A $\frac{x+3}{x-2}$

B $\frac{x-2}{x+1}$

C $\frac{2x-4}{x+3}$

D $\frac{x+3}{x+2}$

Quiz MCQ 3 Concept Quiz · MCQ

In $\frac{3}{x^2 + x} - \frac{3 - x}{x^2 + x}$, the numerator after the minus sign is... Which of the following is correct?:

Choose the correct option.

A $3 - (3 - x)$

B $3 - 3 - x$

C $3 + 3 - x$

D $3 - x$

Classified Practice

Rational Expressions

Quiz MCQ 4 **Concept Quiz · MCQ**

Which statement is safe when simplifying a compound fraction?:

Choose the correct option.

- A Multiply top and bottom by the same inner denominator
- B Add all denominators
- C Cancel terms across +
- D Drop all restrictions

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $\frac{x^2-9}{x^2-2x-3} \times \frac{x^2-4}{x^2+x-6}$ **and state the original restrictions.**

Classified Practice

Rational Expressions

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}$ and state the original restriction.

Quiz WRITTEN 7 Concept Quiz · Written**Simplify**

$$\frac{x^2 - y^2}{x^2 - 2xy + y^2} - \frac{x + y}{x - y} \text{ in } x \text{ and } y.$$

Classified Practice

Rational Expressions

Quiz WRITTEN 8 **Concept Quiz · Written**

Explain why $(x^2-16)/(x-4)=x+4$ is not an identity for every real x .

Answer key

Use the question number shown on each page.

Q001 A	Q024 C	Q043 $\frac{(x-3)(2x-5)}{(x-1)(x+1)}$	Q066 $\frac{x+2}{x+1}$
Q002 B	Q025 D	Q044 $\frac{3}{x+3}$	Q067 $\frac{3x-2}{x(x-1)}$
Q003 C	Q026 D	Q045 $\frac{1}{x}$	Q068 $\frac{2x+15}{(x-1)(x+6)}$
Q004 D	Q027 A	Q046 $-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$	Q069 C
Q005 A	Q028 B	Q047 $\frac{x^2+4x-2}{(x-2)(x+3)}$	Q070 $\frac{2x}{(x-2)(x+2)}$
Q006 B	Q029 C	Q048 $x-1$	Q071 $-\frac{2}{x(x-1)}$
Q007 $\frac{4b^3}{3a^2}$	Q030 $\frac{x-2}{3x+1}$	Q049 C	Q072 $\frac{2}{(x+1)(x+2)}$
Q008 $\frac{3y^2}{10x}$	Q031 $\frac{2x-5}{x+2}$	Q050 D	Q073 $\frac{2x^2-4x+3}{x(x-1)(x+1)}$
Q009 $\frac{3a}{4b^3}$	Q032 $\frac{x}{x-3}$	Q051 A	Q074 $\frac{3x-1}{x(x-1)}$
Q010 $\frac{1}{4ab^2}$	Q033 $\frac{x+2}{x-4}$	Q052 B	Q075 $\frac{2}{(x+2)(x+4)}$
Q011 B	Q034 $\frac{2(x-y)}{x(x+4y)}$	Q053 $\frac{1}{x+1}$	Q076 $\frac{3}{(x+2)(x+3)}$
Q012 A	Q035 $\frac{x^2+x+1}{x+3}$	Q054 A	Q077 B
Q013 A	Q036 $\frac{x}{x-3}$	Q055 B	Q078 $\frac{x^2+2x+4}{x+1}$
Q014 B	Q037 $\frac{x-4}{x(x-2)}$	Q056 C	Q079 $-\frac{x^2-4x+2}{2(x-2)}$
Q015 A	Q038 $\frac{(x-5)(x+3)}{x(x+4)}$	Q057 C	Q080 $\frac{x-1}{x-2}$
Q016 B	Q039 $\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$	Q058 $\frac{x+2}{x(x-1)}$	Q081 $x+1$
Q017 C	Q040 $\frac{(x-3)(x+1)}{(x+2)(x+4)}$	Q059 $\frac{x}{(x+1)(x+2)}$	Q082 D
Q018 D	Q041 $\frac{x-3}{(x-2)(2x-1)}$	Q060 B	Q083 A
Q019 A	Q042 $\frac{3ab^2}{4x^2}$	Q061 C	Q084 B
Q020 B		Q062 D	Q085 C
Q021 C		Q063 A	
Q022 $\frac{3c^2}{a}$		Q064 B	
Q023 $\frac{4a^2c^3}{b^2}$		Q065 C	

Q086 D

Q087 A

Q088 B

Q089 $1/x$

Q090 A

Q091 A

Q092 $-\frac{x}{2}$

Q093 C

Q094 D

Q095 B

Q096 $-\frac{x}{y}$ Q097 $1 - x$ Q098 $x + 1$ Q099 $\frac{x}{y}$ Q100 $-(x + 1)$ Q101 $\frac{2x}{x^2 + 1}$ Q102 $\frac{x + 2}{2x + 3}$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{x + 2}{x + 1}$ Quiz WRITTEN 6 $x - 1$

Quiz WRITTEN 7 0

Quiz WRITTEN 8 $x \neq 4$; removable hole at (4, 8)

Concept 4 | Identities and Proofs

English Student Edition • Focused formulas and guided practice

Formula opener

Proof route

LHS $\longrightarrow$ RHS

Square

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference

$$a^2 - b^2 = (a - b)(a + b)$$

Classified Practice

Identities and Proofs

Q001 **Challenge** $2x^2 + 3x + 1 = (x + 1)(ax + b) + c$. Find **a,b,c**.

Classified Practice

Identities and Proofs

Q002 Written

$$x^2 + 6 = a(x + 1)^2 + b(x + 1) + c$$

Q003 Written

$$x^2 - x - 3 = a(x - 1)^2 + b(x - 1) + c$$

Classified Practice

Identities and Proofs

Q004 MCQ

For the identity $x^2 + 6x + 5 = a(x + 2)^2 + b(x + 2) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A (2, 2, -3)

B (0, 2, -3)

C (1, 2, -3)

D (1, 3, -3)

Q005 MCQ

For the identity $2x^2 + 9x + 8 = a(x + 3)^2 + b(x + 3) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A (3, -3, -1)

B (1, -3, -1)

C (2, -3, -1)

D (2, -2, -1)

Classified Practice

Identities and Proofs

Q006 MCQ

For the identity $3x^2 + 28x + 65 = a(x + 4)^2 + b(x + 4) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A (4, 4, 1)

B (2, 4, 1)

C (3, 5, 1)

D (3, 4, 1)

Q007 MCQ

For the identity $x^2 + 5x + 3 = a(x + 5)^2 + b(x + 5) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A (1, -5, 3)

B (2, -5, 3)

C (0, -5, 3)

D (1, -4, 3)

Classified Practice

Identities and Proofs

Q008 Written

$$a(x + 1) - b(x + 5) = 8x$$

Q009 MCQ

For the identity $a(x + 1) - b(x + 3) = x - 1$, the ordered tuple (a, b) is:

Choose the correct option.

A (2, 1)

B (3, 1)

C (1, 1)

D (2, 2)

Classified Practice

Identities and Proofs

Q010 MCQ

For the identity $a(x + 2) - b(x + 4) = 5x + 14$, the ordered tuple (a, b) is:

Choose the correct option.

A $(4, -2)$

B $(3, -2)$

C $(2, -2)$

D $(3, -1)$

Q011 MCQ

For the identity $a(x + 3) - b(x + 5) = x - 3$, the ordered tuple (a, b) is:

Choose the correct option.

A (5, 3)

B (3, 3)

C (4, 4)

D (4, 3)

Classified Practice

Identities and Proofs

Q012 MCQ

For the identity $a(x + 4) - b(x + 6) = 9x + 44$, the ordered tuple (a, b) is:

Choose the correct option.

A (6, -4)

B (4, -4)

C (5, -3)

D (5, -4)

Q013 Written

$$(x + a)(b + 3) = 4x + 12$$

Classified Practice

Identities and Proofs

Q014 MCQ

For the identity $(x + a)(b + 4) = 3x + 6$, the ordered tuple (a, b) is:

Choose the correct option.

A $(3, -1)$

B $(2, -1)$

C $(1, -1)$

D $(2, 0)$

Q015 MCQ

For the identity $(x + a)(b + 5) = 5x + 15$, the ordered tuple (a, b) is:

Choose the correct option.

A (4, 0)

B (2, 0)

C (3, 0)

D (3, 1)

Classified Practice

Identities and Proofs

Q016 MCQ

For the identity $(x + a)(b + 6) = 7x + 28$, the ordered tuple (a, b) is:

Choose the correct option.

A (5, 1)

B (3, 1)

C (4, 1)

D (4, 2)

Q017 Written

$$2x^2 + 5x + 4 = (x + 1)(ax + b) + c$$

Classified Practice

Identities and Proofs

Q018 Written

$$(ax + b)(x + 1) = 4x^2 + cx + 1$$

Q019 Written

$$2x^2 - 7x + 8 = (x - 3)(ax + b) + c$$

Classified Practice

Identities and Proofs

Q020 MCQ

For the identity $(ax - 1)(x + 1) + c = x^2 - 3$, the ordered tuple (a, c) is:

Choose the correct option.

A $(2, -2)$

B $(1, -2)$

C $(0, -2)$

D $(1, -1)$

Q021 MCQ

For the identity $(ax + 2)(x + 2) + c = 2x^2 + 6x + 3$, the ordered tuple (a, c) is:

Choose the correct option.

A $(3, -1)$

B $(1, -1)$

C $(2, -1)$

D $(2, 0)$

Classified Practice

Identities and Proofs

Q022 MCQ

For the identity $(ax + 3)(x + 3) + c = 3x^2 + 12x + 9$, the ordered tuple (a, c) is:

Choose the correct option.

A (4, 0)

B (3, 0)

C (2, 0)

D (3, 1)

Q023 Written

$$\frac{3x+1}{(x+2)(x-3)} = \frac{a}{x+2} + \frac{b}{x-3}$$

Classified Practice

Identities and Proofs

Q024 Written

$$\frac{x+8}{(x-2)(x+3)} = \frac{a}{x-2} + \frac{b}{x+3}$$

Q025 **Written**

$$\frac{1}{(2x-1)(x+1)} = \frac{a}{2x-1} + \frac{b}{x+1}$$

Classified Practice

Identities and Proofs

Q026 Written

$$\frac{a}{x-1} + \frac{b}{x-5} = \frac{5x-1}{(x-1)(x-5)}$$

Q027 Written

$$\frac{3x - 8}{3x^2 + 5x - 2} = \frac{a}{x + 2} + \frac{b}{3x - 1}$$

Classified Practice

Identities and Proofs

Q028 Written

$$f(t) = \frac{22t + 15}{8t^2 + 10t + 3} = \frac{a}{2t + 1} + \frac{b}{4t + 3}, \quad t > 0$$

Q029 Written**Prove that** $(x + y)^2 - (x - y)^2 = 4xy$.

Classified Practice

Identities and Proofs

Q030 Written

Prove that $(2a + b)^2 - (a + 2b)^2 = 3(a^2 - b^2)$.

Q031 MCQ

For the identity $-(x - 2y + 1)^2 + (x + 2y + 1)^2 = 2y(4x + 4)$, the value of the simplified left-hand side is:

Choose the correct option.

A $8xy + 8y$

B $8xy + 8y + 1$

C $8xy + 8y - 1$

D $-8xy - 8y$

Classified Practice

Identities and Proofs

Q032 MCQ

For the identity $-(x - 3y + 2)^2 + (x + 3y + 2)^2 = 3y(4x + 8)$, the value of the simplified left-hand side is:

Choose the correct option.

- A $12xy + 24y + 1$
 - B $12xy + 24y - 1$
 - C $-12xy - 24y$
 - D $12xy + 24y$
-
-
-

Q033 MCQ

For the identity $-(x - y + 3)^2 + (x + y + 3)^2 = y(4x + 12)$, the value of the simplified left-hand side is:

Choose the correct option.

A $4xy + 12y + 1$

B $4xy + 12y - 1$

C $-4xy - 12y$

D $4xy + 12y$

Classified Practice

Identities and Proofs

Q034 MCQ

For the identity $-(x - 2y + 4)^2 + (x + 2y + 4)^2 = 2y(4x + 16)$, the value of the simplified left-hand side is:

Choose the correct option.

- A $8xy + 32y$
 - B $8xy + 32y + 1$
 - C $8xy + 32y - 1$
 - D $-8xy - 32y$
-
-
-

Q035 Written**Prove that** $(x + y)^2 + (x - y)^2 = 2x^2 + 2y^2$.

Classified Practice

Identities and Proofs

Q036 MCQ

For the identity $(x - 2y + 1)^2 + (x + 2y + 1)^2 = 8y^2 + 2(x + 1)^2$, the value of the simplified left-hand side is:

Choose the correct option.

- A $2x^2 + 4x + 8y^2 + 3$
- B $2x^2 + 4x + 8y^2 + 1$
- C $2x^2 + 4x + 8y^2 + 2$
- D $-2x^2 - 4x - 8y^2 - 2$

Q037 MCQ

For the identity $(x - 3y + 2)^2 + (x + 3y + 2)^2 = 18y^2 + 2(x + 2)^2$, the value of the simplified left-hand side is:

Choose the correct option.

A $2x^2 + 8x + 18y^2 + 9$

B $2x^2 + 8x + 18y^2 + 7$

C $2x^2 + 8x + 18y^2 + 8$

D $-2x^2 - 8x - 18y^2 - 8$

Classified Practice

Identities and Proofs

Q038 MCQ

For the identity $(x - y + 3)^2 + (x + y + 3)^2 = 2y^2 + 2(x + 3)^2$, the value of the simplified left-hand side is:

Choose the correct option.

- A $2x^2 + 12x + 2y^2 + 19$
- B $2x^2 + 12x + 2y^2 + 17$
- C $2x^2 + 12x + 2y^2 + 18$
- D $-2x^2 - 12x - 2y^2 - 18$

Q039 MCQ

For the identity $(x - 2y + 4)^2 + (x + 2y + 4)^2 = 8y^2 + 2(x + 4)^2$, the value of the simplified left-hand side is:

Choose the correct option.

A $2x^2 + 16x + 8y^2 + 33$

B $2x^2 + 16x + 8y^2 + 31$

C $-2x^2 - 16x - 8y^2 - 32$

D $2x^2 + 16x + 8y^2 + 32$

Classified Practice

Identities and Proofs

Q040 Written

Prove that $(a + 2b)^2 - 4ab = (a - 2b)^2 + 4ab$.

Q041 MCQ

For the identity $-(x - 2y + 3)^2 + (x + 2y - 1)^2 = (8x + 8)(y - 1)$, the value of the simplified left-hand side is:

Choose the correct option.

- A $8xy - 8x + 8y - 7$
- B $8xy - 8x + 8y - 9$
- C $-8xy + 8x - 8y + 8$
- D $8xy - 8x + 8y - 8$

Classified Practice

Identities and Proofs

Q042 MCQ

For the identity $-(2x - 3y + 5)^2 + (2x + 3y - 1)^2 = (24x + 24)(y - 1)$, the value of the simplified left-hand side is:

Choose the correct option.

- A $24xy - 24x + 24y - 23$
- B $24xy - 24x + 24y - 25$
- C $24xy - 24x + 24y - 24$
- D $-24xy + 24x - 24y + 24$

Q043 Written**Prove that** $(x^2 + x + 1)(x^2 - x + 1) = x^4 + x^2 + 1$.

Classified Practice

Identities and Proofs

Q044 Written

Prove that $(a^2 - ab + b^2)(a^2 + ab + b^2) = (a^2 + b^2)^2 - (ab)^2$.

Q045 Written**Prove that** $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$.

Classified Practice

Identities and Proofs

Q046 Written

Prove that $(x^2 + 1)(y^2 + 1) = (xy + 1)^2 + (x - y)^2$.

Q047 Written

Prove that $(a^2 - b^2)(c^2 - d^2) = (ac - bd)^2 - (ad - bc)^2$.

Classified Practice

Identities and Proofs

Q048 Written

Prove that $(a + b)^3 - 3ab(a + b) = a^3 + b^3$.

Q049 Written**Prove that** $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$.

Classified Practice

Identities and Proofs

Q050 Challenge

For the integrated identity $(x^2 + 1)(-(x - 2y + 1)^2 + (x + 2y + 1)^2) = 2y(4x + 4)(x^2 + 1)$, the value of the simplified left-hand side is:

Choose the correct option.

- A $8x^3y + 8x^2y + 8xy + 8y + 1$
- B $8x^3y + 8x^2y + 8xy + 8y$
- C $8x^3y + 8x^2y + 8xy + 8y - 1$
- D $-8x^3y - 8x^2y - 8xy - 8y$

Classified Practice

Identities and Proofs

Q051 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 + 1 = 2(x + y - xy)$.

Q052 Written**Under the condition $x + y = 1$, prove that $x^2 + y^2 = 1 - 2xy$.**

Classified Practice

Identities and Proofs

Q053 MCQ

Under the condition $x + y = 2$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2xy + 5$

B $-2xy + 3$

C $-2(xy - 2)$

D $2(xy - 2)$

Q054 MCQ

Under the condition $x + y = 3$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2(xy - 5)$

B $-2(xy - 4)$

C $2xy - 9$

D $-2xy + 9$

Classified Practice

Identities and Proofs

Q055 MCQ

Under the condition $x + y = -1$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2(xy - 1)$

B $-2xy$

C $-2xy + 1$

D $2xy - 1$

Q056 MCQ

Under the condition $x + y = 4$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2(xy - 8)$

B $-2xy + 17$

C $-2xy + 15$

D $2(xy - 8)$

Classified Practice

Identities and Proofs

Q057 Written

Under the condition $x - y = 1$, prove that $x^2 + y^2 = 2xy + 1$.

Q058 MCQ

Under the condition $x - y = 1$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2(xy + 1)$

B $2xy + 1$

C $2xy$

D $-2xy - 1$

Classified Practice

Identities and Proofs

Q059 MCQ

Under the condition $x - y = 2$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2xy + 5$

B $2(xy + 2)$

C $2xy + 3$

D $-2(xy + 2)$

Q060 MCQ**Under the condition $x - y = -2$, the expression $x^2 + y^2$ is equal to:****Choose the correct option.**

A $2xy + 5$

B $2xy + 3$

C $2(xy + 2)$

D $-2(xy + 2)$

Classified Practice

Identities and Proofs

Q061 MCQ

Under the condition $x - y = 3$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2(xy + 5)$

B $2(xy + 4)$

C $-2xy - 9$

D $2xy + 9$

Q062 Written

Under the condition $x + y = 1$, prove that $x^2 - x = y^2 - y$.

Classified Practice

Identities and Proofs

Q063 Written

Under the condition $a + b + c = 0$, prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

Q064 Written

Under the condition $a + b + c = 0$, prove that $a^3 + b^3 + c^3 = 3abc$.

Classified Practice

Identities and Proofs

Q065 Written

Under the condition $a + b + c = 0$, prove that $a^2 + ca = b^2 + bc$.

Q066 Written

Under the condition $a + b + c = 0$, prove that $(a + b)(b + c)(c + a) + abc = 0$.

Classified Practice

Identities and Proofs

Q067 Written

Under the condition $a + b + c = 0$, $abc \neq 0$, prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$.

Classified Practice

Identities and Proofs

Q068 Challenge

Under the condition $x + y = 3$, the expression $x^2 + y^2 + 1$ is equal to:

Choose the correct option.

A $-2xy + 11$

B $-2xy + 9$

C $-2(xy - 5)$

D $2(xy - 5)$

Classified Practice

Identities and Proofs

Q069 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}.$

Q070 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

Classified Practice

Identities and Proofs

Q071 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{c^2 + cd + d^2}{c^2 + d^2}$
- B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$
- C $-\frac{cd}{c^2 + d^2}$
- D $\frac{cd}{c^2 + d^2}$

Q072 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

Classified Practice

Identities and Proofs

Q073 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{c^2 + cd + d^2}{c^2 + d^2}$
- B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$
- C $-\frac{cd}{c^2 + d^2}$
- D $\frac{cd}{c^2 + d^2}$

Classified Practice

Identities and Proofs

Q074 **Challenge**

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{c^2 + cd + d^2}{c^2 + d^2}$
- B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$
- C $-\frac{cd}{c^2 + d^2}$
- D $\frac{cd}{c^2 + d^2}$

Classified Practice

Identities and Proofs

Q075 **Challenge****Under the condition $a/b = c/d$, the required expression is equal to:****Choose the correct option.**

- A $\frac{c^2 + cd + d^2}{c^2 + d^2}$
- B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$
- C $\frac{cd}{c^2 + d^2}$
- D $-\frac{cd}{c^2 + d^2}$

Classified Practice

Identities and Proofs

Q076 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$.

Classified Practice

Identities and Proofs

Q077 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{a^2 + c^2}{(a - c)(a + c)}$

B $\frac{2a^2}{(a - c)(a + c)}$

C $\frac{2c^2}{(a - c)(a + c)}$

D $-\frac{a^2 + c^2}{(a - c)(a + c)}$

Classified Practice

Identities and Proofs

Q078 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{2a^2}{(a-c)(a+c)}$

B $\frac{2c^2}{(a-c)(a+c)}$

C $-\frac{a^2 + c^2}{(a-c)(a+c)}$

D $\frac{a^2 + c^2}{(a-c)(a+c)}$

Classified Practice

Identities and Proofs

Q079 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{2a^2}{(a-c)(a+c)}$

B $\frac{2c^2}{(a-c)(a+c)}$

C $-\frac{a^2 + c^2}{(a-c)(a+c)}$

D $\frac{a^2 + c^2}{(a-c)(a+c)}$

Classified Practice

Identities and Proofs

Q080 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{a^2 + c^2}{(a - c)(a + c)}$

B $\frac{2a^2}{(a - c)(a + c)}$

C $\frac{2c^2}{(a - c)(a + c)}$

D $-\frac{a^2 + c^2}{(a - c)(a + c)}$

Classified Practice

Identities and Proofs

Q081 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Q082 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{2c}{c-d}$
B $\frac{c+d}{c-d}$
C $\frac{2d}{c-d}$
D $-\frac{c+d}{c-d}$

Classified Practice

Identities and Proofs

Q083 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{2c}{c-d}$
B $\frac{c+d}{c-d}$
C $\frac{2d}{c-d}$
D $-\frac{c+d}{c-d}$

Q084 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c+d}{c-d}$

B $\frac{2c}{c-d}$

C $\frac{2d}{c-d}$

D $-\frac{c+d}{c-d}$

Classified Practice

Identities and Proofs

Q085 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c+d}{c-d}$

B $\frac{2c}{c-d}$

C $\frac{2d}{c-d}$

D $-\frac{c+d}{c-d}$

Q086 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$.

Classified Practice

Identities and Proofs

Q087 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{2a - c}{2b - d}$
- B $\frac{2a + 2b - c - d}{2b - d}$
- C $\frac{2a - 2b - c + d}{2b - d}$
- D $-\frac{2a - c}{2b - d}$

Q088 Written

Under the condition $x/a = y/b$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b}, \quad \frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$.

Classified Practice

Identities and Proofs

Q089 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Q090 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$.

Classified Practice

Identities and Proofs

Q091 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x+y+z}{a+b+c} = \frac{x}{a}$.

Q092 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$.

Classified Practice

Identities and Proofs

Q093 Written

Under the condition $x : y : z = 2 : 3 : 5$, prove or find the required expression: $x : y : z = 2 : 3 : 5$, $\frac{2xy+yz+3zx}{x^2+y^2+z^2}$.

Q094 Written

Under the condition $a : b : c = 2 : 3 : 4$, prove or find the required expression: $a : b : c = 2 : 3 : 4$, $\frac{ab+bc+ca}{a^2+b^2+c^2}$.

Classified Practice

Identities and Proofs

Q095 Written

Under the condition $a : b : c = 3 : 5 : 2$, prove or find the required expression: $a : b : c = 3 : 5 : 2$, $\frac{a^2+b^2-c^2}{a^2-b^2+c^2}$.

Q096 Written

Under the condition $x : y : z = 2 : -3 : 4$, prove or find the required expression: $x : y : z = 2 : -3 : 4$, $\frac{x^2+y^2+z^2}{xy+yz+zx}$.

Classified Practice

Identities and Proofs

Q097 Written

Under the condition $x/7 = y/3 \neq 0$, prove or find the required expression: $\frac{x}{7} = \frac{y}{3} \neq 0, \frac{x^2+5y^2}{x^2+xy+y^2}$.

Q098 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Classified Practice

Identities and Proofs

Quiz MCQ 1 **Concept Quiz · MCQ**

If $x^2 + 4x + 1 = (x + 1)(ax + b) + c$, then (a, b, c) is... Which of the following is correct?:

Choose the correct option.

A $(1, 3, -2)$

B $(1, 2, -1)$

C $(2, 3, -2)$

D $(1, -3, 2)$

Quiz MCQ 2 Concept Quiz · MCQ

For the identity $(x^2 - 2x + 2)(x^2 + 2x + 2) = -4x^2 + (x^2 + 2)^2$, the value of the simplified left-hand side is:

Choose the correct option.

- A $x^4 + 4$
- B $x^4 + 5$
- C $x^4 + 3$
- D $-x^4 - 4$

Classified Practice

Identities and Proofs

Quiz MCQ 3 **Concept Quiz · MCQ**

Under the condition $a + b + c = 0$, the expression $3a^2 + 3ac$ is equal to:

Choose the correct option.

A $3b(b + c)$

B $3b^2 + 3bc + 1$

C $3b^2 + 3bc - 1$

D $-3b(b + c)$

Quiz MCQ 4 Concept Quiz · MCQ

Under the condition $x : y : z = 2 : 3 : 5$, the required expression is equal to:

Choose the correct option.

- A $\frac{5}{2}$
- B $\frac{3}{2}$
- C $\frac{1}{2}$
- D $-\frac{3}{2}$

Classified Practice

Identities and Proofs

Quiz WRITTEN 5 Concept Quiz · Written**Find a, b, c if $x^2 - 3x + 5 = (x - 2)(ax + b) + c$.**

Quiz WRITTEN 6 Concept Quiz · Written

Prove that $(x^2 - 3x + 4)(x^2 + 3x + 4) = -9x^2 + (x^2 + 4)^2$.

Classified Practice

Identities and Proofs

Quiz WRITTEN 7 Concept Quiz · Written

Under the condition $a + b + c = 0$, prove that $4a^2 + 4ac = 4b^2 + 4bc$.

Quiz WRITTEN 8 Concept Quiz · Written

Under the condition $x/a = y/b = z/c$, prove that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Answer key

Use the question number shown on each page.

Q001 $a = 2, b = 1, c = 0$

Q002 $a = 1, b = -2, c = 7$

Q003 $a = 1, b = 1, c = -3$

Q004 C

Q005 C

Q006 D

Q007 A

Q008 $a = 10, b = 2$

Q009 A

Q010 B

Q011 D

Q012 D

Q013 $a = 3, b = 1$

Q014 B

Q015 C

Q016 C

Q017 $a = 2, b = 3, c = 1$

Q018 $a = 4, b = 1, c = 5$

Q019 $a = 2, b = -1, c = 5$

Q020 B

Q021 C

Q022 B

Q023 $a = 1, b = 2$

Q024 $a = 2, b = -1$

Q025 $a = \frac{2}{3}, b = -\frac{1}{3}$

Q026 $a = -1, b = 6$

Q027 $a = 2, b = -3$

Q028 $a = 4, b = 3$

Q029 $4xy$

Q030 $3a^2 - 3b^2$

Q031 A

Q032 D

Q033 D

Q034 A

Q035 $2x^2 + 2y^2$

Q036 C

Q037 C

Q038 C

Q039 D

Q040 $a^2 + 4b^2$

Q041 D

Q042 C

Q043 $x^4 + x^2 + 1$

Q044 $a^4 + a^2b^2 + b^4$

Q045 $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Q046 $x^2y^2 + x^2 + y^2 + 1$

Q047 $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$

Q048 $a^3 + b^3$

Q049 $a^3 - 3abc + b^3 + c^3$

Q050 B

Q051 $x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 $x^2 + y^2 = 1 - 2xy$

Q053 C

Q054 D

Q055 C

Q056 A

Q057 $x^2 + y^2 = 2xy + 1$

Q058 B

Q059 B

Q060 C

Q061 D

Q062 $x^2 - x = y^2 - y$

Q063 $a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 $a^3 + b^3 + c^3 = 3abc$

Q065 $a^2 + ca = b^2 + bc$

Q066 $(a + b)(b + c)(c + a) + abc = 0$

Q067 $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$

Q068 C

Q069 $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$

Q070 C

Q071 D

Q072 C

Q073 D

Q074 D

Q075 C

Q076 $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$

Q077 A

Q078 D

Q079 D

Q080 A

Q081 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q082 B

Q083 B

Q084 A

Q085 A

Q086 $\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$

Q087 A

Q088 $\frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$

Q089 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q090 $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$

Q091 $\frac{x+y+z}{a+b+c} = \frac{x}{a}$

Q092 $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$

Q093 $\frac{57}{38}$

Q094 $\frac{26}{29}$

Q095 $-\frac{5}{2}$

Q096 $-\frac{29}{10}$

Q097 $\frac{94}{79}$

Q098 $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Quiz MCQ 1 A**Quiz MCQ 2** A**Quiz MCQ 3** A**Quiz MCQ 4** B**Quiz WRITTEN 5** $a = 1, b = -1, c = 3$ **Quiz WRITTEN 6** $x^4 - x^2 + 16$ **Quiz WRITTEN 7** $4b^2 + 4bc$ **Quiz WRITTEN 8** $(ax + by + cz)^2$

Concept 5 | Inequalities

English Student Edition • Focused formulas and guided practice

Formula opener

Square rule

$$(x - y)^2 \geq 0$$

Order

$$0 < a < b \Rightarrow a^2 < b^2$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Classified Practice

Inequalities

Q001 **Written**

Under the condition $a > b$, $x > y$, prove that $2(ax + by) > (a + b)(x + y)$.

Q002 MCQ**If $x > y$, prove that $3x + y > 2(x + y)$.****If $x > y$, the positive difference between the two sides is:****Choose the correct option.**

A $x - y$

B $x - y + 1$

C $x - y - 1$

D $-x + y$

Classified Practice

Inequalities

Q003 MCQ

If $x > y$, prove that $3x + 2y > 2x + 3y$.

If $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $x - y + 1$

B $x - y - 1$

C $x - y$

D $-x + y$

Q004 MCQ**If $x > y$, prove that $3(x + y) > 2(x + 2y)$.****If $x > y$, the positive difference between the two sides is:****Choose the correct option.**

A $x - y + 1$

B $x - y - 1$

C $-x + y$

D $x - y$

Classified Practice

Inequalities

Q005 MCQ

If $x > y$, prove that $3x + 4y > 2x + 5y$.

If $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $x - y + 1$

B $x - y - 1$

C $-x + y$

D $x - y$

Q006 MCQ**If $x > y$, prove that $3x + 5y > 2(x + 3y)$.****If $x > y$, the positive difference between the two sides is:****Choose the correct option.**

A $x - y + 1$

B $x - y$

C $x - y - 1$

D $-x + y$

Classified Practice

Inequalities

Q007 Written

Under the condition $a > b$, $c > d$, prove that $ac + bd > ad + bc$.

Q008 Written

Under the condition $a > c$, $b > d$, prove that $ab + cd > ad + bc$.

Classified Practice

Inequalities

Q009 MCQ

If $a > b$ and $c > d$, prove that $ac + bd > ad + bc$.

If $a > b$ and $c > d$, the positive difference between the two sides is:

Choose the correct option.

A $ac - ad - bc + bd + 1$

B $ac - ad - bc + bd - 1$

C $-(a - b)(c - d)$

D $(a - b)(c - d)$

Q010 MCQ

If $a > 2b$ and $c > d$, prove that $ac + 2bd > ad + 2bc$.

If $a > 2b$ and $c > d$, the positive difference between the two sides is:

Choose the correct option.

A $ac - ad - 2bc + 2bd + 1$

B $ac - ad - 2bc + 2bd - 1$

C $(a - 2b)(c - d)$

D $-(a - 2b)(c - d)$

Classified Practice

Inequalities

Q011 MCQ

If $a > 3b$ and $c > d$, prove that $ac + 3bd > ad + 3bc$.

If $a > 3b$ and $c > d$, the positive difference between the two sides is:

Choose the correct option.

A $(a - 3b)(c - d)$

B $ac - ad - 3bc + 3bd + 1$

C $ac - ad - 3bc + 3bd - 1$

D $-(a - 3b)(c - d)$

Q012 Written

Under the condition $x > y$, prove that $3x - 4y > 2x - 3y$.

Classified Practice

Inequalities

Q013 Written

Under the condition $x > 3$, $y > -5$, prove that $xy + 5x > 3y + 15$.

Q014 Written

Under the condition $x > 2$, $y > 1$, prove that $xy + 2 > x + 2y$.

Classified Practice

Inequalities

Q015 Written

Under the condition $a > 1$, $b > 1$, prove that $ab + 1 > a + b$.

Q016 Written

Under the condition $a > b > x > y$, prove that $ax + by > ay + bx$.

Classified Practice

Inequalities

Q017 Written

Under the condition $a > b > c > 0$, $x > y > z > 0$, prove that $ax + by + cz > ay + bz + cx$.

Classified Practice

Inequalities

Q018 Challenge

If $a > b$ and $x > y$, prove that $2(ax + by) > 2(ay + bx)$.

If $a > b$ and $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $2ax - 2ay - 2bx + 2by + 1$

B $2ax - 2ay - 2bx + 2by - 1$

C $-2(a - b)(x - y)$

D $2(a - b)(x - y)$

Classified Practice

Inequalities

Q019 Written

Prove that $a^2 + 4b^2 \geq 4ab$.

Q020 MCQ**Under a,b real, prove that $a^2 + 4b^2 \geq 4ab$ select the correct factorisation of LHS – RHS:****Choose the correct option.**

- A $a^2 - 4ab + 4b^2 + 1$
 - B $(a - 2b - 1)(a - 2b + 1)$
 - C $-(a - 2b)^2$
 - D $(a - 2b)^2$
-
-
-

Classified Practice

Inequalities

Q021 MCQ

Under a, b real, prove that $a^2 + 9b^2 \geq 6ab$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $a^2 - 6ab + 9b^2 + 1$
B $(a - 3b - 1)(a - 3b + 1)$
C $(a - 3b)^2$
D $-(a - 3b)^2$
-
-
-

Q022 MCQ**Under a,b real, prove that $a^2 + 16b^2 \geq 8ab$ select the correct factorisation of LHS - RHS:****Choose the correct option.**

A $a^2 - 8ab + 16b^2 + 1$

B $(a - 4b)^2$

C $(a - 4b - 1)(a - 4b + 1)$

D $-(a - 4b)^2$

Classified Practice

Inequalities

Q023 MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $(a - 5b)^2$

B $a^2 - 10ab + 25b^2 + 1$

C $(a - 5b - 1)(a - 5b + 1)$

D $-(a - 5b)^2$

Q024 MCQ

Under a,b real, prove that $a^2 + 36b^2 \geq 12ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $a^2 - 12ab + 36b^2 + 1$

B $(a - 6b - 1)(a - 6b + 1)$

C $(a - 6b)^2$

D $-(a - 6b)^2$

Classified Practice

Inequalities

Q025 MCQ

Under a, b real, prove that $a^2 + 49b^2 \geq 14ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $a^2 - 14ab + 49b^2 + 1$

B $(a - 7b)^2$

C $(a - 7b - 1)(a - 7b + 1)$

D $-(a - 7b)^2$

Q026 Written**Prove that** $a^2 + 5b^2 \geq 4ab$.

Classified Practice

Inequalities

Q027 Written

Prove that $9x^2 \geq y(6x - y)$.

Q028 Written**Prove that** $2(x^2 + y^2) \geq (x + y)^2$.

Classified Practice

Inequalities

Q029 Written

Prove that $(2x - y)^2 \geq 3(x^2 - y^2)$.

Q030 Written**Prove that** $(2a + 3b)^2 \geq 12ab$.

Classified Practice

Inequalities

Q031 MCQ

Under x, y real, prove that $4x^2 \geq y(4x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $(2x - y)^2$

B $4x^2 - 4xy + y^2 + 1$

C $(2x - y - 1)(2x - y + 1)$

D $-(2x - y)^2$

Q032 MCQ

Under x, y real, prove that $9x^2 \geq y(6x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $9x^2 - 6xy + y^2 + 1$
- B $(3x - y - 1)(3x - y + 1)$
- C $(3x - y)^2$
- D $-(3x - y)^2$

Classified Practice

Inequalities

Q033 MCQ

Under x, y real, prove that $16x^2 \geq y(8x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $16x^2 - 8xy + y^2 + 1$

B $(4x - y - 1)(4x - y + 1)$

C $-(4x - y)^2$

D $(4x - y)^2$

Q034 MCQ

Under x, y real, prove that $25x^2 \geq y(10x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $25x^2 - 10xy + y^2 + 1$

B $(5x - y - 1)(5x - y + 1)$

C $-(5x - y)^2$

D $(5x - y)^2$

Classified Practice

Inequalities

Q035 MCQ

Under x, y real, prove that $36x^2 \geq y(12x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $(6x - y)^2$

B $36x^2 - 12xy + y^2 + 1$

C $(6x - y - 1)(6x - y + 1)$

D $-(6x - y)^2$

Q036 MCQ**Under x, y real, prove that $x^2 + y^2 \geq 2xy$ select the correct factorisation of LHS - RHS:****Choose the correct option.**

A $x^2 - 2xy + y^2 + 1$

B $(x - y - 1)(x - y + 1)$

C $(x - y)^2$

D $-(x - y)^2$

Classified Practice

Inequalities

Q037 Written

Prove that $x^2 + y^2 \geq 2(x - y - 1)$.

Q038 Written**Prove that** $x^2 + 2 > 2x$.

Classified Practice

Inequalities

Q039 Written

Prove that $x^2 + y^2 - 6x + 8y + 25 \geq 0$.

Q040 Written**Prove that** $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Classified Practice

Inequalities

Q041 **Written****Prove that** $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$.

Classified Practice

Inequalities

Q042 **Challenge**

For real x, y , prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Classified Practice

Inequalities

Q043 Written

Prove that $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

Classified Practice

Inequalities

Q044 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a-b)^2}{4}$

B $\frac{a^2 - 2ab + b^2 + 4}{4}$

C $\frac{(a-b-2)(a-b+2)}{4}$

D $-\frac{(a-b)^2}{4}$

Classified Practice

Inequalities

Q045 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 2b)^2}{4}$

B $\frac{a^2 - 4ab + 4b^2 + 4}{4}$

C $\frac{(a - 2b - 2)(a - 2b + 2)}{4}$

D $-\frac{(a - 2b)^2}{4}$

Classified Practice

Inequalities

Q046 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{a^2 - 6ab + 9b^2 + 4}{4}$

B $\frac{(a - 3b - 2)(a - 3b + 2)}{4}$

C $\frac{(a - 3b)^2}{4}$

D $-\frac{(a - 3b)^2}{4}$

Classified Practice

Inequalities

Q047 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 4b)^2}{4}$

B $\frac{a^2 - 8ab + 16b^2 + 4}{4}$

C $\frac{(a - 4b - 2)(a - 4b + 2)}{4}$

D $-\frac{(a - 4b)^2}{4}$

Classified Practice

Inequalities

Q048 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

- A $\frac{(a - 5b)^2}{4}$
- B $\frac{a^2 - 10ab + 25b^2 + 4}{4}$
- C $\frac{(a - 5b - 2)(a - 5b + 2)}{4}$
- D $-\frac{(a - 5b)^2}{4}$

Classified Practice

Inequalities

Q049 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{a^2 - 12ab + 36b^2 + 4}{4}$

B $\frac{(a - 6b)^2}{4}$

C $\frac{(a - 6b - 2)(a - 6b + 2)}{4}$

D $-\frac{(a - 6b)^2}{4}$

Classified Practice

Inequalities

Q050 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

- A $\frac{(a - 7b)^2}{4}$
- B $\frac{a^2 - 14ab + 49b^2 + 4}{4}$
- C $\frac{(a - 7b - 2)(a - 7b + 2)}{4}$
- D $-\frac{(a - 7b)^2}{4}$

Classified Practice

Inequalities

Q051 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 8b)^2}{4}$

B $\frac{a^2 - 16ab + 64b^2 + 4}{4}$

C $\frac{(a - 8b - 2)(a - 8b + 2)}{4}$

D $-\frac{(a - 8b)^2}{4}$

Classified Practice

Inequalities

Q052 Written

Prove that $\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$.

Q053 Written**Prove that** $a + 2 \geq \sqrt{3 + 6a}$.

Classified Practice

Inequalities

Q054 Written

Prove that $\sqrt{2(a^2 + b^2)} \geq a + b$.

Q055 Written

Prove that $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a}+\sqrt{b}}{2}$.

Classified Practice

Inequalities

Q056 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $2\sqrt{a}\sqrt{b} + 1$

B $2\sqrt{a}\sqrt{b}$

C $2\sqrt{a}\sqrt{b} - 1$

D $-2\sqrt{a}\sqrt{b}$

Q057 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $4\sqrt{a}\sqrt{b}$

B $4\sqrt{a}\sqrt{b} + 1$

C $4\sqrt{a}\sqrt{b} - 1$

D $-4\sqrt{a}\sqrt{b}$

Classified Practice

Inequalities

Q058 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $6\sqrt{a}\sqrt{b} + 1$

B $6\sqrt{a}\sqrt{b} - 1$

C $6\sqrt{a}\sqrt{b}$

D $-6\sqrt{a}\sqrt{b}$

Q059 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $8\sqrt{a}\sqrt{b} + 1$

B $8\sqrt{a}\sqrt{b}$

C $8\sqrt{a}\sqrt{b} - 1$

D $-8\sqrt{a}\sqrt{b}$

Classified Practice

Inequalities

Q060 Written

Prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Q061 Written**Prove that** $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$.

Classified Practice

Inequalities

Q062 Written

Prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

Q063 Written**Prove that** $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$.

Classified Practice

Inequalities

Q064 Written

Prove that $\sqrt{a-b} > \sqrt{a} - \sqrt{b}$.

Q065 Written

If $a > b > 0$, prove that $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.

Classified Practice

Inequalities

Q066 Challenge

For $a, b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Classified Practice

Inequalities

Q067 Written

Prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Q068 Written**Prove that** $\frac{b}{3a} + \frac{12a}{b} \geq 4$.

Classified Practice

Inequalities

Q069 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{a^2 + b^2}{ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{(a - b)^2}{ab}$

B $\frac{a^2 - ab + b^2}{ab}$

C $\frac{a^2 - 3ab + b^2}{ab}$

D $-\frac{(a - b)^2}{ab}$

Q070 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{4a^2 + b^2}{2ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{4a^2 - 2ab + b^2}{2ab}$

B $\frac{4a^2 - 6ab + b^2}{2ab}$

C $\frac{(2a - b)^2}{2ab}$

D $-\frac{(2a - b)^2}{2ab}$

Classified Practice

Inequalities

Q071 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{9a^2 + b^2}{3ab} \geq 2$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $\frac{9a^2 - 3ab + b^2}{3ab}$

B $\frac{(3a - b)^2}{3ab}$

C $\frac{9a^2 - 9ab + b^2}{3ab}$

D $-\frac{(3a - b)^2}{3ab}$

Q072 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{16a^2 + b^2}{4ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{(4a - b)^2}{4ab}$

B $\frac{16a^2 - 4ab + b^2}{4ab}$

C $\frac{16a^2 - 12ab + b^2}{4ab}$

D $-\frac{(4a - b)^2}{4ab}$

Classified Practice

Inequalities

Q073 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{25a^2 + b^2}{5ab} \geq 2$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $\frac{25a^2 - 5ab + b^2}{5ab}$
- B $\frac{25a^2 - 15ab + b^2}{5ab}$
- C $-\frac{(5a - b)^2}{5ab}$
- D $\frac{(5a - b)^2}{5ab}$

Q074 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{36a^2 + b^2}{6ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{36a^2 - 6ab + b^2}{6ab}$

B $\frac{36a^2 - 18ab + b^2}{6ab}$

C $-\frac{(6a - b)^2}{6ab}$

D $\frac{(6a - b)^2}{6ab}$

Classified Practice

Inequalities

Q075 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{49a^2 + b^2}{7ab} \geq 2$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $\frac{49a^2 - 7ab + b^2}{7ab}$
- B $\frac{49a^2 - 21ab + b^2}{7ab}$
- C $\frac{(7a - b)^2}{7ab}$
- D $-\frac{(7a - b)^2}{7ab}$

Q076 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{64a^2 + b^2}{8ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{64a^2 - 8ab + b^2}{8ab}$

B $\frac{64a^2 - 24ab + b^2}{8ab}$

C $-\frac{(8a - b)^2}{8ab}$

D $\frac{(8a - b)^2}{8ab}$

Classified Practice

Inequalities

Q077 Written

Prove that $(a + b) \left(\frac{1}{a} + \frac{1}{b} \right) \geq 4$.

Q078 Written**Prove that** $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$.

Classified Practice

Inequalities

Q079 Written

Prove that $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$.

Q080 Written**Prove that** $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$.

Classified Practice

Inequalities

Q081 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+b)^2}{ab} \geq 4$ select the correct factorisation of LHS – RHS:

Choose the correct option.

- A $\frac{a^2 - ab + b^2}{ab}$
- B $\frac{a^2 - 3ab + b^2}{ab}$
- C $-\frac{(a-b)^2}{ab}$
- D $\frac{(a-b)^2}{ab}$

Q082 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $\frac{4a^2 - 7ab + 4b^2}{ab}$

B $\frac{4(a-b)^2}{ab}$

C $\frac{4a^2 - 9ab + 4b^2}{ab}$

D $-\frac{4(a-b)^2}{ab}$

Classified Practice

Inequalities

Q083 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+9b)(9a+b)}{ab} \geq 100$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{9a^2 - 17ab + 9b^2}{ab}$

B $\frac{9a^2 - 19ab + 9b^2}{ab}$

C $\frac{9(a-b)^2}{ab}$

D $-\frac{9(a-b)^2}{ab}$

Q084 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+16b)(16a+b)}{ab} \geq 289$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{16a^2 - 31ab + 16b^2}{ab}$

B $\frac{16a^2 - 33ab + 16b^2}{ab}$

C $-\frac{16(a-b)^2}{ab}$

D $\frac{16(a-b)^2}{ab}$

Classified Practice

Inequalities

Q085 Written

Prove that $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$.

Q086 Written**Prove that $a + \frac{4}{a} \geq 4$.**

Classified Practice

Inequalities

Q087 Written

Prove that $3ab + \frac{4}{ab} \geq 4\sqrt{3}$.

Q088 Written**Prove that** $a + b + \frac{9}{a+b} \geq 6$.

Classified Practice

Inequalities

Q089 Written

A rectangle has perimeter 200 cm. Prove that its area $LW \leq 2500 \text{ cm}^2$.

Classified Practice

Inequalities

Q090 **Challenge**

For $a, b > 0$, prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Classified Practice

Inequalities

Quiz MCQ 1 **Concept Quiz · MCQ**

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $x - y$

B $x - y + 1$

C $x - y - 1$

D $-x + y$

Quiz MCQ 2 Concept Quiz · MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $(a - 5b)^2$

B $a^2 - 10ab + 25b^2 + 1$

C $(a - 5b - 1)(a - 5b + 1)$

D $-(a - 5b)^2$

Classified Practice

Inequalities

Quiz MCQ 3 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

- A $\frac{(a - 7b)^2}{4}$
- B $\frac{a^2 - 14ab + 49b^2 + 4}{4}$
- C $\frac{(a - 7b - 2)(a - 7b + 2)}{4}$
- D $-\frac{(a - 7b)^2}{4}$

Classified Practice

Inequalities

Quiz MCQ 4 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $\frac{4a^2 - 7ab + 4b^2}{ab}$

B $\frac{4(a-b)^2}{ab}$

C $\frac{4a^2 - 9ab + 4b^2}{ab}$

D $-\frac{4(a-b)^2}{ab}$

Quiz WRITTEN 5 **Concept Quiz · Written**

If $x > y$, prove that $3x + 4y > 2x + 5y$.

Classified Practice

Inequalities

Quiz WRITTEN 6 Concept Quiz · Written**Under x, y real, prove that $x^2 + y^2 \geq 2xy$.**

Quiz WRITTEN 7 Concept Quiz · Written

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Classified Practice

Inequalities

Quiz WRITTEN 8 Concept Quiz · Written

Under $t > 0$, prove that $\frac{t^2+1}{t} \geq 2$.

Answer key

Use the question number shown on each page.

Q001 $2(ax + by) > (a + b)(x + y)$

Q002 A

Q003 C

Q004 D

Q005 D

Q006 B

Q007 $ac + bd > ad + bc$

Q008 $ab + cd > ad + bc$

Q009 D

Q010 C

Q011 A

Q012 $3x - 4y > 2x - 3y$

Q013 $xy + 5x > 3y + 15$

Q014 $xy + 2 > x + 2y$

Q015 $ab + 1 > a + b$

Q016 $ax + by > ay + bx$

Q017 $ax + by + cz > ay + bz + cx$

Q018 D

Q019 $a^2 + 4b^2 \geq 4ab$

Q020 D

Q021 C

Q022 B

Q023 A

Q024 C

Q025 B

Q026 $a^2 + 5b^2 \geq 4ab$

Q027 $9x^2 \geq y(6x - y)$

Q028 $2(x^2 + y^2) \geq (x + y)^2$

Q029 $(2x - y)^2 \geq 3(x^2 - y^2)$

Q030 $(2a + 3b)^2 \geq 12ab$

Q031 A

Q032 C

Q033 D

Q034 D

Q035 A

Q036 C

Q037 $x^2 + y^2 \geq 2(x - y - 1)$

Q038 $x^2 + 2 > 2x$

Q039 $x^2 + y^2 - 6x + 8y + 25 \geq 0$

Q040 $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$

Q041 $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$

Q042 0

Q043 $\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2}$

Q044 A

Q045 A

Q046 C

Q047 A

Q048 A

Q049 B

Q050 A

Q051 A

Q052 $\sqrt{2(a + b)} \geq \sqrt{a} + \sqrt{b}$

Q053 $a + 2 \geq \sqrt{3 + 6a}$

Q054 $\sqrt{2(a^2 + b^2)} \geq a + b$

Q055 $\sqrt{\frac{a + b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$

Q056 B

Q057 A

Q058 C

Q059 B

Q060 $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$

Q061 $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$

Q062 $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$

Q063 $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$

Q064 $\sqrt{a - b} > \sqrt{a} - \sqrt{b}$

Q065 $\sqrt{a - b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$

Q066 $\sqrt{a + 16b}$

Q067 $\frac{b}{a} + \frac{a}{b} \geq 2$

Q068 $\frac{b}{3a} + \frac{12a}{b} \geq 4$

Q069 A

Q070 C

Q071 B

Q072 A

Q073 D

Q074 D

Q075 C

Q076 D

Q077 $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$

Q078 $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$

Q079 $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$

Q080 $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$

Q081 D

Q082 B

Q083 C

Q084 D

Q085 $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$

Q086 $a + \frac{4}{a} \geq 4$

Q087 $3ab + \frac{4}{ab} \geq 4\sqrt{3}$

Q088 $a + b + \frac{9}{a + b} \geq 6$

Q089 $LW \leq 2500$

Q090 2

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $2x + 5y$

Quiz WRITTEN 6 $2xy$

Quiz WRITTEN 7 $\sqrt{a + 36b}$

Quiz WRITTEN 8 2

Concept 1 | Arithmetic Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Test

$$a_{n+1} - a_n = d$$

Classified Practice

Arithmetic Sequences

Q001 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = 2n - 3$, find the first term through the 3rd term.

Q002 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 7n$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q003 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 5n - 8$, find the first term through the 3rd term.

Q004 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -3n + 1$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q005 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = -2n - 4$, find the first term through the 3rd term.

Q006 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 1$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q007 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + n + 1$, find the first term through the 3rd term.

Q008 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + 2n + 4$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q009 MCQ

For $a_n = 3n - 5$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-2, 1, 5)$

B $(-2, 1, 4)$

C $(-2, 0, 4)$

D $(2, -1, -4)$

Q010 MCQ

For $a_n = 4n + 7$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (11, 15, 20)

B (11, 14, 19)

C (11, 15, 19)

D (-11, -15, -19)

Classified Practice

Arithmetic Sequences

Q011 MCQ

For $a_n = n^2 - 4n + 6$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (3, 2, 4)

B (3, 1, 3)

C (-3, -2, -3)

D (3, 2, 3)

Q012 MCQ

For $a_n = 2n^2 + 3$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (5, 11, 21)

B (5, 11, 22)

C (5, 10, 21)

D (-5, -11, -21)

Classified Practice

Arithmetic Sequences

Q013 MCQ

For $a_n = n^2 + 5n - 2$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (4, 10, 17)

B (4, 10, 16)

C (4, 9, 16)

D (-4, -10, -16)

Q014 MCQ

For $a_n = 5 - 2n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (3, 1, 0)

B (3, 0, -1)

C (3, 1, -1)

D (-3, -1, 1)

Classified Practice

Arithmetic Sequences

Q015 MCQ

For $a_n = 2n^2 - 7n + 9$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (4, 3, 7)

B (4, 2, 6)

C (-4, -3, -6)

D (4, 3, 6)

Q016 MCQ

For $a_n = n^2 - 3n - 1$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-3, -3, -1)$

B $(-3, -3, 0)$

C $(-3, -4, -1)$

D $(3, 3, 1)$

Classified Practice

Arithmetic Sequences

Q017 MCQ

For $a_n = 3n^2 - 2$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (1, 10, 26)

B (1, 10, 25)

C (1, 9, 25)

D (-1, -10, -25)

Q018 MCQ**For $a_n = n^2 + 2$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** (3, 6, 12)**B** (3, 5, 11)**C** (3, 6, 11)**D** (-3, -6, -11)

Classified Practice

Arithmetic Sequences

Q019 MCQ

For $a_n = 7n - 10$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-3, 4, 12)$

B $(-3, 3, 11)$

C $(3, -4, -11)$

D $(-3, 4, 11)$

Q020 MCQ**For $a_n = 2n^2 + n - 1$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** (2, 9, 20)**B** (2, 9, 21)**C** (2, 8, 20)**D** (-2, -9, -20)

Classified Practice

Arithmetic Sequences

Q021 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2 \times 3^n$, find the first term through the 3rd term.

Q022 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3^n - 1$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q023 Written

When the general term of the sequence $\{a_n\}$ is $a_n = (-3)^n$, find the first term through the 3rd term.

Q024 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3 \times (-2)^n$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q025 MCQ

For $a_n = 2^n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (2, 4, 8)

B (2, 4, 9)

C (2, 3, 8)

D (-2, -4, -8)

Q026 MCQ**For $a_n = (-2)^n$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** $(-2, 4, -7)$ **B** $(-2, 4, -8)$ **C** $(-2, 3, -8)$ **D** $(2, -4, 8)$

Classified Practice

Arithmetic Sequences

Q027 MCQ

For $a_n = 2 \times 3^n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (6, 18, 55)

B (6, 17, 54)

C (6, 18, 54)

D (-6, -18, -54)

Q028 MCQ**For $a_n = 5 \times (-3)^n$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** $(-15, 45, -134)$ **B** $(-15, 44, -135)$ **C** $(15, -45, 135)$ **D** $(-15, 45, -135)$

Classified Practice

Arithmetic Sequences

Q029 MCQ

For $a_n = 4^n - 1$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (3, 15, 63)

B (3, 15, 64)

C (3, 14, 63)

D (-3, -15, -63)

Q030 MCQ

For $a_n = 2 \times 3^{n-1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (2, 6, 19)

B (2, 6, 18)

C (2, 5, 18)

D (-2, -6, -18)

Classified Practice

Arithmetic Sequences

Q031 MCQ

For $a_n = (-1)^{n-1}5^n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(5, -25, 126)$

B $(5, -26, 125)$

C $(5, -25, 125)$

D $(-5, 25, -125)$

Q032 MCQ

For $a_n = (-3)^n + 2$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-1, 11, -24)$

B $(-1, 10, -25)$

C $(1, -11, 25)$

D $(-1, 11, -25)$

Classified Practice

Arithmetic Sequences

Q033 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{n+1}$, find the first term through the 3rd term.

Q034 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{2n+1}$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q035 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{1}{3n-2}$, find the first term through the 3rd term.

Q036 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{2n}{n+3}$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q037 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n+1}{2n-1}$, find the first term through the 3rd term.

Q038 MCQ

For $a_n = \frac{n+1}{n+2}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(\frac{2}{3}, \frac{3}{4}, \frac{4}{6})$

B $(\frac{2}{3}, \frac{3}{3}, \frac{4}{5})$

C $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5})$

D $(-\frac{2}{3}, -\frac{3}{4}, -\frac{4}{5})$

Classified Practice

Arithmetic Sequences

Q039 MCQ

For $a_n = \frac{2n-1}{3n+1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(\frac{1}{4}, \frac{3}{7}, \frac{1}{3})$

B $(\frac{1}{4}, \frac{3}{6}, \frac{1}{2})$

C $(-\frac{1}{4}, -\frac{3}{7}, -\frac{1}{2})$

D $(\frac{1}{4}, \frac{3}{7}, \frac{1}{2})$

Q040 MCQ

For $a_n = \frac{n^2+1}{n+1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(1, \frac{5}{3}, \frac{5}{2})$

B $(1, \frac{5}{3}, \frac{5}{3})$

C $(1, \frac{5}{2}, \frac{5}{2})$

D $(-1, -\frac{5}{3}, -\frac{5}{2})$

Classified Practice

Arithmetic Sequences

Q041 MCQ

For $a_n = \frac{3n}{2n-1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(3, 2, \frac{9}{6})$

B $(3, 2, \frac{9}{5})$

C $(3, 1, \frac{9}{5})$

D $(-3, -2, -\frac{9}{5})$

Q042 MCQ

For $a_n = \frac{2n+3}{n+4}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(1, \frac{7}{6}, \frac{9}{8})$

B $(1, \frac{7}{5}, \frac{9}{7})$

C $(1, \frac{7}{6}, \frac{9}{7})$

D $(-1, -\frac{7}{6}, -\frac{9}{7})$

Classified Practice

Arithmetic Sequences

Q043 MCQ

For $a_n = \frac{n}{3n-2}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(1, \frac{1}{4}, \frac{3}{8})$

B $(1, \frac{1}{3}, \frac{3}{7})$

C $(-1, -\frac{1}{4}, -\frac{3}{7})$

D $(1, \frac{1}{4}, \frac{3}{7})$

Q044 Written

For the sequence $\{a_n\}$: 0, 1, 4, 9, 16, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q045 **Written**

For the sequence $\{a_n\} : 0, 1, 8, 27, 64, \dots$, deduce the general term.

Q046 Written

For the sequence $\{a_n\}$: 1, 4, 9, 16, 25, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q047 **Written**

For the sequence $\{a_n\} : 7^2, 8^2, 9^2, 10^2, 11^2, \dots$, deduce the general term.

Q048 MCQ

Which formula gives the general term of $\{a_n\} : 1, 8, 27, 64, \dots$?:

Choose the correct option.

A n^2

B n^3

C n^5

D n^6

Classified Practice

Arithmetic Sequences

Q049 MCQ

Which formula gives the general term of $\{a_n\}$: 9, 16, 25, 36, ...?:

Choose the correct option.

A $(n + 2)^3$

B $(n + 2)^4$

C $(n + 2)^2$

D $(n + 2)^5$

Q050 MCQ

Which formula gives the general term of $\{a_n\} : -1, 0, 1, 8, \dots$?:

Choose the correct option.

A $(n - 2)^2$

B $(n - 2)^5$

C $(n - 2)^6$

D $(n - 2)^3$

Classified Practice

Arithmetic Sequences

Q051 MCQ

Which formula gives the general term of $\{a_n\}$: 36, 49, 64, 81, ...?:

Choose the correct option.

A $(n + 5)^2$

B $(n + 5)^3$

C $(n + 5)^4$

D $(n + 5)^5$

Q052 MCQ

Which formula gives the general term of $\{a_n\}$: 1, 9, 25, 49, ...?:**Choose the correct option.**

A $(2n + 1)^2$

B $(2n - 1)^2$

C $(2n - 1)^4$

D $(2n - 1)^5$

Classified Practice

Arithmetic Sequences

Q053 MCQ

Which formula gives the general term of $\{a_n\} : 0, 1, 16, 81, \dots$?:

Choose the correct option.

A $(n + 1)^4$

B $(n - 1)^6$

C $(n - 1)^4$

D $(n - 1)^7$

Q054 Written

For the sequence $\{a_n\}$: 2, 5, 8, 11, 14, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q055 **Written**

For the sequence $\{a_n\} : 0, 2, 4, 6, 8, \dots$, deduce the general term.

Q056 Written

For the sequence $\{a_n\} : 0, 3, 6, 9, 12, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q057 Written

For the sequence $\{a_n\} : 4, 9, 14, 19, 24, \dots$, deduce the general term.

Q058 Written

For the sequence $\{a_n\}$: 6, 13, 20, 27, 34, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q059 MCQ

Which formula gives the general term of $\{a_n\} : 7, 12, 17, 22, \dots$?:

Choose the correct option.

A $5n + 2$

B $(5n + 2) + 1$

C $(5n + 2) - 1$

D $-(5n + 2)$

Q060 MCQ**Which formula gives the general term of $\{a_n\} : -4, -1, 2, 5, \dots$?:****Choose the correct option.**

A $(3n - 7) + 1$

B $3n - 7$

C $(3n - 7) - 1$

D $-(3n - 7)$

Classified Practice

Arithmetic Sequences

Q061 MCQ

Which formula gives the general term of $\{a_n\}$: 10, 6, 2, -2 , ...?:

Choose the correct option.

A $(14 - 4n) + 1$

B $(14 - 4n) - 1$

C $14 - 4n$

D $-(14 - 4n)$

Q062 MCQ

Which formula gives the general term of $\{a_n\} : \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$?:

Choose the correct option.

A $\left(\frac{n}{2}\right) + 1$

B $\left(\frac{n}{2}\right) - 1$

C $-\left(\frac{n}{2}\right)$

D $\frac{n}{2}$

Classified Practice

Arithmetic Sequences

Q063 Written

For the sequence $\{a_n\} : 2, -4, 6, -8, 10, \dots$, deduce the general term.

Q064 Written

For the sequence $\{a_n\}$: $-3, 6, -9, 12, -15, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q065 **Written**

For the sequence $\{a_n\}$: $-5, 10, -15, 20, -25, \dots$, deduce the general term.

Q066 Written

For the sequence $\{a_n\}$: $-1, 4, -7, 10, -13, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q067 Written

For the sequence $\{a_n\} : 1, \frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{5}{9}, \dots$, deduce the general term.

Q068 Written

For the sequence $\{a_n\} : 1, \frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \frac{9}{25}, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q069 Written

For the sequence $\{a_n\} : 1, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{7}, \frac{1}{9}, \dots$, deduce the general term.

Q070 Written

For the sequence $\{a_n\} : \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q071 Written

For the sequence $\{a_n\}$: 1×3 , 2×9 , 3×27 , 4×81 , 5×243 , ..., deduce the general term.

Q072 Written

For the sequence $\{a_n\} : 1 \times \frac{1}{2}, 2 \times \frac{1}{3}, 3 \times \frac{1}{4}, 4 \times \frac{1}{5}, 5 \times \frac{1}{6}, \dots$, deduce the general term then find the 20th term.

Classified Practice

Arithmetic Sequences

Q073 Written

Find the first term and the common difference of the arithmetic sequences: (a) 2, 5, 8, 11, ...; (b) 3, -2, -7, -12, -17,

Q074 Written**Find the first term and common difference of: (a) 1, 5, 9, 13, 17, ...; (b) 23, 19, 15, 11, 7,**

Classified Practice

Arithmetic Sequences

Q075 MCQ

The listed terms $\{a_n\} = (4, 9, 14, 19, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

- A (4, 4)
- B (4, 5)
- C (5, 4)
- D (4, -5)

Q076 MCQ

The listed terms $\{a_n\} = (-7, -3, 1, 5, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

- A $(-7, 4)$
- B $(-7, -4)$
- C $(7, 4)$
- D $(-3, 4)$

Classified Practice

Arithmetic Sequences

Q077 MCQ

The listed terms $\{a_n\} = (18, 13, 8, 3, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

- A (18, 5)
- B (13, -5)
- C (18, -5)
- D (-18, -5)

Q078 MCQ

The listed terms $\{a_n\} = (12, 8, 4, 0, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(12, 4)$

B $(8, -4)$

C $(-12, -4)$

D $(12, -4)$

Classified Practice

Arithmetic Sequences

Q079 Written

Find the first term and common difference of: (a) 3, 5, 7, 9, 11, ...; (b) $-13, -9, -5, -1, 3, \dots$; (c) $-2, 1, 4, 7, 10, \dots$; (d) $10, 7, 4, 1, -2, \dots$; (e) $6, \frac{16}{3}, \frac{14}{3}, 4, \frac{10}{3}, \dots$; (f) $-4, -\frac{9}{2}, -5, -\frac{11}{2}, -6, \dots$

Classified Practice

Arithmetic Sequences

Q080 MCQ

The listed terms $\{a_n\} = (\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(\frac{1}{2}, \frac{2}{3})$

B $(\frac{1}{2}, \frac{1}{3})$

C $(\frac{1}{3}, \frac{1}{2})$

D $(-\frac{1}{2}, \frac{1}{3})$

Q081 MCQ

The listed terms $\{a_n\} = (-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(-\frac{3}{4}, \frac{1}{2})$

B $(-\frac{3}{4}, -\frac{1}{2})$

C $(\frac{3}{4}, \frac{1}{2})$

D $(-\frac{1}{4}, \frac{1}{2})$

Classified Practice

Arithmetic Sequences

Q082 MCQ

The listed terms $\{a_n\} = (\frac{5}{3}, 2, \frac{7}{3}, \frac{8}{3}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(\frac{5}{3}, \frac{2}{3})$

B $(2, \frac{1}{3})$

C $(\frac{5}{3}, \frac{1}{3})$

D $(-\frac{5}{3}, \frac{1}{3})$

Q083 MCQ

The listed terms $\{a_n\} = (-\frac{5}{2}, -1, \frac{1}{2}, 2, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(-\frac{5}{2}, -\frac{3}{2})$

B $(-\frac{5}{2}, \frac{3}{2})$

C $(\frac{5}{2}, \frac{3}{2})$

D $(-1, \frac{3}{2})$

Classified Practice

Arithmetic Sequences

Q084 **Written**

The sequences are arithmetic. Find the common difference and the numbers that fit in (A) and (B): (a) $A, 2, -1, B, \dots$; (b) $13, A, B, -2, \dots$

Q085 Written

The sequences are arithmetic. Find the common difference and the missing numbers: (a) $2, 9, A, B, \dots$; (b) $A, 17, B, 5, \dots$

Classified Practice

Arithmetic Sequences

Q086 MCQ

The sequence $\{a_n\} = (6, A, 14, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Choose the correct option.

A $d = 4, A = 10, B = 18$

B $d = 8, A = 14, B = 22$

C $d = 4, A = 12, B = 18$

D $d = -4, A = 2, B = 10$

Q087 MCQ

The sequence $\{a_n\} = (A, 5, 9, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Choose the correct option.

A $d = 4, A = 9, B = 13$

B $d = -4, A = 9, B = 1$

C $d = 4, A = 1, B = 13$

D $d = 2, A = 3, B = 11$

Classified Practice

Arithmetic Sequences

Q088 MCQ

The sequence $\{a_n\} = (11, A, B, 23, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Choose the correct option.

A $d = 3, A = 14, B = 17$

B $d = 4, A = 15, B = 19$

C $d = 4, A = 14, B = 17$

D $d = -4, A = 7, B = 3$

Classified Practice

Arithmetic Sequences

Q089 **Written**

Find the common difference and missing terms: (a) $8, 15, A, B, \dots$; (b) $7, 13, A, B, \dots$; (c) $A, 9, 4, B, \dots$; (d) $A, 15, B, 9, \dots$; (e) $11, A, B, 20, \dots$; (f) $17, A, B, 5, C, \dots$

Classified Practice

Arithmetic Sequences

Q090 **Written**

For each arithmetic sequence, find the general term and the 7th term: (a) first term (4), common difference (3); (b) first term (2), common difference -3 .

Q091 Written

For each arithmetic sequence, find the general term and the 7th term: (a) $a = 5, d = 4$; (b) $a = -2, d = 3$; (c) $a = 10, d = -3$; (d) $a = -\frac{1}{2}, d = -\frac{1}{2}$.

Classified Practice

Arithmetic Sequences

Q092 Written

For the arithmetic sequence a_n with first term (8) and common difference -6 , find the general term and the 20th term.

Q093 Written

For each arithmetic sequence, find the general term and the 10th term: (a) $a = -7, d = 3$; (b) $a = -2, d = -3$; (c) $a = 7, d = \frac{1}{2}$; (d) $a = \frac{2}{3}, d = -\frac{1}{3}$.

Classified Practice

Arithmetic Sequences

Q094 Written

In an arithmetic sequence, $a_3 + a_7 = 38$ and $\frac{a_4}{a_{12}} = \frac{2}{5}$. Find the first term and common difference.

Q095 Written

Find the general term of the arithmetic sequence whose first term is -5 and whose 17th term is 75 .

Classified Practice

Arithmetic Sequences

Q096 Written

The first term is 6, and the 4th term is 18. Find a_n .

Q097 Written

The first term is 20, and the 8th term is -1 . Find a_n .

Classified Practice

Arithmetic Sequences

Q098 **Written**

The first term is -8 , and the 8th term is 55 . Find a_n .

Q099 Written

The first term is -18 , and the 10th term is -63 . Find a_n .

Classified Practice

Arithmetic Sequences

Q100 MCQ

An arithmetic sequence has $a = 4$ and $a_7 = 22$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(3n - 1, 3)$

B $(4n, 4)$

C $(3n + 1, 4)$

D $(a_n, d) = (3n + 1, 3)$

Q101 MCQ

An arithmetic sequence has $a = -3$ and $a_9 = 29$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(4n - 3, 4)$

B $(a_n, d) = (4n - 7, 4)$

C $(5n - 8, 5)$

D $(4n - 7, -4)$

Classified Practice

Arithmetic Sequences

Q102 MCQ

An arithmetic sequence has $a = 12$ and $a_5 = -4$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(a_n, d) = (16 - 4n, -4)$

B $(12 - 4n, -4)$

C $(16 + 4n, 4)$

D $(16 - 3n, -3)$

Q103 MCQ

An arithmetic sequence has $a = 5$ and $a_{11} = 45$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(5n, 5)$

B $(a_n, d) = (4n + 1, 4)$

C $(4n - 1, 4)$

D $(4n + 1, -4)$

Classified Practice

Arithmetic Sequences

Q104 **Written**

Find the general term of the arithmetic sequence whose common difference is 3 and whose 16th term is 81.

Q105 Written

The common difference is 2, and the 8th term is 4. Find a_n .

Classified Practice

Arithmetic Sequences

Q106 Written

The common difference is -4 , and the 5th term is -7 . Find a_n .

Q107 Written

The common difference is -7 , and the 6th term is -24 . Find a_n .

Classified Practice

Arithmetic Sequences

Q108 MCQ

An arithmetic sequence has common difference -2 and $a_4 = 1$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(a_n, a) = (9 - 2n, 7)$

B $(7 - 2n, 9)$

C $(1 - 2n, 3)$

D $(9 + 2n, 7)$

Q109 MCQ

An arithmetic sequence has common difference 5 and $a_6 = 17$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(5n - 3, -2)$

B $(17 + 5n, 2)$

C $(13 - 5n, 8)$

D $(a_n, a) = (5n - 13, -8)$

Classified Practice

Arithmetic Sequences

Q110 MCQ

An arithmetic sequence has common difference 4 and $a_{10} = 38$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(4n + 2, -2)$

B $(2n + 4, 6)$

C $(a_n, a) = (4n - 2, 2)$

D $(4n - 2, 4)$

Q111 MCQ

An arithmetic sequence has common difference -3 and $a_8 = -18$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(3 - 3n, 6)$

B $(6 + 3n, 3)$

C $(6 - 2n, 2)$

D $(a_n, a) = (6 - 3n, 3)$

Classified Practice

Arithmetic Sequences

Q112 **Written**

Find the general term of the arithmetic sequence $\{a_n\}$ whose 3rd term is 10 and the 8th term is 50.

Q113 Written

Find the general term of the arithmetic sequence whose 6th term is 2 and whose 15th term is -25 .

Classified Practice

Arithmetic Sequences

Q114 **Written****The 4th term is 14, and the 10th term is 62. Find a_n .**

Q115 Written

The 3rd term is 30, and the 10th term is 2. Find a_n .

Classified Practice

Arithmetic Sequences

Q116 Written

The 10th term is 49, and the 23rd term is 10. Find a_n .

Q117 Written

The 3rd term is 6, and the 10th term is 62. Find a_n .

Classified Practice

Arithmetic Sequences

Q118 MCQ

An arithmetic sequence has $a_3 = 5$ and $a_9 = 17$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(2n + 1, -1)$

B $(n + 3, 2)$

C $(a_n, a) = (2n - 1, 1)$

D $(2n - 1, 5)$

Q119 MCQ

An arithmetic sequence has $a_4 = 7$ and $a_{10} = -5$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(13 - 2n, 15)$

B $(15 + 2n, 5)$

C $(a_n, a) = (15 - 2n, 13)$

D $(2n - 15, -13)$

Classified Practice

Arithmetic Sequences

Q120 MCQ

An arithmetic sequence has $a_7 = 31$ and $a_{12} = 56$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(5n + 4, -1)$

B $(a_n, a) = (5n - 4, 1)$

C $(4n - 5, 2)$

D $(5n - 4, 31)$

Q121 MCQ

An arithmetic sequence has $a_2 = -4$ and $a_8 = 14$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(a_n, a) = (3n - 10, -7)$

B $(3n - 4, -1)$

C $(2n - 7, -5)$

D $(10 - 3n, 7)$

Classified Practice

Arithmetic Sequences

Q122 MCQ

An arithmetic sequence has $a_5 = 3$ and $a_{11} = 9$. Select the correct pair (a_n, a) :

Choose the correct option.

- A $\left(a_n = \frac{n-2}{2}, -\frac{1}{2}\right)$
B $\left(a_n = \frac{n+2}{2}, -\frac{5}{2}\right)$
C $\left(a_n = \frac{n-2}{2}, \frac{1}{2}\right)$
D $\left(a_n = \frac{2n-1}{3}, -\frac{1}{3}\right)$

Q123 MCQ

An arithmetic sequence has $a_2 = 6$ and $a_9 = 20$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(a_n = 2n + 4, 2)$

B $(a_n = 2n + 2, 4)$

C $(a_n = n + 4, 5)$

D $(a_n = 2n - 2, 8)$

Classified Practice

Arithmetic Sequences

Q124 **Written**

The common difference is $\frac{2}{5}$, and the 30th term is $\frac{23}{5}$. Find a_n .

Q125 MCQ

An arithmetic sequence has $d = -\frac{3}{2}$ and $a_5 = -9$. Select the correct pair (a_n, a) :

Choose the correct option.

A $\left(a_n = -\frac{3(n+1)}{2}, -3\right)$

B $\left(a_n = -\frac{3(n-1)}{2}, -6\right)$

C $\left(a_n = \frac{3(n+1)}{2}, 3\right)$

D $\left(a_n = -\frac{n+1}{2}, -3\right)$

Classified Practice

Arithmetic Sequences

Q126 MCQ

An arithmetic sequence has $d = \frac{2}{3}$ and $a_{12} = 10$. Select the correct pair (a_n, a) :

Choose the correct option.

A $\left(a_n = \frac{2n+4}{3}, 2\right)$

B $\left(a_n = \frac{2n-6}{3}, -\frac{8}{3}\right)$

C $\left(a_n = \frac{n+6}{3}, \frac{8}{3}\right)$

D $\left(a_n = \frac{2n+6}{3}, \frac{8}{3}\right)$

Q127 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, which term is 113?

Classified Practice

Arithmetic Sequences

Q128 **Written**

For the arithmetic sequence with first term 1 and common difference 3, determine which term is 40.

Q129 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , determine which term is -52 .

Classified Practice

Arithmetic Sequences

Q130 **Written**

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , determine which term is -110 .

Q131 MCQ**For $a = 3, d = 4$, which term of the arithmetic sequence is 59?:****Choose the correct option.****A $n = 14$** **B $n = 16$** **C $n = 15$** **D $n = 13$**

Classified Practice

Arithmetic Sequences

Q132 MCQ

For $a = -5$, $d = 6$, which term of the arithmetic sequence is 55?:

Choose the correct option.

A $n = 10$

B $n = 12$

C $n = 9$

D $n = 11$

Q133 MCQ

For $a = 20$, $d = -3$, which term of the arithmetic sequence is -16 ?:

Choose the correct option.

A $n = 13$

B $n = 12$

C $n = 14$

D $n = 11$

Classified Practice

Arithmetic Sequences

Q134 MCQ

For $a = 1, d = 5$, which term of the arithmetic sequence is 86?:

Choose the correct option.

A $n = 17$

B $n = 18$

C $n = 19$

D $n = 16$

Q135 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, find the first term greater than 500.

Classified Practice

Arithmetic Sequences

Q136 **Written**

For the arithmetic sequence with first term 1 and common difference 3, find the first term greater than 100.

Q137 MCQ**For $a = 2, d = 7$, find the first term greater than 100:****Choose the correct option.****A $n = 16$** **B $n = 15$** **C $n = 17$** **D $n = 14$**

Classified Practice

Arithmetic Sequences

Q138 MCQ

For $a = -9$, $d = 4$, find the first term greater than 50:**Choose the correct option.**A $n = 15$ B $n = 17$ C $n = 14$ D $n = 16$

Q139 MCQ**For $a = 6$, $d = 3$, find the first term greater than 200:****Choose the correct option.**

A $n = 65$

B $n = 67$

C $n = 64$

D $n = 66$

Classified Practice

Arithmetic Sequences

Q140 MCQ

For $a = -1$, $d = 9$, find the first term greater than 300:

Choose the correct option.

A $n = 34$

B $n = 35$

C $n = 36$

D $n = 33$

Q141 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , find the first term smaller than -300 .

Classified Practice

Arithmetic Sequences

Q142 **Written**

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , find the first term smaller than -500 .

Q143 MCQ**For $a = 10$, $d = -4$, find the first term smaller than -100 :****Choose the correct option.****A $n = 28$** **B $n = 29$** **C $n = 30$** **D $n = 27$**

Classified Practice

Arithmetic Sequences

Q144 MCQ

For $a = 25$, $d = -6$, find the first term smaller than -250 :

Choose the correct option.

A $n = 46$

B $n = 47$

C $n = 48$

D $n = 45$

Q145 MCQ**For $a = -3, d = -5$, find the first term smaller than -200 :****Choose the correct option.**

A $n = 40$

B $n = 42$

C $n = 41$

D $n = 39$

Classified Practice

Arithmetic Sequences

Q146 MCQ

For $a = 4$, $d = 6$, find the first term that is at least 100:

Choose the correct option.

A $n = 17$

B $n = 16$

C $n = 18$

D $n = 15$

Q147 MCQ

For the expression For $a = 5, d = 4$, does the sequence contain the term 42?, the required value is:

Choose the correct option.

- A $n = 9$
- B No term
- C $n = 10$
- D $n = 11$

Classified Practice

Arithmetic Sequences

Q148 **Written**

When $a, 6, 2a$ form an arithmetic sequence in this order, find a .

Q149 Written

When the three numbers $x, 6, x^2$ form an arithmetic sequence in this order, find x .

Classified Practice

Arithmetic Sequences

Q150 **Written**

When x^2 , x , -8 form an arithmetic sequence in this order, find x .

Q151 Written

When $-3, x, x^2$ form an arithmetic sequence in this order, find x .

Classified Practice

Arithmetic Sequences

Q152 Challenge

When $2, x, y, z$ form an arithmetic sequence in this order, and $2x, y + 1, z - y$ also form an arithmetic sequence in this order, find x, y, z .

Classified Practice

Arithmetic Sequences

Q153 **Written****Find the sum (S_7) of a sequence with first term (8), last term (22), and (7) terms.**

Q154 Written

Find (S_{10}) for a sequence with first term (7), last term (13), and (10) terms.

Classified Practice

Arithmetic Sequences

Q155 Written

Find (S_{10}) for a sequence with first term (3), last term (21), and (10) terms.

Q156 Written

Find (S_8) for a sequence with first term (5), last term (33), and (8) terms.

Classified Practice

Arithmetic Sequences

Q157 **Written****Find (S_{10}) for a sequence with first term (2), last term (29), and (10) terms.**

Q158 MCQ**A sequence has first term 6, last term 30, and 7 terms. Find its sum:****Choose the correct option.****A** 105**B** 126**C** 132**D** 168

Classified Practice

Arithmetic Sequences

Q159 MCQ

A sequence has first term -4 , last term 20 , and 9 terms. Find its sum:

Choose the correct option.

A 64 B 72 C 81 D 144

Q160 MCQ**Find the sum of 10 terms whose first term is 15 and last term is 3:****Choose the correct option.**

A 45

B 90

C 150

D 180

Classified Practice

Arithmetic Sequences

Q161 MCQ

The first term is 7, the last term is 42, and there are 8 terms. Find S_8 :

Choose the correct option.

A 168

B 196

C 224

D 392

Q162 MCQ**Find the sum of 6 terms with first term -12 and last term 8 :****Choose the correct option.**A -120 B -12 C 12 D -2

Classified Practice

Arithmetic Sequences

Q163 MCQ

A sequence has $S_9 = 72$ and first term -4 . Find its last term:

Choose the correct option.

A 12

B 16

C 20

D 24

Q164 Written

Find (S_{13}) when the first term is (20), the common difference is (-5) , and the number of terms is (13).

Classified Practice

Arithmetic Sequences

Q165 Written

Find (S_{12}) when the first term is (-6) , the common difference is (3) , and there are (12) terms.

Q166 Written

Find (S_{15}) when the first term is (10), the common difference is (-3) , and there are (15) terms.

Classified Practice

Arithmetic Sequences

Q167 **Written**

Find (S_{11}) when the first term is (-4) , the common difference is (4) , and there are (11) terms.

Q168 Written

Find (S_{18}) when the first term is (3), the common difference is (2), and there are (18) terms.

Classified Practice

Arithmetic Sequences

Q169 MCQ

For $a = 4, d = 3, n = 12$, find S_{12} :**Choose the correct option.**

A 198

B 228

C 246

D 264

Q170 MCQ**For $a = 9, d = -2, n = 10$, find S_{10} :****Choose the correct option.****A** -90 **B** 0 **C** 9 **D** 90

Classified Practice

Arithmetic Sequences

Q171 MCQ

For $a = -5, d = 4, n = 9$, find S_9 :**Choose the correct option.**A -88 B 44 C 88 D 99

Q172 MCQ**For $a = 2, d = -3, n = 15$, find S_{15} :****Choose the correct option.****A** -315 **B** -285 **C** -270 **D** 285

Classified Practice

Arithmetic Sequences

Q173 MCQ

For $a = 11, d = 0, n = 8$, find S_8 :**Choose the correct option.**

A 11

B 44

C 88

D 96

Q174 MCQ**For $a = -3, d = -2, n = 7$, find S_7 :****Choose the correct option.****A** -42 **B** -54 **C** 54 **D** -63

Classified Practice

Arithmetic Sequences

Q175 **Written**

Find the sum (S) of a sequence with first term (2), common difference (3), and last term (47).

Q176 Written

Find (S) for a sequence with first term (5), common difference (3), and last term (44).

Classified Practice

Arithmetic Sequences

Q177 Written

Find (S) for a sequence with first term (3), common difference (4), and last term (75).

Q178 Written

Find (S) for a sequence with first term (2), common difference (5), and last term (57).

Classified Practice

Arithmetic Sequences

Q179 Written

Find (S) for a sequence with first term (4), common difference (3), and last term (100).

Q180 MCQ**A sequence has $a = 5, d = 4, l = 41$. Find its sum:****Choose the correct option.****A** 205**B** 225**C** 230**D** 246

Classified Practice

Arithmetic Sequences

Q181 MCQ

A sequence has $a = -7, d = 5, l = 48$. Find its sum:

Choose the correct option.

A 228

B 240

C 246

D 264

Q182 MCQ**A sequence has $a = 12, d = 2, l = 50$. Find its sum:****Choose the correct option.****A** 580**B** 600**C** 620**D** 640

Classified Practice

Arithmetic Sequences

Q183 MCQ

A sequence has $a = -10$, $d = 6$, $l = 50$. Find its sum:

Choose the correct option.

A 198

B 210

C 220

D 240

Q184 MCQ**A sequence has $a = 18, d = -3, l = -9$. Find its sum:****Choose the correct option.****A** -45 **B** 36 **C** 45 **D** 90

Classified Practice

Arithmetic Sequences

Q185 MCQ

A sequence has $a = 40, d = -4, l = -12$. Find its sum:

Choose the correct option.

A 168

B 196

C 224

D 280

Q186 Written

For the arithmetic sequence with first term (48) and common difference (-2) , find the sum from the 20th term to the 30th term.

Classified Practice

Arithmetic Sequences

Q187 **Written**

For the arithmetic sequence with first term (-53) and common difference (4) , find the sum from the 10th term to the 25th term.

Q188 MCQ**For $a = 3, d = 4$, find the sum from a_5 to a_{12} :****Choose the correct option.****A** 252**B** 264**C** 276**D** 288

Classified Practice

Arithmetic Sequences

Q189 MCQ

For $a = 20$, $d = -3$, find the sum from a_4 to a_{10} :

Choose the correct option.

A -14

B 7

C 14

D 28

Q190 MCQ**For $a = 60, d = -3$, find the sum from a_8 to a_{16} :****Choose the correct option.**

A 216

B 234

C 243

D 270

Classified Practice

Arithmetic Sequences

Q191 MCQ

For $a = 1, d = 7$, find the sum from a_3 to a_9 :**Choose the correct option.**

A 224

B 245

C 252

D 266

Q192 MCQ**For $a = 60$, $d = -3$, find the sum from a_9 to a_{17} :****Choose the correct option.****A 198****B 216****C 225****D 234**

Classified Practice

Arithmetic Sequences

Q193 MCQ

For $a = 12$, $d = 5$, find the sum from a_2 to a_8 :

Choose the correct option.

A 196

B 210

C 224

D 238

Q194 Written

For the arithmetic sequence $(90, 84, 78, \dots)$, find the sum from the 10th term to the 20th term.

Classified Practice

Arithmetic Sequences

Q195 **Written**

For the arithmetic sequence $(-9, -3, 3, \dots)$, find the sum from the 11th term to the 20th term.

Q196 Written

For the arithmetic sequence $(74, 69, 64, \dots)$, find the sum from the 8th term to the 18th term.

Classified Practice

Arithmetic Sequences

Q197 MCQ

For $a = 4$, $d = 5$, find the sum from a_6 to a_{13} :**Choose the correct option.**

A 248

B 336

C 372

D 384

Q198 MCQ**For $a = -25$, $d = 6$, find the sum from a_5 to a_{11} :****Choose the correct option.****A** 105**B** 112**C** 119**D** 126

Classified Practice

Arithmetic Sequences

Q199 **Written****Find (S_n) for a sequence with first term (1) and common difference (4).**

Q200 Written

Find (S_n) for a sequence with first term (2) and common difference (-3) .

Classified Practice

Arithmetic Sequences

Q201 **Written****Find (S_n) for a sequence with first term (0) and common difference (5).**

Q202 Written**Find (S_n) for a sequence with first term (3) and common difference (-4) .**

Classified Practice

Arithmetic Sequences

Q203 Written

Find (S_n) for a sequence with first term (-4) and common difference (-2) .

Classified Practice

Arithmetic Sequences

Q204 Challenge

For the arithmetic sequence $(94, 89, 84, \dots)$, find the sum of the odd-numbered terms from the first term to the 11th term.

Classified Practice

Arithmetic Sequences

Q205 **Written****Find the sum of the natural numbers from 1 to 100.**

Q206 Written**Find the sum of the natural numbers from 1 to 200.**

Classified Practice

Arithmetic Sequences

Q207 **Written****Find the sum of the natural numbers from 1 to 150.**

Q208 Written**Find the sum of the natural numbers from 1 to 70.**

Classified Practice

Arithmetic Sequences

Q209 Written

Find the sum of the natural numbers from 1 to 100.

Q210 Written

Find the sum of the natural numbers from 1 to 400.

Classified Practice

Arithmetic Sequences

Q211 **Written****Find the sum of the natural numbers from 1 to 250.**

Q212 Written**Find the sum of the natural numbers from 1 to 81.**

Classified Practice

Arithmetic Sequences

Q213 **Written****Find the sum of the natural numbers from 1 to 301.**

Q214 Written**Find the sum of the natural numbers from 1 to 99.**

Classified Practice

Arithmetic Sequences

Q215 MCQ

Find the sum of the natural numbers from 1 to 30:**Choose the correct option.**

A 435

B 465

C 495

D 525

Q216 MCQ**Find the sum of the natural numbers from 1 to 45:****Choose the correct option.**

- A 990
 - B 1035
 - C 1080
 - D 1125
-
-
-

Classified Practice

Arithmetic Sequences

Q217 MCQ

Find the sum of the natural numbers from 1 to 60:**Choose the correct option.**

A 1770

B 1830

C 1890

D 1950

Q218 MCQ**Find the sum of the natural numbers from 1 to 75:****Choose the correct option.****A** 2775**B** 2850**C** 2925**D** 3000

Classified Practice

Arithmetic Sequences

Q219 MCQ

Find the sum of the natural numbers from 1 to 90:**Choose the correct option.**

A 4005

B 4095

C 4185

D 4275

Q220 MCQ**Find the sum of the natural numbers from 1 to 125:****Choose the correct option.****A** 7750**B** 7875**C** 8000**D** 8125

Classified Practice

Arithmetic Sequences

Q221 MCQ

Find the sum of the natural numbers from 1 to 180:**Choose the correct option.**

A 16110

B 16290

C 16470

D 16650

Q222 MCQ**Find the sum of the natural numbers from 1 to 220:****Choose the correct option.****A** 24090**B** 24310**C** 24530**D** 24750

Classified Practice

Arithmetic Sequences

Q223 MCQ

Find the sum of the natural numbers from 1 to 275:**Choose the correct option.**

A 37675

B 37950

C 38225

D 38500

Q224 MCQ**Find the sum of the natural numbers from 1 to 350:****Choose the correct option.****A** 61075**B** 61425**C** 61775**D** 62125

Classified Practice

Arithmetic Sequences

Q225 **Written****Find the sum of the natural numbers from 101 to 200.**

Q226 Written**Find the sum of the natural numbers from 201 to 300.**

Classified Practice

Arithmetic Sequences

Q227 **Written****Find the sum of all two-digit natural numbers.**

Q228 MCQ**A sequence has first term 5, last term 35, and 7 terms. Select its sum:****Choose the correct option.****A** 133**B** 140**C** 147**D** 154

Classified Practice

Arithmetic Sequences

Q229 MCQ

A sequence has first term 12, last term 68, and 9 terms. Select its sum:

Choose the correct option.

A 351

B 360

C 369

D 378

Q230 MCQ**A sequence has first term -4, last term 20, and 9 terms. Select its sum:****Choose the correct option.****A** 63**B** 72**C** 81**D** 90

Classified Practice

Arithmetic Sequences

Q231 MCQ

A sequence has first term 7, last term 42, and 8 terms. Select its sum:

Choose the correct option.

A 188

B 196

C 204

D 212

Q232 MCQ**A sequence has first term 15, last term 3, and 10 terms. Select its sum:****Choose the correct option.****A** 80**B** 90**C** 100**D** 110

Classified Practice

Arithmetic Sequences

Q233 MCQ

A sequence has first term 10, last term 70, and 13 terms. Select its sum:

Choose the correct option.

A 507

B 520

C 533

D 546

Q234 MCQ**A sequence has first term 2, last term 57, and 12 terms. Select its sum:****Choose the correct option.****A** 342**B** 354**C** 366**D** 378

Classified Practice

Arithmetic Sequences

Q235 MCQ

A sequence has first term 25, last term 100, and 16 terms. Select its sum:

Choose the correct option.

- A 984
- B 1000
- C 1016
- D 1032

Q236 MCQ**A sequence has first term 101, last term 200, and 100 terms. Select its sum:****Choose the correct option.****A** 14950**B** 15050**C** 15150**D** 15250

Classified Practice

Arithmetic Sequences

Q237 MCQ

A sequence has first term 31, last term 91, and 16 terms. Select its sum:

Choose the correct option.

A 960

B 976

C 992

D 1008

Q238 Written**Find the sum of the multiples of 3 among the two-digit natural numbers.**

Classified Practice

Arithmetic Sequences

Q239 **Written****Find the sum of all two-digit natural numbers that are multiples of 5.**

Q240 Written**Find the sum of the multiples of 4 from 1 to 100.**

Classified Practice

Arithmetic Sequences

Q241 **Written****Find the sum of all multiples of 6 from 10 to 100.**

Q242 MCQ**Select the sum of the multiples of 3 from 3 to 99:****Choose the correct option.****A** 1680**B** 1683**C** 1686**D** 1689

Classified Practice

Arithmetic Sequences

Q243 MCQ

Select the sum of the multiples of 4 from 4 to 100:**Choose the correct option.**

A 1296

B 1300

C 1304

D 1308

Q244 Written

Find the sum of the natural numbers from 1 to 100 that leave a remainder of 1 when divided by 4.

Classified Practice

Arithmetic Sequences

Q245 **Written**

Find the sum of all two-digit natural numbers that leave a remainder of 3 when divided by 5.

Q246 Written**Find the sum of all two-digit odd natural numbers.**

Classified Practice

Arithmetic Sequences

Q247 **Written**

Find the sum of all natural numbers from 10 to 100 that are not divisible by 6.

Classified Practice

Arithmetic Sequences

Q248 **Challenge****Find the sum of the numbers from 1 to 100 which are not divisible by 4.**

Classified Practice

Arithmetic Sequences

Q249 **Written**

In an arithmetic sequence with first term 4 and common difference 3, find how many terms from the first term have sum 50.

Q250 Written

Find n when $a = 3$, $d = 2$ and $S_n = 63$.

Classified Practice

Arithmetic Sequences

Q251 MCQ

Select the correct option: Find n if $S_n = 48$ for $a = 3, d = 2$:**Choose the correct option.**

A 5

B 6

C 7

D 8

Q252 MCQ

Select the correct option: Find n if $S_n = 80$ for $a = 3, d = 2$:

Choose the correct option.

A 7

B 8

C 9

D 10

Classified Practice

Arithmetic Sequences

Q253 MCQ

Select the correct option: Find n if $S_n = 120$ for $a = 3, d = 2$:

Choose the correct option.

A 9

B 10

C 11

D 12

Q254 MCQ

Select the correct option: Find n if $S_n = 168$ for $a = 3, d = 2$:**Choose the correct option.**

A 11

B 12

C 13

D 14

Classified Practice

Arithmetic Sequences

Q255 MCQ

Select the correct option: Find n if $S_n = 255$ for $a = 3, d = 2$:**Choose the correct option.**

A 14

B 15

C 16

D 17

Q256 MCQ**Select the correct option: Find n if $S_n = 360$ for $a = 3, d = 2$:****Choose the correct option.****A** 17**B** 18**C** 19**D** 20

Classified Practice

Arithmetic Sequences

Q257 MCQ

Select the correct option: Find n if $S_n = 440$ for $a = 3, d = 2$:**Choose the correct option.**

A 19

B 20

C 21

D 22

Q258 MCQ

Select the correct option: Find n if $S_n = 624$ for $a = 3, d = 2$:**Choose the correct option.**

A 23

B 24

C 25

D 26

Classified Practice

Arithmetic Sequences

Q259 MCQ

Select the correct option: Find n if $S_n = 840$ for $a = 3, d = 2$:

Choose the correct option.

A 27

B 28

C 29

D 30

Q260 MCQ**Select the correct option: Find n if $S_n = 1088$ for $a = 3, d = 2$:****Choose the correct option.****A** 31**B** 32**C** 33**D** 34

Classified Practice

Arithmetic Sequences

Q261 MCQ

Select the correct option: Find n if $S_n = 1368$ for $a = 3, d = 2$:**Choose the correct option.**

A 35

B 36

C 37

D 38

Q262 MCQ**Select the correct option: Find n if $S_n = 1680$ for $a = 3, d = 2$:****Choose the correct option.****A** 39**B** 40**C** 41**D** 42

Classified Practice

Arithmetic Sequences

Q263 **Written**

For first term 17 and common difference -2 , find how many terms have sum -63 .

Q264 Written

Find n when $a = 24$, $d = -4$ and $S_n = -28$.

Classified Practice

Arithmetic Sequences

Q265 MCQ

Select the correct option: For $a = 10$, $d = 3$, find n when $S_n = 80$:**Choose the correct option.**

A 3

B 4

C 5

D 6

Q266 MCQ**Select the correct option: For $a = 10, d = 3$, find n when $S_n = 133$:****Choose the correct option.****A** 5**B** 6**C** 7**D** 8

Classified Practice

Arithmetic Sequences

Q267 MCQ

Select the correct option: For $a = 10$, $d = 3$, find n when $S_n = 198$:**Choose the correct option.**

A 7

B 8

C 9

D 10

Q268 MCQ**Select the correct option: For $a = 10, d = 3$, find n when $S_n = 275$:****Choose the correct option.****A** 9**B** 10**C** 11**D** 12

Classified Practice

Arithmetic Sequences

Q269 Written

Find the general term of $\{a_n\}$ with first term -60 and sum of the first 15 terms -60 .

Q270 Written

Find the general term with first term 8 and sum of the first 13 terms -130 .

Classified Practice

Arithmetic Sequences

Q271 **Written****Find the general term with first term 4 and $S_6 = 114$.**

Q272 Written**Find the general term when $a_3 = 10$ and $S_{10} = 175$.**

Classified Practice

Arithmetic Sequences

Q273 **Written**

For first term 65 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Q274 Written

For first term 70 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Classified Practice

Arithmetic Sequences

Q275 **Written****For $a = 20$, $d = -6$, find n giving the maximum sum and the sum.**

Q276 Written

For $a = 126, d = -13$, find n giving the maximum sum and the sum.

Classified Practice

Arithmetic Sequences

Q277 Written

Find the general term where $S_{10} = 10$ and $S_{20} = 40$.

Q278 Written**Find the general term where $S_6 = 3$ and $S_{10} = 65$.**

Classified Practice

Arithmetic Sequences

Q279 **Written****Find the general term where $S_{20} = 400$ and $S_{30} = 900$.**

Classified Practice

Arithmetic Sequences

Q280 Challenge

The 15th term is 33 and the 50th term is 103. Find the sum of the first 79 terms.

Classified Practice

Arithmetic Sequences

Quiz MCQ 1 **Concept Quiz · MCQ**

For $a_n = \frac{2n+1}{n+2}$, what is a_5 ?:

Choose the correct option.

- A $\frac{11}{7}$
- B $\frac{10}{7}$
- C $\frac{11}{9}$
- D $\frac{9}{7}$

Quiz MCQ 2 Concept Quiz · MCQ

An arithmetic sequence has $a_3 = 10$ and $a_8 = 30$. Select its general term:

Choose the correct option.

A $a_n = 4n + 2$

B $a_n = 4n - 2$

C $a_n = 5n - 5$

D $a_n = 2n + 4$

Classified Practice

Arithmetic Sequences

Quiz MCQ 3 [Concept Quiz](#) · MCQ

For $a = 12$, $d = -2$, select the sum of odd-numbered terms through a_9 :

Choose the correct option.

A 16

B 18

C 20

D 24

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: If $S_{10} = 10$ and $S_{20} = 40$, find d :

Choose the correct option.

- A $\frac{1}{10}$
- B $\frac{1}{5}$
- C $\frac{5}{2}$
- D $\frac{1}{2}$

Classified Practice

Arithmetic Sequences

Quiz WRITTEN 5 **Concept Quiz · Written**

If $a_n = 3n^2 - 2n + 1$, find a_1, a_2, a_3 .

Quiz WRITTEN 6 Concept Quiz · Written

Find a_n for an arithmetic sequence with $a_4 = 14$ and $a_{10} = 62$.

Classified Practice

Arithmetic Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For $a = 9, d = 4$, find the sum of odd-numbered terms from a_1 to a_{15} .

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term when $S_{20} = 400$ and $S_{30} = 900$.

Answer key

Use the question number shown on each page.

Q001 $a_1 = -1, a_2 = 1, a_3 = 3$

Q002 $a_1 = -6, a_2 = -10, a_3 = -12$

Q003 $a_1 = -3, a_2 = 2, a_3 = 7$

Q004 $a_1 = -2, a_2 = -5, a_3 = -8$

Q005 $a_1 = -6, a_2 = -8, a_3 = -10$

Q006 $a_1 = 0, a_2 = 3, a_3 = 8$

Q007 $a_1 = 3, a_2 = 7, a_3 = 13$

Q008 $a_1 = 7, a_2 = 12, a_3 = 19$

Q009 B

Q010 C

Q011 D

Q012 A

Q013 B

Q014 C

Q015 D

Q016 A

Q017 B

Q018 C

Q019 D

Q020 A

Q021 $a_1 = 6, a_2 = 18, a_3 = 54$

Q022 $a_1 = 2, a_2 = 8, a_3 = 26$

Q023 $a_1 = -3, a_2 = 9, a_3 = -27$

Q024 $a_1 = -6, a_2 = 12, a_3 = -24$

Q025 A

Q026 B

Q027 C

Q028 D

Q029 A

Q030 B

Q031 C

Q032 D

Q033 $a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$

Q034 $a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$

Q035 $a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$

Q036 $a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$

Q037 $a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$

Q038 C

Q039 D

Q040 A

Q041 B

Q042 C

Q043 D

Q044 $a_n = (n-1)^2$

Q045 $a_n = (n-1)^3$

Q046 $a_n = n^2$

Q047 $a_n = (n+6)^2$

Q048 B

Q049 C

Q050 D

Q051 A

Q052 B

Q053 C

Q054 $a_n = 3n - 1$

Q055 $a_n = 2n - 2$

Q056 $a_n = 3n - 3$

Q057 $a_n = 5n - 1$

Q058 $a_n = 7n - 1$

Q059 A

Q060 B

Q061 C

Q062 D

Q063 $a_n = (-1)^{n-1} 2n$

Q064 $a_n = (-1)^n 3n$

Q065 $a_n = (-1)^n 5n$

Q066 $a_n = (-1)^n (3n - 2)$

Q067 $a_n = \frac{n}{2n-1}$

Q068 $a_n = \frac{2n-1}{n^2}$

Q069 $a_n = \frac{(-1)^{n-1}}{2n-1}$

Q070 $a_n = \frac{n}{n+1}$

Q071 $a_n = n 3^n$

Q072 $a_n = \frac{n}{n+1}; a_{20} = \frac{20}{21}$

Q073 (a) $a = 2, d = 3(b) a = 3, d = -5$

Q074 (a) $a = 1, d = 4(b) a = 23, d = -4$

Q075 B

Q076 A

Q077 C

Q078 D

Q079 (a)(3, 2); (b)(-13, 4); (c)(-2, 3); (d)(10, -3); (e)(6, $-\frac{2}{3}$); (f)(-4, $-\frac{1}{2}$).

Q080 B

Q081 A

Q082 C

Q083 B

Q084 (a) $d = -3, A = 5, B = -4(b) d = -5, A = 8, B = 3$

Q085 (a) $d = 7, A = 16, B = 23(b) d = -6, A = 23, B = 11$

Q086 A

Q087 C

Q088 B

Q089 (a) $d = 7, A = 22, B = 28; (b) d = 6, A = 19, B = 25; (c) d = -5, A = 14, B = -1; (d) d = -2, A = 11, B = 13; (e) d = 5, A = 14, B = 17; (f) d = -4, A = 13, B = 9, C = 1.$

Q090 (a) $a_n = 3n + 1, a_7 = 22(b) a_n = 5 - 3n, a_7 = -16$

Q091 (a) $a_n = 4n + 1, a_7 = 29; (b) a_n = 3n - 5, a_7 = 16; (c) a_n = 13 - 3n, a_7 = -8; (d) a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}.$

Q092 $a_n = 8 + (n-1)(-6) = 14 - 6n; a_{20} = -106$

Q093 (a) $a_n = 3n - 10, a_{10} = 20; (b) a_n = 1 - 3n, a_{10} = -29; (c) a_n = \frac{n+13}{2}, a_{10} = \frac{23}{2}; (d) a_n = \frac{3-n}{3}, a_{10} = -\frac{7}{3}.$

Q094 $a = 7, d = 3.$

Q095 $a_n = 5n - 10$

Q096 $a_n = 4n + 2$

Q097 $a_n = 23 - 3n$

Q098 $a_n = 9n - 17$

Q099 $a_n = -5n - 13$

Q100 D

Q101 B	Q133 A	Q165 144	Q197 C
Q102 A	Q134 B	Q166 -165	Q198 C
Q103 B	Q135 $n = 125$, the 125th term	Q167 176	Q199 $2n^2 - n$
Q104 $a_n = 3n + 33$	Q136 $n = 35$, the 35th term	Q168 360	Q200 $\frac{n(7-3n)}{2}$
Q105 $a_n = 2n - 12$	Q137 A	Q169 C	Q201 $\frac{5n(n-1)}{2}$
Q106 $a_n = 13 - 4n$	Q138 D	Q170 B	Q202 $5n - 2n^2$
Q107 $a_n = 18 - 7n$	Q139 D	Q171 C	Q203 $-n^2 - 3n$
Q108 A	Q140 B	Q172 B	Q204 414
Q109 D	Q141 $n = 72$, the 72nd term	Q173 C	Q205 5050
Q110 C	Q142 $n = 171$, the 171st term	Q174 B	Q206 20100
Q111 D	Q143 B	Q175 392	Q207 11325
Q112 $a_n = 8n - 14$	Q144 B	Q176 343	Q208 2485
Q113 $a_n = 20 - 3n$	Q145 C	Q177 741	Q209 5050
Q114 $a_n = 8n - 18$	Q146 A	Q178 354	Q210 80200
Q115 $a_n = 42 - 4n$	Q147 B	Q179 1716	Q211 31375
Q116 $a_n = 79 - 3n$	Q148 $a = 4$	Q180 C	Q212 8085
Q117 $a_n = 8n - 18$	Q149 $x = -4$ or $x = 3$	Q181 C	Q213 24535
Q118 C	Q150 $x = -2$ or $x = 4$	Q182 C	Q214 4950
Q119 C	Q151 $x = -1$ or $x = 3$	Q183 C	Q215 B
Q120 B	Q152 $x = 0$, $y = -2$, $z = -4$	Q184 C	Q216 B
Q121 A	Q153 105	Q185 B	Q217 B
Q122 A	Q154 100	Q186 0	Q218 B
Q123 B	Q155 120	Q187 208	Q219 B
Q124 $a_n = \frac{2n-37}{5}$	Q156 152	Q188 B	Q220 B
Q125 A	Q157 155	Q189 C	Q221 B
Q126 D	Q158 B	Q190 C	Q222 B
Q127 $n = 28$, the 28th term	Q159 B	Q191 C	Q223 B
Q128 $n = 14$, the 14th term	Q160 B	Q192 B	Q224 B
Q129 $n = 22$, the 22nd term	Q161 B	Q193 C	Q225 15050
Q130 $n = 40$, the 40th term	Q162 B	Q194 66	Q226 25050
Q131 C	Q163 C	Q195 750	Q227 4905
Q132 D	Q164 -130	Q196 154	

Q228 B	Q244 1225	Q260 B	Q276 $n = 10, S = 675$
Q229 B	Q245 999	Q261 B	Q277 $a_n = \frac{2n-1}{10}$
Q230 B	Q246 2475	Q262 B	Q278 $a_n = 3n - 10$
Q231 B	Q247 4195	Q263 21	Q279 $a_n = 2n - 1$
Q232 B	Q248 3750	Q264 14	Q280 6557
Q233 B	Q249 5	Q265 C	Quiz MCQ 1 A
Q234 B	Q250 7	Q266 C	Quiz MCQ 2 B
Q235 B	Q251 B	Q267 C	Quiz MCQ 3 C
Q236 B	Q252 B	Q268 C	Quiz MCQ 4 B
Q237 B	Q253 B	Q269 $a_n = 8n - 68$	Quiz WRITTEN 5 $a_1 = 2, a_2 = 9, a_3 = 22$
Q238 1665	Q254 B	Q270 $a_n = 11 - 3n$	Quiz WRITTEN 6 $a_n = 8n - 18$
Q239 945	Q255 B	Q271 $a_n = 6n - 2$	Quiz WRITTEN 7 296
Q240 1300	Q256 B	Q272 $a_n = 3n + 1$	Quiz WRITTEN 8 $a_n = 2n - 1$
Q241 810	Q257 B	Q273 $n = 17, S = 561$	
Q242 B	Q258 B	Q274 $n = 18, S = 648$	
Q243 B	Q259 B	Q275 $n = 4, S = 44$	

Concept 2 | Geometric Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1}$$

Infinite sum

$$S_{\infty} = \frac{a}{1-r} \quad (|r| < 1)$$

Classified Practice

Geometric Sequences

Q001 **Written**

Find the general term of a geometric sequence whose first term is 2 and common ratio is 3.

Q002 Written

Find the general term when the first term is 3 and the common ratio is 4.

Classified Practice

Geometric Sequences

Q003 Written

Find the general term of the geometric sequence in part a. The first term is 5, common ratio 2.

Q004 Written

Find the general term of the geometric sequence in part b. The first term is 2, common ratio 5.

Classified Practice

Geometric Sequences

Q005 **Written**

Find the general term of the geometric sequence in part c. The first term is 7, common ratio 4.

Q006 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = 3$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 2(3)^4$

B $a_n = 2(3)^5$

C $a_n = 2(3)^3$

D $a_n = 2 + (3)^4$

Classified Practice

Geometric Sequences

Q007 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = 2$, and $n = 6$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 3(2)^5$

B $a_n = 3(2)^6$

C $a_n = 3(2)^4$

D $a_n = 3 + (2)^5$

Q008 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 4$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 5(4)^3$

B $a_n = 5(4)^4$

C $a_n = 5(4)^2$

D $a_n = 5 + (4)^3$

Classified Practice

Geometric Sequences

Q009 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = 5$, and $n = 3$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 4(5)^2$

B $a_n = 4(5)^3$

C $a_n = 4(5)^1$

D $a_n = 4 + (5)^2$

Q010 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 2$, and $n = 7$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 5(2)^6$

B $a_n = 5(2)^7$

C $a_n = 5(2)^5$

D $a_n = 5 + (2)^6$

Classified Practice

Geometric Sequences

Q011 MCQ

Select the correct option: For first term $a = 7$, common ratio $r = 4$, and $n = 3$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 7(4)^2$

B $a_n = 7(4)^3$

C $a_n = 7(4)^1$

D $a_n = 7 + (4)^2$

Q012 Written

Find the general term when the first term is 5 and the common ratio is -2 .

Classified Practice

Geometric Sequences

Q013 **Written**

Find the general term of the geometric sequence in part d. The first term is 2, common ratio -3 .

Q014 Written

Find the general term of the geometric sequence in part e. The first term is 3, common ratio -5 .

Classified Practice

Geometric Sequences

Q015 Written

Find the general term of the geometric sequence in part f. The first term is 4, common ratio -2 .

Q016 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = -3$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 2(-3)^4$

B $a_n = 2(-3)^5$

C $a_n = 2(-3)^3$

D $a_n = 2 + (-3)^4$

Classified Practice

Geometric Sequences

Q017 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -2$, and $n = 6$, which expression gives a_n ?

Choose the correct option.

A $a_n = 3(-2)^5$

B $a_n = 3(-2)^6$

C $a_n = 3(-2)^4$

D $a_n = 3 + (-2)^5$

Q018 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = -1$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 6(-1)^3$

B $a_n = 6(-1)^4$

C $a_n = 6(-1)^2$

D $a_n = 6 + (-1)^3$

Classified Practice

Geometric Sequences

Q019 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -2$, and $n = 3$, which expression gives a_n ?

Choose the correct option.

A $a_n = 8(-2)^2$

B $a_n = 8(-2)^3$

C $a_n = 8(-2)^1$

D $a_n = 8 + (-2)^2$

Q020 MCQ

Select the correct option: For first term $a = 9$, common ratio $r = -2$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 9(-2)^4$

B $a_n = 9(-2)^5$

C $a_n = 9(-2)^3$

D $a_n = 9 + (-2)^4$

Classified Practice

Geometric Sequences

Q021 Written

Find the general term when the first term is 8 and the common ratio is $-\frac{1}{3}$.

Q022 Written

Find the general term of the geometric sequence in part g. The first term is 4, common ratio $\frac{1}{3}$.

Classified Practice

Geometric Sequences

Q023 Written

Find the general term of the geometric sequence in part h. The first term is 3, common ratio $-\frac{1}{2}$.

Q024 Written

Find the general term of the geometric sequence in part i. The first term is 6, common ratio $\frac{1}{2}$.

Classified Practice

Geometric Sequences

Q025 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = \frac{1}{3}$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 4\left(\frac{1}{3}\right)^4$

B $a_n = 4\left(\frac{1}{3}\right)^5$

C $a_n = 4\left(\frac{1}{3}\right)^3$

D $a_n = 4 + \left(\frac{1}{3}\right)^4$

Q026 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -\frac{1}{2}$, and $n = 6$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 3\left(-\frac{1}{2}\right)^5$

B $a_n = 3\left(-\frac{1}{2}\right)^6$

C $a_n = 3\left(-\frac{1}{2}\right)^4$

D $a_n = 3 + \left(-\frac{1}{2}\right)^5$

Classified Practice

Geometric Sequences

Q027 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = \frac{1}{2}$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 6 \left(\frac{1}{2}\right)^3$

B $a_n = 6 \left(\frac{1}{2}\right)^4$

C $a_n = 6\left(\frac{1}{2}\right)^2$

D $a_n = 6 + \left(\frac{1}{2}\right)^3$

Q028 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = \frac{3}{2}$, and $n = 5$, which expression gives a_n ?

Choose the correct option.

A $a_n = 2 \left(\frac{3}{2}\right)^4$

B $a_n = 2 \left(\frac{3}{2}\right)^5$

C $a_n = 2 \left(\frac{3}{2}\right)^3$

D $a_n = 2 + \left(\frac{3}{2}\right)^4$

Classified Practice

Geometric Sequences

Q029 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -\frac{3}{2}$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 8 \left(-\frac{3}{2}\right)^3$

B $a_n = 8 \left(-\frac{3}{2}\right)^4$

C $a_n = 8 \left(-\frac{3}{2}\right)^2$

D $a_n = 8 + \left(-\frac{3}{2}\right)^3$

Q030 Written**Find the general term of 5, 10, 20, 40,**

Classified Practice

Geometric Sequences

Q031 **Written****Find the general term of 2, 6, 18, 54,**

Q032 Written

Find the general term of the geometric sequence in part j. The sequence is 3, 6, 12, 24,

Classified Practice

Geometric Sequences

Q033 MCQ

Select the correct option: Find the general term of the geometric sequence 3, 6, 12, 24, ...:**Choose the correct option.**

A $3 \times 2^{n-1}$

B 3×2^n

C $3 \times 2^{n-2}$

D 0

Q034 MCQ

Select the correct option: Find the general term of the geometric sequence 2, 10, 50, 250, ...:**Choose the correct option.**

A $2 \times 5^{n-1}$

B 2×5^n

C $2 \times 5^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q035 MCQ

Select the correct option: Find the general term of the geometric sequence 4, 8, 16, 32, ...:**Choose the correct option.**

A $4 \times 2^{n-1}$

B 4×2^n

C $4 \times 2^{n-2}$

D 0

Q036 MCQ**Select the correct option: Find the general term of the geometric sequence 7, 21, 63, 189, ...:****Choose the correct option.**

A $7 \times 3^{n-1}$

B 7×3^n

C $7 \times 3^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q037 MCQ

Select the correct option: Find the general term of the geometric sequence 1, 4, 16, 64, ...:**Choose the correct option.**

A 4^{n-1}

B 4^n

C 4^{n-2}

D 0

Q038 Written

Find the general term of the geometric sequence in part k. The sequence is $2, -6, 18, -54, \dots$

Classified Practice

Geometric Sequences

Q039 Written

Find the general term of the geometric sequence in part I. The sequence is $5, -10, 20, -40, \dots$

Q040 MCQ**Select the correct option: Find the general term of the geometric sequence $5, -10, 20, -40, \dots$:****Choose the correct option.**

A $5(-2)^{n-1}$

B $5(-2)^n$

C $5(-2)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q041 MCQ

Select the correct option: Find the general term of the geometric sequence $2, -6, 18, -54, \dots$:**Choose the correct option.**

A $2(-3)^{n-1}$

B $2(-3)^n$

C $2(-3)^{n-2}$

D 0

Q042 MCQ**Select the correct option: Find the general term of the geometric sequence $-4, 8, -16, 32, \dots$:****Choose the correct option.**

A $-4(-2)^{n-1}$

B $-4(-2)^n$

C $-4(-2)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q043 MCQ

Select the correct option: Find the general term of the geometric sequence $6, -18, 54, -162, \dots$:**Choose the correct option.**

A $6(-3)^{n-1}$

B $6(-3)^n$

C $6(-3)^{n-2}$

D 0

Q044 Written**Find the general term of $-4, 6, -9, \frac{27}{2}, \dots$**

Classified Practice

Geometric Sequences

Q045 **Written****Find the general term of $-8, 12, -18, 27, \dots$**

Q046 Written**Find the general term of $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots$**

Classified Practice

Geometric Sequences

Q047 Written

Find the general term of the geometric sequence in part m. The sequence is 48, 24, 12, 6,

Q048 Written

Find the general term of the geometric sequence in part n. The sequence is $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \dots$

Classified Practice

Geometric Sequences

Q049 Written

Find the general term of the geometric sequence in part o. The sequence is $8, 16\sqrt{2}, 64, 128\sqrt{2}, \dots$

Q050 Written

Find the general term of the geometric sequence in part p. The sequence is $3, -\frac{15}{2}, \frac{75}{4}, -\frac{375}{8}, \dots$

Classified Practice

Geometric Sequences

Q051 Written

Find the general term of the geometric sequence in part q. The sequence is $9, -3\sqrt{3}, 3, -\sqrt{3}, \dots$

Q052 Written

Find the general term of the geometric sequence in part r. The sequence is $5, -\frac{10}{3}, \frac{20}{9}, -\frac{40}{27}, \dots$

Classified Practice

Geometric Sequences

Q053 MCQ

Select the correct option: Find the general term of the geometric sequence $-8, 12, -18, 27, \dots$:**Choose the correct option.**

A $-8 \left(-\frac{3}{2}\right)^{n-1}$

B $-8 \left(-\frac{3}{2}\right)^n$

C $-8 \left(-\frac{3}{2}\right)^{n-2}$

D 0

Q054 MCQ

Select the correct option: Find the general term of the geometric sequence 48, 24, 12, 6, ...:**Choose the correct option.**

A $48 \left(\frac{1}{2}\right)^{n-1}$

B $48 \left(\frac{1}{2}\right)^n$

C $48 \left(\frac{1}{2}\right)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q055 MCQ

Select the correct option: Find the general term of the geometric sequence $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \dots$:

Choose the correct option.

- A $\frac{1}{6} \left(\frac{3}{2}\right)^{n-1}$
B $\frac{1}{6} \left(\frac{3}{2}\right)^n$
C $\frac{1}{6} \left(\frac{3}{2}\right)^{n-2}$
D 0

Q056 MCQ

Select the correct option: Find the general term of the geometric sequence $8, 16\sqrt{2}, 64, \dots$:

Choose the correct option.

A $8(2\sqrt{2})^{n-1}$

B $8(2\sqrt{2})^n$

C $8(2\sqrt{2})^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q057 MCQ

Select the correct option: Find the general term of the geometric sequence $9, -3\sqrt{3}, 3, \dots$:

Choose the correct option.

A $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

B $9\left(-\frac{\sqrt{3}}{3}\right)^n$

C $9\left(-\frac{\sqrt{3}}{3}\right)^{n-2}$

D 0

Q058 MCQ

Select the correct option: Find the general term of the geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \dots$:

Choose the correct option.

A $3\left(-\frac{1}{2}\right)^{n-1}$

B $3\left(-\frac{1}{2}\right)^n$

C $3\left(-\frac{1}{2}\right)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q059 MCQ

Select the correct option: Find the general term of the geometric sequence $10, 5, \frac{5}{2}, \dots$:

Choose the correct option.

- A $10\left(\frac{1}{2}\right)^{n-1}$
- B $10\left(\frac{1}{2}\right)^n$
- C $10\left(\frac{1}{2}\right)^{n-2}$
- D 0

Q060 Written**Find the common ratio and missing values in $A, -18, 6, B, \dots$**

Classified Practice

Geometric Sequences

Q061 MCQ

Select the correct option: The sequence $A, 12, 6, B, \dots$ is geometric. Find r, A, B :

Choose the correct option.

A $r = \frac{1}{2}, A = 24, B = 3$

B $r = \frac{2}{3}, A = 1, B = 2$

C $r = -1, A = 0, B = 0$

D $r = 3, A = 3, B = 9$

Q062 Written

Find the common ratio and the missing values in $A, 24, 12, B, \dots$

Classified Practice

Geometric Sequences

Q063 Written

Find the common ratio and missing values in $A, -3, 9, B$.

Q064 Written

Find the common ratio and missing values in $A, \frac{1}{3}, \frac{1}{9}, B$.

Classified Practice

Geometric Sequences

Q065 Written

Find the common ratio and missing values in $A, -\frac{1}{4}, \frac{1}{8}, B$.

Classified Practice

Geometric Sequences

Q066 Challenge

In an increasing geometric sequence, if $a_3 + a_5 = 80$ and $a_4 = 24$, find a_7 .

Classified Practice

Geometric Sequences

Q067 **Written**

Find the general term of a geometric sequence whose 4th term is 24 and 6th term is 96.

Q068 Written**Find the general term when the 3rd term is 18 and the 5th term is 162.**

Classified Practice

Geometric Sequences

Q069 **Written**

Find the requested values. The 2nd term is 6 and the 4th term is 54.

Q070 Written

Find the requested values. The 3rd term is 27 and the 5th term is 243.

Classified Practice

Geometric Sequences

Q071 Written

Find the requested values. The 2nd term is 14 and the 5th term is 112.

Q072 Written**Find the general term when the 3rd term is 12 and the 6th term is -96 .**

Classified Practice

Geometric Sequences

Q073 Written

Find the general term when the 2nd term is -6 and the 5th term is 48 .

Q074 Written**Find the general term when the 3rd term is 12 and the 7th term is 192.**

Classified Practice

Geometric Sequences

Q075 Written

Find the requested values. The 3rd term is -28 and the 5th term is -112 .

Q076 Written

Find the requested values. The 4th term is 16 and the 7th term is -128 .

Classified Practice

Geometric Sequences

Q077 Written

Find the requested values. The 3rd term is -20 and the 6th term is 160 .

Q078 Written

Find the requested values. The 2nd term is 3 and the 6th term is 768.

Classified Practice

Geometric Sequences

Q079 Written

Find the requested values. The 3rd term is 9 and the 7th term is $729/16$.

Q080 Written

Find the requested values. The 4th term is $\frac{2}{9}$ and the 8th term is 18.

Classified Practice

Geometric Sequences

Q081 MCQ

Select the correct option: For $a = 2$, $r = 3$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 2(3)^{n-1}$

B $a_n = 2(3)^n$

C $a_n = 2(3)^{n+1}$

D $a_n = 2 + 3^{n-1}$

Q082 MCQ

Select the correct option: For $a = 3$, $r = 2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 3(2)^{n-1}$

B $a_n = 3(2)^n$

C $a_n = 3(2)^{n+1}$

D $a_n = 3 + 2^{n-1}$

Classified Practice

Geometric Sequences

Q083 MCQ

Select the correct option: For $a = 5$, $r = 4$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 5(4)^{n-1}$

B $a_n = 5(4)^n$

C $a_n = 5(4)^{n+1}$

D $a_n = 5 + 4^{n-1}$

Q084 MCQ

Select the correct option: For $a = 4$, $r = 5$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 4(5)^{n-1}$

B $a_n = 4(5)^n$

C $a_n = 4(5)^{n+1}$

D $a_n = 4 + 5^{n-1}$

Classified Practice

Geometric Sequences

Q085 MCQ

Select the correct option: For $a = 2$, $r = -3$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 2(-3)^{n-1}$

B $a_n = 2(-3)^n$

C $a_n = 2(-3)^{n+1}$

D $a_n = 2 - 3^{n-1}$

Q086 MCQ

Select the correct option: For $a = 3$, $r = -2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 3(-2)^{n-1}$

B $a_n = 3(-2)^n$

C $a_n = 3(-2)^{n+1}$

D $a_n = 3 - 2^{n-1}$

Classified Practice

Geometric Sequences

Q087 MCQ

Select the correct option: For $a = 6$, $r = -1$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 6(-1)^{n-1}$

B $a_n = 6(-1)^n$

C $a_n = 6(-1)^{n+1}$

D $a_n = 6 - 1^{n-1}$

Q088 MCQ**Select the correct option: For $a = 8$, $r = -2$, which expression gives the general term a_n :****Choose the correct option.**

A $a_n = 8(-2)^{n-1}$

B $a_n = 8(-2)^n$

C $a_n = 8(-2)^{n+1}$

D $a_n = 8 - 2^{n-1}$

Classified Practice

Geometric Sequences

Q089 MCQ

Select the correct option: For $a = 4$, $r = \frac{1}{3}$, which expression gives the general term a_n :

Choose the correct option.

- A $a_n = 4\left(\frac{1}{3}\right)^{n-1}$
- B $a_n = 4\left(\frac{1}{3}\right)^n$
- C $a_n = 4\left(\frac{1}{3}\right)^{n+1}$
- D $a_n = 4 + \frac{1}{3}^{n-1}$

Q090 MCQ

Select the correct option: For $a = 3$, $r = -\frac{1}{2}$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 3 \left(-\frac{1}{2}\right)^{n-1}$

B $a_n = 3 \left(-\frac{1}{2}\right)^n$

C $a_n = 3 \left(-\frac{1}{2}\right)^{n+1}$

D $a_n = 3 - \frac{1}{2}^{n-1}$

Classified Practice

Geometric Sequences

Q091 MCQ

Select the correct option: For $a = 6$, $r = \frac{1}{2}$, which expression gives the general term a_n :

Choose the correct option.

- A $a_n = 6 \left(\frac{1}{2}\right)^{n-1}$
- B $a_n = 6 \left(\frac{1}{2}\right)^n$
- C $a_n = 6 \left(\frac{1}{2}\right)^{n+1}$
- D $a_n = 6 + \frac{1}{2}^{n-1}$

Q092 MCQ

Select the correct option: For $a = 2$, $r = \frac{3}{2}$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 2 \left(\frac{3}{2}\right)^{n-1}$

B $a_n = 2 \left(\frac{3}{2}\right)^n$

C $a_n = 2 \left(\frac{3}{2}\right)^{n+1}$

D $a_n = 2 + \frac{3^{n-1}}{2}$

Classified Practice

Geometric Sequences

Q093 MCQ

Select the correct option: For $a = 8$, $r = -\frac{3}{2}$, which expression gives the general term a_n :

Choose the correct option.

- A $a_n = 8 \left(-\frac{3}{2}\right)^{n-1}$
- B $a_n = 8 \left(-\frac{3}{2}\right)^n$
- C $a_n = 8 \left(-\frac{3}{2}\right)^{n+1}$
- D $a_n = 8 - \frac{3^{n-1}}{2}$

Q094 MCQ

Select the correct option: For $a = 5$, $r = 2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 5(2)^{n-1}$

B $a_n = 5(2)^n$

C $a_n = 5(2)^{n+1}$

D $a_n = 5 + 2^{n-1}$

Classified Practice

Geometric Sequences

Q095 MCQ

Select the correct option: For $a = 7$, $r = 4$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 7(4)^{n-1}$

B $a_n = 7(4)^n$

C $a_n = 7(4)^{n+1}$

D $a_n = 7 + 4^{n-1}$

Q096 MCQ

Select the correct option: For $a = 9$, $r = -2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 9(-2)^{n-1}$

B $a_n = 9(-2)^n$

C $a_n = 9(-2)^{n+1}$

D $a_n = 9 - 2^{n-1}$

Classified Practice

Geometric Sequences

Q097 Written

When 2, x , 8 form a geometric sequence in this order, find x .

Q098 Written**Find the requested values. When 1, x , 25 form a geometric sequence, find x .**

Classified Practice

Geometric Sequences

Q099 Written

Find the requested values. When $1/2, x, 8$ form a geometric sequence, find x .

Q100 Written

When 1, x , 12 form a geometric sequence in this order, find x .

Classified Practice

Geometric Sequences

Q101 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 8$, find x :

Choose the correct option.

A ± 4

B $+4$

C -4

D 0

Q102 MCQ**Select the correct option: If x is the middle term of the geometric triple $1, x, 25$, find x :****Choose the correct option.**A ± 5 B $+5$ C -5 D 0

Classified Practice

Geometric Sequences

Q103 MCQ

Select the correct option: If x is the middle term of the geometric triple $\frac{1}{2}, x, 8$, find x :

Choose the correct option.

- A ± 2
- B $+2$
- C -2
- D 0

Q104 MCQ

Select the correct option: If x is the middle term of the geometric triple $3, x, 12$, find x :

Choose the correct option.

A ± 6

B $+6$

C -6

D 0

Classified Practice

Geometric Sequences

Q105 MCQ

Select the correct option: If x is the middle term of the geometric triple 4, x , 9, find x :

Choose the correct option.

A ± 6

B $+6$

C -6

D 0

Q106 MCQ**Select the correct option: If x is the middle term of the geometric triple $2, x, 18$, find x :****Choose the correct option.**A ± 6 B $+6$ C -6 D 0

Classified Practice

Geometric Sequences

Q107 Written

The numbers $6, a, b$ form an arithmetic sequence and $a, b, 16$ form a geometric sequence. Find a, b .

Q108 Written

Find the requested values. The numbers $a, b, 2$ form an arithmetic sequence and $a, 2, b$ form a geometric sequence. Find a, b .

Classified Practice

Geometric Sequences

Q109 Written

The numbers $-2, a, b$ form an arithmetic sequence and $a, b, 18$ form a geometric sequence. Find a, b .

Classified Practice

Geometric Sequences

Q110 Challenge

A planning model gives $-5, a, b$ as an arithmetic sequence, while $a, b, 45$ form a geometric sequence. Find a, b .

Classified Practice

Geometric Sequences

Q111 Written

Find the sum S of a geometric sequence with first term 4, common ratio 3, and 5 terms.

Q112 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 5 terms.

Classified Practice

Geometric Sequences

Q113 Written

Find the sum S of a geometric sequence with first term 5, common ratio 3, and 4 terms.

Q114 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 10 terms.

Classified Practice

Geometric Sequences

Q115 MCQ

For a geometric sequence with $a = 4$, $r = 3$, and $n = 4$, calculate S_n :

Choose the correct option.

A 161

B 159

C 162

D 160

Q116 MCQ

For a geometric sequence with $a = 5$, $r = 2$, and $n = 6$, calculate S_n :

Choose the correct option.

A 316

B 314

C 315

D 317

Classified Practice

Geometric Sequences

Q117 MCQ

For a geometric sequence with $a = 2$, $r = 4$, and $n = 5$, calculate S_n :

Choose the correct option.

A 683

B 682

C 681

D 684

Q118 MCQ**For a geometric sequence with $a = 3$, $r = 5$, and $n = 4$, calculate S_n :****Choose the correct option.****A** 468**B** 469**C** 467**D** 470

Classified Practice

Geometric Sequences

Q119 MCQ

For a geometric sequence with $a = 7$, $r = 2$, and $n = 5$, calculate S_n :

Choose the correct option.

A 218

B 216

C 219

D 217

Q120 MCQ**For a geometric sequence with $a = 6$, $r = 3$, and $n = 3$, calculate S_n :****Choose the correct option.****A** 79**B** 77**C** 78**D** 80

Classified Practice

Geometric Sequences

Q121 Written

Find the sum S of a geometric sequence with first term 8, common ratio $\frac{1}{2}$, and 6 terms.

Q122 Written

Find the sum S of a geometric sequence with first term 6, common ratio $\frac{1}{2}$, and 6 terms.

Classified Practice

Geometric Sequences

Q123 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{3}$, and 3 terms.

Q124 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{2}$, and 8 terms.

Classified Practice

Geometric Sequences

Q125 MCQ

For a geometric sequence with $a = 8$, $r = \frac{1}{2}$, and $n = 4$, calculate S_n :

Choose the correct option.

A 16

B 15

C 14

D 17

Q126 MCQ

For a geometric sequence with $a = 9$, $r = \frac{1}{3}$, and $n = 3$, calculate S_n :

Choose the correct option.

- A 13
- B 14
- C 12
- D 15

Classified Practice

Geometric Sequences

Q127 MCQ

For a geometric sequence with $a = 12$, $r = \frac{1}{2}$, and $n = 5$, calculate S_n :

Choose the correct option.

A $\frac{47}{2}$

B $\frac{23}{95}$

C $\frac{4}{93}$

D $\frac{93}{4}$

Q128 MCQ

For a geometric sequence with $a = 5$, $r = \frac{1}{4}$, and $n = 3$, calculate S_n :

Choose the correct option.

A $\frac{53}{8}$

B $\frac{13}{2}$

C $\frac{105}{16}$

D $\frac{107}{16}$

Classified Practice

Geometric Sequences

Q129 MCQ

For a geometric sequence with $a = 10$, $r = \frac{1}{2}$, and $n = 6$, calculate S_n :

Choose the correct option.

A $\frac{79}{4}$

B $\frac{315}{16}$

C $\frac{157}{8}$

D $\frac{317}{16}$

Q130 MCQ

For a geometric sequence with $a = 4$, $r = \frac{1}{3}$, and $n = 4$, calculate S_n :

Choose the correct option.

- A $\frac{160}{27}$
- B $\frac{161}{27}$
- C $\frac{53}{9}$
- D 6

Classified Practice

Geometric Sequences

Q131 Written

Find the sum S of a geometric sequence with first term 6, common ratio -2 , and 4 terms.

Q132 Written

Find the sum S of a geometric sequence with first term 1, common ratio -3 , and 4 terms.

Classified Practice

Geometric Sequences

Q133 Written

Find the sum S of a geometric sequence with first term 2, common ratio -3 , and 6 terms.

Q134 MCQ**For a geometric sequence with $a = 4$, $r = -2$, and $n = 5$, calculate S_n :****Choose the correct option.****A** 45**B** 43**C** 46**D** 44

Classified Practice

Geometric Sequences

Q135 MCQ

For a geometric sequence with $a = 3$, $r = -3$, and $n = 4$, calculate S_n :

Choose the correct option.

A -59

B -61

C -60

D -58

Q136 MCQ

For a geometric sequence with $a = 2$, $r = -1$, and $n = 6$, calculate S_n :

Choose the correct option.

A 1

B 0

C -1

D 2

Classified Practice

Geometric Sequences

Q137 MCQ

For a geometric sequence with $a = 5$, $r = -2$, and $n = 6$, calculate S_n :

Choose the correct option.

A -105 B -104 C -106 D -103

Q138 MCQ

For a geometric sequence with $a = 7$, $r = -1$, and $n = 5$, calculate S_n :

Choose the correct option.

A 8

B 6

C 9

D 7

Classified Practice

Geometric Sequences

Q139 MCQ

For a geometric sequence with $a = 6$, $r = -3$, and $n = 3$, calculate S_n :

Choose the correct option.

A 43

B 41

C 42

D 44

Q140 Written

Find S_n from the first term to the n -th term when the first term is 3 and the common ratio is -2 .

Classified Practice

Geometric Sequences

Q141 **Written**

Find S_n from the first term to the n -th term when the first term is 7 and the common ratio is 2.

Q142 Written

Find S_n from the first term to the n -th term when the first term is 2 and the common ratio is $\frac{1}{3}$.

Classified Practice

Geometric Sequences

Q143 **Written**

Find S_n from the first term to the n -th term when the first term is 4 and the common ratio is 2.

Q144 MCQ

For the geometric sequence with $a = 3, r = 2$, find the sum S_n from the first term to the n -th term:

Choose the correct option.

A $3(2^{n-1} - 1)$

B $6(2^n - 1)$

C $3(2^n - 1)$

D $3(1 - 2^n)$

Classified Practice

Geometric Sequences

Q145 MCQ

For the geometric sequence with $a = 5, r = \frac{1}{2}$, find the sum S_n from the first term to the n -th term:

Choose the correct option.

A $5\{1 - (\frac{1}{2})^n\}$

B $10\{1 - (\frac{1}{2})^{n-1}\}$

C $10\{1 - (\frac{1}{2})^n\}$

D $10\{1 - (\frac{1}{2})^n\}$

Q146 MCQ

For the geometric sequence with $a = 4, r = -2$, find the sum S_n from the first term to the n -th term:

Choose the correct option.

A $\frac{4\{1 - (-2)^n\}}{3}$

B $\frac{4\{1 - (-2)^{n-1}\}}{3}$

C $\frac{4\{1 + (-2)^n\}}{3}$

D $\frac{4\{1 - 2^n\}}{3}$

Classified Practice

Geometric Sequences

Q147 **Written****Find S_n from the first term to the n -th term of 2, 6, 18, 54,**

Q148 Written

Find S_n from the first term to the n -th term of 2, -6 , 18, -54 ,

Classified Practice

Geometric Sequences

Q149 Written

Find S_n from the first term to the n -th term of $8, -4, 2, -1, \dots$

Q150 Written

Find S_n from the first term to the n -th term of $1, -3, 9, -27, \dots$

Classified Practice

Geometric Sequences

Q151 Written

Find S_n from the first term to the n -th term when the first term is 16 and the common ratio is $-\frac{1}{2}$.

Q152 Written

Find S_n from the first term to the n -th term of $1, -2, 4, -8, \dots$

Classified Practice

Geometric Sequences

Q153 MCQ

A culture grows by 3, 9, 27, 81 units. Find the four-stage total:

Choose the correct option.

A 120

B 121

C 119

D 122

Classified Practice

Geometric Sequences

Q154 Challenge

A filtration scale is geometric: $3, 1, \frac{1}{3}, \frac{1}{9}, \dots$ mg/L removed per stage. Calculate the total removed in 4 stages.

Classified Practice

Geometric Sequences

Q155 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is 28, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (2, 2)$

B $(a, r) = (1, 3)$

C $(a, r) = \left(\frac{10}{3}, -2\right)$

D $(a, r) = (1, 2)$ or $(a, r) = \left(\frac{7}{3}, -2\right)$

Classified Practice

Geometric Sequences

Q156 MCQ

Given $S_3 = 26$ and the sum from the 3rd to the 5th terms is 234, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, 3)$

B $(a, r) = (2, 4)$

C $(a, r) = (2, 3)$ or $(a, r) = \left(\frac{26}{7}, -3\right)$

D $(a, r) = \left(\frac{33}{7}, -3\right)$

Classified Practice

Geometric Sequences

Q157 MCQ

Given $S_3 = 21$ and the sum from the 3rd to the 5th terms is 84, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (4, 2)$

B $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

C $(a, r) = (3, 3)$

D $(a, r) = (8, -2)$

Classified Practice

Geometric Sequences

Q158 MCQ

Given $S_3 = 28$ and the sum from the 3rd to the 5th terms is 112, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (4, 2)$ or $(a, r) = \left(\frac{28}{3}, -2\right)$

B $(a, r) = (5, 2)$

C $(a, r) = (4, 3)$

D $(a, r) = \left(\frac{31}{3}, -2\right)$

Classified Practice

Geometric Sequences

Q159 MCQ

Given $S_3 = 13$ and the sum from the 3rd to the 5th terms is 117, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (2, 3)$

B $(a, r) = (1, 4)$

C $(a, r) = \left(\frac{20}{7}, -3\right)$

D $(a, r) = (1, 3)$ or $(a, r) = \left(\frac{13}{7}, -3\right)$

Classified Practice

Geometric Sequences

Q160 MCQ

Given $S_3 = 35$ and the sum from the 3rd to the 5th terms is 140, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (6, 2)$

B $(a, r) = (5, 3)$

C $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$

D $(a, r) = \left(\frac{38}{3}, -2\right)$

Classified Practice

Geometric Sequences

Q161 MCQ

Given $S_3 = \frac{19}{2}$ and the sum from the 3rd to the 5th terms is $\frac{171}{8}$, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = \left(3, \frac{3}{2}\right)$

B $(a, r) = \left(2, \frac{5}{2}\right)$

C $(a, r) = \left(\frac{45}{7}, -\frac{3}{2}\right)$

D $(a, r) = \left(2, \frac{3}{2}\right)$ or $(a, r) = \left(\frac{38}{7}, -\frac{3}{2}\right)$

Classified Practice

Geometric Sequences

Q162 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 4th to the 6th term is -72 .

Classified Practice

Geometric Sequences

Q163 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is $\frac{7}{4}$, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = \left(5, \frac{1}{2}\right)$

B $(a, r) = \left(4, \frac{3}{2}\right)$

C $(a, r) = \left(4, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{28}{3}, -\frac{1}{2}\right)$

D $(a, r) = \left(\frac{31}{3}, -\frac{1}{2}\right)$

Classified Practice

Geometric Sequences

Q164 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 35$ and the sum from the 3rd to the 5th term is 140.

Q165 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and the sum from the 3rd to the 5th term is 84.

Classified Practice

Geometric Sequences

Q166 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 3rd to the 5th term is 36.

Q167 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 78$ and the sum from the 3rd to the 5th term is 702.

Classified Practice

Geometric Sequences

Q168 MCQ

Given $S_3 = 6$ and the sum from the 3rd to the 5th terms is 24, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, -2)$

B $(a, r) = (2, -2)$ or $(a, r) = \left(\frac{6}{7}, 2\right)$

C $(a, r) = (2, -1)$

D $(a, r) = \left(\frac{13}{7}, 2\right)$

Classified Practice

Geometric Sequences

Q169 MCQ

Given $S_3 = 9$ and the sum from the 3rd to the 5th terms is 36, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, -2)$ or $(a, r) = \left(\frac{9}{7}, 2\right)$

B $(a, r) = (4, -2)$

C $(a, r) = (3, -1)$

D $(a, r) = \left(\frac{16}{7}, 2\right)$

Classified Practice

Geometric Sequences

Q170 **Written**

Find the first term and common ratio of a geometric sequence if the 2nd term is 3 and $S_3 = 13$.

Q171 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 6, and $S_3 = 21$.

Classified Practice

Geometric Sequences

Q172 MCQ

The second term is 6 and $S_3 = 21$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (4, 2)$

B $(a, r) = (3, 3)$

C $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

D $(a, r) = \left(13, \frac{1}{2}\right)$

Classified Practice

Geometric Sequences

Q173 MCQ

The second term is 6 and $S_3 = 26$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, 3)$

B $(a, r) = (2, 4)$

C $(a, r) = \left(19, \frac{1}{3}\right)$

D $(a, r) = (2, 3)$ or $(a, r) = \left(18, \frac{1}{3}\right)$

Classified Practice

Geometric Sequences

Q174 MCQ

The second term is 8 and $S_3 = 28$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (5, 2)$

B $(a, r) = (4, 2)$ or $(a, r) = \left(16, \frac{1}{2}\right)$

C $(a, r) = (4, 3)$

D $(a, r) = \left(17, \frac{1}{2}\right)$

Classified Practice

Geometric Sequences

Q175 MCQ

The second term is 3 and $S_3 = 13$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (2, 3)$

B $(a, r) = (1, 4)$

C $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

D $(a, r) = \left(10, \frac{1}{3}\right)$

Classified Practice

Geometric Sequences

Q176 MCQ

The second term is $\frac{5}{2}$ and $S_3 = \frac{35}{4}$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = \left(6, \frac{1}{2}\right)$

B $(a, r) = \left(5, \frac{3}{2}\right)$

C $(a, r) = \left(\frac{9}{4}, 2\right)$

D $(a, r) = \left(5, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{5}{4}, 2\right)$

Classified Practice

Geometric Sequences

Q177 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 4 and $S_3 = 7$.

Q178 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 12, and $S_3 = 9$.

Classified Practice

Geometric Sequences

Q179 **Written**

Find the first term and common ratio of a geometric sequence if the 3rd term is 20, and $S_3 = 35$.

Q180 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and $S_6 = 189$.

Classified Practice

Geometric Sequences

Q181 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 26$ and $S_6 = 728$.

Q182 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 65$ and $S_6 = 1820$.

Classified Practice

Geometric Sequences

Q183 **Written**

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is 3, and $S_4 = 51$.

Q184 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is -8 , and $S_4 = -80$.

Classified Practice

Geometric Sequences

Q185 MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Choose the correct option.

A 537

B 487

C 562

D 512

Classified Practice

Geometric Sequences

Q186 Challenge

If S_n is the sum of the first n terms of a geometric sequence, $S_8 = 255$, $S_6 = 63$, and $a_8 - a_7 = 64$, find a_{10} .

Classified Practice

Geometric Sequences

Quiz MCQ 1 [Concept Quiz](#) · MCQ

Select the correct option: For $a = 3, r = 2$, find a_6 :

Choose the correct option.

A 48

B 96

C 192

D 64

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If 2, x , 8 are consecutive geometric terms, find x :

Choose the correct option.

- A 4
- B -4
- C ± 4
- D 16

Classified Practice

Geometric Sequences

Quiz MCQ 3 Concept Quiz · MCQ

For 1, 3, 9, ..., find S_n :

Choose the correct option.

A $\frac{3^{n-1} - 1}{2}$

B $\frac{3^n - 1}{2}$

C $\frac{3^n + 1}{2}$

D $\frac{3^n - 1}{3}$

Quiz MCQ 4 Concept Quiz · MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Choose the correct option.

A 528

B 496

C 544

D 512

Classified Practice

Geometric Sequences

Quiz WRITTEN 5 Concept Quiz · Written**Find the general term of $-4, 8, -16, 32, \dots$**

Quiz WRITTEN 6 Concept Quiz · Written

If 1, x , 25 are consecutive geometric terms, find x .

Classified Practice

Geometric Sequences

Quiz WRITTEN 7 Concept Quiz · Written

A four-stage process has outputs 5, 15, 45, 135. Find the total output.

Quiz WRITTEN 8 Concept Quiz · Written

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} .

Answer key

Use the question number shown on each page.

Q001 $2 \times 3^{n-1}$

Q002 $3 \times 4^{n-1}$

Q003 $5 \times 2^{n-1}$

Q004 $2 \times 5^{n-1}$

Q005 $7 \times 4^{n-1}$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 $5(-2)^{n-1}$

Q013 $2(-3)^{n-1}$

Q014 $3(-5)^{n-1}$

Q015 $4(-2)^{n-1}$

Q016 A

Q017 A

Q018 A

Q019 A

Q020 A

Q021 $8\left(-\frac{1}{3}\right)^{n-1}$

Q022 $4\left(\frac{1}{3}\right)^{n-1}$

Q023 $3\left(-\frac{1}{2}\right)^{n-1}$

Q024 $6\left(\frac{1}{2}\right)^{n-1}$

Q025 A

Q026 A

Q027 A

Q028 A

Q029 A

Q030 $5 \times 2^{n-1}$

Q031 $2 \times 3^{n-1}$

Q032 $3 \times 2^{n-1}$

Q033 A

Q034 A

Q035 A

Q036 A

Q037 A

Q038 $2(-3)^{n-1}$

Q039 $5(-2)^{n-1}$

Q040 A

Q041 A

Q042 A

Q043 A

Q044 $-4\left(-\frac{3}{2}\right)^{n-1}$

Q045 $-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 $2\left(\frac{1}{3}\right)^{n-1}$

Q047 $48\left(\frac{1}{2}\right)^{n-1}$

Q048 $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q049 $8(2\sqrt{2})^{n-1}$

Q050 $3\left(-\frac{5}{2}\right)^{n-1}$

Q051 $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q052 $5\left(-\frac{2}{3}\right)^{n-1}$

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $r = -\frac{1}{3}, A = 54, B = -2$

Q061 A

Q062 $r = \frac{1}{2}, A = 48, B = 6$

Q063 $r = -3, A = 1, B = -27$

Q064 $r = \frac{1}{3}, A = 1, B = \frac{1}{27}$

Q065 $r = -\frac{1}{2}, A = \frac{1}{2}, B = -\frac{1}{16}$

Q066 $a_7 = 648$

Q067 $a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 $a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Q069 $a_n = 2(3)^{n-1}$ or $-2(-3)^{n-1}$

Q070 $a_n = 3(3)^{n-1}$ or $3(-3)^{n-1}$

Q071 $a_n = 7(2)^{n-1}$

Q072 $a_n = 3(-2)^{n-1}$

Q073 $a_n = 3(-2)^{n-1}$

Q074 $a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Q075 $a_n = -7(2)^{n-1}$ or $-7(-2)^{n-1}$

Q076 $a_n = -2(-2)^{n-1}$

Q077 $a_n = -5(-2)^{n-1}$

Q078 $a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Q079 $a_n = 4\left(\frac{3}{2}\right)^{n-1}$ or $4\left(-\frac{3}{2}\right)^{n-1}$

Q080 $a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Q081 A

Q082 A

Q083 A

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A

Q090 A

Q091 A

Q092 A

Q093 A

Q094 A

Q095 A	Q122 $\frac{189}{16}$	Q148 $\frac{1 - (-3)^n}{2}$	Q171 $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$
Q096 A	Q123 $\frac{13}{9}$	Q149 $\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$	Q172 C
Q097 $x = \pm 4$	Q124 $\frac{255}{128}$	Q150 $\frac{1 - (-3)^n}{4}$	Q173 D
Q098 $x = \pm 5$	Q125 B	Q151 $\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$	Q174 B
Q099 $x = \pm 2$	Q126 A	Q152 $\frac{1 - (-2)^n}{3}$	Q175 C
Q100 $x = \pm 2\sqrt{3}$	Q127 D	Q153 A	Q176 D
Q101 A	Q128 C	Q154 $\frac{40}{9}$	Q177 $(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$
Q102 A	Q129 B	Q155 D	Q178 $(a, r) = (3, -2)$
Q103 A	Q130 A	Q156 C	Q179 $(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$
Q104 A	Q131 -30	Q157 B	Q180 $(a, r) = (3, 2)$
Q105 A	Q132 -20	Q158 A	Q181 $(a, r) = (2, 3)$
Q106 A	Q133 -364	Q159 D	Q182 $(a, r) = (5, 3)$
Q107 $(a, b) = (9, 12)$ or $(1, -4)$	Q134 D	Q160 C	Q183 $(a, r) = \left(\frac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$
Q108 $(a, b) = (2, 2)$ or $(-4, -1)$	Q135 C	Q161 D	Q184 $(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$
Q109 $(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$	Q136 B	Q162 $(a, r) = (3, -2)$	Q185 D
Q110 $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$	Q137 A	Q163 C	Q186 $a_7 = 64, a_8 = 128, r = 2, a_{10} = 512$
Q111 484	Q138 D	Q164 $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$	Quiz MCQ 1 B
Q112 93	Q139 C	Q165 $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$	Quiz MCQ 2 C
Q113 200	Q140 $1 - (-2)^n$	Q166 $(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$	Quiz MCQ 3 B
Q114 3069	Q141 $7(2^n - 1)$	Q167 $(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$	Quiz MCQ 4 D
Q115 D	Q142 $3\{1 - (\frac{1}{3})^n\}$	Q168 B	Quiz WRITTEN 5 $-4(-2)^{n-1}$
Q116 C	Q143 $4(2^n - 1)$	Q169 A	Quiz WRITTEN 6 $x = \pm 5$
Q117 B	Q144 C	Q170 $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$	Quiz WRITTEN 7 200
Q118 A	Q145 D		Quiz WRITTEN 8 512
Q119 D	Q146 A		
Q120 C	Q147 $3^n - 1$		
Q121 $\frac{63}{4}$			

Concept 3 | Summation

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$\sum (au_k + bv_k) = a \sum u_k + b \sum v_k$$

Standard sums

$$\sum k = \frac{n(n+1)}{2}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Classified Practice

Summation

Q001 Written

Write $\sum_{k=1}^5 2k$ as a sum of terms.

Q002 Written

Write $\sum_{k=1}^5 (2k + 1)$ as a sum of terms.

Classified Practice

Summation

Q003 Written

Write $\sum_{k=1}^5 3k$ as a sum of terms.

Q004 Written

Write $\sum_{k=1}^4 (4k - 2)$ as a sum of terms.

Classified Practice

Summation

Q005 Written

Write $\sum_{k=1}^5 (2k - 3)$ as a sum of terms.

Q006 MCQ

Select the correct option: Expand $\sum_{k=1}^3 4k$ into a sum of terms:

Choose the correct option.

A $4 + 8 + 12$

B $4 + 8 + 16$

C $8 + 12 + 16$

D $4 + 12 + 20$

Classified Practice

Summation

Q007 MCQ

Select the correct option: Expand $\sum_{k=1}^4 2^k$ into a sum of terms:

Choose the correct option.

A $2 + 4 + 8 + 16$

B $1 + 2 + 4 + 8$

C $2 + 6 + 18 + 54$

D $4 + 8 + 16 + 32$

Q008 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (5^k - 1)$ into a sum of terms:

Choose the correct option.

A $4 + 24 + 124$

B $5 + 25 + 125$

C $4 + 25 + 124$

D $1 + 4 + 24$

Classified Practice

Summation

Q009 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (k + 4)$ into a sum of terms:

Choose the correct option.

A $5 + 6 + 7 + 8 + 9$

B $4 + 5 + 6 + 7 + 8$

C $1 + 2 + 3 + 4 + 5$

D $6 + 7 + 8 + 9 + 10$

Q010 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (2k + 3)$ into a sum of terms:

Choose the correct option.

A $7 + 9 + 11 + 13$

B $5 + 7 + 9 + 11$

C $4 + 7 + 10 + 13$

D $8 + 10 + 12 + 14$

Classified Practice

Summation

Q011 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2^k + 1)$ into a sum of terms:

Choose the correct option.

- A $3 + 5 + 9$
- B $2 + 4 + 8$
- C $1 + 3 + 5$
- D $4 + 6 + 10$

Q012 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 3k)$ into a sum of terms:

Choose the correct option.

A $4 + 10 + 18$

B $3 + 8 + 15$

C $1 + 4 + 9$

D $6 + 12 + 18$

Classified Practice

Summation

Q013 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (k^2 + 1)$ into a sum of terms:

Choose the correct option.

A $5 + 10 + 17 + 26$

B $2 + 5 + 10 + 17$

C $4 + 9 + 16 + 25$

D $6 + 11 + 18 + 27$

Q014 Written

Write $\sum_{k=1}^3 (k^2 - 3)$ as a sum of terms.

Classified Practice

Summation

Q015 Written

Write $\sum_{k=1}^3 (3k^2 + 2)$ as a sum of terms.

Q016 Written

Write $\sum_{k=1}^4 k^2$ as a sum of terms.

Classified Practice

Summation

Q017 Written

Write $\sum_{k=1}^3 (2k^2 + 3)$ as a sum of terms.

Q018 Written

Write $\sum_{k=1}^4 (k^2 - 2k)$ as a sum of terms.

Classified Practice

Summation

Q019 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k + 1)$ into a sum of terms:

Choose the correct option.

A $3 + 5 + 7 + 9$

B $2 + 4 + 6 + 8$

C $3 + 6 + 9 + 12$

D $1 + 3 + 5 + 7$

Q020 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (3k - 1)$ into a sum of terms:

Choose the correct option.

A $5 + 8 + 11 + 14$

B $2 + 5 + 8 + 11$

C $5 + 9 + 13 + 17$

D $4 + 7 + 10 + 13$

Classified Practice

Summation

Q021 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k^2 - 3)$ into a sum of terms:

Choose the correct option.

A $-1 + 5 + 15 + 29$

B $-1 + 1 + 5 + 13$

C $2 + 5 + 10 + 17$

D $-3 + 5 + 13 + 29$

Q022 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k^2 + 1)$ into a sum of terms:

Choose the correct option.

A $4 + 13 + 28 + 49$

B $3 + 12 + 27 + 48$

C $4 + 10 + 19 + 28$

D $1 + 4 + 9 + 16$

Classified Practice

Summation

Q023 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (k^2 - k)$ into a sum of terms:

Choose the correct option.

A $0 + 2 + 6 + 12$

B $1 + 4 + 9 + 16$

C $0 + 1 + 4 + 9$

D $2 + 6 + 12 + 20$

Q024 MCQ

Select the correct option: Expand $\sum_{k=2}^4 5k$ into a sum of terms:

Choose the correct option.

A $10 + 15 + 20$

B $5 + 10 + 15$

C $2 + 4 + 6$

D $15 + 20 + 25$

Classified Practice

Summation

Q025 MCQ

Select the correct option: Expand $\sum_{k=1}^4 3$ into a sum of terms:

Choose the correct option.

A $3 + 3 + 3 + 3$

B $3 + 6 + 9 + 12$

C $1 + 1 + 1 + 1$

D $4 + 4 + 4 + 4$

Q026 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (5k + 1)$ into a sum of terms:

Choose the correct option.

A $6 + 11 + 16 + 21$

B $5 + 10 + 15 + 20$

C $1 + 6 + 11 + 16$

D $7 + 12 + 17 + 22$

Classified Practice

Summation

Q027 Written

Write $\sum_{k=1}^4 3^k$ as a sum of terms.

Q028 Written

Write $\sum_{k=1}^3 2$ as a sum of terms.

Classified Practice

Summation

Q029 Written

Write $\sum_{k=1}^4 2^k$ as a sum of terms.

Q030 Written

Write $\sum_{k=1}^6 3$ as a sum of terms.

Classified Practice

Summation

Q031 Written

Write $\sum_{k=1}^3 3^k$ as a sum of terms.

Q032 Written

Write $\sum_{k=1}^3 (5^k + 3)$ as a sum of terms.

Classified Practice

Summation

Q033 Written

Write $\sum_{k=1}^4 (2^k - 3)$ as a sum of terms.

Q034 Written

Write $\sum_{k=1}^4 5$ as a sum of terms.

Classified Practice

Summation

Q035 Written

Write $\sum_{k=1}^3 3$ as a sum of terms.

Q036 Written

Write $\sum_{k=1}^5 4$ as a sum of terms.

Classified Practice

Summation

Q037 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 2)$ into a sum of terms:

Choose the correct option.

A $3 + 6 + 11$

B $2 + 5 + 10$

C $3 + 5 + 9$

D $1 + 4 + 9$

Q038 MCQ

Select the correct option: Expand $\sum_{k=1}^5 7$ into a sum of terms:

Choose the correct option.

A $7 + 7 + 7 + 7 + 7$

B $7 + 14 + 21 + 28 + 35$

C $7 + 7 + 7 + 7$

D $5 + 5 + 5 + 5 + 5$

Classified Practice

Summation

Q039 MCQ

Select the correct option: Expand $\sum_{k=3}^6 k^2$ into a sum of terms:

Choose the correct option.

A $9 + 16 + 25 + 36$

B $1 + 4 + 9 + 16$

C $9 + 25 + 49 + 81$

D $3 + 4 + 5 + 6$

Q040 MCQ

Select the correct option: Expand $\sum_{k=1}^3 6^k$ into a sum of terms:

Choose the correct option.

A $6 + 36 + 216$

B $1 + 6 + 36$

C $6 + 12 + 18$

D $36 + 72 + 108$

Classified Practice

Summation

Q041 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (4k - 3)$ into a sum of terms:

Choose the correct option.

A $1 + 5 + 9 + 13 + 17$

B $4 + 8 + 12 + 16 + 20$

C $1 + 4 + 7 + 10 + 13$

D $-1 + 3 + 7 + 11 + 15$

Q042 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k - 5)$ into a sum of terms:

Choose the correct option.

A $-2 + 1 + 4 + 7$

B $-2 + 0 + 2 + 4$

C $3 + 6 + 9 + 12$

D $-5 - 2 + 1 + 4$

Classified Practice

Summation

Q043 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (4^k - 2)$ into a sum of terms:

Choose the correct option.

A $2 + 14 + 62$

B $4 + 16 + 64$

C $2 + 12 + 60$

D $-2 + 2 + 14$

Q044 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2k^2 - k)$ into a sum of terms:

Choose the correct option.

A $1 + 6 + 15$

B $2 + 7 + 18$

C $1 + 3 + 5$

D $0 + 6 + 12$

Classified Practice

Summation

Q045 Written

Express $3 + 7 + 11 + \cdots + (4n - 1)$ using the summation symbol Σ .

Q046 Written**Express $5 + 8 + 11 + \dots + (3n + 2)$ using Σ .**

Classified Practice

Summation

Q047 Written

Write the sum of the first n terms of the sequence $a_n = 2n + 1$ using Σ .

Q048 Written**Express $1 + 3 + 5 + \cdots + (2n - 1)$ using Σ .**

Classified Practice

Summation

Q049 Written

Write the sum of the first n terms of $a_n = 7n - 1$ using Σ .

Q050 Written**Write the sum of the first n terms of $a_n = 4n + 5$ using Σ .**

Classified Practice

Summation

Q051 Written

Write the sum of the first n terms of $a_n = 3n - 2$ using Σ .

Q052 MCQ

Select the correct option: Express $2 + 5 + 8 + \dots + (3n - 1)$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=1}^n (3k - 1)$
 - B $\sum_{k=0}^n (3k - 1)$
 - C $\sum_{k=1}^n (3k + 1)$
 - D $\sum_{k=1}^n (2k - 1)$
-
-
-

Classified Practice

Summation

Q053 MCQ

Select the correct option: Express $7 + 11 + 15 + \dots + 43$ using the summation symbol Σ :**Choose the correct option.**

A $\sum_{k=1}^{10} (4k + 3)$

B $\sum_{k=1}^9 (4k + 3)$

C $\sum_{k=0}^{10} (4k + 3)$

D $\sum_{k=1}^{10} (4k - 3)$

Q054 MCQ

Select the correct option: Express $6 + 10 + 14 + \dots + (4n + 2)$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=1}^n (4k + 2)$
- B $\sum_{k=0}^n (4k + 2)$
- C $\sum_{k=1}^n (4k - 2)$
- D $\sum_{k=1}^n (3k + 2)$

Classified Practice

Summation

Q055 MCQ

Select the correct option: Express $8 + 13 + 18 + \dots + 103$ using the summation symbol Σ :**Choose the correct option.**

A $\sum_{k=1}^{20} (5k + 3)$

B $\sum_{k=1}^{19} (5k + 3)$

C $\sum_{k=0}^{20} (5k + 3)$

D $\sum_{k=1}^{20} (5k - 3)$

Q056 MCQ

Select the correct option: Express $11^2 + 12^2 + \dots + 19^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=11}^{19} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=11}^{19} k$

D $\sum_{k=12}^{19} k^2$

Classified Practice

Summation

Q057 MCQ

Select the correct option: Express $2^2 + 3^2 + \dots + 10^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=2}^{10} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=2}^{10} k$

D $\sum_{k=3}^{10} k^2$

Q058 MCQ

Select the correct option: Express $1 + 3 + 9 + \dots + 3^{n-1}$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=0}^{n-1} 3^k$

B $\sum_{k=1}^n 3^k$

C $\sum_{k=1}^{n-1} 3^k$

D $\sum_{k=0}^n 3^k$

Classified Practice

Summation

Q059 MCQ

Select the correct option: Express $2 + 7 + 12 + \dots + (5n - 3)$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^n (5k - 3)$

B $\sum_{k=0}^n (5k - 3)$

C $\sum_{k=1}^n (5k + 3)$

D $\sum_{k=1}^n (4k - 3)$

Q060 Written

Express $6^2 + 7^2 + 8^2 + \dots + 15^2$ using Σ .

Classified Practice

Summation

Q061 Written

Express $4^2 + 5^2 + 6^2 + \dots + 13^2$ using Σ .

Q062 Written

Express $1 + 7 + 25 + 79 + \dots + (3^n - 2)$ using Σ .

Classified Practice

Summation

Q063 Written

Express $1 + 2 + 4 + 8 + \dots + 2^{n-1}$ using Σ .

Q064 Written**Express $7^2 + 8^2 + \dots + 21^2$ using Σ .**

Classified Practice

Summation

Q065 Written

Express $12^2 + 13^2 + \dots + 18^2$ using Σ .

Q066 Written**Express $10^2 + 11^2 + \dots + 20^2$ using Σ .**

Classified Practice

Summation

Q067 MCQ

Select the correct option: Express $4^2 + 5^2 + \dots + 12^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=4}^{12} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=4}^{12} k$

D $\sum_{k=5}^{12} k^2$

Q068 MCQ

Select the correct option: Express $3^1 + 3^2 + \dots + 3^n$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^n 3^k$

B $\sum_{k=0}^n 3^k$

C $\sum_{k=1}^n 3^{k-1}$

D $\sum_{k=1}^n k^3$

Classified Practice

Summation

Q069 MCQ

Select the correct option: Express $2 + 4 + 8 + \dots + 2^n$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=1}^n 2^k$
- B $\sum_{k=0}^n 2^k$
- C $\sum_{k=1}^n 2^{k-1}$
- D $\sum_{k=1}^n k^2$

Q070 MCQ

Select the correct option: Express $1 + 2^1 + 2^2 + \dots + 2^{n-1}$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=0}^{n-1} 2^k$
- B $\sum_{k=1}^n 2^k$
- C $\sum_{k=1}^{n-1} 2^k$
- D $\sum_{k=0}^n 2^k$

Classified Practice

Summation

Q071 MCQ

Select the correct option: Express $7 + 14 + 21 + \dots + 7n$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^n 7k$

B $\sum_{k=0}^n 7k$

C $\sum_{k=1}^n (7k + 1)$

D $\sum_{k=1}^n k^7$

Q072 MCQ

Select the correct option: Express $10 + 15 + 20 + \dots + 5n$ using the summation symbol Σ :**Choose the correct option.**

A $\sum_{k=2}^n 5k$

B $\sum_{k=1}^n 5k$

C $\sum_{k=2}^n (5k + 1)$

D $\sum_{k=0}^n 5k$

Classified Practice

Summation

Q073 MCQ

Select the correct option: Express $6^2 + 7^2 + \dots + 16^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=6}^{16} k^2$

B $\sum_{k=1}^{11} k^2$

C $\sum_{k=6}^{16} k$

D $\sum_{k=7}^{16} k^2$

Q074 Written**Express $2 + 5 + 8 + \cdots + 29$ using Σ .**

Classified Practice

Summation

Q075 Written

Express $8 + 16 + 24 + \cdots + 56$ using Σ .

Q076 Written**Express $7 + 13 + 19 + \cdots + 73$ using Σ .**

Classified Practice

Summation

Q077 Written

Express $4 + 9 + 14 + \cdots + 119$ using Σ .

Classified Practice

Summation

Q078 Challenge

A green-tech startup installs 1 solar panel in Week 1, 4 in Week 2, 7 in Week 3, and $3n - 2$ in Week n . Write one expression using Σ for the total panels installed from Week 1 through Week n .

Classified Practice

Summation

Q079 Written

Find $\sum_{k=1}^n (-3)$.

Q080 Written**Find** $\sum_{k=1}^n 3.$

Classified Practice

Summation

Q081 Written

Find $\sum_{k=1}^n 6$.

Q082 Written

Find $\sum_{k=1}^n (-4)$.

Classified Practice

Summation

Q083 Written

Find $\sum_{k=1}^n 1$.

Q084 MCQ

Select the correct option: Find $\sum_{k=1}^n 9$:

Choose the correct option.

A $9n$

B $n + 9$

C $9n^2$

D $\frac{9n(n+1)}{2}$

Classified Practice

Summation

Q085 MCQ

Select the correct option: Find $\sum_{k=1}^n (-2)$:

Choose the correct option.

A $-2n$

B $2n$

C $-n - 2$

D $-n^2$

Q086 MCQ

Select the correct option: Find $\sum_{k=1}^7 4$:

Choose the correct option.

A 28

B 11

C 16

D 49

Classified Practice

Summation

Q087 MCQ

Select the correct option: Find $\sum_{k=1}^{12} (-5)$:

Choose the correct option.

A -60

B 60

C -17

D -144

Q088 MCQ

Select the correct option: Find $\sum_{k=1}^n k$:

Choose the correct option.

A $\frac{n(n+1)}{2}$

B $n(n+1)$

C $\frac{n(n-1)}{2}$

D n^2

Classified Practice

Summation

Q089 MCQ

Select the correct option: Find $\sum_{k=1}^n 4k$:

Choose the correct option.

A $2n(n+1)$

B $4n(n+1)$

C $2n(n-1)$

D $4n$

Q090 MCQ

Select the correct option: Find $\sum_{k=1}^{10} k$:

Choose the correct option.

A 55

B 45

C 100

D 110

Classified Practice

Summation

Q091 MCQ

Select the correct option: Find $\sum_{k=1}^8 3k$:

Choose the correct option.

A 108

B 72

C 96

D 120

Q092 MCQ

Select the correct option: Find $\sum_{k=1}^6 5k$:

Choose the correct option.

A 105

B 75

C 90

D 120

Classified Practice

Summation

Q093 MCQ

Select the correct option: Find $\sum_{k=1}^{20} k$:

Choose the correct option.

- A 210
- B 190
- C 200
- D 420

Q094 MCQ

Select the correct option: Find $\sum_{k=1}^n k^2$:

Choose the correct option.

A $\frac{n(n+1)(2n+1)}{6}$

B $\frac{n(n+1)^6}{2}$

C $\frac{n(n+1)(2n-1)}{6}$

D $\frac{n^2(n+1)^2}{4}$

Classified Practice

Summation

Q095 MCQ

Select the correct option: Find $\sum_{k=1}^n 6k^2$:

Choose the correct option.

A $n(n+1)(2n+1)$

B $3n(n+1)$

C $n(n+1)(2n-1)$

D $6n^2$

Q096 MCQ

Select the correct option: Find $\sum_{k=1}^5 k^2$:

Choose the correct option.

A 55

B 25

C 30

D 50

Classified Practice

Summation

Q097 MCQ

Select the correct option: Find $\sum_{k=1}^4 2k^2$:

Choose the correct option.

A 60

B 30

C 40

D 80

Q098 MCQ

Select the correct option: Find $\sum_{k=1}^3 4k^2$:

Choose the correct option.

A 56

B 36

C 48

D 72

Classified Practice

Summation

Q099 MCQ

Select the correct option: Find $\sum_{k=1}^7 k^2$:

Choose the correct option.

A 140

B 49

C 91

D 196

Q100 Written

Find $\sum_{k=1}^n (5k + 4)$.

Classified Practice

Summation

Q101 Written

Find $\sum_{k=1}^n (2k - 3)$.

Q102 Written

Find $\sum_{k=1}^n (4k - 1)$.

Classified Practice

Summation

Q103 Written

Find $\sum_{k=1}^n (3k + 2)$.

Q104 Written

Find $\sum_{k=1}^n (6k - 3)$.

Classified Practice

Summation

Q105 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k - 1)$:

Choose the correct option.

A $\frac{n(3n+1)}{2}$

B $\frac{n(3n-1)}{2}$

C $n(3n + 1)$

D $\frac{3n(n+1)}{2}$

Q106 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k + 5)$:

Choose the correct option.

A $n(n + 6)$

B $n(n + 5)$

C $n(n + 1) + 5$

D $2n(n + 6)$

Classified Practice

Summation

Q107 MCQ

Select the correct option: Find $\sum_{k=1}^n (8k - 4)$:

Choose the correct option.

A $4n^2$

B $4n(n + 1)$

C $8n^2$

D $4n(n - 1)$

Q108 MCQ

Select the correct option: Find $\sum_{k=1}^n (5k + 3)$:

Choose the correct option.

A $\frac{n(5n+11)}{2}$

B $\frac{n(5n+3)}{2}$

C $\frac{n(5n+5)}{2}$

D $n(5n + 11)$

Classified Practice

Summation

Q109 Written

Find $\sum_{k=1}^n (k-1)^2$.

Q110 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 + k)$:

Choose the correct option.

A $\frac{n(n+1)(n+2)}{6}$

B $\frac{n(n+1)(2n+1)}{6}$

C $\frac{n(n+1)(n+2)}{6}$

D $n(n+1)(n+2)$

Classified Practice

Summation

Q111 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k^2 - k)$:

Choose the correct option.

A $\frac{n(n+1)(4n-1)}{6}$

B $\frac{n(n+1)(4n+1)}{6}$

C $\frac{n(n+1)(2n-1)}{6}$

D $\frac{n(n-1)(4n-1)}{6}$

Q112 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 - 1)$:

Choose the correct option.

A $\frac{n(2n^2+3n-5)}{6}$

B $\frac{n(2n^2+3n-1)}{6}$

C $\frac{n(2n^2-3n+1)}{6}$

D $\frac{n(n+1)(2n+1)}{6}$

Classified Practice

Summation

Q113 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k^2 + 2k)$:

Choose the correct option.

A $\frac{n(n+1)(2n+3)}{2}$

B $\frac{n(n+1)(2n+1)}{2}$

C $n(n+1)(2n+3)$

D $\frac{n(n-1)(2n+3)}{2}$

Q114 MCQ

Select the correct option: Find $\sum_{k=1}^n (4k^2 - 3k + 2)$:

Choose the correct option.

A $\frac{n(8n^2+3n+7)}{6}$

B $\frac{n(8n^2+9n+7)}{6}$

C $\frac{n(4n^2+3n+2)}{6}$

D $\frac{n(8n^2-3n+7)}{6}$

Classified Practice

Summation

Q115 Written

Find $1 \cdot 3 + 2 \cdot 8 + 3 \cdot 13 + \cdots + n(5n - 2)$.

Q116 Written

Find $\sum_{k=1}^n (k-1)(3k-1)$.

Classified Practice

Summation

Q117 Written

Find $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + n(2n + 1)$.

Q118 Written

Find $\sum_{k=1}^n k(k-2)$.

Classified Practice

Summation

Q119 Written

Find $\sum_{k=1}^n k(2k - 2)$.

Q120 Written

Find $1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 + \cdots + n(n + 3)$.

Classified Practice

Summation

Q121 Written

Find $1 \cdot 5 + 2 \cdot 8 + 3 \cdot 11 + \cdots + n(3n + 2)$.

Q122 Written

Find $2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 + \cdots + (n + 1)(n + 4)$.

Classified Practice

Summation

Q123 Written

Find $\sum_{k=1}^4 (2k - 1)$.

Q124 Written

Find $\sum_{k=1}^5 (3k - 2)$.

Classified Practice

Summation

Q125 Written

Find $\sum_{k=1}^{20} (2k - 7)$.

Q126 Written

Find $\sum_{k=1}^8 (4k + 5)$.

Classified Practice

Summation

Q127 Written

Find $\sum_{k=1}^{10} (2k - 1)$.

Classified Practice

Summation

Q128 **Challenge**

Find $\sum_{r=4}^8 (r^2 + r + 5)$.

Classified Practice

Summation

Q129 Written

Find the sum $\sum_{k=1}^n k^2(k+1)$.

Q130 Written

Find the sum $\sum_{k=1}^n k^2(k-2)$.

Classified Practice

Summation

Q131 Written

Find the sum $\sum_{k=1}^n k^2(k+2)$.

Q132 Written

Find the sum $\sum_{k=1}^n k^2(k-3)$.

Classified Practice

Summation

Q133 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+1)$:

Choose the correct option.

A $\frac{n(n+1)(n+2)(3n+1)}{12}$

B $\frac{n(n+1)(n+2)(3n-1)}{12}$

C $\frac{n(n+1)(3n+1)}{12}$

D $\frac{n(n+1)(n+2)(3n+2)}{6}$

Q134 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-2)$:

Choose the correct option.

- A $\frac{n(n+1)(3n^2-5n-4)}{12}$
- B $\frac{n(n+1)(3n^2+5n-4)}{12}$
- C $\frac{n(n+1)(3n^2-5n+4)}{12}$
- D $\frac{n(n+1)(3n^2-4n-5)}{12}$

Classified Practice

Summation

Q135 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+2)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+11n+4)}{12}$

B $\frac{n(n+1)(3n^2+11n-4)}{12}$

C $\frac{n(n+1)(3n^2+9n+4)}{12}$

D $\frac{n(n+1)(3n^2+4n+11)}{12}$

Q136 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-3)$:

Choose the correct option.

A $\frac{n(n+1)(n^2-3n-2)}{4}$

B $\frac{n(n+1)(n^2+3n-2)}{4}$

C $\frac{n(n+1)(n^2-3n+2)}{4}$

D $\frac{n(n+1)(n^2-2n-3)}{4}$

Classified Practice

Summation

Q137 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+3)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+5n+2)}{4}$

B $\frac{n(n+1)(n^2+5n-2)}{4}$

C $\frac{n(n+1)(n^2+3n+2)}{4}$

D $\frac{n(n+1)(n^2+2n+5)}{4}$

Q138 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-4)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2-13n-8)}{12}$

B $\frac{n(n+1)(3n^2-13n+8)}{12}$

C $\frac{n(n+1)(3n^2-11n-8)}{12}$

D $\frac{n(n+1)(3n^2-8n-13)}{12}$

Classified Practice

Summation

Q139 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+4)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+19n+8)}{12}$

B $\frac{n(n+1)(3n^2+19n-8)}{12}$

C $\frac{n(n+1)(3n^2+17n+8)}{12}$

D $\frac{n(n+1)(3n^2+8n+19)}{12}$

Q140 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-1)$:

Choose the correct option.

- A $\frac{n(n+1)(3n^2-5n-4)}{12}$
- B $\frac{n(n+1)(3n^2-5n+4)}{12}$
- C $\frac{n(n+1)(3n^2-4n-5)}{12}$
- D $\frac{n(n+1)(3n^2-n-4)}{12}$

Classified Practice

Summation

Q141 Written

Find the sum $\sum_{k=1}^n k(k^2 - 1)$.

Q142 Written

Find the sum $\sum_{k=1}^n k(k^2 + 2)$.

Classified Practice

Summation

Q143 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Choose the correct option.

A $\frac{n(n+1)(n-1)(n+2)}{4}$

B $\frac{n(n+1)(n+1)(n+2)}{4}$

C $\frac{n(n-1)(n+2)}{4}$

D $\frac{n(n+1)(n-2)(n+3)}{4}$

Q144 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 2)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+4)}{4}$
- B $\frac{n(n+1)(n^2+n-4)}{4}$
- C $\frac{n(n+1)(n^2+2n+4)}{4}$
- D $\frac{n(n+1)(n^2+n+2)}{4}$

Classified Practice

Summation

Q145 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 3)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+6)}{4}$
- B $\frac{n(n+1)(n^2+n-6)}{4}$
- C $\frac{n(n+1)(n^2+3n+6)}{4}$
- D $\frac{n(n+1)(n^2+n+3)}{4}$

Q146 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 2)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-4)}{4}$

B $\frac{n(n+1)(n^2+n+4)}{4}$

C $\frac{n(n+1)(n^2-2n-4)}{4}$

D $\frac{n(n+1)(n^2+n-2)}{4}$

Classified Practice

Summation

Q147 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 4)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+8)}{4}$
- B $\frac{n(n+1)(n^2+n-8)}{4}$
- C $\frac{n(n+1)(n^2+4n+8)}{4}$
- D $\frac{n(n+1)(n^2+n+4)}{4}$

Q148 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 3)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n-6)}{4}$
- B $\frac{n(n+1)(n^2+n+6)}{4}$
- C $\frac{n(n+1)(n^2-3n-6)}{4}$
- D $\frac{n(n+1)(n^2+n-3)}{4}$

Classified Practice

Summation

Q149 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+5)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+23n+10)}{12}$

B $\frac{n(n+1)(3n^2+23n-10)}{12}$

C $\frac{n(n+1)(3n^2+21n+10)}{12}$

D $\frac{n(n+1)(3n^2+10n+23)}{12}$

Q150 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-5)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2-17n-10)}{12}$

B $\frac{n(n+1)(3n^2-17n+10)}{12}$

C $\frac{n(n+1)(3n^2-15n-10)}{12}$

D $\frac{n(n+1)(3n^2-10n-17)}{12}$

Classified Practice

Summation

Q151 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 5)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n+10)}{4}$

B $\frac{n(n+1)(n^2+n-10)}{4}$

C $\frac{n(n+1)(n^2+5n+10)}{4}$

D $\frac{n(n+1)(n^2+n+5)}{4}$

Q152 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 5)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-10)}{4}$

B $\frac{n(n+1)(n^2+n+10)}{4}$

C $\frac{n(n+1)(n^2-5n-10)}{4}$

D $\frac{n(n+1)(n^2+n-5)}{4}$

Classified Practice

Summation

Q153 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+6)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+27n+12)}{12}$

B $\frac{n(n+1)(3n^2+27n-12)}{12}$

C $\frac{n(n+1)(3n^2+25n+12)}{12}$

D $\frac{n(n+1)(3n^2+12n+27)}{12}$

Q154 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 6)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+12)}{4}$
- B $\frac{n(n+1)(n^2+n-12)}{4}$
- C $\frac{n(n+1)(n^2+6n+12)}{4}$
- D $\frac{n(n+1)(n^2+n+6)}{4}$

Classified Practice

Summation

Q155 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-6)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2-25n-12)}{12}$

B $\frac{n(n+1)(3n^2-25n+12)}{12}$

C $\frac{n(n+1)(3n^2-23n-12)}{12}$

D $\frac{n(n+1)(3n^2-12n-25)}{12}$

Q156 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 6)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-12)}{4}$

B $\frac{n(n+1)(n^2+n+12)}{4}$

C $\frac{n(n+1)(n^2-6n-12)}{4}$

D $\frac{n(n+1)(n^2+n-6)}{4}$

Classified Practice

Summation

Q157 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+7)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+31n+14)}{12}$

B $\frac{n(n+1)(3n^2+31n-14)}{12}$

C $\frac{n(n+1)(3n^2+29n+14)}{12}$

D $\frac{n(n+1)(3n^2+14n+31)}{12}$

Q158 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+8)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+35n+16)}{12}$

B $\frac{n(n+1)(3n^2+35n-16)}{12}$

C $\frac{n(n+1)(3n^2+33n+16)}{12}$

D $\frac{n(n+1)(3n^2+16n+35)}{12}$

Classified Practice

Summation

Q159 Written

Find the sum $\sum_{k=1}^n 6 \times 3^{k-1}$.

Q160 Written

Find the sum $\sum_{k=1}^n 4 \times 2^{k-1}$.

Classified Practice

Summation

Q161 Written

Find the sum $\sum_{k=1}^n 2^k$.

Q162 Written

Find the sum $\sum_{k=1}^n 2 \times (-3)^{k-1}$.

Classified Practice

Summation

Q163 Written

Find the sum $\sum_{k=1}^n 3 \times (-2)^{k-1}$.

Q164 Written

Find the sum $\sum_{k=1}^n (-5)^k$.

Classified Practice

Summation

Q165 Written

Find the sum $\sum_{k=1}^n 2 \times (-1)^{k-1}$.

Q166 Written

Find the sum $\sum_{k=1}^n 8 \times (-3)^{k-1}$.

Classified Practice

Summation

Q167 Written

Find the sum $\sum_{k=1}^n (-2)^{k-1}$.

Q168 Written

Find the sum $\sum_{k=1}^n (-3)^k$.

Classified Practice

Summation

Q169 Written

Find the sum $\sum_{k=1}^n 4^k$.

Q170 Written

Find the sum $\sum_{k=1}^n 4^{k-1}$.

Classified Practice

Summation

Q171 Written

Find the sum $\sum_{k=1}^n 3^{k-1}$.

Q172 Written

Find the sum $\sum_{k=1}^4 (-2)^{k-1}$.

Classified Practice

Summation

Q173 Written

Find the sum $\sum_{k=1}^3 3^k$.

Q174 Written

Find the sum $\sum_{k=1}^5 2^k$.

Classified Practice

Summation

Q175 Written

Find the sum $\sum_{k=1}^4 (-3)^k$.

Classified Practice

Summation

Q176 Challenge

Find the finite sum $\sum_{r=5}^9 (2r + 7)$.

Classified Practice

Summation

Q177 Written

Find $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)}$.

Q178 Written

Find $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n(n+1)}.$

Classified Practice

Summation

Q179 Written

Find $\frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \cdots + \frac{1}{(n+2)(n+3)}$.

Q180 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 2} + \cdots + \frac{1}{n(n+1)}$:

Choose the correct option.

A $\frac{n}{n+1}$

B $\frac{1}{n+1}$

C $\frac{2(n+1)}{n}$

D $\frac{n+1}{n}$

Classified Practice

Summation

Q181 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 3} + \cdots + \frac{1}{(n+1)(n+2)}$:

Choose the correct option.

A $\frac{n}{2(n+2)}$

B $\frac{1}{2(n+2)}$

C $\frac{n}{n+2}$

D $\frac{n+1}{2(n+2)}$

Q182 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 4} + \cdots + \frac{1}{(n+2)(n+3)}$:

Choose the correct option.

A $\frac{n}{3(n+3)}$

B $\frac{n}{n+3}$

C $\frac{1}{3(n+3)}$

D $\frac{n+1}{3(n+3)}$

Classified Practice

Summation

Q183 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 5} + \cdots + \frac{1}{(n+3)(n+4)}$:

Choose the correct option.

A $\frac{n}{4(n+4)}$

B $\frac{n}{3(n+4)}$

C $\frac{n+1}{4(n+4)}$

D $\frac{1}{4(n+4)}$

Q184 MCQ

Select the correct option: Find the sum $\frac{1}{6 \times 7} + \cdots + \frac{1}{(n+5)(n+6)}$:

Choose the correct option.

A $\frac{n}{6(n+6)}$

B $\frac{n}{5(n+6)}$

C $\frac{n+1}{6(n+6)}$

D $\frac{1}{6(n+6)}$

Classified Practice

Summation

Q185 MCQ

Select the correct option: Find the sum $\frac{1}{7 \times 8} + \cdots + \frac{1}{(n+6)(n+7)}$:

Choose the correct option.

A $\frac{n}{7(n+7)}$

B $\frac{n}{6(n+7)}$

C $\frac{n+1}{7(n+7)}$

D $\frac{1}{7(n+7)}$

Q186 MCQ

Select the correct option: Find the sum $\frac{1}{9 \times 10} + \cdots + \frac{1}{(n+8)(n+9)}$:

Choose the correct option.

A $\frac{n}{9(n+9)}$

B $\frac{n}{8(n+9)}$

C $\frac{n+1}{9(n+9)}$

D $\frac{1}{9(n+9)}$

Classified Practice

Summation

Q187 MCQ

Select the correct option: Find the sum $\frac{1}{10 \times 11} + \cdots + \frac{1}{(n+9)(n+10)}$:

Choose the correct option.

A $\frac{n}{10(n+10)}$

B $\frac{n}{9(n+10)}$

C $\frac{n+1}{10(n+10)}$

D $\frac{1}{10(n+10)}$

Q188 MCQ

Select the correct option: Find the sum $\frac{1}{11 \times 12} + \cdots + \frac{1}{(n+10)(n+11)}$:

Choose the correct option.

A $\frac{n}{11(n+11)}$

B $\frac{n}{10(n+11)}$

C $\frac{n+1}{11(n+11)}$

D $\frac{1}{11(n+11)}$

Classified Practice

Summation

Q189 Written

Find the following sum: $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \cdots + \frac{1}{(3n-1)(3n+2)}$.

Q190 Written

Find $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)}$.

Classified Practice

Summation

Q191 Written

Find $\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \cdots + \frac{1}{n(n+2)}$.

Q192 Written

Find $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \cdots + \frac{1}{(3n-2)(3n+1)}$.

Classified Practice

Summation

Q193 Written

Find $\frac{1}{1 \times 7} + \frac{1}{4 \times 10} + \frac{1}{7 \times 13} + \cdots + \frac{1}{(3n-2)(3n+4)}$.

Q194 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 3} + \cdots + \frac{1}{(2n-1)(2n+1)}$:

Choose the correct option.

- A $\frac{n}{2n+1}$
B $\frac{n}{2(2n+1)}$
C $\frac{1}{2n+1}$
D $\frac{n+1}{2n+1}$

Classified Practice

Summation

Q195 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 4} + \cdots + \frac{1}{(3n-2)(3n+1)}$:

Choose the correct option.

- A $\frac{n}{3n+1}$
- B $\frac{n}{3(3n+1)}$
- C $\frac{1}{3n+1}$
- D $\frac{n+1}{3n+1}$

Q196 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 7} + \cdots + \frac{1}{(3n-2)(3n+4)}$:

Choose the correct option.

A $\frac{1}{6} \left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4} \right)$

B $\frac{1}{6} \left(1 - \frac{1}{3n+4} \right)$

C $\frac{1}{2(3n+4)}$

D $\frac{n}{2(3n+4)}$

Classified Practice

Summation

Q197 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 6} + \dots + \frac{1}{(4n-2)(4n+2)}$:

Choose the correct option.

A $\frac{n}{2(4n+2)}$

B $\frac{n}{4(4n+2)}$

C $\frac{1}{4n+2}$

D $\frac{n+1}{2(4n+2)}$

Q198 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 8} + \cdots + \frac{1}{(5n-2)(5n+3)}$:

Choose the correct option.

A $\frac{n}{3(5n+3)}$

B $\frac{n}{5(5n+3)}$

C $\frac{n+1}{3(5n+3)}$

D $\frac{1}{5n+3}$

Classified Practice

Summation

Q199 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 5} + \cdots + \frac{1}{(4n-3)(4n+1)}$:

Choose the correct option.

- A $\frac{n}{4n+1}$
- B $\frac{n}{4(4n+1)}$
- C $\frac{1}{4n+1}$
- D $\frac{n+1}{4n+1}$

Q200 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 7} + \cdots + \frac{1}{(5n-3)(5n+2)}$:

Choose the correct option.

A $\frac{n}{2(5n+2)}$

B $\frac{n}{5(5n+2)}$

C $\frac{n+1}{2(5n+2)}$

D $\frac{1}{5n+2}$

Classified Practice

Summation

Q201 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 10} + \cdots + \frac{1}{(6n-2)(6n+4)}$:

Choose the correct option.

A $\frac{n}{4(6n+4)}$

B $\frac{n}{6(6n+4)}$

C $\frac{n+1}{4(6n+4)}$

D $\frac{1}{6n+4}$

Q202 Written

Find $\frac{1}{4 \times 6} + \frac{1}{5 \times 7} + \frac{1}{6 \times 8} + \cdots + \frac{1}{(n+3)(n+5)}$.

Classified Practice

Summation

Q203 Written

Find $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \cdots + \frac{1}{2n(2n+2)}$.

Q204 Written

Find the following sum: $\sum_{k=1}^n \frac{1}{\sqrt{k+2} + \sqrt{k+1}}.$

Classified Practice

Summation

Q205 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}}.$

Q206 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k-1}}.$

Classified Practice

Summation

Q207 Written

Find $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{n} + \sqrt{n+1}}$.

Q208 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{2k-1} + \sqrt{2k+1}}.$

Classified Practice

Summation

Q209 Written

Find $\frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{n+1} + \sqrt{n+2}}.$

Classified Practice

Summation

Q210 Challenge

Find the sum, showing your steps: $\sum_{r=1}^{30} \frac{1}{r(r+2)}.$

Classified Practice

Summation

Q211 Written

Find the following sum $S = 3 \times 1 + 5 \times 3 + 7 \times 3^2 + 9 \times 3^3 + \cdots + (2n + 1) \times 3^{n-1}$.

Q212 Written

Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$.

Classified Practice

Summation

Q213 Written

Find the sum $\sum_{k=1}^4 (2k + 1) \times (3)^{k-1}$.

Q214 Written

Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$.

Classified Practice

Summation

Q215 Written

Find the sum $\sum_{k=1}^6 (2k - 1) \times (4)^{k-1}$.

Q216 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$:

Choose the correct option.

A $3^n n$

B $-(3^n n)$

C $(3^n n) + 1$

D $(3^n n) - 1$

Classified Practice

Summation

Q217 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$:

Choose the correct option.

A $\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}$

B $-\left(\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}\right)$

C $\left(\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}\right) + 1$

D $\left(\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}\right) - 1$

Q218 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 4) \times (2)^{k-1}$:

Choose the correct option.

A $2^n n + 3 \cdot 2^n - 3$

B $-(2^n n + 3 \cdot 2^n - 3)$

C $(2^n n + 3 \cdot 2^n - 3) + 1$

D $(2^n n + 3 \cdot 2^n - 3) - 1$

Classified Practice

Summation

Q219 Written

Find the following sum $S = 2 \times 1 + 3 \times 3 + 4 \times 3^2 + 5 \times 3^3 + \cdots + (n + 1) \times 3^{n-1}$.

Q220 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+1) \times (3)^{k-1}$:

Choose the correct option.

A $\frac{2 \cdot 3^n n + 3^n - 1}{4}$

B $-\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right)$

C $\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right) + 1$

D $\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right) - 1$

Classified Practice

Summation

Q221 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k-2) \times (3)^{k-1}$:

Choose the correct option.

A $\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}$

B $-\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right)$

C $\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right) + 1$

D $\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right) - 1$

Q222 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 3) \times (3)^{k-1}$:

Choose the correct option.

A $3^n n - 2 \cdot 3^n + 2$

B $-(3^n n - 2 \cdot 3^n + 2)$

C $(3^n n - 2 \cdot 3^n + 2) + 1$

D $(3^n n - 2 \cdot 3^n + 2) - 1$

Classified Practice

Summation

Q223 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (-2)^{k-1}$:

Choose the correct option.

A $-\frac{6(-2)^n n + 5(-2)^n - 5}{9}$

B $-\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right)$

C $\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right) + 1$

D $\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right) - 1$

Classified Practice

Summation

Q224 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+2) \times (-3)^{k-1}$:

Choose the correct option.

- A $-\frac{4(-3)^n n + 9(-3)^n - 9}{16}$
- B $-\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right)$
- C $\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right) + 1$
- D $\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right) - 1$

Classified Practice

Summation

Q225 Written

Find the following sum $S = 1 \times 1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \cdots + n \times 2^{n-1}$.

Q226 Written**Find the following sum** $S = 1 \times 3 + 3 \times 3^2 + 5 \times 3^3 + 7 \times 3^4 + \cdots + (2n - 1) \times 3^n$.

Classified Practice

Summation

Q227 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (3k) \times (3)^{k-1}$:

Choose the correct option.

A $\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}$

B $-\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right)$

C $\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right) + 1$

D $\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right) - 1$

Classified Practice

Summation

Q228 **Challenge**

Find the sum of the partial sums $5 + (5 + 9) + (5 + 9 + 13) + \cdots + [5 + 9 + \cdots + (4k + 1)]$.

Classified Practice

Summation

Q229 Written

For the sequence $3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, \dots$, find the k -th term.

Q230 Written

For the sequence $1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, \dots$, find the k -th term.

Classified Practice

Summation

Q231 **Written**

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, find the k -th term.

Q232 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, find the k -th term.

Classified Practice

Summation

Q233 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, what is a_k :

Choose the correct option.

A $k(k + 1)$

B $k(k + 3)$

C $2k^2$

D $k(k + 2)$

Q234 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 6, 3 + 6 + 9, 3 + 6 + 9 + 12, \dots$, what is a_k :

Choose the correct option.

A $\frac{3k(k+1)}{2}$

B $\frac{3k(k+3)}{2}$

C $3k^2$

D $\frac{3k(k+2)}{2}$

Classified Practice

Summation

Q235 MCQ

Select the correct option: For the cumulative sequence $4, 4 + 6, 4 + 6 + 8, 4 + 6 + 8 + 10, \dots$, what is a_k :

Choose the correct option.

A $k(k + 3)$

B $k(k + 5)$

C $2k(k + 1)$

D $k(k + 4)$

Q236 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 6, 2 + 6 + 10, 2 + 6 + 10 + 14, \dots$, what is a_k :

Choose the correct option.

A $2k^2$

B $2k(k + 2)$

C $2k(2k - 1)$

D $2k(k + 1)$

Classified Practice

Summation

Q237 MCQ

Select the correct option: For the cumulative sequence $5, 5 + 8, 5 + 8 + 11, 5 + 8 + 11 + 14, \dots$, what is a_k :

Choose the correct option.

A $\frac{k(3k+7)}{2}$

B $\frac{k(3k+13)}{2}$

C $k(3k+2)$

D $\frac{k(3k+10)}{2}$

Q238 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 8, 3 + 8 + 13, 3 + 8 + 13 + 18, \dots$, what is a_k :

Choose the correct option.

A $\frac{k(5k+1)}{2}$

B $\frac{k(5k+11)}{2}$

C $k(5k - 2)$

D $\frac{k(5k+6)}{2}$

Classified Practice

Summation

Q239 MCQ

Select the correct option: For the cumulative sequence $6, 6 + 8, 6 + 8 + 10, 6 + 8 + 10 + 12, \dots$, what is a_k :

Choose the correct option.

A $k(k + 5)$

B $k(k + 7)$

C $2k(k + 2)$

D $k(k + 6)$

Q240 MCQ

Select the correct option: For the cumulative sequence $4, 4 + 8, 4 + 8 + 12, 4 + 8 + 12 + 16, \dots$, what is a_k :

Choose the correct option.

A $2k(k + 1)$

B $2k(k + 3)$

C $4k^2$

D $2k(k + 2)$

Classified Practice

Summation

Q241 Written

For the sequence $3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, \dots$, find the sum from the first term to the n -th term.

Q242 Written

For the sequence $1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, \dots$, find the sum from the first term to the n -th term.

Classified Practice

Summation

Q243 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, find the sum from the first term to the n -th term.

Q244 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, find the sum from the first term to the n -th term.

Classified Practice

Summation

Q245 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 5, 2 + 5 + 8, 2 + 5 + 8 + 11, \dots$, find a_5 :

Choose the correct option.

A 40

B 46

C 34

D 52

Classified Practice

Summation

Q246 **Challenge**

The sum of the first n terms of $\{a_n\}$ is $S_n = 2n^2 + 5n$. Find the general term.

Classified Practice

Summation

Q247 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 3)$.

Classified Practice

Summation

Q248 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(2n + 5)$.

Classified Practice

Summation

Q249 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2$.

Classified Practice

Summation

Q250 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 4)$.

Classified Practice

Summation

Q251 MCQ

Select the correct option: If $S_n = n(n + 2)$, the general term a_n is:

Choose the correct option.

A $2n + 3$

B $2n - 1$

C $2(n + 2)$

D $a_n = 2n + 1$

Classified Practice

Summation

Q252 MCQ

Select the correct option: If $S_n = 2n(n + 2)$, the general term a_n is:

Choose the correct option.

A $4n + 1$

B $2(2n + 3)$

C $a_n = 2(2n + 1)$

D $4n + 5$

Classified Practice

Summation

Q253 MCQ

Select the correct option: If $S_n = 3n(n - 1)$, the general term a_n is:

Choose the correct option.

A $6n$

B $a_n = 6(n - 1)$

C $6n - 5$

D $2(3n - 4)$

Classified Practice

Summation

Q254 MCQ

Select the correct option: If $S_n = 2n(2n - 1)$, the general term a_n is:

Choose the correct option.

A $a_n = 2(4n - 3)$

B $4(2n - 1)$

C $8n - 7$

D $2(4n + 1)$

Classified Practice

Summation

Q255 MCQ

Select the correct option: If $S_n = 5n(n + 1)$, the general term a_n is:

Choose the correct option.

A $10n + 3$

B $2(5n - 1)$

C $10(n + 1)$

D $a_n = 10n$

Classified Practice

Summation

Q256 MCQ

Select the correct option: If $S_n = n(2n + 3)$, the general term a_n is:

Choose the correct option.

A $4n - 1$

B $4n + 5$

C $a_n = 4n + 1$

D $4n + 3$

Classified Practice

Summation

Q257 MCQ

Select the correct option: If $S_n = n(3n + 1)$, the general term a_n is:

Choose the correct option.

A $3(2n - 1)$

B $2(3n + 2)$

C $a_n = 2(3n - 1)$

D $6n + 1$

Classified Practice

Summation

Q258 MCQ

Select the correct option: If $S_n = n(n^2 + 2n + 3)$, the general term a_n is:

Choose the correct option.

A $3n^2 + n + 1$

B $3n^2 + 7n + 6$

C $a_n = 3n^2 + n + 2$

D $3n^2 + n + 3$

Classified Practice

Summation

Q259 MCQ

Select the correct option: If $S_n = n(n+1)(2n-1)$, the general term a_n is:

Choose the correct option.

A $2(n+1)(3n+1)$

B $a_n = 2n(3n-2)$

C $2(3n^2 - 2n + 1)$

D $2(n-1)(3n+1)$

Classified Practice

Summation

Q260 MCQ

Select the correct option: If $S_n = n(3n^2 - 2n + 4)$, the general term a_n is:

Choose the correct option.

A $a_n = 9n^2 - 13n + 9$

B $9n^2 - 13n + 12$

C $9n^2 - 13n + 8$

D $9n^2 + 5n + 5$

Classified Practice

Summation

Q261 MCQ

Select the correct option: If $S_n = n(n^2 - n + 2)$, the general term a_n is:

Choose the correct option.

A $3n^2 - 5n + 5$

B $(n - 1)(3n - 2)$

C $3n^2 + n + 2$

D $a_n = 3n^2 - 5n + 4$

Classified Practice

Summation

Q262 MCQ

Select the correct option: If $S_n = n(n+1)(2n+1)$, the general term a_n is:

Choose the correct option.

A $2(3n^2 + 1)$

B $6n^2 - 1$

C $6(n+1)^2$

D $a_n = 6n^2$

Classified Practice

Summation

Q263 MCQ

Select the correct option: If $S_n = n(n+1)(n+2)$, the general term a_n is:

Choose the correct option.

A $3(n+1)(n+2)$

B $a_n = 3n(n+1)$

C $3n^2 + 3n + 1$

D $3n^2 + 3n - 2$

Classified Practice

Summation

Q264 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^3 + 2n + 6$.

Classified Practice

Summation

Q265 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2 + 2n - 1$.

Classified Practice

Summation

Q266 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = (n-1)(n^2 + n + 1)$.

Classified Practice

Summation

Q267 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 1$.

Classified Practice

Summation

Q268 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 3^n - 1$.

Classified Practice

Summation

Q269 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 2^n - 1$.

Classified Practice

Summation

Q270 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 7^n - 1$.

Classified Practice

Summation

Q271 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 2$.

Classified Practice

Summation

Q272 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 4 \cdot 2^n - 3$.

Classified Practice

Summation

Q273 MCQ

Select the correct option: If $S_n = n(n + 4)$, find a_5 :**Choose the correct option.**

A -13

B 13

C 14

D 12

Classified Practice

Summation

Q274 **Challenge**

Find the general term of the sequence $\{a_n\}$ where S_n is given by $S_n = 3 \cdot 3^n - 2n + 1$.

Classified Practice

Summation

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Select the correct option: Expand $\sum_{k=1}^4 (3k - 1)$ into a sum of terms:

Choose the correct option.

- A $2 + 5 + 8 + 11$
- B $3 + 6 + 9 + 12$
- C $1 + 4 + 7 + 10$
- D $2 + 6 + 10 + 14$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Choose the correct option.

- A $\frac{n(n+1)(n-1)(n+2)}{4}$
- B $\frac{n(n+1)(n+1)(n+2)}{4}$
- C $\frac{n(n-1)(n+2)}{4}$
- D $\frac{n(n+1)(n-2)(n+3)}{4}$

Classified Practice

Summation

Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k \times 2^{k-1}$:

Choose the correct option.

A $(n-1) \times 2^n + 1$

B $-((n-1) \times 2^n + 1)$

C $((n-1) \times 2^n + 1) + 1$

D $((n-1) \times 2^n + 1) - 1$

Classified Practice

Summation

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: If $S_n = 2^n - 3$, find a_6 :

Choose the correct option.

A 30

B -32

C 32

D 34

Classified Practice

Summation

Quiz WRITTEN 5 Concept Quiz · Written

Expand $\sum_{k=1}^5 (k^2 - 2)$ into a sum of terms.

Quiz WRITTEN 6 Concept Quiz · Written

Find the sum $\sum_{k=1}^n k^2(k+5)$.

Classified Practice

Summation

Quiz WRITTEN 7 Concept Quiz · Written**Find the sum** $\sum_{k=1}^n (4k - 1) \times 3^k$.

Classified Practice

Summation

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term of the sequence $\{a_n\}$ when S_n is $2 \cdot 2^n - 2n + 1$.

Answer key

Use the question number shown on each page.

Q001 $2 + 4 + 6 + 8 + 10$

Q002 $3 + 5 + 7 + 9 + 11$

Q003 $3 + 6 + 9 + 12 + 15$

Q004 $2 + 6 + 10 + 14$

Q005 $-1 + 1 + 3 + 5 + 7$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 A

Q013 A

Q014 $-2 + 1 + 6$

Q015 $5 + 14 + 29$

Q016 $1 + 4 + 9 + 16$

Q017 $5 + 11 + 21$

Q018 $-1 + 0 + 3 + 8$

Q019 A

Q020 A

Q021 A

Q022 A

Q023 A

Q024 A

Q025 A

Q026 A

Q027 $3 + 9 + 27 + 81$

Q028 $2 + 2 + 2$

Q029 $2 + 4 + 8 + 16$

Q030 $3 + 3 + 3 + 3 + 3 + 3$

Q031 $3 + 9 + 27$

Q032 $8 + 28 + 128$

Q033 $-1 + 1 + 5 + 13$

Q034 $5 + 5 + 5 + 5$

Q035 $3 + 3 + 3$

Q036 $4 + 4 + 4 + 4 + 4$

Q037 A

Q038 A

Q039 A

Q040 A

Q041 A

Q042 A

Q043 A

Q044 A

Q045 $\sum_{k=1}^n (4k - 1)$

Q046 $\sum_{k=1}^n (3k + 2)$

Q047 $\sum_{k=1}^n (2k + 1)$

Q048 $\sum_{k=1}^n (2k - 1)$

Q049 $\sum_{k=1}^n (7k - 1)$

Q050 $\sum_{k=1}^n (4k + 5)$

Q051 $\sum_{k=1}^n (3k - 2)$

Q052 A

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $\sum_{k=6}^{15} k^2$

Q061 $\sum_{k=4}^{13} k^2$

Q062 $\sum_{k=1}^n (3^k - 2)$

Q063 $\sum_{k=1}^n 2^{k-1}$

Q064 $\sum_{k=7}^{21} k^2$

Q065 $\sum_{k=12}^{18} k^2$

Q066 $\sum_{k=10}^{20} k^2$

Q067 A

Q068 A

Q069 A

Q070 A

Q071 A

Q072 A

Q073 A

Q074 $\sum_{k=1}^{10} (3k - 1)$

Q075 $\sum_{k=1}^7 8k$

Q076 $\sum_{k=1}^{12} (6k + 1)$

Q077 $\sum_{k=1}^{24} (5k - 1)$

Q078 $\sum_{k=1}^n (3k - 2)$

Q079 $-3n$

Q080 $3n$

Q081 $6n$

Q082 $-4n$

Q083 n

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A	Q118 $\frac{n(n+1)(2n-5)}{6}$	Q145 A	Q174 62
Q090 A	Q119 $\frac{2n(n+1)(n-1)}{3}$	Q146 A	Q175 60
Q091 A	Q120 $\frac{n(n+1)(n+5)}{3}$	Q147 A	Q176 105
Q092 A	Q121 $\frac{n(n+1)(2n+3)}{2}$	Q148 A	Q177 $\frac{n}{2(n+2)}$
Q093 A	Q122 $\frac{n(n+4)(n+5)}{3}$	Q149 A	Q178 $\frac{n}{n+1}$
Q094 A	Q123 16	Q150 A	Q179 $\frac{n}{3(n+3)}$
Q095 A	Q124 35	Q151 A	Q180 A
Q096 A	Q125 280	Q152 A	Q181 A
Q097 A	Q126 184	Q153 A	Q182 A
Q098 A	Q127 100	Q154 A	Q183 A
Q099 A	Q128 245	Q155 A	Q184 A
Q100 $\frac{n(5n+13)}{2}$	Q129 $\frac{n(n+1)(n+2)(3n+1)}{12}$	Q156 A	Q185 A
Q101 $n(n-2)$	Q130 $\frac{n(n+1)(3n^2-5n-4)}{12}$	Q157 A	Q186 A
Q102 $n(2n+1)$	Q131 $\frac{n(n+1)(3n^2+11n+4)}{12}$	Q158 A	Q187 A
Q103 $\frac{n(3n+7)}{2}$	Q132 $\frac{n(n+1)(n^2-3n-2)}{4}$	Q159 $3(3^n-1)$	Q188 A
Q104 $3n^2$	Q133 A	Q160 $4(2^n-1)$	Q189 $\frac{n}{2(3n+2)}$
Q105 A	Q134 A	Q161 $2(2^n-1)$	Q190 $\frac{n}{2n+1}$
Q106 A	Q135 A	Q162 $\frac{1-(-3)^n}{2}$	Q191 $\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$
Q107 A	Q136 A	Q163 $1-(-2)^n$	Q192 $\frac{n}{3n+1}$
Q108 A	Q137 A	Q164 $\frac{5((-5)^n-1)}{6}$	Q193 $\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$
Q109 $\frac{n(n-1)(2n-1)}{6}$	Q138 A	Q165 $1-(-1)^n$	Q194 A
Q110 A	Q139 A	Q166 $2(1-(-3)^n)$	Q195 A
Q111 A	Q140 A	Q167 $\frac{1-(-2)^n}{3}$	Q196 A
Q112 A	Q141 $\frac{n(n+1)(n-1)(n+2)}{4}$	Q168 $\frac{3((-3)^n-1)}{4}$	Q197 A
Q113 A	Q142 $\frac{n(n+1)(n^2+n+4)}{4}$	Q169 $\frac{4(4^n-1)}{3}$	Q198 A
Q114 A	Q143 A	Q170 $\frac{4^n-1}{3}$	Q199 A
Q115 $\frac{n(n+1)(10n-1)}{6}$	Q144 A	Q171 $\frac{3^n-1}{2}$	Q200 A
Q116 $\frac{n(n-1)(2n+1)}{2}$		Q172 -5	Q201 A
Q117 $\frac{n(n+1)(4n+5)}{6}$		Q173 39	Q202 $\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$

Q203 $\frac{n}{4(n+1)}$

Q204 $\sqrt{n+2} - \sqrt{2}$

Q205 $\sqrt{n+1} - 1$

Q206 $\sqrt{n}$

Q207 $\sqrt{n+1} - 1$

Q208 $\frac{\sqrt{2n+1} - 1}{2}$

Q209 $\sqrt{n+2} - \sqrt{2}$

Q210 $\frac{1425}{1984}$

Q211 $n \times 3^n$

Q212 $3^n n$

Q213 324

Q214 $\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n} + 5}{9}$

Q215 14109

Q216 A

Q217 A

Q218 A

Q219 $\frac{(2n+1) \times 3^n - 1}{4}$

Q220 A

Q221 A

Q222 A

Q223 A

Q224 A

Q225 $(n-1) \times 2^n + 1$

Q226 $(n-1) \times 3^{n+1} + 3$

Q227 A

Q228 $\frac{k(k+1)(4k+11)}{6}$

Q229 $a_k = k(k+2)$

Q230 $a_k = k^2$

Q231 $a_k = k(k+1)$

Q232 $a_k = k(2k-1)$

Q233 A

Q234 A

Q235 A

Q236 A

Q237 A

Q238 A

Q239 A

Q240 A

Q241 $S_n = \frac{n(n+1)(2n+7)}{6}$

Q242 $S_n = \frac{n(n+1)(2n+1)}{6}$

Q243 $S_n = \frac{n(n+1)(4n-1)}{6}$

Q244 $S_n = \frac{n(n+1)(n+2)}{3}$

Q245 A

Q246 $a_n = 4n + 3$

Q247 $a_n = 2(n-2)$

Q248 $a_n = 4n + 3$

Q249 $a_n = 2n - 1$

Q250 $a_n = 2n - 5$

Q251 D

Q252 C

Q253 B

Q254 A

Q255 D

Q256 C

Q257 C

Q258 C

Q259 B

Q260 A

Q261 D

Q262 D

Q263 B

Q264 $a_1 = 9, a_n = 3(n^2 - n + 1) (n \geq 2)$

Q265 $a_1 = 2, a_n = 2n + 1 (n \geq 2)$

Q266 $a_1 = 0, a_n = 3n^2 - 3n + 1 (n \geq 2)$

Q267 $a_n = 4 \cdot 5^{n-1}$

Q268 $a_n = 2 \cdot 3^{n-1}$

Q269 $a_n = 2^{n-1}$

Q270 $a_n = 6 \cdot 7^{n-1}$

Q271 $a_1 = 3, a_n = 4 \cdot 5^{n-1} (n \geq 2)$

Q272 $a_1 = 5, a_n = 2^{n+1} (n \geq 2)$

Q273 B

Q274 $a_1 = 8, a_n = 2(3^n - 1) (n \geq 2)$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 C

Quiz WRITTEN 5 $-1 + 2 + 7 + 14 + 23$

Quiz WRITTEN 6 $\frac{n(n+1)(3n^2 + 23n + 10)}{12}$

Quiz WRITTEN 7 $\frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$

Quiz WRITTEN 8 $a_1 = 3, a_n = 2^n - 2 (n \geq 2)$

Concept 4 | Advanced Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Differences

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

odd/even index $\rightarrow$ separate rules

Classified Practice

Advanced Sequences

Q001 **Written****Find the general term of the sequence $\{2, 5, 12, 23, 38, 57, \dots\}$.**

Classified Practice

Advanced Sequences

Q002 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, then a_n is:

Choose the correct option.

A $n^2 - 3$

B $(n + 1)^2$

C $a_n = n^2$

D $n^2 + 3$

Q003 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, then a_n is:

Choose the correct option.

A $\frac{3n^2 - n + 4}{2}$

B $a_n = \frac{3n^2 - 7n + 8}{2}$

C $\frac{3n^2 - 7n + 16}{2}$

D $\frac{3n^2 - 7n + 6}{2}$

Classified Practice

Advanced Sequences

Q004 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, then a_n is:

Choose the correct option.

A $a_n = 3n^2 - 8n + 12$

B $3n^2 - 8n + 13$

C $3n^2 - 8n + 9$

D $3n^2 - 2n + 7$

Classified Practice

Advanced Sequences

Q005 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, find a_5 :

Choose the correct option.

A -25

B 25

C 30

D 23

Classified Practice

Advanced Sequences

Q006 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, find a_6 :**Choose the correct option.**

A 37

B 38

C 34

D -37

Classified Practice

Advanced Sequences

Q007 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, find a_6 :

Choose the correct option.

A 73

B 70

C -72

D 72

Classified Practice

Advanced Sequences

Q008 **Written****Find the general term of the sequence $\{2, 10, 24, 44, 70, 102, \dots\}$.**

Classified Practice

Advanced Sequences

Q009 Written

Find the general term of the sequence $\{1, 6, 15, 28, 45, 66, \dots\}$.

Classified Practice

Advanced Sequences

Q010 MCQ

Select the correct option: If $a_1 = 3$ and $b_k = 5k + 4$, then a_n is:

Choose the correct option.

A $a_n = \frac{(n+1)(5n-2)}{2}$

B $\frac{n(5n+3)}{2}$

C $\frac{5n^2+3n-6}{2}$

D $\frac{(n+2)(5n+3)}{2}$

Q011 MCQ

Select the correct option: If $a_1 = 4$ and $b_k = 7k - 3$, then a_n is:

Choose the correct option.

A $\frac{7n^2 - 13n + 18}{2}$

B $\frac{7n^2 - 13n + 8}{2}$

C $\frac{7n^2 + n + 8}{2}$

D $a_n = \frac{7n^2 - 13n + 14}{2}$

Classified Practice

Advanced Sequences

Q012 MCQ

Select the correct option: If $a_1 = 6$ and $b_k = 2(2k + 3)$, then a_n is:

Choose the correct option.

A $2(n + 1)(n + 3)$

B $a_n = 2n(n + 2)$

C $2(n^2 + 2n + 2)$

D $2(n^2 + 2n - 1)$

Classified Practice

Advanced Sequences

Q013 Written

Find the general term of the sequence $\{5, 8, 9, 8, 5, 0, \dots\}$.

Classified Practice

Advanced Sequences

Q014 **Written****Find the general term of the sequence $\{2, 3, 6, 15, 42, 123, \dots\}$.**

Classified Practice

Advanced Sequences

Q015 Written

Find the general term of the sequence $\{1, 2, 5, 14, 41, 122, \dots\}$.

Classified Practice

Advanced Sequences

Q016 **Written****Find the general term of the sequence $\{5, 7, 11, 19, 35, \dots\}$.**

Classified Practice

Advanced Sequences

Q017 Written

Find the general term of the sequence $\{2, 3, 1, 5, -3, 13, \dots\}$.

Classified Practice

Advanced Sequences

Q018 **Written****Find the general term of the sequence $\{5, 4, 7, -2, 25, -56, \dots\}$.**

Classified Practice

Advanced Sequences

Q019 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = k^2$, then a_n is:

Choose the correct option.

A $a_n = \frac{(n+1)(2n^2-5n+6)}{6}$

B $\frac{2n^3-3n^2+n+12}{6}$

C $\frac{2n^3-3n^2+n-12}{6}$

D $\frac{(n+2)(2n^2-n+3)}{6}$

Classified Practice

Advanced Sequences

Q020 Challenge

An oncology team records tissue volumes 1, 3, 7, 13, 21, 31, ... in mm^3 . Find the general formula a_n and calculate a_{12} to decide whether the 130 mm^3 threshold is exceeded.

Classified Practice

Advanced Sequences

Q021 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Q022 Written

For groups with $a_j = 1 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Classified Practice

Advanced Sequences

Q023 Written

For groups with $a_j = 1 + (j - 1)1$ and $c_j = n$ terms in group j , find the first term of group 10.

Q024 MCQ**Select the correct option: find the first term of group 3:****Choose the correct option.**

A 6

B 4

C 5

D 3

Classified Practice

Advanced Sequences

Q025 MCQ

Select the correct option: find the first term of group 6:**Choose the correct option.**

- A 77
- B 108
- C 78
- D 79

Q026 MCQ**Select the correct option: find the first term of group 9:****Choose the correct option.**

A 38

B 36

C 45

D 37

Classified Practice

Advanced Sequences

Q027 MCQ

Select the correct option: find the first term of group 4:**Choose the correct option.**

A 48

B 30

C 31

D 29

Q028 MCQ**Select the correct option: find the first term of group 7:****Choose the correct option.**

A 21

B 28

C 22

D 23

Classified Practice

Advanced Sequences

Q029 MCQ

Select the correct option: find the first term of group 10:**Choose the correct option.**

A 247

B 245

C 300

D 246

Q030 MCQ**Select the correct option: find the first term of group 4:****Choose the correct option.**

A 20

B 21

C 19

D 32

Classified Practice

Advanced Sequences

Q031 MCQ

Select the correct option: find the first term of group 7:**Choose the correct option.**

A 138

B 110

C 111

D 109

Q032 MCQ**Select the correct option: find the first term of group 10:****Choose the correct option.**

A 163

B 200

C 164

D 165

Classified Practice

Advanced Sequences

Q033 MCQ

Select the correct option: find the first term of group 5:**Choose the correct option.**

A 58

B 59

C 57

D 78

Q034 MCQ**Select the correct option: find the first term of group 8:****Choose the correct option.**

A 128

B 100

C 101

D 99

Classified Practice

Advanced Sequences

Q035 **Written**

For groups with $a_j = 1 + (j - 1)3$ and $c_j = 3n - 1$ terms in group j , find the first term of group 10.

Q036 MCQ**Select the correct option: find the first term of group 5:****Choose the correct option.**

A 30

B 28

C 39

D 29

Classified Practice

Advanced Sequences

Q037 MCQ

Select the correct option: find the first term of group 8:**Choose the correct option.**

A 89

B 90

C 88

D 110

Q038 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , derive the sum of the terms in group n .

Classified Practice

Advanced Sequences

Q039 **Written****For the same grouped sequence, find the sum of the terms in group 10.**

Q040 Written

For the same grouped sequence, find the sum of the terms in group 10.

Classified Practice

Advanced Sequences

Q041 **Written****For the same grouped sequence, find the sum of the terms in group 10.**

Q042 Written

In the grouped arithmetic sequence, locate the value 88: state its group and its position within that group.

Classified Practice

Advanced Sequences

Q043 **Written**

In the grouped arithmetic sequence, locate the value 101: state its group and its position within that group.

Q044 Written

In the grouped arithmetic sequence, locate the value 124: state its group and its position within that group.

Classified Practice

Advanced Sequences

Q045 **Written**

In the grouped arithmetic sequence, locate the value 100: state its group and its position within that group.

Classified Practice

Advanced Sequences

Q046 Challenge

For the grouped sequence $2, 6 \mid 10, 14, 18 \mid 22, 26, 30, 34 \mid \dots$, solve: (a) the first term of Group 12; (b) the sum in Group n ; (c) the group and position of 202; (d) the total through Group 12.

Classified Practice

Advanced Sequences

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: If $a_1 = 2$ and $b_k = 3k + 1$, then a_n is:

Choose the correct option.

A $a_n = \frac{3n^2 - n + 2}{2}$

B $\frac{3n^2 - n + 4}{2}$

C $\frac{n(3n-1)}{2}$

D $\frac{3n^2 + 5n + 4}{2}$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 3$ and $b_k = (-2)^{k-1}$, then a_n is:

Choose the correct option.

A $\frac{(-2)^n + 2}{6}$

B $-\frac{(-2)^n - 10}{3}$

C $a_n = \frac{(-2)^n + 20}{6}$

D $\frac{(-2)^n + 38}{6}$

Classified Practice

Advanced Sequences

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

Select the correct option: find the first term of group 9:

Choose the correct option.

A 195

B 196

C 194

D 243

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: the value 29 is in:

Choose the correct option.

- A Group 5, position 2
- B Group 5, position 1
- C Group 6, position 1
- D Group 4, position 2

Classified Practice

Advanced Sequences

Quiz WRITTEN 5 Concept Quiz · Written

A sequence has $a_1 = 1$ and difference sequence $b_k = 4k - 1$. Find the required general term a_n .

Quiz WRITTEN 6 Concept Quiz · Written

A sequence has $a_1 = 2$ and difference sequence $b_k = (-2)^{k-1}$. Find the required general term a_n .

Classified Practice

Advanced Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For groups with $a_j = 5 + (j - 1)3$ and $c_j = n$ terms in group j , find the first term of group 12.

Quiz WRITTEN 8 Concept Quiz · Written

In the grouped arithmetic sequence, locate the value 66: state its group and its position within that group.

Answer key

Use the question number shown on each page.

Q001 $a_n = 2n^2 - 3n + 3$

Q002 C

Q003 B

Q004 A

Q005 B

Q006 A

Q007 D

Q008 $a_n = n(3n - 1)$

Q009 $a_n = n(2n - 1)$

Q010 A

Q011 D

Q012 B

Q013 $a_n = -n(n - 6)$

Q014 $a_n = \frac{3^n + 9}{6}$

Q015 $a_n = \frac{3^n + 3}{6}$

Q016 $a_n = 2^n + 3$

Q017 $a_n = \frac{(-2)^n + 14}{6}$

Q018 $a_n = -\frac{(-3)^n - 57}{12}$

Q019 A

Q020 $a_n = n^2 - n + 1$, $a_{12} = 133$; surgery required

Q021 164

Q022 163

Q023 46

Q024 B

Q025 C

Q026 D

Q027 B

Q028 C

Q029 D

Q030 A

Q031 B

Q032 C

Q033 A

Q034 B

Q035 379

Q036 D

Q037 A

Q038 $2(2n - 1)(n^2 - n + 1)$

Q039 3439

Q040 505

Q041 12209

Q042 Group 7, position 8

Q043 Group 8, position 2

Q044 Group 16, position 4

Q045 Group 5, position 8

Q046 (a) 310; (b) $2(n + 1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a_n = 2n^2 - 3n + 2$

Quiz WRITTEN 6 $a_n = \frac{(-2)^n + 14}{6}$

Quiz WRITTEN 7 203

Quiz WRITTEN 8 Group 5, position 3

Concept 5 | Recurrence and Induction

English Student Edition • Focused formulas and guided practice

Formula opener

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1), P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = mq'$$

Classified Practice

Recurrence and Induction

Q001 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (2)$. Find the 2nd through the 4th terms.

Q002 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 2a_n + (3)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q003 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 1a_n + (n(n-2))$. Find the 2nd through the 4th terms.

Q004 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (-n)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q005 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (2n)$. Find the 2nd through the 4th terms.

Q006 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3^n)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q007 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The fourth term is:

Choose the correct option.

A -81

B 81

C 82

D 80

Q008 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The fourth term is:

Choose the correct option.

A 133

B 127

C -128

D 128

Classified Practice

Recurrence and Induction

Q009 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (n + 1)$. The fourth term is:

Choose the correct option.

A 34

B 37

C 33

D -34

Q010 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 3a_n + (2n)$. The fourth term is:

Choose the correct option.

A -171

B 171

C 172

D 170

Classified Practice

Recurrence and Induction

Q011 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The fourth term is:

Choose the correct option.

A 17

B 11

C -13

D 13

Q012 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = -2a_n + (5)$. The fourth term is:

Choose the correct option.

A -17

B -15

C -19

D 17

Classified Practice

Recurrence and Induction

Q013 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -3a_n + (2)$. The fourth term is:

Choose the correct option.

A -15

B 13

C -13

D -12

Q014 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The fourth term is:

Choose the correct option.

A 22

B 16

C -18

D 18

Classified Practice

Recurrence and Induction

Q015 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (1)$. The fourth term is:

Choose the correct option.

A 20

B -23

C 23

D 28

Q016 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The fourth term is:

Choose the correct option.

A -1

B -7

C 4

D -4

Classified Practice

Recurrence and Induction

Q017 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The fourth term is:

Choose the correct option.

A -9

B 9

C 11

D 6

Q018 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 2a_n + (-n)$. The fourth term is:

Choose the correct option.

A -6

B 3

C -3

D 2

Classified Practice

Recurrence and Induction

Q019 Written

For the sequence $\{a_n\}$ with $a_1 = -1$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Q020 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Classified Practice

Recurrence and Induction

Q021 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-2)$. Find the general term.

Q022 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (6)$. Find the general term.

Classified Practice

Recurrence and Induction

Q023 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. Find the general term.

Q024 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Classified Practice

Recurrence and Induction

Q025 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-4)$. Find the general term.

Q026 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The general term is:

Choose the correct option.

A $3^n + 2$

B $3^n - 1$

C $3^n + 3$

D $a_n = 3^n$

Classified Practice

Recurrence and Induction

Q027 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The general term is:

Choose the correct option.

A $a_n = 2^{2n-1}$

B $\frac{2^{2n+2}}{2}$

C $\frac{2^{2n-2}}{2}$

D $\frac{2^{2n+4}}{2}$

Q028 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 5a_n + (0)$. The general term is:

Choose the correct option.

A $a_n = 4 \cdot 5^{n-1}$

B $\frac{4 \cdot 5^{n+5}}{5}$

C $\frac{4 \cdot 5^n - 5}{5}$

D $\frac{2(2 \cdot 5^n + 5)}{5}$

Classified Practice

Recurrence and Induction

Q029 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 6$ and recurrence $a_{n+1} = 2a_n + (0)$. The general term is:

Choose the correct option.

A $a_n = 3 \cdot 2^n$

B $3 \cdot 2^n + 1$

C $3 \cdot 2^n - 1$

D $3 \cdot 2^n + 2$

Q030 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q031 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 5a_n + (0)$. Find the general term.

Q032 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q033 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 4a_n + (0)$. Find the general term.

Q034 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Choose the correct option.

A $4n - 1$

B $a_n = 4n - 3$

C $4n + 1$

D $4n - 5$

Classified Practice

Recurrence and Induction

Q035 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The general term is:

Choose the correct option.

A $\frac{2n^3 - 3n^2 + n + 36}{6}$

B $a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$

C $\frac{2n^3 - 3n^2 + n + 48}{6}$

D $\frac{2n^3 - 3n^2 + n + 12}{6}$

Q036 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -4a_n + (0)$. The general term is:

Choose the correct option.

A $-\frac{(-4)^n - 8}{4}$

B $-\frac{(-4)^n + 8}{4}$

C $-\frac{(-4)^n - 12}{4}$

D $a_n = (-4)^{n-1}$

Classified Practice

Recurrence and Induction

Q037 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (3^n)$. The general term is:

Choose the correct option.

A $a_n = \frac{3^{n+7}}{2}$

B $\frac{3^{n+9}}{2}$

C $\frac{3^{n^2+3}}{2}$

D $\frac{3^{n^2+11}}{2}$

Q038 Written

For the sequence $\{a_n\}$ with $a_1 = -3$ and recurrence $a_{n+1} = -2a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q039 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = -5a_n + (0)$. Find the general term.

Q040 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = -4a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q041 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The general term is:

Choose the correct option.

A $10 - 3n$

B $a_n = 8 - 3n$

C $-3(n - 4)$

D $5 - 3n$

Q042 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The general term is:

Choose the correct option.

A $\frac{n(n-1)}{2}$

B $\frac{n^2-n+10}{2}$

C $a_n = \frac{n^2-n+6}{2}$

D $\frac{n^2-n+12}{2}$

Classified Practice

Recurrence and Induction

Q043 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. The general term is:

Choose the correct option.

A $-5(n - 2)$

B $a_n = 8 - 5n$

C $12 - 5n$

D $-5(n - 1)$

Q044 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (6)$. The general term is:

Choose the correct option.

A $6n - 7$

B $2(3n - 1)$

C $a_n = 2(3n - 2)$

D $6n - 1$

Classified Practice

Recurrence and Induction

Q045 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Q046 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -2a_n + (5)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q047 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q048 Challenge

Solve the third-order recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$ with $a_0 = 0, a_1 = 1, a_2 = 6$.

Classified Practice

Recurrence and Induction

Q049 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2n - 3)$.

Q050 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (3n)$.

Classified Practice

Recurrence and Induction

Q051 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (2n)$.

Q052 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$.

Classified Practice

Recurrence and Induction

Q053 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n + 1)$.

Q054 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n(2n + 1))$.

Classified Practice

Recurrence and Induction

Q055 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$, then a_n is:

Choose the correct option.

A $\frac{(n+1)(2n^2-5n+6)}{6}$

B $\frac{(n+2)(2n^2-7n+15)}{6}$

C $a_n = \frac{2n^3-3n^2+n+18}{6}$

D $\frac{2n^3-3n^2+n+36}{6}$

Q056 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$, then a_n is:

Choose the correct option.

A $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

B $\frac{(n+1)(2n^2 - 5n + 6)}{2}$

C $\frac{n(n-1)(2n-1)}{2}$

D $\frac{2n^3 - 3n^2 + n + 8}{2}$

Classified Practice

Recurrence and Induction

Q057 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$, then a_n is:

Choose the correct option.

A $\frac{(n-1)(2n^2-n+6)}{6}$

B $\frac{2n^3-3n^2+7n+18}{6}$

C $a_n = \frac{2n^3-3n^2+7n+6}{6}$

D $\frac{(2n+3)(n^2-3n+8)}{6}$

Q058 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$, then a_n is:

Choose the correct option.

A $\frac{8n^3 - 15n^2 + 7n + 42}{6}$

B $\frac{8n^3 - 15n^2 + 7n + 18}{6}$

C $\frac{8n^3 - 15n^2 + 7n + 48}{6}$

D $a_n = \frac{(n+1)(8n^2 - 23n + 30)}{6}$

Classified Practice

Recurrence and Induction

Q059 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (8n)$, then a_n is:

Choose the correct option.

A $4n(n-1)$

B $4(n^2 - n + 1)$

C $a_n = 2(2n^2 - 2n + 1)$

D $4n^2 - 4n + 5$

Q060 MCQ**Select the correct option: find a_4 when $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$:****Choose the correct option.**

A 17

B 19

C 15

D -17

Classified Practice

Recurrence and Induction

Q061 MCQ

Select the correct option: find a_5 when $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$:

Choose the correct option.

A 96

B 90

C -92

D 92

Q062 MCQ

Select the correct option: find a_3 when $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$:

Choose the correct option.

A 9

B 11

C 7

D -9

Classified Practice

Recurrence and Induction

Q063 MCQ

Select the correct option: find a_6 when $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$:

Choose the correct option.

A -210

B 210

C 215

D 208

Q064 MCQ

Select the correct option: find a_2 when $a_1 = 2$ and $a_{n+1} = a_n + (8n)$:

Choose the correct option.

A 10

B 12

C 8

D -10

Classified Practice

Recurrence and Induction

Q065 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n^2)$.

Q066 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = a_n + (2n + 1)$, then a_n is:

Choose the correct option.

A $a_n = n^2$

B $n^2 + 1$

C $n^2 - 3$

D $n^2 + 2$

Classified Practice

Recurrence and Induction

Q067 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2^n)$.

Q068 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (-3^n)$.

Classified Practice

Recurrence and Induction

Q069 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 4$ and $a_{n+1} = a_n + (-4^n)$.

Classified Practice

Recurrence and Induction

Q070 Challenge

A technology startup has a reserve of 400,000 LE, represented by $a_1 = 4$, and burns $0.04n$ hundred-thousand LE each quarter: $a_{n+1} = a_n - 0.04n$. Find an explicit formula for a_n .

Classified Practice

Recurrence and Induction

Q071 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 3a_n + (4)$.

Classified Practice

Recurrence and Induction

Q072 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Classified Practice

Recurrence and Induction

Q073 Written

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = 2a_n + (-1)$.

Classified Practice

Recurrence and Induction

Q074 **Written**

Find the general term of $\{a_n\}$ given $a_1 = -3$ and $a_{n+1} = 4a_n + (3)$.

Classified Practice

Recurrence and Induction

Q075 Written

Find the general term of $\{a_n\}$ given $a_1 = 3$ and $a_{n+1} = 3a_n + (-2)$.

Classified Practice

Recurrence and Induction

Q076 Written

Find the general term of $\{a_n\}$ given $a_1 = -2$ and $a_{n+1} = -3a_n + (4)$.

Classified Practice

Recurrence and Induction

Q077 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = \frac{1}{3}a_n + (2)$.

Classified Practice

Recurrence and Induction

Q078 Written

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = \frac{2}{3}a_n + (-\frac{1}{2})$.

Classified Practice

Recurrence and Induction

Q079 Written

Find the general term of $\{a_n\}$ given $a_1 = 4$ and $a_{n+1} = -2a_n + (3)$.

Classified Practice

Recurrence and Induction

Q080 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Choose the correct option.

A $-\frac{2(-2)^n - 11}{3}$

B $a_n = -\frac{2(-2)^n - 5}{3}$

C $-\frac{2(-2)^n - 17}{3}$

D $-\frac{2((-2)^n + 2)}{3}$

Classified Practice

Recurrence and Induction

Q081 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{2}{3}a_n + (-2)$, the general term is:

Choose the correct option.

A $\frac{3 \cdot 3^{-n}(7 \cdot 2^n - 6 \cdot 3^n)}{2}$

B $\frac{3^{-n}(21 \cdot 2^n - 8 \cdot 3^n)}{2}$

C $a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 4 \cdot 3^n)}{2}$

D $\frac{3 \cdot 3^{-n}(7 \cdot 2^n - 2 \cdot 3^n)}{2}$

Classified Practice

Recurrence and Induction

Q082 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = 7a_n + (-4)$, the general term is:

Choose the correct option.

A $\frac{4(7^n+14)}{21}$

B $\frac{4 \cdot 7^n - 49}{21}$

C $\frac{4 \cdot 7^n + 77}{21}$

D $a_n = \frac{2(2 \cdot 7^n + 7)}{21}$

Classified Practice

Recurrence and Induction

Q083 MCQ

Select the correct option: If $a_1 = 6$ and $a_{n+1} = 2a_n + (-5)$, the general term is:

Choose the correct option.

A $\frac{2^n+14}{2}$

B $a_n = \frac{2^n+10}{2}$

C $\frac{2^n+18}{2}$

D $\frac{2^n+4}{2}$

Classified Practice

Recurrence and Induction

Q084 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = -2a_n + (7)$, the general term is:

Choose the correct option.

A $-\frac{2(2(-2)^n+1)}{3}$

B $-\frac{4(-2)^n-13}{3}$

C $a_n = -\frac{4(-2)^n-7}{3}$

D $-\frac{4((-2)^n-4)}{3}$

Classified Practice

Recurrence and Induction

Q085 MCQ

Select the correct option: If $a_1 = 4$ and $a_{n+1} = -5a_n + (-2)$, the general term is:

Choose the correct option.

A $-\frac{13(-5)^n - 25}{15}$

B $-\frac{13(-5)^n + 50}{15}$

C $-\frac{13(-5)^n - 40}{15}$

D $a_n = -\frac{13(-5)^n + 5}{15}$

Classified Practice

Recurrence and Induction

Q086 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -3a_n + (5)$, the general term is:

Choose the correct option.

A $-\frac{(-3)^n - 13}{4}$

B $a_n = -\frac{(-3)^n - 5}{4}$

C $-\frac{(-3)^n - 21}{4}$

D $-\frac{(-3)^n + 7}{4}$

Classified Practice

Recurrence and Induction

Q087 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{1}{2}a_n + (7)$, the general term is:

Choose the correct option.

A $2^{-n} (11 \cdot 2^n - 26)$

B $2 \cdot 2^{-n} (8 \cdot 2^n - 13)$

C $a_n = 2 \cdot 2^{-n} (7 \cdot 2^n - 13)$

D $2^{-n} (17 \cdot 2^n - 26)$

Classified Practice

Recurrence and Induction

Q088 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = 2a_n + (-1)$, the general term is:

Choose the correct option.

A 3

B 0

C 4

D $a_n = 1$

Classified Practice

Recurrence and Induction

Q089 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Choose the correct option.

A $-\frac{2(-2)^n - 11}{3}$

B $-\frac{2((-2)^n + 2)}{3}$

C $-\frac{2((-2)^n - 7)}{3}$

D $a_n = -\frac{2(-2)^n - 5}{3}$

Classified Practice

Recurrence and Induction

Q090 Challenge

A sequence satisfies $a_{n+1} = 4a_n - 9$ with $a_1 = 2$. Find the equilibrium value, the explicit formula, and a_5 .

Classified Practice

Recurrence and Induction

Q091 Written

Use mathematical induction to prove $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

Classified Practice

Recurrence and Induction

Q092 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1)^2 = \frac{n(2n-1)(2n+1)}{3}$.

Classified Practice

Recurrence and Induction

Q093 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Classified Practice

Recurrence and Induction

Q094 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Choose the correct option.

A $\frac{n(n+1)(n+2)}{3}$

B $\frac{n^3+3n^2+2n+3}{3}$

C $\frac{n^3+3n^2+2n-3}{3}$

D $\frac{(n+3)(n^2+2)}{3}$

Classified Practice

Recurrence and Induction

Q095 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Choose the correct option.

A $4^n + 1$

B $4^n - 1$

C $4^n + 3$

D $4^n - 2$

Classified Practice

Recurrence and Induction

Q096 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Choose the correct option.

A $n^2 + 4n - 1$

B $n^2 + 4n + 2$

C $n(n + 4)$

D $(n + 1)(n + 3)$

Classified Practice

Recurrence and Induction

Q097 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i^2 + 3i$ is:

Choose the correct option.

A $\frac{4n^3+15n^2+11n+12}{6}$

B $\frac{n(n+1)(4n+11)}{6}$

C $\frac{4n^3+15n^2+11n+24}{6}$

D $\frac{(n+2)(4n^2+7n-3)}{6}$

Classified Practice

Recurrence and Induction

Q098 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3i^2 + 2i$ is:

Choose the correct option.

A $\frac{2n^3 + 5n^2 + 3n - 2}{2}$

B $\frac{2n^3 + 5n^2 + 3n + 4}{2}$

C $\frac{n(n+1)(2n+3)}{2}$

D $\frac{2n^3 + 5n^2 + 3n + 6}{2}$

Classified Practice

Recurrence and Induction

Q099 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 7 \cdot 2^{i-1}$ is:

Choose the correct option.

A $7(2^n - 1)$

B $72^n - 6$

C $72^n - 8$

D $72^n - 5$

Classified Practice

Recurrence and Induction

Q100 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Choose the correct option.

A $\frac{n^3+3n^2+2n-3}{3}$

B $\frac{(n+3)(n^2+2)}{3}$

C $\frac{n(n+1)(n+2)}{3}$

D $\frac{n^3+3n^2+2n+9}{3}$

Classified Practice

Recurrence and Induction

Q101 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Choose the correct option.

A $4^n - 1$

B 4^n

C $4^n - 2$

D $4^n + 1$

Classified Practice

Recurrence and Induction

Q102 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Choose the correct option.

A $n^2 + 4n + 2$

B $n(n + 4)$

C $(n + 2)^2$

D $n^2 + 4n - 1$

Classified Practice

Recurrence and Induction

Q103 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1) = n^2$.

Classified Practice

Recurrence and Induction

Q104 **Written**

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Classified Practice

Recurrence and Induction

Q105 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+2) = \frac{n(n+1)(2n+7)}{6}$.

Classified Practice

Recurrence and Induction

Q106 Written

Use mathematical induction to prove $\sum_{i=1}^n 3 \cdot 4^{i-1} = 4^n - 1$.

Classified Practice

Recurrence and Induction

Q107 Written

Use mathematical induction to prove $\sum_{i=0}^n 10^i = \frac{10^{n+1}-10}{9}$.

Classified Practice

Recurrence and Induction

Q108 Challenge

Use mathematical induction to prove $\sum_{i=1}^n (i^2 + i + 1) = \frac{n(n^2 + 3n + 5)}{3}$.

Classified Practice

Recurrence and Induction

Q109 Written

Prove by mathematical induction that $2^n > 3n$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q110 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 3$.

Classified Practice

Recurrence and Induction

Q111 **Written****Prove by mathematical induction that $3^n > 2n$ for $n \geq 1$.**

Classified Practice

Recurrence and Induction

Q112 Written

Prove by mathematical induction that $2^n > 1n + 2$ for $n \geq 3$.

Classified Practice

Recurrence and Induction

Q113 **Written****Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 4$.**

Classified Practice

Recurrence and Induction

Q114 **Written****Prove by mathematical induction that $3^n > 5n + 2$ for $n \geq 4$.**

Classified Practice

Recurrence and Induction

Q115 Written

Prove by mathematical induction that $4^n > 2n + 3$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q116 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $2^n \geq 1n + 2$

B $2^n > 2n + 2$

C $2^n + 1 > 1n + 2$

D $2^n > 1n + 2$

Classified Practice

Recurrence and Induction

Q117 MCQ

Select the correct option: which inequality is proved for $n \geq 2$:**Choose the correct option.**

A $3^n > 2n$

B $3^n \geq 2n + 0$

C $3^n > 3n + 0$

D $3^n + 1 > 2n + 0$

Classified Practice

Recurrence and Induction

Q118 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Choose the correct option.**

A $3^n + 1 > 5n + 1$

B $3^n > 5n + 1$

C $3^n \geq 5n + 1$

D $3^n > 6n + 1$

Classified Practice

Recurrence and Induction

Q119 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $3^n \geq 3n + 0$

B $3^n > 4n + 0$

C $3^n + 1 > 3n + 0$

D $3^n > 3n$

Classified Practice

Recurrence and Induction

Q120 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Choose the correct option.**

A $3^n > 5n + 2$

B $3^n \geq 5n + 2$

C $3^n > 6n + 2$

D $3^n + 1 > 5n + 2$

Classified Practice

Recurrence and Induction

Q121 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:

Choose the correct option.

A $4^n + 1 > 3n + 1$

B $4^n > 3n + 1$

C $4^n \geq 3n + 1$

D $4^n > 4n + 1$

Classified Practice

Recurrence and Induction

Q122 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Choose the correct option.**

A $4^n \geq 2n + 3$

B $4^n > 3n + 3$

C $4^n + 1 > 2n + 3$

D $4^n > 2n + 3$

Classified Practice

Recurrence and Induction

Q123 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?**Choose the correct option.**

A $4^n > 4n + 1$

B $4^n \geq 4n + 1$

C $4^n > 5n + 1$

D $4^n + 1 > 4n + 1$

Classified Practice

Recurrence and Induction

Q124 Written

Prove by mathematical induction that $2^n > 3n + 1$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q125 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $4^n + 1 > 3n + 0$

B $4^n > 3n$

C $4^n \geq 3n + 0$

D $4^n > 4n + 0$

Classified Practice

Recurrence and Induction

Q126 **Challenge****When $h > 0$ and $n \geq 3$, prove $(1 + h)^n > 1 + nh^2$.**

Classified Practice

Recurrence and Induction

Q127 Written

Prove that $n^3 + (n + 1)^3 + (n + 2)^3$ is a multiple of 9 for every natural number n , using mathematical induction.

Classified Practice

Recurrence and Induction

Q128 **Written**

Prove that $n^3 + 2n$ is a multiple of 3 for every natural number n , using mathematical induction.

Classified Practice

Recurrence and Induction

Q129 Written

Prove by mathematical induction that $n^2 + n$ is a multiple of 2.

Classified Practice

Recurrence and Induction

Q130 Written

Prove by mathematical induction that $n^2 + 5n + 8$ is a multiple of 2.

Classified Practice

Recurrence and Induction

Q131 **Written****Prove by mathematical induction that $n^2 + 7n + 12$ is a multiple of 2.**

Classified Practice

Recurrence and Induction

Q132 **Written****Prove by mathematical induction that $n^2 + 11n + 32$ is a multiple of 2.**

Classified Practice

Recurrence and Induction

Q133 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n$ for every natural number n ?:

Choose the correct option.

A 2

B 3

C 4

D 5

Classified Practice

Recurrence and Induction

Q134 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 5n + 8$ for every natural number n ?:

Choose the correct option.

A 3

B 4

C 5

D 2

Classified Practice

Recurrence and Induction

Q135 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 7n + 12$ for every natural number n ?:

Choose the correct option.

A 4

B 5

C 2

D 3

Classified Practice

Recurrence and Induction

Q136 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 11n + 32$ for every natural number n ?:

Choose the correct option.

A 4

B 5

C 2

D 3

Classified Practice

Recurrence and Induction

Q137 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 3n^2 + 2n$ for every natural number n ?:

Choose the correct option.

A 9

B 6

C 7

D 8

Classified Practice

Recurrence and Induction

Q138 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 9n^2 + 26n + 30$ for every natural number n ?:

Choose the correct option.

A 6

B 7

C 8

D 9

Classified Practice

Recurrence and Induction

Q139 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 12n^2 + 47n + 60$ for every natural number n ?:

Choose the correct option.

A 7

B 8

C 9

D 6

Classified Practice

Recurrence and Induction

Q140 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 18n^2 + 107n + 216$ for every natural number n ?:

Choose the correct option.

A 7

B 8

C 9

D 6

Classified Practice

Recurrence and Induction

Q141 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n + 2$ for every natural number n ?:

Choose the correct option.

A 3

B 4

C 5

D 2

Classified Practice

Recurrence and Induction

Q142 Written

Prove by mathematical induction that $2n^3 - 3n^2 + n$ is a multiple of 6.

Classified Practice

Recurrence and Induction

Q143 **Written****Prove by mathematical induction that $4n^3 - n$ is a multiple of 3.**

Classified Practice

Recurrence and Induction

Q144 Challenge

For any integer $n \geq 1$, prove that $(n + 1)^n - 1$ is a multiple of n^2 .

Classified Practice

Recurrence and Induction

Q145 Written

For the sequence determined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, find a_2, a_3, a_4 and estimate the general term.

Q146 Written

For the sequence determined by $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, find a_2, a_3, a_4 and estimate the general term.

Classified Practice

Recurrence and Induction

Q147 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, find a_2, a_3, a_4 and estimate the general term.

Q148 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, find a_2, a_3, a_4 and estimate the general term.

Classified Practice

Recurrence and Induction

Q149 Written

For a sequence with $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, prove by mathematical induction that $a_n = n + 2$.

Classified Practice

Recurrence and Induction

Q150 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2n$

B $a_n = 2(n - 1)$

C $a_n = 1 - 2n$

D $a_n = 2n - 1$

Classified Practice

Recurrence and Induction

Q151 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2n - 1$

B $a_n = -2n$

C $a_n = 2n$

D $a_n = 2n + 1$

Classified Practice

Recurrence and Induction

Q152 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = -2n - 1$

B $a_n = 2n + 1$

C $a_n = 2(n + 1)$

D $a_n = 2n$

Classified Practice

Recurrence and Induction

Q153 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2(n + 1)$

B $a_n = 2n + 3$

C $a_n = 2n + 1$

D $a_n = -2(n + 1)$

Classified Practice

Recurrence and Induction

Q154 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = 3n - 1$

B $a_n = 3(n - 1)$

C $a_n = 2 - 3n$

D $a_n = 3n - 2$

Classified Practice

Recurrence and Induction

Q155 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = 3n - 2$

B $a_n = 1 - 3n$

C $a_n = 3n - 1$

D $a_n = 3n$

Classified Practice

Recurrence and Induction

Q156 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = -3n$

B $a_n = 3n$

C $a_n = 3n + 1$

D $a_n = 3n - 1$

Classified Practice

Recurrence and Induction

Q157 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = 3n + 1$

B $a_n = 3n + 2$

C $a_n = 3n$

D $a_n = -3n - 1$

Classified Practice

Recurrence and Induction

Q158 **Written**

For a sequence with $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, prove by mathematical induction that $a_n = n^2$.

Classified Practice

Recurrence and Induction

Q159 Written

For a sequence with $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, prove by mathematical induction that $a_n = 1 - 2n$.

Classified Practice

Recurrence and Induction

Q160 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = 2a_n + 0$, the general term is:

Choose the correct option.

A $a_n = \frac{2^n - 2}{2}$

B $a_n = -2^{n-1}$

C $a_n = 2^{n-1}$

D $a_n = \frac{2^n + 2}{2}$

Classified Practice

Recurrence and Induction

Q161 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, prove by mathematical induction that $a_n = \frac{2n-1}{n}$.

Classified Practice

Recurrence and Induction

Q162 Challenge

Three numbers A, B, C in that order form a geometric sequence with common ratio r . If $A, 2B, C$ in that order form an arithmetic sequence, determine the possible values of r .

Classified Practice

Recurrence and Induction

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Choose the correct option.

A $4n$

B $4n - 5$

C $4n + 1$

D $a_n = 2(2n - 1)$

Classified Practice

Recurrence and Induction

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -2a_n + (3)$, the general term is:

Choose the correct option.

A $a_n = -\frac{(-2)^n - 2}{2}$

B $-\frac{(-2)^n - 4}{2}$

C $-\frac{(-2)^n + 4}{2}$

D $-\frac{(-2)^n - 6}{2}$

Classified Practice

Recurrence and Induction

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: which inequality is proved for $n \geq 4$?

Choose the correct option.

A $2^n + 1 > 3n + 1$

B $2^n > 3n + 1$

C $2^n \geq 3n + 1$

D $2^n > 4n + 1$

Classified Practice

Recurrence and Induction

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2n + 1$

B $a_n = -2(n + 1)$

C $a_n = 2(n + 1)$

D $a_n = 2n + 3$

Classified Practice

Recurrence and Induction

Quiz WRITTEN 5 Concept Quiz · Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Classified Practice

Recurrence and Induction

Quiz WRITTEN 6 Concept Quiz · Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Classified Practice

Recurrence and Induction

Quiz WRITTEN 7 Concept Quiz · Written

Prove by mathematical induction that $2^n > 4n + 1$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Quiz WRITTEN 8 Concept Quiz · Written

For a sequence with $a_1 = 5$ and $a_{n+1} = a_n + 1(2n + 1) + 2$, prove by mathematical induction that $a_n = n^2 + 2n + 2$.

Answer key

Use the question number shown on each page.

Q001 $a_2 = 8, a_3 = 26, a_4 = 80$

Q002 $a_2 = -9, a_3 = -15, a_4 = -27$

Q003 $a_2 = -7, a_3 = -7, a_4 = -4$

Q004 $a_2 = -3, a_3 = -5, a_4 = -8$

Q005 $a_2 = 4, a_3 = 8, a_4 = 14$

Q006 $a_2 = 1, a_3 = 10, a_4 = 37$

Q007 B

Q008 D

Q009 A

Q010 B

Q011 D

Q012 A

Q013 C

Q014 D

Q015 C

Q016 D

Q017 B

Q018 C

Q019 $a_n = 5n - 6$

Q020 $a_n = 3n - 2$

Q021 $a_n = -2(n - 3)$

Q022 $a_n = 6n - 5$

Q023 $a_n = 8 - 5n$

Q024 $a_n = 3n - 5$

Q025 $a_n = -4(n - 2)$

Q026 D

Q027 A

Q028 A

Q029 A

Q030 $a_n = 2 \cdot 3^{n-1}$

Q031 $a_n = 3 \cdot 5^{n-1}$

Q032 $a_n = 4 \cdot 3^{n-1}$

Q033 $a_n = 2^{2n-2}$

Q034 B

Q035 B

Q036 D

Q037 A

Q038 $a_n = \frac{3(-2)^n}{2}$

Q039 $a_n = 2(-5)^{n-1}$

Q040 $a_n = \frac{(-4)^n}{2}$

Q041 B

Q042 C

Q043 B

Q044 C

Q045 $a_2 = 5, a_3 = 13, a_4 = 36$

Q046 $a_2 = 3, a_3 = -1, a_4 = 7$

Q047 $a_2 = 8, a_3 = 22, a_4 = 63$

Q048 $a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$

Q049 $a_n = (n - 2)^2$

Q050 $a_n = \frac{3n^2 - 3n + 2}{2}$

Q051 $a_n = n^2 - n + 2$

Q052 $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

Q053 $a_n = \frac{3n^2 - n + 2}{2}$

Q054 $a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$

Q055 C

Q056 A

Q057 C

Q058 D

Q059 C

Q060 A

Q061 D

Q062 A

Q063 B

Q064 A

Q065 $a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$

Q066 A

Q067 $a_n = 2^n - 1$

Q068 $a_n = -\frac{3^n - 7}{2}$

Q069 $a_n = -\frac{4^n - 16}{3}$

Q070 $a_n = -\frac{n^2 - n - 200}{50}$

Q071 $a_n = 3^n - 2$

Q072 $a_n = \frac{3 \cdot 5^n - 10}{5}$

Q073 $a_n = \frac{2^n + 2}{2}$

Q074 $a_n = -\frac{4^n + 2}{2}$

Q075 $a_n = \frac{2 \cdot 3^n + 3}{3}$

Q076 $a_n = (-3)^n + 1$

Q077 $a_n = 3 \cdot 3^{-n} (3^n - 2)$

Q078 $a_n = \frac{3 \cdot 3^{-n} (7 \cdot 2^n - 2 \cdot 3^n)}{4}$

Q079 $a_n = -\frac{3(-2)^n - 2}{2}$

Q080 B

Q081 C

Q082 D

Q083 B

Q084 C

Q085 D

Q086 B

Q087 C

Q088 D

Q089 D

Q090 $c = 3, a_n = -\frac{2^{2n} - 12}{4}, a_5 = -253$

Q091 $\frac{n(n+1)}{2}$

Q092 $\frac{n(2n-1)(2n+1)}{3}$

Q093 $\frac{n(n+1)(n+2)}{3}$

Q094 A

Q095 B	Q113 $3^n > 5n + 1$	Q133 A	Q152 B
Q096 C	Q114 $3^n > 5n + 2$	Q134 D	Q153 A
Q097 B	Q115 $4^n > 2n + 3$	Q135 C	Q154 D
Q098 C	Q116 D	Q136 C	Q155 C
Q099 A	Q117 A	Q137 B	Q156 B
Q100 C	Q118 B	Q138 A	Q157 A
Q101 A	Q119 D	Q139 D	Q158 $a_n = n^2$
Q102 B	Q120 A	Q140 D	Q159 $a_n = 1 - 2n$
Q103 n^2	Q121 B	Q141 D	Q160 C
Q104 $\frac{n(n+1)(n+2)}{3}$	Q122 D	Q142 $6 \mid (2n^3 - 3n^2 + n)$	Q161 $a_n = \frac{2n-1}{n}$
Q105 $\frac{n(n+1)(2n+7)}{6}$	Q123 A	Q143 $3 \mid (4n^3 - n)$	Q162 $r = 2 \pm \sqrt{3}$
Q106 $4^n - 1$	Q124 $2^n > 3n + 1$	Q144 $n^2 \mid [(n+1)^n - 1]$	Quiz MCQ 1 D
Q107 $\frac{10 \cdot 10^n - 1}{9}$	Q125 B	Q145 $a_2 = 4, a_3 = 5, a_4 = 6; a_n = n + 2$	Quiz MCQ 2 A
Q108 $\frac{n(n^2 + 3n + 5)}{3}$	Q126 $(1+h)^n > 1 + nh^2$	Q146 $a_2 = -3, a_3 = -5, a_4 = -7; a_n = 1 - 2n$	Quiz MCQ 3 B
Q109 $2^n > 3n$	Q127 $9 \mid [n^3 + (n+1)^3 + (n+2)^3]$	Q147 $a_2 = 4, a_3 = 9, a_4 = 16; a_n = n^2$	Quiz MCQ 4 C
Q110 $3^n > 5n + 1$	Q128 $3 \mid (n^3 + 2n)$	Q148 $a_2 = \frac{3}{2}, a_3 = \frac{5}{3}, a_4 = \frac{7}{4}; a_n = \frac{2n-1}{n}$	Quiz WRITTEN 5 $a_n = 5n - 3$
Q111 $3^n > 2n$	Q129 $2 \mid (n^2 + n)$	Q149 $a_n = n + 2$	Quiz WRITTEN 6 $a_n = \frac{3 \cdot 5^n - 10}{5}$
Q112 $2^n > 1n + 2$	Q130 $2 \mid (n^2 + 5n + 8)$	Q150 D	Quiz WRITTEN 7 $2^n > 4n + 1$
	Q131 $2 \mid (n^2 + 7n + 12)$	Q151 C	Quiz WRITTEN 8 $a_n = n^2 + 2n + 2$
	Q132 $2 \mid (n^2 + 11n + 32)$		

Concept 1 | Complex Numbers

English Student Edition • Focused formulas and guided practice

Formula opener

Standard form

$$z = a + bi$$

Conjugate

$$\overline{a + bi} = a - bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Classified Practice

Complex Numbers

Q001 **Written****State the real and imaginary parts of $3 - 2i$.**

Q002 Written

State the real and imaginary parts of $(4 - \sqrt{5}i)/3$.

Classified Practice

Complex Numbers

Q003 **Written****State the real and imaginary parts of -8 .**

Q004 Written**State the real and imaginary parts of $2i$.**

Classified Practice

Complex Numbers

Q005 Written

Find the real values of x and y satisfying $(2 + i)x + (1 + i)y = 2 + 3i$, x, y real.

Q006 Written**State the real and imaginary parts of $-3 + 5i$.**

Classified Practice

Complex Numbers

Q007 Written

State the real and imaginary parts of $(-1 - \sqrt{3}i)/2$.

Q008 Written**State the real and imaginary parts of 1.**

Classified Practice

Complex Numbers

Q009 Written

State the real and imaginary parts of $3i$.

Q010 Written

Find the real values of x and y satisfying $(x - 6) + (x + 3y)i = 0$, x, y real.

Classified Practice

Complex Numbers

Q011 Written

Find the real values of x and y satisfying $(3 + 2i)x + (4 - i)y = 6 - 7i$, x, y real.

Q012 Written**State the real and imaginary parts of $2 + 5i$.**

Classified Practice

Complex Numbers

Q013 Written

State the real and imaginary parts of $-25 + 13i$.

Q014 Written**State the real and imaginary parts of $(2 - \sqrt{5}i)/3$.**

Classified Practice

Complex Numbers

Q015 Written

State the real and imaginary parts of $(7 - \sqrt{2}i)/2$.

Q016 Written**State the real and imaginary parts of -4 .**

Classified Practice

Complex Numbers

Q017 **Written****State the real and imaginary parts of $1/4$.**

Q018 Written**State the real and imaginary parts of $\sqrt{3}i$.**

Classified Practice

Complex Numbers

Q019 Written

State the real and imaginary parts of $-i$.

Q020 Written**Find the real values of x and y satisfying $(2x + y) + (3x - y)i = 8 + 7i$.**

Classified Practice

Complex Numbers

Q021 Written

Find the real values of x and y satisfying $(x + 1) - 5i = 4 + (2y + 1)i$.

Q022 Written**Find the real values of x and y satisfying $(x - 2) + (x + y)i = 0$.**

Classified Practice

Complex Numbers

Q023 Written

Find the real values of x and y satisfying $(3 + 2i)x + (1 - i)y = 7 + 3i$.

Q024 Written**Find the real values of x and y satisfying $(1 - i)(x + yi) = 2 + i$.**

Classified Practice

Complex Numbers

Q025 Written

Solve the complex-number equation and find the exact real values of x and y : $(1 + 2i)(x + i) = x(y + i)$.

Q026 Written**Calculate $(3 - 2i) - (2 + 5i)$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q027 Written

Calculate $(i - 2)^2$ and give the answer in standard form.

Q028 Written**Calculate $i^6 + i^{27} + i^{17}$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q029 Written

Calculate $(4 + 6i) + (-3 + 2i)$ and give the answer in standard form.

Q030 Written

Calculate $(2 + 3i)(1 - 2i)$ and give the answer in standard form.

Classified Practice

Complex Numbers

Q031 Written

Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.

Q032 Written

Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.

Classified Practice

Complex Numbers

Q033 Written

Calculate $(-4 + 3i) + (5 - 6i)$ and give the answer in standard form.

Q034 Written**Calculate $(3 - 3i) - (5 - 3i)$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q035 Written

Calculate $(\sqrt{2} + 3i)(\sqrt{2} + i)$ and give the answer in standard form.

Q036 Written**Calculate $(2 + 3i) + (5 + 8i)$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q037 Written

Calculate $(\frac{2}{5} - \frac{1}{2}i) - (\frac{1}{3} + \frac{3}{7}i)$ and give the answer in standard form.

Q038 Written**Calculate $(2i - 1)^2$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q039 Written

Calculate $(2/i + i)(2/i - i)$ and give the answer in standard form.

Q040 Written

Calculate $(2 + i)^3 + (2 - i)^3$ and give the answer in standard form.

Classified Practice

Complex Numbers

Q041 Written

Calculate $i^5 + i^3 + i$ and give the answer in standard form.

Q042 Written**Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q043 Written

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

Q044 Written**Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2i^{40}$.**

Classified Practice

Complex Numbers

Q045 Written

State the complex conjugate of $2 + 3i$.

Q046 Written**State the complex conjugate of $7 - i\sqrt{11}$.**

Classified Practice

Complex Numbers

Q047 Written

State the complex conjugate of 6.

Q048 Written**State the complex conjugate of $-5i$.**

Classified Practice

Complex Numbers

Q049 **Written****Calculate $(2 + i)/(2 - i)$ and give the answer in the form $a+bi$.**

Q050 Written**Calculate $(3 - i)/(2i)$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q051 Written

State the complex conjugate of $1 - 4i$.

Q052 Written**State the complex conjugate of $\sqrt{2} + i\sqrt{10}$.**

Classified Practice

Complex Numbers

Q053 Written

State the complex conjugate of 2.

Q054 Written**State the complex conjugate of $-8i$.**

Classified Practice

Complex Numbers

Q055 **Written****Calculate $(3 + i)/(1 + 2i)$ and give the answer in the form $a+bi$.**

Q056 Written**Calculate $(1 - i)/i$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q057 Written

State the complex conjugate of $3 + 4i$.

Q058 Written**State the complex conjugate of $9 + 8i$.**

Classified Practice

Complex Numbers

Q059 Written

State the complex conjugate of $8 + i\sqrt{3}$.

Q060 Written**State the complex conjugate of 5.**

Classified Practice

Complex Numbers

Q061 Written

State the complex conjugate of -15 .

Q062 Written**State the complex conjugate of $\sqrt{7}$.**

Classified Practice

Complex Numbers

Q063 Written

State the complex conjugate of i .

Q064 Written**State the complex conjugate of $-50i$.**

Classified Practice

Complex Numbers

Q065 Written

State the complex conjugate of $i\sqrt{6}$.

Q066 Written**Calculate $(1 + i)/(1 - i)$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q067 Written

Calculate $(3 + i)/(2 + i)$ and give the answer in the form $a+bi$.

Q068 Written

Calculate $(\sqrt{5} + i)/(\sqrt{5} - i)$ and give the answer in the form $a+bi$.

Classified Practice

Complex Numbers

Q069 Written

Calculate $5i/(1 + 2i)$ and give the answer in the form $a+bi$.

Q070 Written**Calculate $5i/(4 + 3i)$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q071 Written

Calculate $(1 - 3i)/(3 - 2i)$ and give the answer in the form $a+bi$.

Q072 Written**Calculate $1/i$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q073 Written

Calculate $(2 + i)/(3i)$ and give the answer in the form $a+bi$.

Q074 Written

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)} \right)^{2027}$.

Classified Practice

Complex Numbers

Q075 Written

Express the square roots of -6 in terms of i .

Q076 Written**Express $\sqrt{-12}\sqrt{-6}$ in terms of i.**

Classified Practice

Complex Numbers

Q077 Written

Express $\sqrt{8}/\sqrt{-18}$ in terms of i .

Q078 Written**Express $\sqrt{-7}$ in terms of i .**

Classified Practice

Complex Numbers

Q079 Written

Express $\sqrt{-27}$ in terms of i .

Q080 Written**Express $-\sqrt{-12}$ in terms of i .**

Classified Practice

Complex Numbers

Q081 **Written****Express the square roots of -12 in terms of i .**

Q082 Written**Express $\sqrt{-3}\sqrt{-6}$ in terms of i .**

Classified Practice

Complex Numbers

Q083 Written

Express $\sqrt{27}/\sqrt{-3}$ in terms of i.

Q084 Written**Express $(3 + \sqrt{-2})^2$ in terms of i .**

Classified Practice

Complex Numbers

Q085 Written

Express $\sqrt{-2}$ in terms of i .

Q086 Written**Express $\sqrt{-9}$ in terms of i .**

Classified Practice

Complex Numbers

Q087 Written

Express $\sqrt{-45}$ in terms of i .

Q088 Written**Express $-\sqrt{-24}$ in terms of i .**

Classified Practice

Complex Numbers

Q089 Written

Express $-\sqrt{-36}$ in terms of i .

Q090 Written**Express the square roots of -2 in terms of i .**

Classified Practice

Complex Numbers

Q091 **Written****Express the square roots of -18 in terms of i .**

Q092 Written**Express the square roots of -16 in terms of i .**

Classified Practice

Complex Numbers

Q093 Written

Express $\sqrt{-2}\sqrt{-6}$ in terms of i .

Q094 Written**Express $\sqrt{-28}\sqrt{-35}$ in terms of i .**

Classified Practice

Complex Numbers

Q095 Written

Express $\sqrt{8}/\sqrt{-2}$ in terms of i .

Q096 Written**Express $\sqrt{27}/\sqrt{-12}$ in terms of i .**

Classified Practice

Complex Numbers

Q097 Written

Express $\sqrt{-32} + \sqrt{-98} + \sqrt{-8}$ in terms of i .

Q098 Written**Express $\sqrt{-98} - \sqrt{-72} + \sqrt{-50}$ in terms of i .**

Classified Practice

Complex Numbers

Q099 Written

Express $(\sqrt{3} + \sqrt{-5})^2$ in terms of i .

Q100 Written

Express $(2 + \sqrt{-3})(2 - \sqrt{-3})$ in terms of i .

Classified Practice

Complex Numbers

Q101 Written

Solve over the complex numbers: $(2x - 8)^2 = -10$.

Q102 MCQ

For $2 + 3i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A (2, 4)

B (3, 2)

C (2, 3)

D (3, 3)

Classified Practice

Complex Numbers

Q103 MCQ

For $-4 + 5i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(-3, 5)$

B $(-4, 6)$

C $(5, -4)$

D $(-4, 5)$

Q104 MCQ

For $7 - 2i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A (8, -2)

B (7, -1)

C (-2, 7)

D (7, -2)

Classified Practice

Complex Numbers

Q105 MCQ

For $(1 - \sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(-\frac{\sqrt{2}}{3}, \frac{1}{3}\right)$

B $\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$

C $\left(\frac{4}{3}, -\frac{\sqrt{2}}{3}\right)$

D $\left(\frac{1}{3}, 1 - \frac{\sqrt{2}}{3}\right)$

Q106 MCQ

For -6 , select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(-5, 0)$

B $(-6, 1)$

C $(0, -6)$

D $(-6, 0)$

Classified Practice

Complex Numbers

Q107 MCQ

For $4i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A (0, 4)

B (1, 4)

C (0, 5)

D (4, 0)

Q108 MCQ**For $-i$, select the ordered pair (real part, imaginary part):****Choose the correct option.****A** $(-1, 0)$ **B** $(0, -1)$ **C** $(1, -1)$ **D** $(0, 0)$

Classified Practice

Complex Numbers

Q109 MCQ

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(0, 1 + \sqrt{5})$

B $(\sqrt{5}, 0)$

C $(0, \sqrt{5})$

D $(1, \sqrt{5})$

Q110 MCQ

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(\sqrt{3}, \frac{3}{2})$

B $(\frac{3}{2}, \sqrt{3})$

C $(\frac{5}{2}, \sqrt{3})$

D $(\frac{3}{2}, 1 + \sqrt{3})$

Classified Practice

Complex Numbers

Q111 MCQ

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):**Choose the correct option.**

A $\left(\frac{3}{4}, -2\right)$

B $\left(-\frac{1}{4}, -1\right)$

C $\left(-2, -\frac{1}{4}\right)$

D $\left(-\frac{1}{4}, -2\right)$

Q112 MCQ**For $5/3$, select the ordered pair (real part, imaginary part):****Choose the correct option.**

A $\left(\frac{8}{3}, 0\right)$

B $\left(\frac{5}{3}, 1\right)$

C $\left(0, \frac{5}{3}\right)$

D $\left(\frac{5}{3}, 0\right)$

Classified Practice

Complex Numbers

Q113 MCQ

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(0, 1 - \sqrt{7})$

B $(-\sqrt{7}, 0)$

C $(0, -\sqrt{7})$

D $(1, -\sqrt{7})$

Q114 MCQ

For $-9 + \sqrt{2}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(-8, \sqrt{2})$

B $(-9, 1 + \sqrt{2})$

C $(\sqrt{2}, -9)$

D $(-9, \sqrt{2})$

Classified Practice

Complex Numbers

Q115 MCQ

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(-\frac{3}{4}, \frac{2}{5}\right)$

B $\left(\frac{2}{5}, -\frac{3}{4}\right)$

C $\left(\frac{7}{5}, -\frac{3}{4}\right)$

D $\left(\frac{2}{5}, \frac{1}{4}\right)$

Q116 MCQ

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(0, -\sqrt{11})$

B $(-\sqrt{11}, 0)$

C $(1 - \sqrt{11}, 0)$

D $(-\sqrt{11}, 1)$

Classified Practice

Complex Numbers

Q117 MCQ

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(0, 1 + \sqrt{13})$

B $(\sqrt{13}, 0)$

C $(0, \sqrt{13})$

D $(1, \sqrt{13})$

Q118 MCQ

For $(2 + i)x + (1 - i)y = 3 + 3i$, select the ordered pair (x,y) :

Choose the correct option.

A $(3, -1)$

B $(2, -2)$

C $(-1, 2)$

D $(2, -1)$

Classified Practice

Complex Numbers

Q119 MCQ

For $(1 + 2i)x + (3 - i)y = 5 - 4i$, select the ordered pair (x,y):

Choose the correct option.

A $(-1, 2)$

B $(0, 2)$

C $(-1, 1)$

D $(2, -1)$

Q120 MCQ

For $(3 - 2i)x + (1 + i)y = 6 + i$, select the ordered pair (x,y):

Choose the correct option.

A (3, 1)

B (1, 3)

C (2, 3)

D (1, 2)

Classified Practice

Complex Numbers

Q121 MCQ

For $(2 - i)x + (4 + 2i)y = 4i$, select the ordered pair (x,y):

Choose the correct option.

A (1, -2)

B (-2, 1)

C (-1, 1)

D (-2, 0)

Q122 MCQ

For $(1 + i)x + (2 + 3i)y = 1$, select the ordered pair (x,y):

Choose the correct option.

A (3, -2)

B (-1, 3)

C (3, -1)

D (4, -1)

Classified Practice

Complex Numbers

Q123 MCQ

For $(4 + i)x + (1 - 2i)y = 2 + 5i$, select the ordered pair (x,y):

Choose the correct option.

A (2, -2)

B (1, -3)

C (-2, 1)

D (1, -2)

Q124 MCQ

For $(2 + 3i)x + (3 + i)y = -8 - 5i$, select the ordered pair (x,y):

Choose the correct option.

A (0, -2)

B (-1, -3)

C (-2, -1)

D (-1, -2)

Classified Practice

Complex Numbers

Q125 MCQ

For $(1 - i)x + (2 - i)y = 6 - 4i$, select the ordered pair (x,y):

Choose the correct option.

A (2, 1)

B 3

C (2, 2)

D (3, 2)

Q126 MCQ**For $(3 + 4i) - (1 - 2i)$, select the simplified standard form:****Choose the correct option.**

A $1 + 6i$

B $-2 - 6i$

C $2 + 6i$

D $3 + 6i$

Classified Practice

Complex Numbers

Q127 MCQ

For $(-2 + i) + (5 - 3i)$, select the simplified standard form:

Choose the correct option.

A $2 - 2i$

B $-3 + 2i$

C $3 - 2i$

D $4 - 2i$

Q128 MCQ**For $(1 + 2i)(3 - i)$, select the simplified standard form:****Choose the correct option.**

A $6 + 5i$

B $4 + 5i$

C $-5 - 5i$

D $5 + 5i$

Classified Practice

Complex Numbers

Q129 MCQ

For $(\sqrt{3} + i)(\sqrt{3} - i)$, select the simplified standard form:

Choose the correct option.

- A 4
- B 5
- C 3
- D -4

Q130 MCQ

For $(2 - i)^2$, select the simplified standard form:

Choose the correct option.

A $2 - 4i$

B $-3 + 4i$

C $(2 - i)^2$

D $4 - 4i$

Classified Practice

Complex Numbers

Q131 MCQ

For $(1 + 3i)^2$, select the simplified standard form:**Choose the correct option.**

A $-8 + 6i$

B $-7 + 6i$

C $-9 + 6i$

D $8 - 6i$

Q132 MCQ**For i^{37} , select the simplified standard form:****Choose the correct option.****A** $-i$ **B** i **C** $1 + i$ **D** $-1 + i$

Classified Practice

Complex Numbers

Q133 MCQ

For $i^{22} + i^9$, select the simplified standard form:

Choose the correct option.

A $1 - i$

B $-1 + i$

C i

D $-2 + i$

Q134 MCQ**For $i^{15} - i^6$, select the simplified standard form:****Choose the correct option.****A** $-1 + i$ **B** $1 - i$ **C** $2 - i$ **D** $-i$

Classified Practice

Complex Numbers

Q135 MCQ

For $(4 - 2i) + (1 + 5i)$, select the simplified standard form:

Choose the correct option.

A $4 + 3i$

B $-5 - 3i$

C $5 + 3i$

D $6 + 3i$

Q136 MCQ**For $(2 + i)(2 - i)$, select the simplified standard form:****Choose the correct option.****A 5****B 6****C 4****D -5**

Classified Practice

Complex Numbers

Q137 MCQ

For $(\sqrt{2} + 2i)^2$, select the simplified standard form:

Choose the correct option.

A $-3 + 4\sqrt{2}i$

B $2 - 4\sqrt{2}i$

C $(\sqrt{2} + 2i)^2$

D $-1 + 4\sqrt{2}i$

Q138 MCQ**For $i^8 + i^{11} + i^{14}$, select the simplified standard form:****Choose the correct option.****A** $-1 - i$ **B** i **C** $-i$ **D** $1 - i$

Classified Practice

Complex Numbers

Q139 MCQ

For $(3 - i)^3$, select the simplified standard form:**Choose the correct option.**

A $19 - 26i$

B $17 - 26i$

C $-18 + 26i$

D $(3 - i)^3$

Q140 MCQ

For $(1 - 2i)(-2 + i)$, select the simplified standard form:

Choose the correct option.

A $1 + 5i$

B $-1 + 5i$

C $-5i$

D $5i$

Classified Practice

Complex Numbers

Q141 MCQ

For $i^5 + i^{10}$, select the simplified standard form:

Choose the correct option.

A i

B $-2 + i$

C $1 - i$

D $-1 + i$

Q142 MCQ

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:

Choose the correct option.

A $3 + 6i$

B $1 + 6i$

C $-2 - 6i$

D $2 + 6i$

Classified Practice

Complex Numbers

Q143 MCQ

For $(2 - 3i)^2$, select the simplified standard form:**Choose the correct option.**A $-6 - 12i$ B $5 + 12i$ C $(2 - 3i)^2$ D $-4 - 12i$

Q144 MCQ

For the expression State the complex conjugate of $3 - 2i$., the required value is:

Choose the correct option.

A $4 + 2i$

B $3 - 2i$

C $3 + 2i$

D $-3 - 2i$

Classified Practice

Complex Numbers

Q145 MCQ

For the expression State the complex conjugate of $-4 + \sqrt{3}i$, the required value is:

Choose the correct option.

A $-4 - \sqrt{3}i$

B $4 + \sqrt{3}i$

C $-3 - \sqrt{3}i$

D $-4 + \sqrt{3}i$

Q146 MCQ**For the expression State the complex conjugate of $7i$., the required value is:****Choose the correct option.****A** $1 - 7i$ **B** 3 **C** $-7i$ **D** $7i$

Classified Practice

Complex Numbers

Q147 MCQ

For the expression State the complex conjugate of $-\sqrt{5}$; the required value is:

Choose the correct option.

A $1 - \sqrt{5}$

B 3

C $-\sqrt{5}$

D $\sqrt{5}$

Q148 MCQ

For the expression State the complex conjugate of $\frac{1}{2} - 3i$, the required value is:

Choose the correct option.

A $\frac{1}{2} - 3i$

B $\frac{1}{2} + 3i$

C $-\frac{1}{2} - 3i$

D $\frac{3}{2} + 3i$

Classified Practice

Complex Numbers

Q149 MCQ

For $(2 + i)/(1 - i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{1}{2} + \frac{3i}{2}$

B $\frac{3}{2} + \frac{3i}{2}$

C $-\frac{1}{2} + \frac{3i}{2}$

D $-\frac{1}{2} - \frac{3i}{2}$

Q150 MCQ

For $(4 - 3i)/(2 + i)$, select the value in the form $a+bi$:

Choose the correct option.

A $1 - 2i$

B $2 - 2i$

C $-2i$

D $-1 + 2i$

Classified Practice

Complex Numbers

Q151 MCQ

For $(1 + 2i)/(3 - i)$, select the value in the form $a+bi$:**Choose the correct option.**

A $-\frac{1}{10} - \frac{7i}{10}$

B $\frac{1}{10} + \frac{7i}{10}$

C $\frac{11}{10} + \frac{7i}{10}$

D $-\frac{9}{10} + \frac{7i}{10}$

Q152 MCQ

For $5i/(2 - i)$, select the value in the form $a+bi$:

Choose the correct option.

A $-2 + 2i$

B $1 - 2i$

C $i(2 + i)$

D $2i$

Classified Practice

Complex Numbers

Q153 MCQ

For $(3 - i)/(1 + 3i)$, select the value in the form $a+bi$:

Choose the correct option.

A $1 - i$

B $-1 - i$

C i

D $-i$

Q154 MCQ**For $1/i$, select the value in the form $a+bi$:****Choose the correct option.****A** $1 - i$ **B** $-1 - i$ **C** i **D** $-i$

Classified Practice

Complex Numbers

Q155 MCQ

For $(2 + 3i)/(2 - 3i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{5}{13} - \frac{12i}{13}$

B $-\frac{5}{13} + \frac{12i}{13}$

C $\frac{8}{13} + \frac{12i}{13}$

D $-\frac{18}{13} + \frac{12i}{13}$

Q156 MCQ

For $(\sqrt{2} + i)/(\sqrt{2} - i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{-\sqrt{2} - i}{\sqrt{2} - i}$

B $\frac{\sqrt{2} + i}{\sqrt{2} - i}$

C $\frac{2\sqrt{2}}{\sqrt{2} - i}$

D $\frac{2i}{\sqrt{2} - i}$

Classified Practice

Complex Numbers

Q157 MCQ

For $(1 - 4i)/(3 + 2i)$, select the value in the form $a+bi$:

Choose the correct option.

- A $\frac{8}{13} - \frac{14i}{13}$
B $-\frac{18}{13} - \frac{14i}{13}$
C $\frac{5}{13} + \frac{14i}{13}$
D $-\frac{5}{13} - \frac{14i}{13}$

Q158 MCQ

For $7i/(1 - i)$, select the value in the form $a+bi$:

Choose the correct option.

A $-\frac{9}{2} + \frac{7i}{2}$

B $\frac{7}{2} - \frac{7i}{2}$

C $-\frac{7}{2} + \frac{7i}{2}$

D $-\frac{5}{2} + \frac{7i}{2}$

Classified Practice

Complex Numbers

Q159 MCQ

For the expression State the complex conjugate of $-2 + 5i$, the required value is:

Choose the correct option.

A $-2 + 5i$

B $-2 - 5i$

C $2 + 5i$

D $-1 - 5i$

Q160 MCQ

For the expression State the complex conjugate of $6 - \sqrt{7}i$., the required value is:

Choose the correct option.

- A $-6 - \sqrt{7}i$
 - B $7 + \sqrt{7}i$
 - C $6 - \sqrt{7}i$
 - D $6 + \sqrt{7}i$
-
-
-

Classified Practice

Complex Numbers

Q161 MCQ

For the expression State the complex conjugate of $-i\sqrt{2}$; the required value is:

Choose the correct option.

- A 3
- B $\sqrt{2}i$
- C $-\sqrt{2}i$
- D $1 + \sqrt{2}i$

Q162 MCQ**For the expression State the complex conjugate of 9;, the required value is:****Choose the correct option.**

A 10

B 3

C 9

D -9

Classified Practice

Complex Numbers

Q163 MCQ

For the expression State the complex conjugate of $-3/4 + 2i$, the required value is:

Choose the correct option.

A $-\frac{3}{4} - 2i$

B $\frac{3}{4} + 2i$

C $\frac{1}{4} - 2i$

D $-\frac{3}{4} + 2i$

Q164 MCQ

For $(5 + 2i)/(1 - 2i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{6}{5} + \frac{12i}{5}$

B $-\frac{4}{5} + \frac{12i}{5}$

C $-\frac{1}{5} - \frac{12i}{5}$

D $\frac{1}{5} + \frac{12i}{5}$

Classified Practice

Complex Numbers

Q165 MCQ

For $(2 - i)/(4 + i)$, select the value in the form $a+bi$:**Choose the correct option.**

A $\frac{7}{17} - \frac{6i}{17}$

B $\frac{24}{17} - \frac{6i}{17}$

C $-\frac{10}{17} - \frac{6i}{17}$

D $-\frac{7}{17} + \frac{6i}{17}$

Q166 MCQ

For $3i/(2 + 3i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{9}{13} + \frac{6i}{13}$

B $\frac{22}{13} + \frac{6i}{13}$

C $-\frac{4}{13} + \frac{6i}{13}$

D $-\frac{9}{13} - \frac{6i}{13}$

Classified Practice

Complex Numbers

Q167 MCQ

For $(\sqrt{3} - i)/(\sqrt{3} + i)$, select the value in the form $a+bi$:

Choose the correct option.

- A $-\frac{2i}{\sqrt{3} + i}$
- B $\frac{-\sqrt{3} + i}{\sqrt{3} + i}$
- C $\frac{\sqrt{3} - i}{\sqrt{3} + i}$
- D $\frac{2\sqrt{3}}{\sqrt{3} + i}$

Q168 MCQ

For $(1 + i)/(2i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{1}{2} - \frac{i}{2}$

B $\frac{3}{2} - \frac{i}{2}$

C $-\frac{1}{2} - \frac{i}{2}$

D $-\frac{1}{2} + \frac{i}{2}$

Classified Practice

Complex Numbers

Q169 MCQ

For the expression State the complex conjugate of $4 - 7i$., the required value is:

Choose the correct option.

A $-4 - 7i$

B $5 + 7i$

C $4 - 7i$

D $4 + 7i$

Q170 MCQ

For the expression State the complex conjugate of $\sqrt{10} + 2i$., the required value is:

Choose the correct option.

- A $\sqrt{10} - 2i$
- B $-\sqrt{10} + 2i$
- C $1 + \sqrt{10} - 2i$
- D $\sqrt{10} + 2i$

Classified Practice

Complex Numbers

Q171 MCQ

For the expression State the complex conjugate of $-8i$, the required value is:

Choose the correct option.

A 3

B $8i$ C $-8i$ D $1 + 8i$

Q172 MCQ

For $(3 + 2i)/(3 - 2i)$, select the value in the form $a+bi$:

Choose the correct option.

A $-\frac{5}{13} - \frac{12i}{13}$

B $\frac{5}{13} + \frac{12i}{13}$

C $\frac{18}{13} + \frac{12i}{13}$

D $-\frac{8}{13} + \frac{12i}{13}$

Classified Practice

Complex Numbers

Q173 MCQ

For $\sqrt{-5}$, select the simplified value in terms of i :

Choose the correct option.

- A $5i$
- B $i\sqrt{5}$
- C $-i\sqrt{5}$
- D $\sqrt{5}$

Q174 MCQ

For $\sqrt{-20}$, select the simplified value in terms of i :

Choose the correct option.

- A $2i\sqrt{5}$
 - B $-2i\sqrt{5}$
 - C $4i\sqrt{5}$
 - D $i\sqrt{20}$
-
-
-

Classified Practice

Complex Numbers

Q175 MCQ

For $-\sqrt{-18}$, select the simplified value in terms of i :

Choose the correct option.

A $-3i\sqrt{2}$

B $3i\sqrt{2}$

C $-6i$

D $-i\sqrt{18}$

Q176 MCQ**For square roots of -8 , select the simplified value in terms of i :****Choose the correct option.**

A $2i\sqrt{2}$

B $\pm 8i$

C $\pm 2i\sqrt{2}$

D $\pm i\sqrt{8}$

Classified Practice

Complex Numbers

Q177 MCQ

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i :

Choose the correct option.

A 3

B -3 C $3i$ D $\sqrt{9}i$

Q178 MCQ

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i :

Choose the correct option.

A 2

B -4

C 4

D $-4i$

Classified Practice

Complex Numbers

Q179 MCQ

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i :

Choose the correct option.

- A -2
- B $i\sqrt{4}$
- C $-2i$
- D $2i$

Q180 MCQ

For $(2 + \sqrt{-3})^2$, select the simplified value in terms of i :

Choose the correct option.

- A $1 - 4i\sqrt{3}$
 - B $7 + 4i$
 - C $4 + 2i\sqrt{3}$
 - D $1 + 4i\sqrt{3}$
-
-
-

Classified Practice

Complex Numbers

Q181 MCQ

For $\sqrt{-72}$, select the simplified value in terms of i :

Choose the correct option.

A $-6i\sqrt{2}$

B $12i$

C $i\sqrt{72}$

D $6i\sqrt{2}$

Q182 MCQ

For square roots of -50 , select the simplified value in terms of i :

Choose the correct option.

- A $\pm i\sqrt{50}$
 - B $5i\sqrt{2}$
 - C $\pm 10i$
 - D $\pm 5i\sqrt{2}$
-
-
-

Classified Practice

Complex Numbers

Q183 MCQ

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i :

Choose the correct option.

A -6

B 6

C $-6i$

D 3

Q184 MCQ

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i :

Choose the correct option.

A $2i$

B -2

C $-i\sqrt{2}$

D $-2i$

Classified Practice

Complex Numbers

Q185 MCQ

For $\sqrt{-32} + \sqrt{-8}$, select the simplified value in terms of i :

Choose the correct option.

- A $6\sqrt{2}$
 - B $6i\sqrt{2}$
 - C $2i\sqrt{2}$
 - D $-6i\sqrt{2}$
-
-
-

Q186 MCQ

For $\sqrt{-98} - \sqrt{-18}$, select the simplified value in terms of i :

Choose the correct option.

A $4\sqrt{2}$

B $4i\sqrt{2}$

C $8i\sqrt{2}$

D $-4i\sqrt{2}$

Classified Practice

Complex Numbers

Q187 MCQ

For $(\sqrt{5} + \sqrt{-2})^2$, select the simplified value in terms of i :

Choose the correct option.

A $3 + 2i\sqrt{10}$

B $7 + 2i\sqrt{10}$

C $3 - 2i\sqrt{10}$

D $5 + 2i\sqrt{2}$

Q188 MCQ

For $(3 + \sqrt{-5})(3 - \sqrt{-5})$, select the simplified value in terms of i :

Choose the correct option.

A 4

B $14i$

C $9 - 5i$

D 14

Classified Practice

Complex Numbers

Q189 MCQ

For $\sqrt{-7}$, select the simplified value in terms of i :

Choose the correct option.

- A $7i$
- B $\sqrt{7}$
- C $i\sqrt{7}$
- D $-i\sqrt{7}$

Q190 MCQ

For $\sqrt{-28}$, select the simplified value in terms of i :

Choose the correct option.

- A $2i\sqrt{7}$
- B $-2i\sqrt{7}$
- C $4i\sqrt{7}$
- D $i\sqrt{28}$

Classified Practice

Complex Numbers

Q191 MCQ

For $\sqrt{-12}$, select the simplified value in terms of i :

Choose the correct option.

A $\pm 2i\sqrt{3}$

B $\pm i\sqrt{12}$

C $2i\sqrt{3}$

D $\pm 4i$

Q192 MCQ

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i :

Choose the correct option.

A $-3i$

B $3i$

C -3

D $i\sqrt{3}$

Classified Practice

Complex Numbers

Q193 MCQ

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i :

Choose the correct option.

- A 6
- B $-6i$
- C -3
- D -6

Q194 MCQ

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i :

Choose the correct option.

- A -4
- B $-i\sqrt{4}$
- C $-4i$
- D $4i$

Classified Practice

Complex Numbers

Q195 MCQ

For $\sqrt{-8} + \sqrt{-18}$, select the simplified value in terms of i :

Choose the correct option.

A $-5i\sqrt{2}$

B $5\sqrt{2}$

C $5i\sqrt{2}$

D $i\sqrt{26}$

Q196 MCQ

For $(1 + \sqrt{-2})^2$, select the simplified value in terms of i :

Choose the correct option.

- A $-1 + 2i\sqrt{2}$
- B $3 + 2i\sqrt{2}$
- C $-1 - 2i\sqrt{2}$
- D $1 + 2i$

Classified Practice

Complex Numbers

Q197 MCQ

For $(4 + \sqrt{-3})(4 - \sqrt{-3})$, select the simplified value in terms of i :

Choose the correct option.

A $16 - 3i$

B 19

C 13

D $19i$

Q198 MCQ

For $\sqrt{-75}$, select the simplified value in terms of i :

Choose the correct option.

A $-5i\sqrt{3}$

B $15i$

C $i\sqrt{75}$

D $5i\sqrt{3}$

Classified Practice

Complex Numbers

Q199 MCQ

Solve for the real values x and y : $(1 + 3i)(x + i) = x(y + i)$:**Choose the correct option.**

A $\left(-\frac{1}{2}, -7\right)$

B $\left(-\frac{1}{2}, 7\right)$

C $\left(-\frac{1}{2}, 6\right)$

D $\left(\frac{1}{2}, 7\right)$

Q200 MCQ**Given x is real, solve: $(4x - 4i)(x^2 + xi) = 10x^2i^{32}$:****Choose the correct option.****A** $\{0, 1, 2\}$ **B** $\left\{2, \frac{1}{2}\right\}$ **C** $\left\{-2, 2, \frac{1}{2}\right\}$ **D** $\left\{0, 2, \frac{1}{2}\right\}$

Classified Practice

Complex Numbers

Q201 MCQ

Simplify exactly: $(i(8 - 6i)/((1 + 2i)(3 - i)))^{2027}$:**Choose the correct option.**

A $((1 + 7i)/5)^{2027}$

B $((7 + i)/5)^{2027}$

C $((7 - i)/5)^{2027}$

D $(7 + i)^{2027}$

Q202 MCQ

Solve over the complex numbers: $(3x - 6)^2 = -8$:**Choose the correct option.**

A $x = 2 \pm 2i\sqrt{2}$

B $x = 2 \pm (2/3)i\sqrt{2}$

C $x = 2 \pm i\sqrt{2}$

D $x = 3 \pm i\sqrt{2}$

Classified Practice

Complex Numbers

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

For $4-3i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(4, -3)$

B $(4, 3)$

C $(-4, -3)$

D $(-3, 4)$

Quiz MCQ 2 Concept Quiz · MCQ

For $(1+2i)(2-i)$, select the simplified value:

Choose the correct option.

A $4 + 3i$

B 1

C 2

D $-4 - 3i$

Classified Practice

Complex Numbers

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

For $(3+i)/(1-i)$, select the value in the form $a+bi$:

Choose the correct option.

A 1

B 2

C $-1 - 2i$

D $1 + 2i$

Quiz MCQ 4 Concept Quiz · MCQ

For $\sqrt{-27}$, select the expression in terms of i :

Choose the correct option.

- A $3i\sqrt{3}$
 - B $-3i\sqrt{3}$
 - C $9i$
 - D $i\sqrt{27}$
-
-
-

Classified Practice

Complex Numbers

Quiz WRITTEN 5 Concept Quiz · Written

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

Quiz WRITTEN 6 Concept Quiz · Written

Solve for real x, y : $(2 + i)x + (1 - i)y = 3 + i$.

Classified Practice

Complex Numbers

Quiz WRITTEN 7 Concept Quiz · Written

Calculate $i^{18} + i^7 - i^3$ in standard form.

Quiz WRITTEN 8 Concept Quiz · Written

Solve over the complex numbers: $(x - 3)^2 = -12$.

Answer key

Use the question number shown on each page.

Q001 $(3, -2)$	Q027 $3 - 4i$	Q053 2	Q079 $3i\sqrt{3}$
Q002 $(4/3, -\sqrt{5}/3)$	Q028 -1	Q054 $8i$	Q080 $-2i\sqrt{3}$
Q003 $(-8, 0)$	Q029 $1 + 8i$	Q055 $1 - i$	Q081 $\pm 2i\sqrt{3}$
Q004 $(0, 2)$	Q030 $8 - i$	Q056 $-1 - i$	Q082 $-3\sqrt{2}$
Q005 $x = -1, y = 4$	Q031 0	Q057 $3 - 4i$	Q083 $-3i$
Q006 $(-3, 5)$	Q032 $4 - i$	Q058 $9 - 8i$	Q084 $7 + 6\sqrt{2}i$
Q007 $(-1/2, -\sqrt{3}/2)$	Q033 $1 - 3i$	Q059 $8 - i\sqrt{3}$	Q085 $i\sqrt{2}$
Q008 $(1, 0)$	Q034 -2	Q060 5	Q086 $3i$
Q009 $(0, 3)$	Q035 $-1 + 4\sqrt{2}i$	Q061 -15	Q087 $3i\sqrt{5}$
Q010 $x = 6, y = -2$	Q036 $7 + 11i$	Q062 $\sqrt{7}$	Q088 $-2i\sqrt{6}$
Q011 $x = -2, y = 3$	Q037 $1/15 - 13/14i$	Q063 $-i$	Q089 $-6i$
Q012 $(2, 5)$	Q038 $-3 - 4i$	Q064 $50i$	Q090 $\pm i\sqrt{2}$
Q013 $(-25, 13)$	Q039 -3	Q065 $-i\sqrt{6}$	Q091 $\pm 3i\sqrt{2}$
Q014 $(2/3, -\sqrt{5}/3)$	Q040 4	Q066 i	Q092 $\pm 4i$
Q015 $(7/2, -\sqrt{2}/2)$	Q041 i	Q067 $7/5 - 1/5i$	Q093 $-2\sqrt{3}$
Q016 $(-4, 0)$	Q042 -1	Q068 $2/3 + \sqrt{5}/3i$	Q094 $-14\sqrt{5}$
Q017 $(1/4, 0)$	Q043 $-i$	Q069 $2 + i$	Q095 $-2i$
Q018 $(0, \sqrt{3})$	Q044 $x = 0$ or $x = 1$	Q070 $3/5 + 4/5i$	Q096 $-(3/2)i$
Q019 $(0, -1)$	Q045 $2 - 3i$	Q071 $9/13 - 7/13i$	Q097 $13i\sqrt{2}$
Q020 $x = 3, y = 2$	Q046 $7 + i\sqrt{11}$	Q072 $-i$	Q098 $6i\sqrt{2}$
Q021 $x = 3, y = -3$	Q047 6	Q073 $1/3 - 2/3i$	Q099 $-2 + 2i\sqrt{15}$
Q022 $x = 2, y = -2$	Q048 $5i$	Q074 $((-2 - 144i)/85)^{2027}$	Q100 7
Q023 $x = 2, y = 1$	Q049 $3/5 + 4/5i$	Q075 $\pm i\sqrt{6}$	Q101 $x = 4 \pm (\sqrt{10}/2)i$
Q024 $x = 1/2, y = 3/2$	Q050 $-1/2 - 3/2i$	Q076 $-6\sqrt{2}$	Q102 C
Q025 $x = -1, y = 3$	Q051 $1 + 4i$	Q077 $-(2/3)i$	Q103 D
Q026 $1 - 7i$	Q052 $\sqrt{2} - i\sqrt{10}$	Q078 $i\sqrt{7}$	Q104 D

Q105 B
Q106 D
Q107 A
Q108 B
Q109 C
Q110 B
Q111 D
Q112 D
Q113 C
Q114 D
Q115 B
Q116 B
Q117 C
Q118 D
Q119 A
Q120 B
Q121 B
Q122 C
Q123 D
Q124 D
Q125 C
Q126 C
Q127 C
Q128 D
Q129 A
Q130 C
Q131 A

Q132 B
Q133 B
Q134 B
Q135 C
Q136 A
Q137 C
Q138 C
Q139 D
Q140 D
Q141 D
Q142 D
Q143 C
Q144 C
Q145 A
Q146 C
Q147 C
Q148 B
Q149 A
Q150 A
Q151 B
Q152 C
Q153 D
Q154 D
Q155 B
Q156 B
Q157 D
Q158 C

Q159 B
Q160 D
Q161 B
Q162 C
Q163 A
Q164 D
Q165 A
Q166 A
Q167 C
Q168 A
Q169 D
Q170 A
Q171 B
Q172 B
Q173 B
Q174 A
Q175 A
Q176 C
Q177 A
Q178 B
Q179 C
Q180 D
Q181 D
Q182 D
Q183 A
Q184 D
Q185 B

Q186 B
Q187 A
Q188 D
Q189 C
Q190 A
Q191 A
Q192 A
Q193 D
Q194 C
Q195 C
Q196 A
Q197 B
Q198 D
Q199 B
Q200 D
Q201 B
Q202 B
Quiz MCQ 1 A
Quiz MCQ 2 A
Quiz MCQ 3 D
Quiz MCQ 4 A
Quiz WRITTEN 5 $(5/2, -\sqrt{5}/2)$
Quiz WRITTEN 6 $x = 4/3, y = 1/3$
Quiz WRITTEN 7 -1
Quiz WRITTEN 8 $x = 3 \pm 2i\sqrt{3}$

Concept 2 | Quadratic Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum/product

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Classified Practice

Quadratic Equations

Q001 Written

Solve $x^2 + 8 = 0$.

Q002 Written**Solve** $x^2 + 4 = 0$.

Classified Practice

Quadratic Equations

Q003 Written

Solve $x^2 + 49 = 0$.

Q004 Written**Solve** $9x^2 - 4 = 0$.

Classified Practice

Quadratic Equations

Q005 Written

Solve $x^2 - 12 = 0$.

Q006 Written**Solve** $4x^2 + 3 = 0$.

Classified Practice

Quadratic Equations

Q007 MCQ

Solve $x^2 + 8 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \pm 2i\sqrt{2}$

B $x = 2i\sqrt{2}$

C $x = -2i\sqrt{2}$

D $x = +2i\sqrt{2}$

Q008 MCQ**Solve $x^2 + 4 = 0$ Select the correct answer:****Choose the correct option.**

A $x = \pm 2i$

B $x = 2i$

C $x = -2i$

D $x = +2i$

Classified Practice

Quadratic Equations

Q009 MCQ

Solve $x^2 + 49 = 0$ Select the correct answer:**Choose the correct option.**

A $x = -7i$

B $x = +7i$

C $x = \pm 7i$

D $x = 7i$

Q010 MCQ**Solve $9x^2 - 4 = 0$ Select the correct answer:****Choose the correct option.**

A $x = -\frac{2}{3}$

B $x = +\frac{2}{3}$

C $x = \pm\frac{2}{3}$

D $x = \frac{2}{3}$

Classified Practice

Quadratic Equations

Q011 MCQ

Solve $x^2 - 12 = 0$ Select the correct answer:**Choose the correct option.**

A $x = +2\sqrt{3}$

B $x = \pm 2\sqrt{3}$

C $x = 2\sqrt{3}$

D $x = -2\sqrt{3}$

Q012 MCQ

Solve $4x^2 + 3 = 0$ Select the correct answer:**Choose the correct option.**

A $x = +\frac{i\sqrt{3}}{2}$

B $x = \pm\frac{i\sqrt{3}}{2}$

C $x = \frac{i\sqrt{3}}{2}$

D $x = -\frac{i\sqrt{3}}{2}$

Classified Practice

Quadratic Equations

Q013 Written

Solve $(x - 3)^2 = -8$.

Q014 Written**Solve** $(x - 2)^2 = 5$.

Classified Practice

Quadratic Equations

Q015 Written

Solve $(x + 6)^2 = -6$.

Q016 Written**Solve** $(x + 4)^2 = -18$.

Classified Practice

Quadratic Equations

Q017 Written

Solve $(2x - 3)^2 = 7$.

Q018 MCQ**Solve $(x - 3)^2 = -8$ Select the correct answer:****Choose the correct option.**

A $x = 3 \pm 2i\sqrt{2}$

B $x = 3 \pm 2i\sqrt{2}$

C $x = 3 - 2i\sqrt{2}$

D $x = 3 + 2i\sqrt{2}$

Classified Practice

Quadratic Equations

Q019 MCQ

Solve $(x - 2)^2 = 5$ Select the correct answer:**Choose the correct option.**

A $x = 2 + \sqrt{5}$

B $x = 2 \pm \sqrt{5}$

C $x = 2\sqrt{5}$

D $x = 2 - \sqrt{5}$

Q020 MCQ**Solve $(x + 6)^2 = -6$ Select the correct answer:****Choose the correct option.**

- A $x = -6 - i\sqrt{6}$
 - B $x = -6 + i\sqrt{6}$
 - C $x = -6 \pm i\sqrt{6}$
 - D $x = -6i\sqrt{6}$
-
-
-

Classified Practice

Quadratic Equations

Q021 MCQ

Solve $(x + 4)^2 = -18$ Select the correct answer:**Choose the correct option.**

A $x = -4 - 3i\sqrt{2}$

B $x = -4 + 3i\sqrt{2}$

C $x = -4 \pm 3i\sqrt{2}$

D $x = -43i\sqrt{2}$

Q022 MCQ

Solve $(2x - 3)^2 = 7$ Select the correct answer:**Choose the correct option.**

A $x = \frac{3\sqrt{7}}{2}$

B $x = \frac{3 - \sqrt{7}}{2}$

C $x = \frac{3 + \sqrt{7}}{2}$

D $x = \frac{3 \pm \sqrt{7}}{2}$

Classified Practice

Quadratic Equations

Q023 Written

Solve $2x^2 - 7x - 9 = 0$.

Q024 Written**Solve** $2x^2 + 5x - 3 = 0$.

Classified Practice

Quadratic Equations

Q025 Written

Solve $2x^2 + 7x = 0$.

Q026 Written**Solve** $2x^2 + 5x - 12 = 0$.

Classified Practice

Quadratic Equations

Q027 Written

Solve $x^2 - 8x + 15 = 0$.

Q028 MCQ

Solve $2x^2 - 7x - 9 = 0$ Select the correct answer:**Choose the correct option.**

- A $x = -1, \frac{9}{2}$
- B $-(x = -1, \frac{9}{2})$
- C $x = -1, \frac{9}{2} + 1$
- D $x = -1, \frac{9}{2} - 1$

Classified Practice

Quadratic Equations

Q029 MCQ

Solve $2x^2 + 5x - 3 = 0$ Select the correct answer:**Choose the correct option.**

A $-(x = \frac{1}{2}, -3)$

B $x = \frac{1}{2}, -3 + 1$

C $x = \frac{1}{2}, -3 - 1$

D $x = \frac{1}{2}, -3$

Q030 MCQ**Solve $2x^2 + 7x = 0$ Select the correct answer:****Choose the correct option.**

A $-(x = 0, -\frac{7}{2})$

B $x = 0, -\frac{7}{2} + 1$

C $x = 0, -\frac{7}{2} - 1$

D $x = 0, -\frac{7}{2}$

Classified Practice

Quadratic Equations

Q031 MCQ

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{3}{2}, -4$

B $-(x = \frac{3}{2}, -4)$

C $x = \frac{3}{2}, -4 + 1$

D $x = \frac{3}{2}, -4 - 1$

Q032 MCQ**Solve $x^2 - 8x + 15 = 0$ Select the correct answer:****Choose the correct option.****A** $x = 3, 5$ **B** $x = 3, 6$ **C** $x = 3, 5 - 1$ **D** $x = 3, 5$ (sign error)

Classified Practice

Quadratic Equations

Q033 Written

Solve $3x^2 - 4x - 1 = 0$.

Q034 MCQ

Solve $3x^2 - 4x - 1 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{2 + \sqrt{7}}{3}$

B $x = \frac{2 \pm \sqrt{7}}{3}$

C $x = \frac{2\sqrt{7}}{3}$

D $x = \frac{2 - \sqrt{7}}{3}$

Classified Practice

Quadratic Equations

Q035 Written

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$.

Q036 Written**Solve** $2x^2 + 2\sqrt{3}x + 5 = 0$.

Classified Practice

Quadratic Equations

Q037 Written

Solve $x^2 + x + 1 = 0$.

Q038 Written**Solve** $3x^2 - 5x + 3 = 0$.

Classified Practice

Quadratic Equations

Q039 Written

Solve $x^2 - 2\sqrt{3}x + 5 = 0$.

Q040 Written**Solve** $3x^2 + 2\sqrt{2}x + 1 = 0$.

Classified Practice

Quadratic Equations

Q041 Written

Solve $x^2 - x + 2 = 0$.

Q042 Written**Solve** $3x^2 - \sqrt{5}x + 1 = 0$.

Classified Practice

Quadratic Equations

Q043 Written

Solve $6x^2 - 12x + 15 = 0$.

Q044 MCQ

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = \frac{-\sqrt{6} + 4i}{2}$

B $x = \frac{-\sqrt{6} \pm 4i}{2}$

C $x = \frac{-\sqrt{6}4i}{2}$

D $x = \frac{-\sqrt{6} - 4i}{2}$

Classified Practice

Quadratic Equations

Q045 MCQ

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = \frac{-\sqrt{3} - i\sqrt{7}}{2}$

B $x = \frac{-\sqrt{3} + i\sqrt{7}}{2}$

C $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

D $x = \frac{-\sqrt{3}i\sqrt{7}}{2}$

Q046 MCQ

Solve $x^2 + x + 1 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{-1i\sqrt{3}}{2}$

B $x = \frac{-1 - i\sqrt{3}}{2}$

C $x = \frac{-1 + i\sqrt{3}}{2}$

D $x = \frac{-1 \pm i\sqrt{3}}{2}$

Classified Practice

Quadratic Equations

Q047 MCQ

Solve $3x^2 - 5x + 3 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{5 + i\sqrt{11}}{6}$

B $x = \frac{5 \pm i\sqrt{11}}{6}$

C $x = \frac{5i\sqrt{11}}{6}$

D $x = \frac{5 - i\sqrt{11}}{6}$

Q048 MCQ**Solve $x^2 - 2\sqrt{3}x + 5 = 0$ Select the correct answer:****Choose the correct option.**

- A $x = \sqrt{3}i\sqrt{2}$
 - B $x = \sqrt{3} - i\sqrt{2}$
 - C $x = \sqrt{3} + i\sqrt{2}$
 - D $x = \sqrt{3} \pm i\sqrt{2}$
-
-
-

Classified Practice

Quadratic Equations

Q049 MCQ

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = \frac{-\sqrt{2} + i}{3}$

B $x = \frac{-\sqrt{2} \pm i}{3}$

C $x = \frac{-\sqrt{2}i}{3}$

D $x = \frac{-\sqrt{2} - i}{3}$

Q050 MCQ

Solve $x^2 - x + 2 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{1 - i\sqrt{7}}{2}$

B $x = \frac{1 + i\sqrt{7}}{2}$

C $x = \frac{1 \pm i\sqrt{7}}{2}$

D $x = \frac{1i\sqrt{7}}{2}$

Classified Practice

Quadratic Equations

Q051 MCQ

Solve $3x^2 - \sqrt{5}x + 1 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = \frac{\sqrt{5} - i\sqrt{7}}{6}$

B $x = \frac{\sqrt{5} + i\sqrt{7}}{6}$

C $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

D $x = \frac{\sqrt{5}i\sqrt{7}}{6}$

Q052 MCQ

Solve $6x^2 - 12x + 15 = 0$ Select the correct answer:**Choose the correct option.**

A $x = 1 + \frac{i\sqrt{6}}{2}$

B $x = 1 \pm \frac{i\sqrt{6}}{2}$

C $x = 1 \frac{i\sqrt{6}}{2}$

D $x = 1 - \frac{i\sqrt{6}}{2}$

Classified Practice

Quadratic Equations

Q053 Written

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.

Q054 Written

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.

Classified Practice

Quadratic Equations

Q055 Written

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

Q056 Written**For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.**

Classified Practice

Quadratic Equations

Q057 **Written****For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.**

Q058 Written

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.

Classified Practice

Quadratic Equations

Q059 Written

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.

Q060 Written**Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.**

Classified Practice

Quadratic Equations

Q061 Written

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.

Q062 Written**Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.**

Classified Practice

Quadratic Equations

Q063 Written

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

Q064 Written**Determine the nature of the solutions of $x^2 - x - 2 = 0$.**

Classified Practice

Quadratic Equations

Q065 MCQ

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct imaginary (non-real complex) solutions
- B Two repeated imaginary (non-real complex) solutions
- C Two distinct imaginary (non-imaginary complex) solutions
- D Two distinct real (non-real complex) solutions

Classified Practice

Quadratic Equations

Q066 MCQ

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions Select the correct answer:

Choose the correct option.

A $-2 \geq m \geq 6$

B $+2 \leq m \leq 6$

C $-2 \leq m \leq 7$

D $-2 \leq m \leq 6$

Classified Practice

Quadratic Equations

Q067 MCQ

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Choose the correct option.

- A $k < -6$ or $k > 2$: distinct imaginary; $k = -6, 2$: repeated imaginary; $-6 < k < 2$: distinct imaginary
- B $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct real
- C $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary
- D $k < -6$ or $k > 2$: repeated real; $k = -6, 2$: repeated real; $-6 < k < 2$: repeated imaginary

Classified Practice

Quadratic Equations

Q068 MCQ

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions Select the correct answer:

Choose the correct option.

- A Two distinct imaginary (non-imaginary complex) solutions
- B Two distinct real (non-real complex) solutions
- C Two distinct imaginary (non-real complex) solutions
- D Two repeated imaginary (non-real complex) solutions

Classified Practice

Quadratic Equations

Q069 MCQ

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions Select the correct answer:

Choose the correct option.

- A Two distinct real solutions
- B Two repeated real solutions
- C Two distinct imaginary solutions
- D Two distinct real solutions;(sign error)

Classified Practice

Quadratic Equations

Q070 MCQ

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions Select the correct answer:

Choose the correct option.

A One repeated real solution

B One repeated real solution;(sign error)

C 0

D 1

Classified Practice

Quadratic Equations

Q071 MCQ

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct imaginary solutions
- B Two distinct real solutions;(sign error)
- C Two distinct real solutions
- D Two repeated real solutions

Classified Practice

Quadratic Equations

Q072 MCQ

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:**Choose the correct option.**

A 0

B 1

C One repeated real solution

D One repeated real solution;(sign error)

Classified Practice

Quadratic Equations

Q073 MCQ

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct real (non-real complex) solutions
- B Two distinct imaginary (non-real complex) solutions
- C Two repeated imaginary (non-real complex) solutions
- D Two distinct imaginary (non-imaginary complex) solutions

Classified Practice

Quadratic Equations

Q074 MCQ

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:**Choose the correct option.**

A One repeated real solution

B One repeated real solution;(sign error)

C 0

D 1

Classified Practice

Quadratic Equations

Q075 MCQ

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct imaginary (non-real complex) solutions
- B Two repeated imaginary (non-real complex) solutions
- C Two distinct imaginary (non-imaginary complex) solutions
- D Two distinct real (non-real complex) solutions

Classified Practice

Quadratic Equations

Q076 MCQ

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct real solutions
- B Two repeated real solutions
- C Two distinct imaginary solutions
- D Two distinct real solutions;(sign error)

Classified Practice

Quadratic Equations

Q077 Written

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

Q078 Written

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

Classified Practice

Quadratic Equations

Q079 Written

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution.

Q080 Written

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions.

Classified Practice

Quadratic Equations

Q081 Written

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

Q082 MCQ

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions Select the correct answer:

Choose the correct option.

A $+1 < m < 2$

B $-1 < m < 3$

C $-1 < m < 2$

D $-1 > m > 2$

Classified Practice

Quadratic Equations

Q083 MCQ

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Choose the correct option.

A $a < 2$ or $a > 6$: distinct imaginary; $a = 2, 6$: repeated imaginary; $2 < a < 6$: distinct imaginary

B $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct real

C $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

D $a < 2$ or $a > 6$: repeated real; $a = 2, 6$: repeated real; $2 < a < 6$: repeated imaginary

Classified Practice

Quadratic Equations

Q084 MCQ

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution. Select the correct answer:

Choose the correct option.

A $a = 3$ or $a = -5$

B $a = 3$ or $a = -4$

C $a = 3$ or $a = -5 - 1$

D $a = 3$ or $a = -5$ (sign error)

Q085 MCQ

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions Select the correct answer:

Choose the correct option.

A $-18 > m > -2$

B $+18 < m < -2$

C $-18 < m < -1$

D $-18 < m < -2$

Classified Practice

Quadratic Equations

Q086 MCQ

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions Select the correct answer:

Choose the correct option.

A $0 \leq m \leq 4$

B $0 \leq m \leq 3 - 1$

C $0 \leq m \leq 3$

D $0 \geq m \geq 3$

Q087 Written**Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.**

Classified Practice

Quadratic Equations

Q088 Written

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.

Q089 Written**Find the sum and product of the roots of $x^2 + x + 2 = 0$.**

Classified Practice

Quadratic Equations

Q090 Written

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.

Q091 MCQ

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ Select the correct answer:

Choose the correct option.

- A $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3} + 1$
- B $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3} - 1$
- C $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$
- D $-(\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3})$

Classified Practice

Quadratic Equations

Q092 MCQ

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:**Choose the correct option.**

- A $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$
- B $-(\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4})$
- C $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4} + 1$
- D $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4} - 1$

Q093 MCQ

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:

Choose the correct option.

A $S = -1$, $P = 2$ (sign error)

B $S = -1$, $P = 2$

C $S = -1$, $P = 3$

D $S = -1$, $P = 2 - 1$

Classified Practice

Quadratic Equations

Q094 MCQ

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:**Choose the correct option.**

A $S = \frac{7}{3}, P = -2$

B $-(S = \frac{7}{3}, P = -2)$

C $S = \frac{7}{3}, P = -2 + 1$

D $S = \frac{7}{3}, P = -2 - 1$

Q095 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice

Quadratic Equations

Q096 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

Q097 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice

Quadratic Equations

Q098 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Q099 Written**If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.**

Classified Practice

Quadratic Equations

Q100 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

Q101 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice

Quadratic Equations

Q102 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

Q103 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A 12

B 13

C $12 - 1$

D 12 (sign error)

Classified Practice

Quadratic Equations

Q104 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A $\frac{81}{2} - 1$

B $\frac{81}{2}$

C $-(\frac{81}{2})$

D $\frac{81}{2} + 1$

Q105 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A 1

B 2

C $1 - 1$

D 1 (sign error)

Classified Practice

Quadratic Equations

Q106 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A -1 (sign error)

B -1

C 0

D $-1 - 1$

Q107 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A -1

B -2 - 1

C -2 (sign error)

D -2

Classified Practice

Quadratic Equations

Q108 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A -10 (sign error)

B -10

C -9

D $-10 - 1$

Q109 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A $\frac{14}{3} - 1$

B $\frac{14}{3}$

C $-(\frac{14}{3})$

D $\frac{14}{3} + 1$

Classified Practice

Quadratic Equations

Q110 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A -9

B $-10 - 1$

C -10 (sign error)

D -10

Q111 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

Classified Practice

Quadratic Equations

Q112 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

Q113 Written

Find the sum and product of the roots of $5x^2 - 3 = 0$.

Classified Practice

Quadratic Equations

Q114 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Q115 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Classified Practice

Quadratic Equations

Q116 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

Choose the correct option.

A $-4 - 1$

B -4 (sign error)

C -4

D -3

Q117 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$ Select the correct answer:

Choose the correct option.

A $-(\frac{2}{3})$

B $\frac{2}{3} + 1$

C $\frac{2}{3} - 1$

D $\frac{2}{3}$

Classified Practice

Quadratic Equations

Q118 MCQ

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:**Choose the correct option.**

A $-(S = 0, \quad P = -\frac{3}{5})$

B $S = 0, \quad P = -\frac{3}{5} + 1$

C $S = 0, \quad P = -\frac{3}{5} - 1$

D $S = 0, \quad P = -\frac{3}{5}$

Q119 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Choose the correct option.

A $-(\frac{2}{3})$

B $\frac{2}{3} + 1$

C $\frac{2}{3} - 1$

D $\frac{2}{3}$

Classified Practice

Quadratic Equations

Q120 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Choose the correct option.

A 7

B $6 - 1$

C 6 (sign error)

D 6

Q121 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

Classified Practice

Quadratic Equations

Q122 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

Q123 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

Classified Practice

Quadratic Equations

Q124 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

Q125 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

Choose the correct option.

- A $\frac{5}{2}$
B $-(\frac{5}{2})$
C $\frac{5}{2} + 1$
D $\frac{5}{2} - 1$
-
-
-

Classified Practice

Quadratic Equations

Q126 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

Choose the correct option.

A $\frac{1}{2} + 1$

B $\frac{1}{2} - 1$

C $\frac{1}{2}$

D $-(\frac{1}{2})$

Q127 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

Choose the correct option.

A 12

B $11 - 1$

C 11 (sign error)

D 11

Classified Practice

Quadratic Equations

Q128 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

Choose the correct option.

A $-(\frac{23}{3})$

B $\frac{23}{3} + 1$

C $\frac{23}{3} - 1$

D $\frac{23}{3}$

Q129 Written

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

Classified Practice

Quadratic Equations

Q130 **Written**

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

Q131 Written

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

Classified Practice

Quadratic Equations

Q132 MCQ

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots Select the correct answer:

Choose the correct option.

A $k = 6, \quad x = 2, 6$

B $k = 6, \quad x = 2, 7$

C $k = 6, \quad x = 2, 6 - 1$

D $k = 6, \quad x = 2, 6$ (sign error)

Q133 MCQ

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots Select the correct answer:

Choose the correct option.

- A $m = 8, \quad x = -2, -4$
- B $m = 8, \quad x = -2, -3$
- C $m = 8, \quad x = -2, -4 - 1$
- D $m = 8, \quad x = -2, -4$ (sign error)

Classified Practice

Quadratic Equations

Q134 MCQ

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots Select the correct answer:

Choose the correct option.

- A $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{3})$
- B $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2}) - 1$
- C $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$ (sign error)
- D $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Classified Practice

Quadratic Equations

Q135 Written

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots.

Q136 Written

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots.

Classified Practice

Quadratic Equations

Q137 Written

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

Q138 Written

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

Classified Practice

Quadratic Equations

Q139 MCQ

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots Select the correct answer:

Choose the correct option.

- A $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 3)$
- B $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2) - 1$
- C $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$ (sign error)
- D $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Classified Practice

Quadratic Equations

Q140 MCQ

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots Select the correct answer:

Choose the correct option.

- A $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$
- B $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, 0)$
- C $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1) - 1$
- D $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$ (sign error)

Classified Practice

Quadratic Equations

Q141 MCQ

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots Select the correct answer:

Choose the correct option.

A $m = 5$, $x = -5, -1$ (sign error)

B $m = 5$, $x = -5, -1$

C $m = 5$, $x = -5, 0$

D $m = 5$, $x = -5, -1 - 1$

Classified Practice

Quadratic Equations

Q142 MCQ

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots Select the correct answer:

Choose the correct option.

A $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$ (sign error)

B $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

C $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

D $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2}) - 1$

Classified Practice

Quadratic Equations

Q143 **Written**

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots.

Q144 MCQ

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots Select the correct answer:

Choose the correct option.

- A $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$
- B $-(m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5})$
- C $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5} + 1$
- D $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5} - 1$

Classified Practice

Quadratic Equations

Q145 Written

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

Classified Practice

Quadratic Equations

Q146 MCQ

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots. Select the correct answer:

Choose the correct option.

A $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$ (sign error)

B $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

C $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 10)$

D $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9) - 1$

Classified Practice

Quadratic Equations

Q147 **Written****Factorise $2x^2 - 3x + 2$ over the complex numbers.**

Q148 Written**Factorise $x^2 + 2x + 5$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q149 **Written****Factorise $2x^2 - 6x + 1$ over the complex numbers.**

Q150 Written**Factorise $x^2 + 4x + 5$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q151 Written

Factorise $x^2 + 4x + 2$ over the complex numbers.

Q152 Written**Factorise $x^2 - 4x + 7$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q153 **Written****Factorise $2x^2 - x + 1$ over the complex numbers.**

Q154 Written**Factorise $3x^2 - 5x + 3$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q155 Written

Factorise $2x^2 - 2x - 1$ over the complex numbers.

Classified Practice

Quadratic Equations

Q156 MCQ

Factorise $2x^2 - 3x + 2$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $2\left(x - \frac{3 - i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$

B $-(2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right))$

C $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$

D $2\left(x - \frac{3 + \sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$

Classified Practice

Quadratic Equations

Q157 MCQ

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $(x + 1 - 2i)(x + 1 + 2i)$

B $(x + 1 - 2)(x + 1 + 2i)$

C $(x + 1 - 2 - i)(x + 1 + 2i)$

D $(x + 1 - 2i)(x + 13i)$

Classified Practice

Quadratic Equations

Q158 MCQ

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $-(2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right))$

B $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right) + 1$

C $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right) - 1$

D $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$

Classified Practice

Quadratic Equations

Q159 MCQ

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $(x + 2 - i)(x + 2 + i)$

B $(x + 2 + i)(x + 2 + i)$

C $(x + 2 - i)(x + 2 + i)$

D $(x + 2 - i)(x + 2 + i)$

Q160 MCQ

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:

Choose the correct option.

- A $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$ (sign error)
 - B $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$
 - C $(x + 2 - \sqrt{2})(x + 2 + \sqrt{3})$
 - D $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2}) - 1$
-
-
-

Classified Practice

Quadratic Equations

Q161 MCQ

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

B $(x - 2 - \sqrt{3})(x - 2 + i\sqrt{3})$

C $(x - 2 + i\sqrt{3})(x - 2 + i\sqrt{3})$

D $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{4})$

Classified Practice

Quadratic Equations

Q162 MCQ

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $2\left(x - \frac{1 - i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$

B $-(2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right))$

C $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$

D $2\left(x - \frac{1 + \sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$

Classified Practice

Quadratic Equations

Q163 MCQ

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

B $3\left(x - \frac{5 + \sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

C $3\left(x - \frac{5 - i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

D $-(3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right))$

Classified Practice

Quadratic Equations

Q164 MCQ

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right) - 1$

B $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$

C $-(2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right))$

D $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right) + 1$

Classified Practice

Quadratic Equations

Q165 **Written****Factorise $x^4 + 2x^2 - 3$ over the complex numbers.**

Q166 Written**Factorise $x^4 - x^2 - 6$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q167 **Written****Factorise $x^4 - 9$ over the complex numbers.**

Q168 Written**Factorise $x^4 - 3x^2 + 2$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q169 MCQ

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{4})$

B $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$

C $(x - 1)(x + 1)(x - \sqrt{3})(x + i\sqrt{3})$

D $(x - 1)(x + 1)(x + i\sqrt{3})(x + i\sqrt{3})$

Q170 MCQ

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{3})$

B $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$

C $(x - \sqrt{3})(x + \sqrt{3})(x - \sqrt{2})(x + i\sqrt{2})$

D $(x - \sqrt{3})(x + \sqrt{3})(x + i\sqrt{2})(x + i\sqrt{2})$

Classified Practice

Quadratic Equations

Q171 MCQ

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:**Choose the correct option.**

- A $(x - \sqrt{3})(x + \sqrt{3})(x - \sqrt{3})(x + i\sqrt{3})$
B $(x - \sqrt{3})(x + \sqrt{3})(x + i\sqrt{3})(x + i\sqrt{3})$
C $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{4})$
D $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$
-
-
-

Q172 MCQ

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:

Choose the correct option.

- A $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2}) - 1$
 - B $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$ (sign error)
 - C $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$
 - D $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{3})$
-
-
-

Classified Practice

Quadratic Equations

Q173 Written

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

Q174 Written**Find two numbers whose sum is -1 and product is 2 .**

Classified Practice

Quadratic Equations

Q175 Written

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

Q176 Written

Find two numbers whose sum and product are both 1.

Classified Practice

Quadratic Equations

Q177 Written

Create a quadratic equation with roots -1 and 2 .

Q178 Written

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.

Classified Practice

Quadratic Equations

Q179 Written

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

Q180 Written

Find two numbers whose sum is 2 and product is 5.

Classified Practice

Quadratic Equations

Q181 **Written**

Find two numbers whose sum is 4 and product is 2.

Q182 MCQ

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots Select the correct answer:

Choose the correct option.

A $2x^2 - 2x + 1 = 0$ (sign error)

B $2x^2 - 2x + 1 = 0$

C $2x^2 - 2x + 1 = 1$

D $2x^2 - 2x + 1 = 0 - 1$

Classified Practice

Quadratic Equations

Q183 MCQ

Find two numbers whose sum is -1 and product is 2 Select the correct answer:**Choose the correct option.**

A $-\left(\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}\right)$

B $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$

C $\frac{-1 + \sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$

D $\frac{-1 - i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$

Q184 MCQ

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$ Select the correct answer:

Choose the correct option.

A $x^2 - 6x + 13 = 0$ (sign error)

B $x^2 - 6x + 13 = 0$

C $x^2 - 6x + 13 = 1$

D $x^2 - 6x + 13 = 0 - 1$

Classified Practice

Quadratic Equations

Q185 MCQ

Find two numbers whose sum and product are both 1 Select the correct answer:**Choose the correct option.**

- A $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
- B $\frac{1+\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
- C $\frac{1-i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
- D $-(\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2})$

Q186 MCQ

Create a quadratic equation with roots -1 and 2 Select the correct answer:

Choose the correct option.

A $x^2 - x - 2 = 0$

B $x^2 - x - 2 = 1$

C $x^2 - x - 2 = 0 - 1$

D $x^2 - x - 2 = 0$ (sign error)

Classified Practice

Quadratic Equations

Q187 MCQ

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$ Select the correct answer:**Choose the correct option.**

A $x^2 - 2x + 17 = 1$

B $x^2 - 2x + 17 = 0 - 1$

C $x^2 - 2x + 17 = 0$ (sign error)

D $x^2 - 2x + 17 = 0$

Q188 MCQ

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$. Select the correct answer:

Choose the correct option.

A $3x^2 - 10x + 6 = 0$

B $3x^2 - 10x + 6 = 1$

C $3x^2 - 10x + 6 = 0 - 1$

D $3x^2 - 10x + 6 = 0$ (sign error)

Classified Practice

Quadratic Equations

Q189 MCQ

Find two numbers whose sum is 2 and product is 5 Select the correct answer:**Choose the correct option.**A $1 + 2i, 1 - 1i$ B $1 + 2i, 1 - 2i$ C $1 + 2, 1 - 2i$ D $1 + 2 - i, 1 - 2i$

Q190 MCQ

Find two numbers whose sum is 4 and product is 2 Select the correct answer:**Choose the correct option.**A $2 + \sqrt{2}$, $2 - \sqrt{2}$ (sign error)B $2 + \sqrt{2}$, $2 - \sqrt{2}$ C $2 + \sqrt{2}$, $2 - \sqrt{3}$ D $2 + \sqrt{2}$, $2 - \sqrt{2} - 1$

Classified Practice

Quadratic Equations

Q191 Written

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

Q192 MCQ

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$ Select the correct answer:

Choose the correct option.

- A $x^2 - 2x - 4 = 0 - 1$
- B $x^2 - 2x - 4 = 0$ (sign error)
- C $x^2 - 2x - 4 = 0$
- D $x^2 - 2x - 4 = 1$

Classified Practice

Quadratic Equations

Q193 Written

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 .

Q194 Written

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 .

Classified Practice

Quadratic Equations

Q195 MCQ

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 Select the correct answer:

Choose the correct option.

A $x^2 - 7x + 9 = 1$

B $x^2 - 7x + 9 = 0 - 1$

C $x^2 - 7x + 9 = 0$ (sign error)

D $x^2 - 7x + 9 = 0$

Q196 MCQ

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 Select the correct answer:

Choose the correct option.

A $x^2 + 10x + 49 = 0 - 1$

B $x^2 + 10x + 49 = 0$ (sign error)

C $x^2 + 10x + 49 = 0$

D $x^2 + 10x + 49 = 1$

Classified Practice

Quadratic Equations

Q197 Written

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

Q198 MCQ

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$ Select the correct answer:

Choose the correct option.

- A $4x^2 + 3x + 2 = 0$
- B $4x^2 + 3x + 2 = 1$
- C $4x^2 + 3x + 2 = 0 - 1$
- D $4x^2 + 3x + 2 = 0$ (sign error)

Classified Practice

Quadratic Equations

Q199 Written

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

Q200 MCQ

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$ Select the correct answer:

Choose the correct option.

A $x^2 + 8x + 24 = 1$

B $x^2 + 8x + 24 = 0 - 1$

C $x^2 + 8x + 24 = 0$ (sign error)

D $x^2 + 8x + 24 = 0$

Classified Practice

Quadratic Equations

Q201 **Written**

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m .

Classified Practice

Quadratic Equations

Q202 Written

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m .

Classified Practice

Quadratic Equations

Q203 MCQ

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Choose the correct option.

A $-2 < m < -1$

B $-2 > m > -1$

C $+2 < m < -1$

D $-2 < m < 0$

Q204 MCQ

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Choose the correct option.

A $-1 < m < -\frac{1}{2} - 1$

B $-1 < m < -\frac{1}{2}$

C $-(-1 < m < -\frac{1}{2})$

D $-1 < m < -\frac{1}{2} + 1$

Classified Practice

Quadratic Equations

Q205 Written

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m .

Classified Practice

Quadratic Equations

Q206 **Written**

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k .

Classified Practice

Quadratic Equations

Q207 MCQ

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m Select the correct answer:

Choose the correct option.

A $+6 < m < -2$

B $-6 < m < -1$

C $-6 < m < -2$

D $-6 > m > -2$

Q208 MCQ

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

Choose the correct option.

A $k > 10$ (sign error)

B $k > 10$

C $k > 11$

D $k > 10 - 1$

Classified Practice

Quadratic Equations

Q209 Written

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m .

Q210 Written

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k .

Classified Practice

Quadratic Equations

Q211 MCQ

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m Select the correct answer:

Choose the correct option.

A $m > -4$

B $m > -3$

C $m > -4 - 1$

D $m > -4$ (sign error)

Q212 MCQ

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

Choose the correct option.

A $k > -2$

B $k > -3 - 1$

C $k > -3$ (sign error)

D $k > -3$

Classified Practice

Quadratic Equations

Q213 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

Classified Practice

Quadratic Equations

Q214 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies.

Classified Practice

Quadratic Equations

Q215 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

Classified Practice

Quadratic Equations

Q216 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

Classified Practice

Quadratic Equations

Q217 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

Classified Practice

Quadratic Equations

Q218 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

Classified Practice

Quadratic Equations

Q219 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

Classified Practice

Quadratic Equations

Q220 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

Classified Practice

Quadratic Equations

Q221 **Challenge**

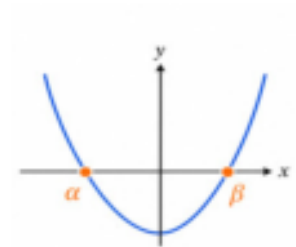
For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

Classified Practice

Quadratic Equations

Q222 Challenge

In the opposite figure, α, β are the two roots. If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.



Classified Practice

Quadratic Equations

Q223 **Challenge**

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

Classified Practice

Quadratic Equations

Q224 **Challenge**

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

Classified Practice

Quadratic Equations

Q225 **Challenge**

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

Classified Practice

Quadratic Equations

Q226 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

Classified Practice

Quadratic Equations

Q227 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Choose the correct option.

- A $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct real
- B $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary
- C $a < -4$ or $a > 0$: repeated real; $a = -4, 0$: repeated real; $-4 < a < 0$: repeated imaginary
- D $a < -4$ or $a > 0$: distinct imaginary; $a = -4, 0$: repeated imaginary; $-4 < a < 0$: distinct imaginary

Classified Practice

Quadratic Equations

Q228 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Choose the correct option.

- A $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary – 1
- B $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary
- C $-(\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary)
- D $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary + 1

Classified Practice

Quadratic Equations

Q229 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Choose the correct option.

- A $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct real
B $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary
C $a > 3$: repeated real; $a = 3$: repeated real; $a < 3$: repeated imaginary
D $a > 3$: distinct imaginary; $a = 3$: repeated imaginary; $a < 3$: distinct imaginary

Classified Practice

Quadratic Equations

Q230 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$. Select the correct answer:

Choose the correct option.

A $\frac{9}{2} - 1$

B $\frac{9}{2}$

C $-(\frac{9}{2})$

D $\frac{9}{2} + 1$

Classified Practice

Quadratic Equations

Q231 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

Choose the correct option.

A 4

B $3 - 1$

C 3 (sign error)

D 3

Classified Practice

Quadratic Equations

Q232 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A $\frac{27}{2} - 1$

B $\frac{27}{2}$

C $-(\frac{27}{2})$

D $\frac{27}{2} + 1$

Classified Practice

Quadratic Equations

Q233 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

Choose the correct option.

A 9

B 10

C $9 - 1$

D 9 (sign error)

Classified Practice

Quadratic Equations

Q234 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$ Select the correct answer:

Choose the correct option.

- A Common root c; other roots a,b; equation $x^2 + cx + ab = 0$ (equation = -1).
- B Common root c; other roots a,b; equation $x^2 + cx + ab = 0$ (sign error).
- C Common root c; other roots a,b; equation $x^2 + cx + ab = 0$.
- D Common root c; other roots a,b; equation $x^2 + cx + ab = 0$ (equation = 1).

Classified Practice

Quadratic Equations

Q235 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model. Select the correct answer:

Choose the correct option.

A $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$

B $x = \frac{\sqrt{6}}{2}$, $x = \frac{i\sqrt{3}}{3}$; real: $\sqrt{6}/2$

C $x = -\frac{\sqrt{6}}{2}$, $x = -\frac{i\sqrt{3}}{3}$; real: $-\sqrt{6}/2$

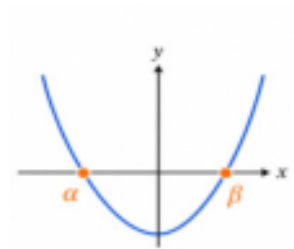
D $x = +\frac{\sqrt{6}}{2}$, $x = +\frac{i\sqrt{3}}{3}$; real: $+\sqrt{6}/2$

Classified Practice

Quadratic Equations

Q236 Challenge

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:



Choose the correct option.

- A If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then $Px^2 - Sx + 1 = 0$.
- B If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then $Px^2 - Sx + 1 = 1$.
- C If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then $Px^2 - Sx + 1 = 0$. -1
- D If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then $Px^2 - Sx + 1 = 0$.; (sign error)

Classified Practice

Quadratic Equations

Q237 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive. Select the correct answer:

Choose the correct option.

- A $+4 < m < -3$
- B $-4 < m < -2$
- C $-4 < m < -3$
- D $-4 > m > -3$

Classified Practice

Quadratic Equations

Q238 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative. Select the correct answer:

Choose the correct option.

A $m > 6$

B $m > 5 - 1$

C $m > 5$ (sign error)

D $m > 5$

Classified Practice

Quadratic Equations

Q239 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs. Select the correct answer:

Choose the correct option.

A $m > -4$

B $m < +4$

C $m < -3$

D $m < -4$

Classified Practice

Quadratic Equations

Q240 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result. Select the correct answer:

Choose the correct option.

- A $x = \frac{-\sqrt{3}i\sqrt{3}}{3}$; no real equilibrium
- B $x = \frac{-\sqrt{3} - i\sqrt{3}}{3}$; no real equilibrium
- C $x = \frac{-\sqrt{3} + i\sqrt{3}}{3}$; no real equilibrium
- D $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Classified Practice

Quadratic Equations

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $x^2 - 5x + 6 = 0$ Select the correct answer:

Choose the correct option.

A 2, 3 (sign error)

B 2, 3

C 2, 4

D 2, 3 - 1

Classified Practice

Quadratic Equations

Quiz MCQ 2 **Concept Quiz · MCQ**

For $x^2 - 4x + 7 = 0$, determine the nature of the roots Select the correct answer:

Choose the correct option.

- A Two distinct non-real complex roots
- B Two repeated non-real complex roots
- C Two distinct non-imaginary complex roots
- D Two distinct non-real complex roots;(sign error)

Classified Practice

Quadratic Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

If α, β are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A $7 - 1$

B 7 (sign error)

C 7

D 8

Quiz MCQ 4 Concept Quiz · MCQ

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

Choose the correct option.

A $x^2 - 4x + 5 = 0$

B $x^2 - 4x + 5 = 1$

C $x^2 - 4x + 5 = 0 - 1$

D $x^2 - 4x + 5 = 0$ (sign error)

Classified Practice

Quadratic Equations

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $2x^2 + 3x - 2 = 0$.

Quiz WRITTEN 6 Concept Quiz · Written

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.

Classified Practice

Quadratic Equations

Quiz WRITTEN 7 Concept Quiz · Written

If α, β are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Quiz WRITTEN 8 Concept Quiz · Written

If α, β are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2 .

Answer key

Use the question number shown on each page.

Q001 $x = \pm 2i\sqrt{2}$

Q002 $x = \pm 2i$

Q003 $x = \pm 7i$

Q004 $x = \pm \frac{2}{3}$

Q005 $x = \pm 2\sqrt{3}$

Q006 $x = \pm \frac{i\sqrt{3}}{2}$

Q007 A

Q008 A

Q009 C

Q010 C

Q011 B

Q012 B

Q013 $x = 3 \pm 2i\sqrt{2}$

Q014 $x = 2 \pm \sqrt{5}$

Q015 $x = -6 \pm i\sqrt{6}$

Q016 $x = -4 \pm 3i\sqrt{2}$

Q017 $x = \frac{3 \pm \sqrt{7}}{2}$

Q018 A

Q019 B

Q020 C

Q021 C

Q022 D

Q023 $x = -1, \frac{9}{2}$

Q024 $x = \frac{1}{2}, -3$

Q025 $x = 0, -\frac{7}{2}$

Q026 $x = \frac{3}{2}, -4$

Q027 $x = 3, 5$

Q028 A

Q029 D

Q030 D

Q031 A

Q032 A

Q033 $x = \frac{2 \pm \sqrt{7}}{3}$

Q034 B

Q035 $x = \frac{-\sqrt{6} \pm 4i}{2}$

Q036 $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

Q037 $x = \frac{-1 \pm i\sqrt{3}}{2}$

Q038 $x = \frac{5 \pm i\sqrt{11}}{6}$

Q039 $x = \sqrt{3} \pm i\sqrt{2}$

Q040 $x = \frac{-\sqrt{2} \pm i}{3}$

Q041 $x = \frac{1 \pm i\sqrt{7}}{2}$

Q042 $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

Q043 $x = 1 \pm \frac{i\sqrt{6}}{2}$

Q044 B

Q045 C

Q046 D

Q047 B

Q048 D

Q049 B

Q050 C

Q051 C

Q052 B

Q053 Two distinct imaginary (non-real complex) solutions

Q054 $-2 \leq m \leq 6$

Q055 $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Two distinct imaginary (non-real complex) solutions

Q057 Two distinct real solutions

Q058 One repeated real solution

Q059 Two distinct real solutions

Q060 One repeated real solution

Q061 Two distinct imaginary (non-real complex) solutions

Q062 One repeated real solution

Q063 Two distinct imaginary (non-real complex) solutions

Q064 Two distinct real solutions

Q065 A

Q066 D

Q067 C

Q068 C

Q069 A

Q070 A

Q071 C

Q072 C

Q073 B

Q074 A

Q075 A

Q076 A

Q077 $-1 < m < 2$

Q078 $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Q079 $a = 3$ or $a = -5$

Q080 $-18 < m < -2$

Q081 $0 \leq m \leq 3$

Q082 C

Q083 C

Q084 A

Q085 D

Q086 C

Q087 $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$

Q088 $\alpha + \beta = -\frac{1}{2}, \alpha\beta = -\frac{7}{4}$

Q089 $S = -1, P = 2$

Q090 $S = \frac{7}{3}, P = -2$

Q091 C

Q092 A

Q093 B

Q094 A

Q095 12

Q096 $\frac{81}{2}$

Q097 1

Q098 -1

Q099 -2

Q100 -10

Q101 $\frac{14}{3}$

Q102 -10

Q103 A

Q104 B

Q105 A

Q106 B

Q107 D

Q108 B

Q109 B

Q110 D

Q111 -4

Q112 $\frac{2}{3}$

Q113 $S = 0, P = -\frac{3}{5}$

Q114 $\frac{2}{3}$

Q115 6

Q116 C

Q117 D

Q118 D

Q119 D

Q120 D

Q121 $\frac{5}{2}$

Q122 $\frac{1}{2}$

Q123 11

Q124 $\frac{23}{3}$

Q125 A

Q126 C

Q127 D

Q128 D

Q129 $k = 6, x = 2, 6$

Q130 $m = 8, x = -2, -4$

Q131 $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Q132 A

Q133 A

Q134 D

Q135 $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Q136 $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Q137 $m = 5, x = -5, -1$

Q138 $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

Q139 D

Q140 A

Q141 B

Q142 B

Q143 $m = \frac{24}{25}, x = \frac{4}{5}, \frac{6}{5}$

Q144 A

Q145 $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

Q146 B

Q147 $2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$

Q148 $(x+1-2i)(x+1+2i)$

Q149 $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$

Q150 $(x+2-i)(x+2+i)$

Q151 $(x+2-\sqrt{2})(x+2+\sqrt{2})$

Q152 $(x-2-i\sqrt{3})(x-2+i\sqrt{3})$

Q153 $2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$

Q154 $3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$

Q155 $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$

Q156 C

Q157 A

Q158 D

Q159 D

Q160 B

Q161 A

Q162 C

Q163 A

Q164 B

Q165 $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$

Q166 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$

Q167 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{3})$

Q168 $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$

Q169 B

Q170 B

Q171 D

Q172 C

Q173 $2x^2 - 2x + 1 = 0$

Q174 $\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$

Q175 $x^2 - 6x + 13 = 0$

Q176 $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

Q177 $x^2 - x - 2 = 0$

Q178 $x^2 - 2x + 17 = 0$

Q179 $3x^2 - 10x + 6 = 0$

Q180 $1+2i, 1-2i$

Q181 $2+\sqrt{2}, 2-\sqrt{2}$

Q182 B

Q183 B

Q184 B

Q185 A

Q186 A

Q187 D

Q188 A

Q189 B

Q190 B

Q191 $x^2 - 2x - 4 = 0$

Q192 C

Q193 $x^2 - 7x + 9 = 0$

Q194 $x^2 + 10x + 49 = 0$

Q195 D

Q196 C

Q197 $4x^2 + 3x + 2 = 0$

Q198 A

Q199 $x^2 + 8x + 24 = 0$

Q200 D

Q201 $-2 < m < -1$

Q202 $-1 < m < -\frac{1}{2}$

Q203 A

Q204 B

Q205 $-6 < m < -2$

Q206 $k > 10$

Q207 C

Q208 B

Q209 $m > -4$

Q210 $k > -3$

Q211 A

Q212 D

Q213 $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Q214 $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Q215 $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Q216 $\frac{9}{2}$

Q217 3

Q218 $\frac{27}{2}$

Q219 9

Q220 Common root c ; other roots a and b .Q221 $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$ Q222 For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$ Q223 $-4 < m < -3$ Q224 $m > 5$ Q225 $m < -4$ Q226 $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Q227 B

Q228 B

Q229 B

Q230 B

Q231 D

Q232 B

Q233 A

Q234 C

Q235 A

Q236 A

Q237 C

Q238 D

Q239 D

Q240 D

Quiz MCQ 1 B

Quiz MCQ 2 A

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $x = \frac{1}{2}, -2$ Quiz WRITTEN 6 $-1 < m < 2$

Quiz WRITTEN 7 28

Quiz WRITTEN 8 $x^2 + 6x + 25 = 0$

Concept 3 | Polynomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Remainder

$$P(a) = R \text{ when divisor} = x - a$$

Factor

$$P(a) = 0 \Rightarrow (x - a) \mid P(x)$$

Roots

$$P(x) = a \prod (x - r_i)$$

Classified Practice

Polynomial Theorems

Q001 Written

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.

Q002 Written**Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.**

Classified Practice

Polynomial Theorems

Q003 MCQ

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

Choose the correct option.

A $2 - 1$

B 2 (sign error)

C 2

D 3

Q004 MCQ**Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:****Choose the correct option.****A** $6 - 1$ **B** 6 (sign error)**C** 6**D** 7

Classified Practice

Polynomial Theorems

Q005 Written

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.

Q006 Written**Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.**

Classified Practice

Polynomial Theorems

Q007 Written

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.

Q008 Written**Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.**

Classified Practice

Polynomial Theorems

Q009 MCQ

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:

Choose the correct option.

A 9

B $8 - 1$

C 8 (sign error)

D 8

Q010 MCQ**Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:****Choose the correct option.**

A $-\frac{1}{8}$

B $-(-\frac{1}{8})$

C $-\frac{1}{8} + 1$

D $-\frac{1}{8} - 1$

Classified Practice

Polynomial Theorems

Q011 MCQ

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

Choose the correct option.

A 1

B $0 - 1$

C 0 (sign error)

D 0

Q012 MCQ**Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:****Choose the correct option.****A** 4**B** 5**C** $4 - 1$ **D** 4 (sign error)

Classified Practice

Polynomial Theorems

Q013 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a .

Q014 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a .

Classified Practice

Polynomial Theorems

Q015 Written

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a .

Q016 Written

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a .

Classified Practice

Polynomial Theorems

Q017 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

Choose the correct option.

A $a = 13/3$ (sign error)

B $a = 13/3$

C $a = 13/3 + 1$

D $a = 13/3 - 1$

Q018 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

Choose the correct option.

A $a = -4$ (sign error)

B $a = -4$

C $a = -3$

D $a = -4 - 1$

Classified Practice

Polynomial Theorems

Q019 MCQ

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a Select the correct answer:

Choose the correct option.

A $a = -4 - 1$

B $a = -4$ (sign error)

C $a = -4$

D $a = -3$

Q020 MCQ

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a Select the correct answer:

Choose the correct option.

A $a = -16$

B $a = -15$

C $a = -16 - 1$

D $a = -16$ (sign error)

Classified Practice

Polynomial Theorems

Q021 Written

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a .

Q022 MCQ

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

Choose the correct option.

A $a = -4$

B $a = -3$

C $a = -4 - 1$

D $a = -4$ (sign error)

Classified Practice

Polynomial Theorems

Q023 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Q024 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Classified Practice

Polynomial Theorems

Q025 Written

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$.

Q026 Written

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Classified Practice

Polynomial Theorems

Q027 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$ Select the correct answer:

Choose the correct option.

A $2x + 1$

B $2x^2$

C $2x + 1 - 1$

D $2x + 1$ (sign error)

Q028 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Choose the correct option.

A $-10x + 13$

B $-10x14$

C $-10x + 13 - 1$

D $-10x + 13$ (sign error)

Classified Practice

Polynomial Theorems

Q029 MCQ

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$ Select the correct answer:

Choose the correct option.

A $2x + 3$

B $2x^4$

C $2x + 3 - 1$

D $2x + 3$ (sign error)

Q030 MCQ

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Choose the correct option.

A $-15x + 19$

B $-15x20$

C $-15x + 19 - 1$

D $-15x + 19$ (sign error)

Classified Practice

Polynomial Theorems

Q031 Written

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$.

Q032 Written

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$.

Classified Practice

Polynomial Theorems

Q033 Written

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$.

Q034 Written

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$.

Classified Practice

Polynomial Theorems

Q035 MCQ

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$. Select the correct answer:

Choose the correct option.

A $3x - 2$ (sign error)

B $3x - 2$

C $3x - 1$

D $3x - 2 - 1$

Q036 MCQ

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$ Select the correct answer:

Choose the correct option.

A 1

B x

C x (sign error)

D 0

Classified Practice

Polynomial Theorems

Q037 MCQ

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$. Select the correct answer:

Choose the correct option.

- A $3x - 1$
- B $3x0$
- C $3x - 1 - 1$
- D $3x - 1$ (sign error)

Q038 MCQ

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$ Select the correct answer:

Choose the correct option.

- A $7x - 11$
 - B $7x - 12 - 1$
 - C $7x - 12$ (sign error)
 - D $7x - 12$
-
-
-

Classified Practice

Polynomial Theorems

Q039 Written

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$.

Q040 MCQ

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$ Select the correct answer:

Choose the correct option.

A $x^2 + 3x - 4$

B $x^2 + 3x - 3$

C $x^2 + 3x - 4 - 1$

D $x^2 + 3x - 4$ (sign error)

Classified Practice

Polynomial Theorems

Q041 Written

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

Q042 MCQ

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$ Select the correct answer:

Choose the correct option.

A $Q = x^2 - 2x + 9, R = -7$

B $Q = x^2 - 2x + 9, R = -6$

C $Q = x^2 - 2x + 9, R = -7 - 1$

D $Q = x^2 - 2x + 9, R = -7$ (sign error)

Classified Practice

Polynomial Theorems

Q043 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

Q044 Written

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

Classified Practice

Polynomial Theorems

Q045 Written

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

Q046 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$.

Classified Practice

Polynomial Theorems

Q047 **Written**

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

Q048 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$ Select the correct answer:

Choose the correct option.

A $Q = x^2 + x + 6, R = 0$ (sign error)

B $Q = x^2 + x + 6, R = 0$

C $Q = x^2 + x + 6, R = 1$

D $Q = x^2 + x + 6, R = 0 - 1$

Classified Practice

Polynomial Theorems

Q049 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$. Select the correct answer:

Choose the correct option.

A $Q = 3x^2 + 4x + 9, R = 27$

B $Q = 3x^2 + 4x + 9, R = 26 - 1$

C $Q = 3x^2 + 4x + 9, R = 26$ (sign error)

D $Q = 3x^2 + 4x + 9, R = 26$

Q050 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$ Select the correct answer:

Choose the correct option.

A $Q = x^2 + 2x - 1, R = 5$

B $Q = x^2 + 2x - 1, R = 4 - 1$

C $Q = x^2 + 2x - 1, R = 4$ (sign error)

D $Q = x^2 + 2x - 1, R = 4$

Classified Practice

Polynomial Theorems

Q051 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$. Select the correct answer:

Choose the correct option.

A $Q = x^2 + 3x + 2, R = -1$

B $Q = x^2 + 3x + 2, R = 0$

C $Q = x^2 + 3x + 2, R = -1 - 1$

D $Q = x^2 + 3x + 2, R = -1$ (sign error)

Q052 MCQ

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$ Select the correct answer:

Choose the correct option.

A $Q = 4x + 2, R = 4$ (sign error)

B $Q = 4x + 2, R = 4$

C $Q = 4x + 2, R = 5$

D $Q = 4x + 2, R = 4 - 1$

Classified Practice

Polynomial Theorems

Q053 Written

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

Q054 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$ Select the correct answer:

Choose the correct option.

A $Q = 3x^3 - x^2 + 2x - 6, R = 1$

B $Q = 3x^3 - x^2 + 2x - 6, R = 0 - 1$

C $Q = 3x^3 - x^2 + 2x - 6, R = 0$ (sign error)

D $Q = 3x^3 - x^2 + 2x - 6, R = 0$

Classified Practice

Polynomial Theorems

Q055 Written

Factorise $x^3 + 2x^2 - x - 2$.

Q056 Written**Factorise** $x^3 - 2x^2 - 5x + 6$.

Classified Practice

Polynomial Theorems

Q057 **Written****Factorise** $x^3 - x^2 - 16x - 20$.

Q058 Written**Factorise** $2x^3 - x^2 - 5x - 2$.

Classified Practice

Polynomial Theorems

Q059 Written

Factorise $x^3 - 7x - 6$.

Q060 Written**Factorise** $x^3 + 4x^2 + x - 6$.

Classified Practice

Polynomial Theorems

Q061 Written

Factorise $x^3 + 6x^2 + 11x + 6$.

Q062 Written**Factorise** $x^3 - 3x + 2$.

Classified Practice

Polynomial Theorems

Q063 Written

Factorise $x^3 + x^2 - 8x - 12$.

Q064 Written**Factorise** $x^3 - 3x^2 - 9x - 5$.

Classified Practice

Polynomial Theorems

Q065 **Written****Factorise** $2x^3 - 3x^2 - 3x + 2$.

Q066 Written**Factorise** $3x^3 + x^2 - 8x + 4$.

Classified Practice

Polynomial Theorems

Q067 **Written****Factorise** $2x^3 + 7x^2 + 8x + 3$.

Q068 MCQ**Factorise $x^3 + 2x^2 - x - 2$ Select the correct answer:****Choose the correct option.**

A $(x - 1)(x + 1)(x^3)$

B $(x - 1)(x + 1)(x + 2) - 1$

C $(x - 1)(x + 1)(x + 2)$ (sign error)

D $(x - 1)(x + 1)(x + 2)$

Classified Practice

Polynomial Theorems

Q069 MCQ

Factorise $x^3 - 2x^2 - 5x + 6$ Select the correct answer:**Choose the correct option.**

A $(x - 1)(x - 3)(x3)$

B $(x - 1)(x - 3)(x + 2) - 1$

C $(x - 1)(x - 3)(x + 2)$ (sign error)

D $(x - 1)(x - 3)(x + 2)$

Q070 MCQ**Factorise $x^3 - x^2 - 16x - 20$ Select the correct answer:****Choose the correct option.**

- A $(x + 2)^2(x - 5) - 1$
 - B $(x + 2)^2(x - 5)$ (sign error)
 - C $(x + 2)^2(x - 5)$
 - D $(x + 2)^2(x - 4)$
-
-
-

Classified Practice

Polynomial Theorems

Q071 MCQ

Factorise $2x^3 - x^2 - 5x - 2$ Select the correct answer:**Choose the correct option.**A $(x - 2)(2x + 1)(x + 1)$ (sign error)B $(x - 2)(2x + 1)(x + 1)$ C $(x - 2)(2x + 1)(x^2)$ D $(x - 2)(2x + 1)(x + 1) - 1$

Q072 MCQ

Factorise $x^3 - 7x - 6$ Select the correct answer:**Choose the correct option.**A $(x - 3)(x + 1)(x + 2)$ (sign error)B $(x - 3)(x + 1)(x + 2)$ C $(x - 3)(x + 1)(x3)$ D $(x - 3)(x + 1)(x + 2) - 1$

Classified Practice

Polynomial Theorems

Q073 MCQ

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:**Choose the correct option.**

A $(x - 1)(x + 2)(x + 3) - 1$

B $(x - 1)(x + 2)(x + 3)$ (sign error)

C $(x - 1)(x + 2)(x + 3)$

D $(x - 1)(x + 2)(x^4)$

Q074 MCQ

Factorise $x^3 + 6x^2 + 11x + 6$ Select the correct answer:**Choose the correct option.**A $(x + 1)(x + 2)(x + 3)$ (sign error)B $(x + 1)(x + 2)(x + 3)$ C $(x + 1)(x + 2)(x + 4)$ D $(x + 1)(x + 2)(x + 3) - 1$

Classified Practice

Polynomial Theorems

Q075 MCQ

Factorise $x^3 - 3x + 2$ Select the correct answer:**Choose the correct option.**

A $(x - 1)^2(x + 3)$

B $(x - 1)^2(x + 2) - 1$

C $(x - 1)^2(x + 2)$ (sign error)

D $(x - 1)^2(x + 2)$

Q076 MCQ**Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:****Choose the correct option.**

A $(x - 3)(x + 2)^3$

B $(x - 3)(x + 2)^2 - 1$

C $(x - 3)(x + 2)^2$ (sign error)

D $(x - 3)(x + 2)^2$

Classified Practice

Polynomial Theorems

Q077 MCQ

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:**Choose the correct option.**

A $(x - 5)(x + 1)^3$

B $(x - 5)(x + 1)^2 - 1$

C $(x - 5)(x + 1)^2$ (sign error)

D $(x - 5)(x + 1)^2$

Q078 MCQ**Factorise $2x^3 - 3x^2 - 3x + 2$ Select the correct answer:****Choose the correct option.**

A $(x - 2)(x + 1)(2x - 1)$

B $(x - 2)(x + 1)(2x0)$

C $(x - 2)(x + 1)(2x - 1) - 1$

D $(x - 2)(x + 1)(2x - 1)$ (sign error)

Classified Practice

Polynomial Theorems

Q079 MCQ

Factorise $3x^3 + x^2 - 8x + 4$ Select the correct answer:**Choose the correct option.**

A $(x - 1)(x + 2)(3x - 2) - 1$

B $(x - 1)(x + 2)(3x - 2)$ (sign error)

C $(x - 1)(x + 2)(3x - 2)$

D $(x - 1)(x + 2)(3x - 1)$

Q080 MCQ**Factorise $2x^3 + 7x^2 + 8x + 3$ Select the correct answer:****Choose the correct option.**

A $(x + 1)^2(2x + 4)$

B $(x + 1)^2(2x + 3) - 1$

C $(x + 1)^2(2x + 3)$ (sign error)

D $(x + 1)^2(2x + 3)$

Classified Practice

Polynomial Theorems

Q081 **Written****Factorise** $x^4 - 6x^3 + 8x^2 + 8x - 16$.

Q082 Written**Factorise** $x^4 + 3x^3 - 27x^2 + 13x + 42$.

Classified Practice

Polynomial Theorems

Q083 Written

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$.

Q084 Written**Factorise** $x^4 - 10x^3 + 35x^2 - 50x + 24$.

Classified Practice

Polynomial Theorems

Q085 **Written****Factorise** $x^4 - 5x^3 - 3x^2 + 17x - 10$.

Q086 Written**Factorise** $2x^4 - 5x^3 - 20x^2 + 20x + 48$.

Classified Practice

Polynomial Theorems

Q087 Written

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$.

Q088 Written**Factorise** $x^4 - 4x^3 + 3x^2 + 2x - 6$.

Classified Practice

Polynomial Theorems

Q089 Written

Factorise $x^4 - x^3 + x^2 + x - 2$.

Q090 MCQ

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ Select the correct answer:**Choose the correct option.**

A $(x - 2)^2(x^2 - 2x - 3)$

B $(x - 2)^2(x^2 - 2x - 4) - 1$

C $(x - 2)^2(x^2 - 2x - 4)$ (sign error)

D $(x - 2)^2(x^2 - 2x - 4)$

Classified Practice

Polynomial Theorems

Q091 MCQ

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ Select the correct answer:**Choose the correct option.**

A $(x - 3)(x - 2)(x + 1)(x + 7)$

B $(x - 3)(x - 2)(x + 1)(x8)$

C $(x - 3)(x - 2)(x + 1)(x + 7) - 1$

D $(x - 3)(x - 2)(x + 1)(x + 7)$ (sign error)

Q092 MCQ

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ Select the correct answer:**Choose the correct option.**

A $(x - 1)^2(x^2 - 2x + 4)$

B $(x - 1)^2(x^2 - 2x + 3) - 1$

C $(x - 1)^2(x^2 - 2x + 3)$ (sign error)

D $(x - 1)^2(x^2 - 2x + 3)$

Classified Practice

Polynomial Theorems

Q093 MCQ

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ Select the correct answer:**Choose the correct option.**

A $(x - 4)(x - 3)(x - 2)(x - 1)$

B $(x - 4)(x - 3)(x - 2)(x - 1) - 1$

C $(x - 4)(x - 3)(x - 2)(x - 1)$ (sign error)

D $(x - 4)(x - 3)(x - 2)(x - 1)$

Q094 MCQ

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ Select the correct answer:**Choose the correct option.**

- A $(x - 5)(x - 1)^2(x + 2) - 1$
B $(x - 5)(x - 1)^2(x + 2)$ (sign error)
C $(x - 5)(x - 1)^2(x + 2)$
D $(x - 5)(x - 1)^2(x3)$
-
-
-

Classified Practice

Polynomial Theorems

Q095 MCQ

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ Select the correct answer:**Choose the correct option.**

A $(x - 4)(x - 2)(x + 2)(2x + 4)$

B $(x - 4)(x - 2)(x + 2)(2x + 3) - 1$

C $(x - 4)(x - 2)(x + 2)(2x + 3)$ (sign error)

D $(x - 4)(x - 2)(x + 2)(2x + 3)$

Q096 MCQ

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ Select the correct answer:**Choose the correct option.**

A $(x - 1)(x + 2)(x^2 - 5x + 3)$

B $(x - 1)(x + 2)(x^2 - 5x + 4)$

C $(x - 1)(x + 2)(x^2 - 5x + 3) - 1$

D $(x - 1)(x + 2)(x^2 - 5x + 3)$ (sign error)

Classified Practice

Polynomial Theorems

Q097 MCQ

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ Select the correct answer:**Choose the correct option.**A $(x - 3)(x + 1)(x^2 - 2x + 2)$ (sign error)B $(x - 3)(x + 1)(x^2 - 2x + 2)$ C $(x - 3)(x + 1)(x^2 - 2x3)$ D $(x - 3)(x + 1)(x^2 - 2x + 2) - 1$

Q098 MCQ

Factorise $x^4 - x^3 + x^2 + x - 2$ Select the correct answer:**Choose the correct option.**

A $(x - 1)(x + 1)(x^2 - x + 2) - 1$

B $(x - 1)(x + 1)(x^2 - x + 2)$ (sign error)

C $(x - 1)(x + 1)(x^2 - x + 2)$

D $(x - 1)(x + 1)(x^2 - x^3)$

Classified Practice

Polynomial Theorems

Q099 Written

Factorise $2x^3 - 3x^2 - x - 2$.

Q100 Written**Factorise** $2x^3 - 7x^2 + 3x + 6$.

Classified Practice

Polynomial Theorems

Q101 **Written****Factorise** $x^3 + 4x^2 + 2x - 4$.

Q102 Written**Factorise** $x^3 - 3x^2 + 2$.

Classified Practice

Polynomial Theorems

Q103 MCQ

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:**Choose the correct option.**

- A $(x - 2)(2x^2 + x + 1) - 1$
B $(x - 2)(2x^2 + x + 1)$ (sign error)
C $(x - 2)(2x^2 + x + 1)$
D $(x - 2)(2x^2 + x2)$
-
-
-

Q104 MCQ

Factorise $2x^3 - 7x^2 + 3x + 6$ Select the correct answer:

Choose the correct option.

A $(x - 2)(2x^2 - 3x - 3)$ (sign error)

B $(x - 2)(2x^2 - 3x - 3)$

C $(x - 2)(2x^2 - 3x - 2)$

D $(x - 2)(2x^2 - 3x - 3) - 1$

Classified Practice

Polynomial Theorems

Q105 MCQ

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:**Choose the correct option.**A $(x + 2)(x^2 + 2x - 2)$ (sign error)B $(x + 2)(x^2 + 2x - 2)$ C $(x + 2)(x^2 + 2x - 1)$ D $(x + 2)(x^2 + 2x - 2) - 1$

Q106 MCQ**Factorise $x^3 - 3x^2 + 2$ Select the correct answer:****Choose the correct option.**

A $(x - 1)(x^2 - 2x - 1)$

B $(x - 1)(x^2 - 2x - 2) - 1$

C $(x - 1)(x^2 - 2x - 2)$ (sign error)

D $(x - 1)(x^2 - 2x - 2)$

Classified Practice

Polynomial Theorems

Q107 **Challenge****Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.**

Classified Practice

Polynomial Theorems

Q108 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

Classified Practice

Polynomial Theorems

Q109 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Classified Practice

Polynomial Theorems

Q110 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

Classified Practice

Polynomial Theorems

Q111 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b .

Classified Practice

Polynomial Theorems

Q112 Challenge

For the polynomial above with $a = -12$, $b = 2$, determine all roots of $P(x) = 0$.

Classified Practice

Polynomial Theorems

Q113 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$ Select the correct answer:

Choose the correct option.

A 56 (sign error)

B 56

C 57

D $56 - 1$

Classified Practice

Polynomial Theorems

Q114 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial. Select the correct answer:

Choose the correct option.

- A $a = 10$; for $x + 2$, $a = -10$; linear factors correspond to roots.
- B $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots (minus 1).
- C $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots (sign error).
- D $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots.

Classified Practice

Polynomial Theorems

Q115 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$. Select the correct answer:

Choose the correct option.

- A $x - 1 - 1$
- B $x - 1$ (sign error)
- C $x - 1$
- D $x0$

Classified Practice

Polynomial Theorems

Q116 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts Select the correct answer:

Choose the correct option.

A $x = \sqrt{3}, -\sqrt{3}$

B $x = \sqrt{3}, -\sqrt{4}$

C $x = \sqrt{3}, -\sqrt{3} - 1$

D $x = \sqrt{3}, -\sqrt{3}$ (sign error)

Classified Practice

Polynomial Theorems

Q117 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b Select the correct answer:

Choose the correct option.

- A $a = -12, b = 2 - 1$
- B $a = -12, b = 2$ (sign error)
- C $a = -12, b = 2$
- D $a = -12, b = 3$

Classified Practice

Polynomial Theorems

Q118 Challenge

For the polynomial above with $a = -12$, $b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

Choose the correct option.

A $x = 1, -2, 2 \pm \sqrt{10}$

B $x = 1, -2, 2\sqrt{10}$

C $x = 1, -2, 2 - \sqrt{10}$

D $x = 1, -2, 2 + \sqrt{10}$

Classified Practice

Polynomial Theorems

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

Choose the correct option.

A $9 - 1$

B 9 (sign error)

C 9

D 10

Quiz MCQ 2 Concept Quiz · MCQ

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

Choose the correct option.

- A $P(-1) = 0 - 1$
- B $P(-1) = 0$ (sign error)
- C $P(-1) = 0$
- D $P(-1) = 1$

Classified Practice

Polynomial Theorems

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

The remainder on division by a quadratic divisor has the form Select the correct answer:

Choose the correct option.

A 1

B $ax + b$

C $ax + b$ (sign error)

D 0

Quiz MCQ 4 Concept Quiz · MCQ

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

Choose the correct option.

A 0

B 1

C *Syntheticdivision*

D *Syntheticdivision* (sign error)

Classified Practice

Polynomial Theorems

Quiz WRITTEN 5 Concept Quiz · Written

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

Quiz WRITTEN 6 Concept Quiz · Written

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

Classified Practice

Polynomial Theorems

Quiz WRITTEN 7 Concept Quiz · Written

Factorise $x^3 - 6x^2 + 11x - 6$.

Quiz WRITTEN 8 Concept Quiz · Written

If $x - 1$ and $x + 2$ are factors of $x^4 - 3x^3 + ax^2 + bx + 12$, find a, b .

Answer key

Use the question number shown on each page.

Q001 2	Q027 A	Q053 $Q = 3x^3 - x^2 + 2x - 6, R = 0$	Q079 C
Q002 6	Q028 A	Q054 D	Q080 D
Q003 C	Q029 A	Q055 $(x - 1)(x + 1)(x + 2)$	Q081 $(x - 2)(x^2 - 2x - 4)$
Q004 C	Q030 A	Q056 $(x - 1)(x - 3)(x + 2)$	Q082 $(x - 3)(x - 2)(x + 1)(x + 7)$
Q005 8	Q031 $3x - 2$	Q057 $(x + 2)^2(x - 5)$	Q083 $(x - 1)^2(x^2 - 2x + 3)$
Q006 $-\frac{1}{8}$	Q032 x	Q058 $(x - 2)(2x + 1)(x + 1)$	Q084 $(x - 4)(x - 3)(x - 2)(x - 1)$
Q007 0	Q033 $3x - 1$	Q059 $(x - 3)(x + 1)(x + 2)$	Q085 $(x - 5)(x - 1)^2(x + 2)$
Q008 4	Q034 $7x - 12$	Q060 $(x - 1)(x + 2)(x + 3)$	Q086 $(x - 4)(x - 2)(x + 2)(2x + 3)$
Q009 D	Q035 B	Q061 $(x + 1)(x + 2)(x + 3)$	Q087 $(x - 1)(x + 2)(x^2 - 5x + 3)$
Q010 A	Q036 B	Q062 $(x - 1)^2(x + 2)$	Q088 $(x - 3)(x + 1)(x^2 - 2x + 2)$
Q011 D	Q037 A	Q063 $(x - 3)(x + 2)^2$	Q089 $(x - 1)(x + 1)(x^2 - x + 2)$
Q012 A	Q038 D	Q064 $(x - 5)(x + 1)^2$	Q090 D
Q013 $a = 13/3$	Q039 $x^2 + 3x - 4$	Q065 $(x - 2)(x + 1)(2x - 1)$	Q091 A
Q014 $a = -4$	Q040 A	Q066 $(x - 1)(x + 2)(3x - 2)$	Q092 D
Q015 $a = -4$	Q041 $Q = x^2 - 2x + 9, R = -7$	Q067 $(x + 1)^2(2x + 3)$	Q093 D
Q016 $a = -16$	Q042 A	Q068 D	Q094 C
Q017 B	Q043 $Q = x^2 + x + 6, R = 0$	Q069 D	Q095 D
Q018 B	Q044 $Q = 3x^2 + 4x + 9, R = 26$	Q070 C	Q096 A
Q019 C	Q045 $Q = x^2 + 2x - 1, R = 4$	Q071 B	Q097 B
Q020 A	Q046 $Q = x^2 + 3x + 2, R = -1$	Q072 B	Q098 C
Q021 $a = -4$	Q047 $Q = 4x + 2, R = 4$	Q073 C	Q099 $(x - 2)(2x^2 + x + 1)$
Q022 A	Q048 B	Q074 B	Q100 $(x - 2)(2x^2 - 3x - 3)$
Q023 $2x + 1$	Q049 D	Q075 D	Q101 $(x + 2)(x^2 + 2x - 2)$
Q024 $-10x + 13$	Q050 D	Q076 D	Q102 $(x - 1)(x^2 - 2x - 2)$
Q025 $2x + 3$	Q051 A	Q077 D	Q103 C
Q026 $-15x + 19$	Q052 B	Q078 A	Q104 B

Q105 B**Q106** D**Q107** 56

$$a = 10 \quad \text{for } 2x + 1$$

Q108 $a = -11$ for $x + 2$ linear factors $\leftrightarrow$ roots**Q109** $x - 1$

Q110 $x = \sqrt{3}, -\sqrt{3}$

Q111 $a = -12, b = 2$

Q112 $x = 1, -2, 2 \pm \sqrt{10}$

Q113 B**Q114** D**Q115** C**Q116** A**Q117** C**Q118** A**Quiz MCQ 1** C**Quiz MCQ 2** C**Quiz MCQ 3** B**Quiz MCQ 4** C**Quiz WRITTEN 5** 4**Quiz WRITTEN 6** $R = 3x$ **Quiz WRITTEN 7** $(x - 1)(x - 2)(x - 3)$ **Quiz WRITTEN 8** $a = -12, b = 2$

Concept 4 | Higher-Degree Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Factor theorem

$$P(a) = 0 \Rightarrow x - a \text{ is a factor}$$

Degree

$$\deg(PQ) = \deg P + \deg Q$$

Conjugate roots

$$a + bi \Rightarrow a - bi$$

Classified Practice

Higher-Degree Equations

Q001 Written

$$x^3 - 8 = 0$$

Q002 Written

$$x^3 = -1$$

Classified Practice

Higher-Degree Equations

Q003 Written

$$x^3 = 27$$

Q004 Written

$$x^3 = -125$$

Classified Practice

Higher-Degree Equations

Q005 Written

$$8x^3 = 1$$

Q006 MCQ

$x^3 - 8 = 0$ **Select the correct answer:**

Choose the correct option.

- A $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{4}i$
 - B $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$
 - C $x = 2, -1 - \sqrt{3}, -1 + \sqrt{3}i$
 - D $x = 2, -1 - \sqrt{3} - i, -1 + \sqrt{3}i$
-
-
-

Classified Practice

Higher-Degree Equations

Q007 MCQ

 $x^3 = -1$ Select the correct answer:

Choose the correct option.

A $x = -1, (1 - \sqrt{3})/2, (1 + \sqrt{3}i)/2$

B $x = -1, (1 - \sqrt{3} - i)/2, (1 + \sqrt{3}i)/2$

C $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/3$

D $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$

Q008 MCQ $x^3 = 27$ **Select the correct answer:****Choose the correct option.**

- A $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/3$
- B $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$
- C $x = 3, (-3 - 3\sqrt{3})/2, (-3 + 3\sqrt{3}i)/2$
- D $x = 3, (-3 - 3\sqrt{3} - i)/2, (-3 + 3\sqrt{3}i)/2$

Classified Practice

Higher-Degree Equations

Q009 MCQ

 $x^3 = -125$ Select the correct answer:

Choose the correct option.

A $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$

B $x = -5, (5 - 5\sqrt{3})/2, (5 + 5\sqrt{3}i)/2$

C $x = -5, (5 - 5\sqrt{3} - i)/2, (5 + 5\sqrt{3}i)/2$

D $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/3$

Q010 MCQ

$8x^3 = 1$ Select the correct answer:

Choose the correct option.

A $x = 1/2, (-1 - \sqrt{3} - i)/4, (-1 + \sqrt{3}i)/4$

B $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/5$

C $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$

D $x = 1/2, (-1 - \sqrt{3})/4, (-1 + \sqrt{3}i)/4$

Classified Practice

Higher-Degree Equations

Q011 Written

$$x^3 - 2x^2 - 5x + 6 = 0$$

Q012 Written

$$3x^3 + x^2 - 8x + 4 = 0$$

Classified Practice

Higher-Degree Equations

Q013 Written

$$2x^3 - 3x^2 - x - 2 = 0$$

Q014 Written

$$x^3 - x^2 - 16x - 20 = 0$$

Classified Practice

Higher-Degree Equations

Q015 Written

$$x^3 + 6x^2 + 11x + 6 = 0$$

Q016 Written

$$x^3 + 3x^2 - 4x - 12 = 0$$

Classified Practice

Higher-Degree Equations

Q017 Written

$$3x^3 - 2x^2 - 3x + 2 = 0$$

Q018 Written

$$x^3 - x^2 - 4 = 0$$

Classified Practice

Higher-Degree Equations

Q019 Written

$$x^3 + 4x^2 + 2x - 4 = 0$$

Q020 Written

$$x^3 - 4x^2 + 9x - 10 = 0$$

Classified Practice

Higher-Degree Equations

Q021 Written

$$x^3 + x^2 - 8x - 12 = 0$$

Q022 Written

$$2x^3 + 7x^2 + 8x + 3 = 0$$

Classified Practice

Higher-Degree Equations

Q023 Written

$$x^3 + 6x^2 + 12x + 8 = 0$$

Q024 MCQ $x^3 - 2x^2 - 5x + 6 = 0$ **Select the correct answer:****Choose the correct option.****A** $x = -2, 1, 3 - 1$ **B** $x = -2, 1, 3$ (sign error)**C** $x = -2, 1, 3$ **D** $x = -2, 1, 4$

Classified Practice

Higher-Degree Equations

Q025 MCQ

 $3x^3 + x^2 - 8x + 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2, 2/3, 2$ B $x = -2, 2/3, 1 - 1$ C $x = -2, 2/3, 1$ (sign error)D $x = -2, 2/3, 1$

Q026 MCQ

$2x^3 - 3x^2 - x - 2 = 0$ Select the correct answer:

Choose the correct option.

- A $x = 2, (-1 - \sqrt{7})/4, (-1 + \sqrt{7}i)/4$
 - B $x = 2, (-1 - \sqrt{7} - i)/4, (-1 + \sqrt{7}i)/4$
 - C $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/5$
 - D $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$
-
-
-

Classified Practice

Higher-Degree Equations

Q027 MCQ

 $x^3 - x^2 - 16x - 20 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2$ (repeated), 5 (sign error)B $x = -2$ (repeated), 5C $x = -2$ (repeated), 6D $x = -2$ (repeated), 5 – 1

Q028 MCQ $x^3 + 6x^2 + 11x + 6 = 0$ **Select the correct answer:****Choose the correct option.**

- A $x = -3, -2, -1 - 1$
 - B $x = -3, -2, -1$ (sign error)
 - C $x = -3, -2, -1$
 - D $x = -3, -2, 0$
-
-
-

Classified Practice

Higher-Degree Equations

Q029 MCQ

 $x^3 + 3x^2 - 4x - 12 = 0$ Select the correct answer:

Choose the correct option.

A $x = -3, -2, 3$ B $x = -3, -2, 2 - 1$ C $x = -3, -2, 2$ (sign error)D $x = -3, -2, 2$

Q030 MCQ $3x^3 - 2x^2 - 3x + 2 = 0$ **Select the correct answer:****Choose the correct option.****A** $x = -1, 2/3, 2$ **B** $x = -1, 2/3, 1 - 1$ **C** $x = -1, 2/3, 1$ (sign error)**D** $x = -1, 2/3, 1$

Classified Practice

Higher-Degree Equations

Q031 MCQ

 $x^3 - x^2 - 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$

B $x = 2, (-1 - \sqrt{7})/2, (-1 + \sqrt{7}i)/2$

C $x = 2, (-1 - \sqrt{7} - i)/2, (-1 + \sqrt{7}i)/2$

D $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/3$

Classified Practice

Higher-Degree Equations

Q032 MCQ

 $x^3 + 4x^2 + 2x - 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2, -1 - \sqrt{3}, -1 + \sqrt{4}$

B $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3} - 1$

C $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$ (sign error)

D $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$

Classified Practice

Higher-Degree Equations

Q033 MCQ

 $x^3 - 4x^2 + 9x - 10 = 0$ Select the correct answer:

Choose the correct option.

A $x = 2, 1 - 2, 1 + 2i$

B $x = 2, 1 - 2 - i, 1 + 2i$

C $x = 2, 1 - 2i, 1 + 3i$

D $x = 2, 1 - 2i, 1 + 2i$

Q034 MCQ

$x^3 + x^2 - 8x - 12 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2$ (repeated), 4

B $x = -2$ (repeated), 3 – 1

C $x = -2$ (repeated), 3 (sign error)

D $x = -2$ (repeated), 3

Classified Practice

Higher-Degree Equations

Q035 MCQ

 $2x^3 + 7x^2 + 8x + 3 = 0$ Select the correct answer:

Choose the correct option.

A $x = -1$ (repeated), $-3/2$ (sign error)B $x = -1$ (repeated), $-3/2$ C $x = -1$ (repeated), $-3/2 + 1$ D $x = -1$ (repeated), $-3/2 - 1$

Q036 MCQ

$x^3 + 6x^2 + 12x + 8 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -1$ (triple)

B $x = -2$ (triple) – 1

C $x = -2$ (triple) (sign error)

D $x = -2$ (triple)

Classified Practice

Higher-Degree Equations

Q037 Written

$$x^4 - 7x^2 - 18 = 0$$

Q038 Written

$$x^4 - 5x^2 + 4 = 0$$

Classified Practice

Higher-Degree Equations

Q039 Written

$$x^4 - 5x^2 - 36 = 0$$

Q040 Written

$$x^4 - 6x^2 - 27 = 0$$

Classified Practice

Higher-Degree Equations

Q041 Written

$$x^4 - x^2 - 2 = 0$$

Q042 MCQ $x^4 - 7x^2 - 18 = 0$ **Select the correct answer:****Choose the correct option.**

- A $x = -3, 3, -\sqrt{2}, \sqrt{2}i$
 - B $x = -3, 3, -\sqrt{2} - i, \sqrt{2}i$
 - C $x = -3, 3, -\sqrt{2}i, \sqrt{3}i$
 - D $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$
-
-
-

Classified Practice

Higher-Degree Equations

Q043 MCQ

 $x^4 - 5x^2 + 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2, -1, 1, 3$ B $x = -2, -1, 1, 2 - 1$ C $x = -2, -1, 1, 2$ (sign error)D $x = -2, -1, 1, 2$

Q044 MCQ $x^4 - 5x^2 - 36 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = -3, 3, -2 - i, 2i$

B $x = -3, 3, -2i, 3i$

C $x = -3, 3, -2i, 2i$

D $x = -3, 3, -2, 2i$

Classified Practice

Higher-Degree Equations

Q045 MCQ

 $x^4 - 6x^2 - 27 = 0$ Select the correct answer:

Choose the correct option.

A $x = -3, 3, -\sqrt{3} - i, \sqrt{3}i$

B $x = -3, 3, -\sqrt{3}i, \sqrt{4}i$

C $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$

D $x = -3, 3, -\sqrt{3}, \sqrt{3}i$

Q046 MCQ $x^4 - x^2 - 2 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = -\sqrt{2}, \sqrt{3}, -i, i$

B $x = -\sqrt{2}, \sqrt{2}, -i, i$

C $x = -\sqrt{2}, \sqrt{2}, -, i$

D $x = -\sqrt{2}, \sqrt{2}, +i, i$

Classified Practice

Higher-Degree Equations

Q047 Written

$$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$$

Q048 Written

$$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$$

Classified Practice

Higher-Degree Equations

Q049 Written

$$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$$

Q050 Written

$$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$$

Classified Practice

Higher-Degree Equations

Q051 MCQ

 $x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ Select the correct answer:

Choose the correct option.

A $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$ (sign error)B $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$ C $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{6}$ D $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5} - 1$

Classified Practice

Higher-Degree Equations

Q052 MCQ

 $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ Select the correct answer:

Choose the correct option.

A $x = 1, 2, 3, 4$ B $x = 1, 2, 3, 5$ C $x = 1, 2, 3, 4 - 1$ D $x = 1, 2, 3, 4$ (sign error)

Q053 MCQ $x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ **Select the correct answer:****Choose the correct option.****A** $x = 1, 2, 3(\text{repeated}) - 1$ **B** $x = 1, 2, 3(\text{repeated})$ (sign error)**C** $x = 1, 2, 3(\text{repeated})$ **D** $x = 1, 2, 4(\text{repeated})$

Classified Practice

Higher-Degree Equations

Q054 MCQ

 $x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ Select the correct answer:

Choose the correct option.

A $x = -1, 3, 1 - i, 1 + i$

B $x = -1, 3, 1 - i, 2 + i$

C $x = -1, 3, 1 - i, 1 + i$

D $x = -1, 3, 1 - i, 1 + i$

Q055 Written

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root.

Classified Practice

Higher-Degree Equations

Q056 Written

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a,b and the other root.

Q057 Written

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a,b and the other root.

Classified Practice

Higher-Degree Equations

Q058 MCQ

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root Select the correct answer:

Choose the correct option.

A $a = 1, b = -6$; other roots: $-2 - 1$

B $a = 1, b = -6$; other roots: -2 (sign error)

C $a = 1, b = -6$; other roots: -2

D $a = 1, b = -6$; other roots: -1

Classified Practice

Higher-Degree Equations

Q059 MCQ

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a,b and the other root Select the correct answer:

Choose the correct option.

A $a = 2, b = 2$; other roots: 1 (sign error)

B $a = 2, b = 2$; other roots: 1

C $a = 2, b = 2$; other roots: 2

D $a = 2, b = 2$; other roots: 1 - 1

Classified Practice

Higher-Degree Equations

Q060 MCQ

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a,b and the other root Select the correct answer:

Choose the correct option.

- A $a = -7, b = 14$; other roots: $4 - 1$
 - B $a = -7, b = 14$; other roots: 4 (sign error)
 - C $a = -7, b = 14$; other roots: 4
 - D $a = -7, b = 14$; other roots: 5
-
-
-

Classified Practice

Higher-Degree Equations

Q061 Written

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots.

Q062 Written

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots.

Classified Practice

Higher-Degree Equations

Q063 Written

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots.

Q064 Written

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots.

Classified Practice

Higher-Degree Equations

Q065 MCQ

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots Select the correct answer:

Choose the correct option.

A $a = -3, b = 1$; other roots: $-1, 2 - i$

B $a = -3, b = 1$; other roots: $-1, 2 -$

C $a = -3, b = 1$; other roots: $-1, 2 + i$

D $a = -3, b = 1$; other roots: $-1, 3 - i$

Q066 MCQ

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots Select the correct answer:

Choose the correct option.

- A $a = -4, b = 6$; other roots: $1 - i, 3$
- B $a = -4, b = 6$; other roots: $1 - i, 2$
- C $a = -4, b = 6$; other roots: $1-, 2$
- D $a = -4, b = 6$; other roots: $1 + i, 2$

Classified Practice

Higher-Degree Equations

Q067 MCQ

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots Select the correct answer:

Choose the correct option.

- A $a = -4, b = 14$; other roots: $1 + 3, 2$
B $a = -4, b = 14$; other roots: $1 + 3 - i, 2$
C $a = -4, b = 14$; other roots: $1 + 3i, 3$
D $a = -4, b = 14$; other roots: $1 + 3i, 2$

Q068 MCQ

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots Select the correct answer:

Choose the correct option.

A $a = 0, b = 1$; other roots: $1 - 2i, -1$

B $a = 0, b = 1$; other roots: $1 - 2i, -2$

C $a = 0, b = 1$; other roots: $1 - 2, -2$

D $a = 0, b = 1$; other roots: $1 - 2 - i, -2$

Classified Practice

Higher-Degree Equations

Q069 Written

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots.

Q070 MCQ

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots Select the correct answer:

Choose the correct option.

A $k = 9$; other roots: 3, 4 (sign error)

B $k = 9$; other roots: 3, 4

C $k = 9$; other roots: 3, 5

D $k = 9$; other roots: 3, 4 - 1

Classified Practice

Higher-Degree Equations

Q071 Written

For an imaginary cube root of unity w , evaluate w^9 .

Q072 Written**For an imaginary cube root of unity w , evaluate w^{2010} .**

Classified Practice

Higher-Degree Equations

Q073 Written

For an imaginary cube root of unity w , evaluate w^3 .

Q074 MCQ

For an imaginary cube root of unity w , evaluate w^9 Select the correct answer:

Choose the correct option.

A 1

B 2

C $1 - 1$

D 1 (sign error)

Classified Practice

Higher-Degree Equations

Q075 MCQ

For an imaginary cube root of unity w , evaluate w^{2010} Select the correct answer:

Choose the correct option.

A 1 (sign error)

B 1

C 2

D $1 - 1$

Q076 MCQ**For an imaginary cube root of unity w , evaluate w^3 Select the correct answer:****Choose the correct option.****A** 2**B** $1 - 1$ **C** 1 (sign error)**D** 1

Classified Practice

Higher-Degree Equations

Q077 **Written**

For an imaginary cube root of unity w , evaluate $w^4 + w^5$.

Q078 Written**For an imaginary cube root of unity w , evaluate $w^2 + w$.**

Classified Practice

Higher-Degree Equations

Q079 Written

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$.

Q080 MCQ**For an imaginary cube root of unity w , evaluate $w^4 + w^5$ Select the correct answer:****Choose the correct option.****A** -1 **B** 0 **C** $-1 - 1$ **D** -1 (sign error)

Classified Practice

Higher-Degree Equations

Q081 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Choose the correct option.

A -1 (sign error)

B -1

C 0

D $-1 - 1$

Q082 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$ Select the correct answer:

Choose the correct option.

A 1

B $0 - 1$

C 0 (sign error)

D 0

Classified Practice

Higher-Degree Equations

Q083 Written

For an imaginary cube root of unity w , evaluate $w + 1/w$.

Q084 Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Classified Practice

Higher-Degree Equations

Q085 Written

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$.

Q086 MCQ**For an imaginary cube root of unity w , evaluate $w + 1/w$ Select the correct answer:****Choose the correct option.****A** 0**B** $-1 - 1$ **C** -1 (sign error)**D** -1

Classified Practice

Higher-Degree Equations

Q087 MCQ

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$ Select the correct answer:

Choose the correct option.

A $1 - 1$

B 1 (sign error)

C 1

D 2

Q088 MCQ**For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$ Select the correct answer:****Choose the correct option.****A** 1**B** $0 - 1$ **C** 0 (sign error)**D** 0

Classified Practice

Higher-Degree Equations

Q089 Written

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root.

Q090 **Written**

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root.

Classified Practice

Higher-Degree Equations

Q091 **Written**

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root.

Q092 **Written**

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root.

Classified Practice

Higher-Degree Equations

Q093 Written

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root.

Q094 MCQ

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = -4$ or $9/4 + 1$

B $a = -4$ or $9/4 - 1$

C $a = -4$ or $9/4$ (sign error)

D $a = -4$ or $9/4$

Classified Practice

Higher-Degree Equations

Q095 MCQ

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = -3$ or 2

B $a = -3$ or $1 - 1$

C $a = -3$ or 1 (sign error)

D $a = -3$ or 1

Q096 MCQ

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = 5$ or 9 (sign error)

B $a = 5$ or 9

C $a = 5$ or 10

D $a = 5$ or $9 - 1$

Classified Practice

Higher-Degree Equations

Q097 MCQ

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = -6, -2$, or 10

B $a = -6, -2$, or 11

C $a = -6, -2$, or $10 - 1$

D $a = -6, -2$, or 10 (sign error)

Q098 MCQ

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = -7, -3,$ or 2

B $a = -7, -3,$ or 3

C $a = -7, -3,$ or $2 - 1$

D $a = -7, -3,$ or 2 (sign error)

Classified Practice

Higher-Degree Equations

Q099 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a,b and the other roots.

Classified Practice

Higher-Degree Equations

Q100 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

Classified Practice

Higher-Degree Equations

Q101 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root.

Classified Practice

Higher-Degree Equations

Q102 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

Classified Practice

Higher-Degree Equations

Q103 **Challenge****Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels.**

Classified Practice

Higher-Degree Equations

Q104 **Challenge****Classify the stable pressure solutions as real or complex.**

Classified Practice

Higher-Degree Equations

Q105 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots Select the correct answer:

Choose the correct option.

A $a = -1, b = 6$; other roots: $1 - \sqrt{2} - i, -2$

B $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -1$

C $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

D $a = -1, b = 6$; other roots: $1 - \sqrt{2}, -2$

Classified Practice

Higher-Degree Equations

Q106 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion. Select the correct answer:

Choose the correct option.

A 5

B $4 - 1$

C 4 (sign error)

D 4

Classified Practice

Higher-Degree Equations

Q107 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = 2/3, 2,$ or 7

B $a = 2/3, 2,$ or $6 - 1$

C $a = 2/3, 2,$ or 6 (sign error)

D $a = 2/3, 2,$ or 6

Classified Practice

Higher-Degree Equations

Q108 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

Choose the correct option.

A $x^4 - x^3 + x^2 + x - 2 = 0$

B $x^4 - x^3 + x^2 + x - 2 = 1$

C $x^4 - x^3 + x^2 + x - 2 = 0 - 1$

D $x^4 - x^3 + x^2 + x - 2 = 0$ (sign error)

Classified Practice

Higher-Degree Equations

Q109 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels Select the correct answer:

Choose the correct option.

A $x = -1, 1, (1 - \sqrt{7})/2, (1 + \sqrt{7}i)/2$

B $x = -1, 1, (1 - \sqrt{7} - i)/2, (1 + \sqrt{7}i)/2$

C $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/3$

D $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$

Classified Practice

Higher-Degree Equations

Q110 Challenge

Classify the stable pressure solutions as real or complex Select the correct answer:

Choose the correct option.

- A Both types occur : $\text{real } x = -1, 1$ and $\text{non-real conjugates } x = (1 \pm \sqrt{7}i)/2$. If pressure is restricted to $x \geq 0$, the real physical level is $x = 2$.
- B Both types occur : $\text{real } x = -1, 1$ and $\text{non-real conjugates } x = (1 \pm \sqrt{7}i)/2$. If pressure is restricted to $x \geq 0$, the real physical level is $x = 1$.
- C Both types occur : $\text{real } x = -1, 1$ and $\text{non-real conjugates } x = (1 \pm \sqrt{7})/2$. If pressure is restricted to $x \geq 0$, the real physical level is $x = 1$.
- D Both types occur : $\text{real } x = -1, 1$ and $\text{non-real conjugates } x = (1 \pm \sqrt{7} - i)/2$. If pressure is restricted to $x \geq 0$, the real physical level is $x = 1$.

Classified Practice

Higher-Degree Equations

Quiz MCQ 1 Concept Quiz · MCQ

Solve $x^3 - 8 = 0$. Select the correct answer:

Choose the correct option.

A $x = 2, -1 - \sqrt{3}, -1 + \sqrt{3}i$

B $x = 2, -1 - \sqrt{3} - i, -1 + \sqrt{3}i$

C $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{4}i$

D $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$

Quiz MCQ 2 Concept Quiz · MCQ

Are a cube root of $1 + 2i$. Which root must also occur? Select the correct answer:

Choose the correct option.

A $1 - 2$

B $1 - 2 - i$

C $1 - 1i$

D $1 - 2i$

Classified Practice

Higher-Degree Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Choose the correct option.

A -1 (sign error)

B -1

C 0

D $-1 - 1$

Quiz MCQ 4 Concept Quiz · MCQ

For the remaining quadratic factor, a repeated root requires Select the correct answer:

Choose the correct option.

A $D = 0$ (sign error)

B $D = 0$

C $D = 1$

D $D = 0 - 1$

Classified Practice

Higher-Degree Equations

Quiz WRITTEN 5 Concept Quiz · Written

Solve $x^4 - 5x^2 + 4 = 0$.

Quiz WRITTEN 6 Concept Quiz · Written

For $x^3 + ax^2 + bx + 10 = 0$ with root $1 + 2i$, find a, b and the other roots.

Classified Practice

Higher-Degree Equations

Quiz WRITTEN 7 Concept Quiz · Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Quiz WRITTEN 8 Concept Quiz · Written

Find all real a for $x^3 - 2x^2 + (a - 3)x + a = 0$ to have a repeated root.

Answer key

Use the question number shown on each page.

Q001 $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$	Q027 B	Q053 C	Q079 0
Q002 $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$	Q028 C	Q054 C	Q080 A
Q003 $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$	Q029 D	Q055 $a = 1, b = -6$; other roots: -2	Q081 B
Q004 $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$	Q030 D	Q056 $a = 2, b = 2$; other roots: 1	Q082 D
Q005 $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$	Q031 A	Q057 $a = -7, b = 14$; other roots: 4	Q083 -1
Q006 B	Q032 D	Q058 C	Q084 1
Q007 D	Q033 D	Q059 B	Q085 0
Q008 B	Q034 D	Q060 C	Q086 D
Q009 A	Q035 B	Q061 $a = -3, b = 1$; other roots: $-1, 2 - i$	Q087 C
Q010 C	Q036 D	Q062 $a = -4, b = 6$; other roots: $1 - i, 2$	Q088 D
Q011 $x = -2, 1, 3$	Q037 $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$	Q063 $a = -4, b = 14$; other roots: $1 + 3i, 2$	Q089 $a = -4$ or $9/4$
Q012 $x = -2, 2/3, 1$	Q038 $x = -2, -1, 1, 2$	Q064 $a = 0, b = 1$; other roots: $1 - 2i, -2$	Q090 $a = -3$ or 1
Q013 $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$	Q039 $x = -3, 3, -2i, 2i$	Q065 A	Q091 $a = 5$ or 9
Q014 $x = -2$ (repeated), 5	Q040 $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$	Q066 B	Q092 $a = -6, -2$, or 10
Q015 $x = -3, -2, -1$	Q041 $x = -\sqrt{2}, \sqrt{2}, -i, i$	Q067 D	Q093 $a = -7, -3$, or 2
Q016 $x = -3, -2, 2$	Q042 D	Q068 B	Q094 D
Q017 $x = -1, 2/3, 1$	Q043 D	Q069 $k = 9$; other roots: $3, 4$	Q095 D
Q018 $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$	Q044 C	Q070 B	Q096 B
Q019 $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$	Q045 C	Q071 1	Q097 A
Q020 $x = 2, 1 - 2i, 1 + 2i$	Q046 B	Q072 1	Q098 A
Q021 $x = -2$ (repeated), 3	Q047 $x = 2$ (repeated), $1 - \sqrt{5}, 1 + \sqrt{5}$	Q073 1	Q099 $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$
Q022 $x = -1$ (repeated), $-3/2$	Q048 $x = 1, 2, 3, 4$	Q074 A	Q100 4
Q023 $x = -2$ (triple)	Q049 $x = 1, 2, 3$ (repeated)	Q075 B	Q101 $a = 2/3, 2$, or 6
Q024 C	Q050 $x = -1, 3, 1 - i, 1 + i$	Q076 D	Q102 $x^4 - x^3 + x^2 + x - 2 = 0$
Q025 D	Q051 B	Q077 -1	Q103 $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$
Q026 D	Q052 A	Q078 -1	

- Both types occur: real $x = -1, 1$
- Q104 non-real conjugates $x = (1 \pm \sqrt{7}i)/2$
if $x \geq 0 : x = 1$

Q105 C

Q106 D
- Q107 D

Q108 A

Q109 D

Q110 B
- Quiz MCQ 1 D

Quiz MCQ 2 D

Quiz MCQ 3 B

Quiz MCQ 4 B
- Quiz WRITTEN 5 $x = -2, -1, 1, 2$

Quiz WRITTEN 6 $a = 0, b = 1$; other roots: $1 - 2i, -2$

Quiz WRITTEN 7 1

Quiz WRITTEN 8 $a = -4$ or $9/4$

Concept 1 | Coordinate Geometry Basics

English Student Edition • Focused formulas and guided practice

Formula opener

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Classified Practice

Coordinate Geometry Basics

Q001 **Written****A (−4), B (8): distance AB**

Q002 Written**A (−5), B (7): distance AB**

Classified Practice

Coordinate Geometry Basics

Q003 **Written****For two points A (-2) and B (6) , find the distance AB.**

Q004 Written

For two points A (4) and B (10), find the distance AB.

Classified Practice

Coordinate Geometry Basics

Q005 MCQ

Find A (−4), B (8): distance AB:**Choose the correct option.**

A 11

B −12

C 12

D 13

Q006 MCQ**Find a (-5) , B (7) : distance AB:****Choose the correct option.**

A 11

B -12

C 12

D 13

Classified Practice

Coordinate Geometry Basics

Q007 MCQ

For two points A (-2) and B (6) , find the distance AB:**Choose the correct option.**

A 7

B -8

C 8

D 9

Q008 MCQ**For two points A (4) and B (10), find the distance AB:****Choose the correct option.**

A 5

B -6

C 6

D 7

Classified Practice

Coordinate Geometry Basics

Q009 **Written****A (−4), B (8): internal point C in ratio 2:1**

Q010 Written**A (-5) , B (7) : internal point C in ratio 2:1**

Classified Practice

Coordinate Geometry Basics

Q011 **Written**

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1.

Q012 Written

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1.

Classified Practice

Coordinate Geometry Basics

Q013 MCQ

Find a (-4) , B (8) : internal point C in ratio 2:1:**Choose the correct option.**A -4 B 4 C 5 D 3

Q014 MCQ**Find a (-5) , B (7) : internal point C in ratio 2:1:****Choose the correct option.**

A 2

B -3

C 3

D 4

Classified Practice

Coordinate Geometry Basics

Q015 MCQ

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1:

Choose the correct option.

A -4

B 4

C 5

D 3

Q016 MCQ

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1:

Choose the correct option.

A -8

B 8

C 9

D 7

Classified Practice

Coordinate Geometry Basics

Q017 **Written****A (−4), B (8): external point D in ratio 2:3**

Q018 Written**A (−5), B (7): external point D in ratio 2:5**

Classified Practice

Coordinate Geometry Basics

Q019 **Written**

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3.

Q020 Written

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3.

Classified Practice

Coordinate Geometry Basics

Q021 MCQ

Find a (-4) , B (8) : external point D in ratio 2:3:**Choose the correct option.**A -28 B -27 C -29 D 28

Q022 MCQ**Find a (-5) , B (7) : external point D in ratio 2:5:****Choose the correct option.**

A 13

B -13

C -12

D -14

Classified Practice

Coordinate Geometry Basics

Q023 MCQ

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3:

Choose the correct option.

A 18

B 19

C 17

D -18

Q024 MCQ

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3:

Choose the correct option.

A -1

B 1

C 2

D 0

Classified Practice

Coordinate Geometry Basics

Q025 **Written****A (−4), B (8): midpoint M**

Q026 Written**A (−5), B (7): midpoint M**

Classified Practice

Coordinate Geometry Basics

Q027 **Written**

For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB.

Q028 Written

For two points A (4) and B (10), find the coordinate of midpoint M of segment AB.

Classified Practice

Coordinate Geometry Basics

Q029 MCQ

Find a (-4) , B (8) : midpoint M:**Choose the correct option.**

A 1

B -2

C 2

D 3

Q030 MCQ**Find a (-5) , B (7) : midpoint M:****Choose the correct option.****A** 1**B** 2**C** 0**D** -1

Classified Practice

Coordinate Geometry Basics

Q031 MCQ

For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB:

Choose the correct option.

A 3

B 1

C -2

D 2

Q032 MCQ**For two points A (4) and B (10), find the coordinate of midpoint M of segment AB:****Choose the correct option.****A** 7**B** 8**C** 6**D** -7

Classified Practice

Coordinate Geometry Basics

Q033 **Challenge****Pods A (-8) and B (-3) : fibre-optic tether length**

Classified Practice

Coordinate Geometry Basics

Q034 **Challenge****Pods A (-8) , B (-3) : internal sensor point in ratio 1:4**

Classified Practice

Coordinate Geometry Basics

Q035 **Challenge****Pods A (-8) , B (-3) : external outpost in ratio 3:4**

Classified Practice

Coordinate Geometry Basics

Q036 **Challenge****Pods A (-8) , B (-3) : midpoint supply hub**

Classified Practice

Coordinate Geometry Basics

Q037 **Challenge****Find pods A (-8) and B (-3) : fibre-optic tether length:****Choose the correct option.**

A 5

B 6

C 4

D -5

Classified Practice

Coordinate Geometry Basics

Q038 **Challenge****Find pods A (-8) , B (-3) : internal sensor point in ratio 1:4:****Choose the correct option.**A -8 B 7 C -7 D -6

Classified Practice

Coordinate Geometry Basics

Q039 **Challenge****Find pods A (-8) , B (-3) : external outpost in ratio 3:4:****Choose the correct option.**

A 23

B -23 C -22 D -24

Classified Practice

Coordinate Geometry Basics

Q040 Challenge**Find pods A (-8) , B (-3) : midpoint supply hub:****Choose the correct option.**A $-13/2$ B $11/2$ C $-11/2$ D $-9/2$

Classified Practice

Coordinate Geometry Basics

Q041 Written

Point a has coordinates (2,3): determine its quadrant

Q042 Written**Point b has coordinates (2,-3): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q043 Written

Point c has coordinates (-2,3): determine its quadrant

Q044 Written**Point d has coordinates $(-2,-3)$: determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q045 Written

Point a has coordinates $(-4,2)$: determine its quadrant

Q046 Written**Point b has coordinates (1,2): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q047 **Written****Point c has coordinates (3,-2): determine its quadrant**

Q048 Written**Point d has coordinates $(-3,-1)$: determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q049 **Written**

Point a has coordinates (-3,1): determine its quadrant

Q050 Written**Point b has coordinates (2,-8): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q051 **Written**

Point c has coordinates (4,4): determine its quadrant

Q052 Written**Point d has coordinates $(-6, -4)$: determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q053 Written

Point e has coordinates (-2,6): determine its quadrant

Q054 Written**Point f has coordinates (3,-3): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q055 Written

Point g has coordinates (-5,-7): determine its quadrant

Q056 Written**Point h has coordinates (1,4): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q057 MCQ

Point a has coordinates (2,3): determine its quadrant:**Choose the correct option.**A *III*B *IV*C *I*D *II*

Q058 MCQ**Point b has coordinates (2,-3): determine its quadrant:****Choose the correct option.****A** *III***B** *IV***C** *I***D** *II*

Classified Practice

Coordinate Geometry Basics

Q059 MCQ

Point c has coordinates (-2,3): determine its quadrant:**Choose the correct option.**A *II*B *I*C *III*D *IV*

Q060 MCQ**Point d has coordinates $(-2,-3)$: determine its quadrant:****Choose the correct option.****A** *II***B** *IV***C** *III***D** *I*

Classified Practice

Coordinate Geometry Basics

Q061 MCQ

Point a has coordinates $(-4,2)$: determine its quadrant:**Choose the correct option.**A *II*B *I*C *III*D *IV*

Q062 MCQ**Point b has coordinates (1,2): determine its quadrant:****Choose the correct option.****A** *III***B** *IV***C** *I***D** *II*

Classified Practice

Coordinate Geometry Basics

Q063 MCQ

Point c has coordinates (3,-2): determine its quadrant:**Choose the correct option.**A *III*B *IV*C *I*D *II*

Q064 MCQ**Point d has coordinates (-3,-1): determine its quadrant:****Choose the correct option.****A** *III***B** *I***C** *II***D** *IV*

Classified Practice

Coordinate Geometry Basics

Q065 MCQ

Point a has coordinates (-3,1): determine its quadrant:**Choose the correct option.**A *IV*B *II*C *I*D *III*

Q066 MCQ**Point b has coordinates (2,-8): determine its quadrant:****Choose the correct option.****A** *IV***B** *I***C** *II***D** *III*

Classified Practice

Coordinate Geometry Basics

Q067 MCQ

Point c has coordinates (4,4): determine its quadrant:**Choose the correct option.**A *I*B *II*C *III*D *IV*

Q068 MCQ**Point d has coordinates $(-6, -4)$: determine its quadrant:****Choose the correct option.****A** *II***B** *IV***C** *III***D** *I*

Classified Practice

Coordinate Geometry Basics

Q069 MCQ

Point e has coordinates (-2,6): determine its quadrant:**Choose the correct option.**A *IV*B *II*C *I*D *III*

Q070 MCQ**Point f has coordinates (3,-3): determine its quadrant:****Choose the correct option.****A** *II***B** *III***C** *IV***D** *I*

Classified Practice

Coordinate Geometry Basics

Q071 MCQ

Point g has coordinates (-5,-7): determine its quadrant:**Choose the correct option.**A *II*B *IV*C *III*D *I*

Q072 MCQ**Point h has coordinates (1,4): determine its quadrant:****Choose the correct option.****A** *II***B** *III***C** *IV***D** *I*

Classified Practice

Coordinate Geometry Basics

Q073 Written

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Q074 Written

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Classified Practice

Coordinate Geometry Basics

Q075 Written

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Q076 Written

For point A (1, -2), find the coordinates of point P symmetric with respect to the x-axis.

Classified Practice

Coordinate Geometry Basics

Q077 Written

For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis.

Q078 MCQ**For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis:****Choose the correct option.**A $(-3, 2)$ B $(3, -2)$ C $(-3, -2)$ D $(-2, -3)$

Classified Practice

Coordinate Geometry Basics

Q079 MCQ

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Choose the correct option.

A $(-2, -1)$

B $(-1, 2)$

C $(1, -2)$

D $(-1, -2)$

Q080 MCQ**For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis:****Choose the correct option.****A $(-4, -2)$** **B $(4, 2)$** **C $(-4, 2)$** **D $(2, -4)$**

Classified Practice

Coordinate Geometry Basics

Q081 MCQ

For point A $(1, -2)$, find the coordinates of point P symmetric with respect to the x-axis:

Choose the correct option.

A $(2, 1)$

B $(1, -2)$

C $(-1, 2)$

D $(1, 2)$

Q082 MCQ**For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis:****Choose the correct option.****A $(-3, 4)$** **B $(3, -4)$** **C $(-3, -4)$** **D $(-4, -3)$**

Classified Practice

Coordinate Geometry Basics

Q083 Written

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Q084 Written

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Classified Practice

Coordinate Geometry Basics

Q085 **Written**

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Q086 Written

For point A (1, -2), find the coordinates of point Q symmetric with respect to the y-axis.

Classified Practice

Coordinate Geometry Basics

Q087 **Written**

For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis.

Q088 MCQ**For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:****Choose the correct option.**A $(3, -2)$ B $(-3, 2)$ C $(3, 2)$ D $(2, 3)$

Classified Practice

Coordinate Geometry Basics

Q089 MCQ

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Choose the correct option.

A $(1, -2)$

B $(-1, 2)$

C $(1, 2)$

D $(2, 1)$

Q090 MCQ**For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:****Choose the correct option.****A $(4, -2)$** **B $(-2, 4)$** **C $(4, 2)$** **D $(-4, -2)$**

Classified Practice

Coordinate Geometry Basics

Q091 MCQ

For point A $(1, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Choose the correct option.

A $(-1, -2)$

B $(-2, -1)$

C $(-1, 2)$

D $(1, -2)$

Q092 MCQ**For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis:****Choose the correct option.****A $(3, 4)$** **B $(4, 3)$** **C $(3, -4)$** **D $(-3, 4)$**

Classified Practice

Coordinate Geometry Basics

Q093 Written

For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin.

Q094 Written

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin.

Classified Practice

Coordinate Geometry Basics

Q095 **Written**

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin.

Q096 Written

For point A (1, -2), find the coordinates of point R symmetric with respect to the origin.

Classified Practice

Coordinate Geometry Basics

Q097 **Written**

For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin.

Q098 MCQ**For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin:****Choose the correct option.**A $(-3, -2)$ B $(3, -2)$ C $(-2, 3)$ D $(3, 2)$

Classified Practice

Coordinate Geometry Basics

Q099 MCQ

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-1, -2)$

B $(1, -2)$

C $(-2, 1)$

D $(1, 2)$

Q100 MCQ

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-4, 2)$

B $(4, 2)$

C $(2, 4)$

D $(4, -2)$

Classified Practice

Coordinate Geometry Basics

Q101 MCQ

For point A $(1, -2)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-1, 2)$

B $(2, -1)$

C $(-1, -2)$

D $(1, 2)$

Q102 MCQ

For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-4, 3)$

B $(3, 4)$

C $(-3, -4)$

D $(3, -4)$

Classified Practice

Coordinate Geometry Basics

Q103 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis.

Classified Practice

Coordinate Geometry Basics

Q104 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis.

Classified Practice

Coordinate Geometry Basics

Q105 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin.

Classified Practice

Coordinate Geometry Basics

Q106 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis:

Choose the correct option.

A $(-3, -4)$

B $(-4, -3)$

C $(-3, 4)$

D $(3, -4)$

Classified Practice

Coordinate Geometry Basics

Q107 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis:

Choose the correct option.

A $(-3, 4)$

B $(3, 4)$

C $(4, 3)$

D $(3, -4)$

Classified Practice

Coordinate Geometry Basics

Q108 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(3, 4)$

B $(-3, -4)$

C $(3, -4)$

D $(-4, 3)$

Classified Practice

Coordinate Geometry Basics

Q109 **Written**

Distance between A (2, 6) and B (5, 10)

Q110 Written**Distance between A (4, 3) and B (1, 2)**

Classified Practice

Coordinate Geometry Basics

Q111 Written

Distance between $A(\sqrt{3}, -1)$ and $B(0, 1)$

Q112 Written**Distance between A (1, 2) and B (11, 7)**

Classified Practice

Coordinate Geometry Basics

Q113 Written

Distance between A (3, -1) and B (-1, 3)

Q114 MCQ**Find distance between A (2, 6) and B (5, 10):****Choose the correct option.****A** 4**B** -5**C** 5**D** 6

Classified Practice

Coordinate Geometry Basics

Q115 MCQ

Find distance between A (4, 3) and B (1, 2):**Choose the correct option.**

A $\sqrt{10}$

B $2\sqrt{10}$

C (10)

D 1

Q116 MCQ

Find distance between A($\sqrt{3}$, -1) and B (0, 1):**Choose the correct option.**

A (7)

B 1

C $\sqrt{7}$ D $2\sqrt{7}$

Classified Practice

Coordinate Geometry Basics

Q117 MCQ

Find distance between A (1, 2) and B (11, 7):**Choose the correct option.**

A 1

B $5\sqrt{5}$ C $52\sqrt{5}$ D $5(5)$

Q118 MCQ**Find distance between A (3, -1) and B (-1, 3):****Choose the correct option.**

A $4\sqrt{2}$

B $42\sqrt{2}$

C $4(2)$

D 1

Classified Practice

Coordinate Geometry Basics

Q119 **Written****A (3, -2), B (x, 1), distance 5: find x**

Q120 Written

A $(-1, 3)$, **B** $(x, 1)$, distance $\sqrt{5}$: find **x**

Classified Practice

Coordinate Geometry Basics

Q121 Written

A (3, -1), B (x, 2), distance 5: find x

Q122 Written

A (1, -3), **B** (-2, y), **distance** $\sqrt{13}$: **find** y

Classified Practice

Coordinate Geometry Basics

Q123 **Written****A $(-4, -2)$, B $(x, 3)$, distance $5\sqrt{5}$: find x**

Q124 MCQ**A (3, -2), B (x, 1), distance 5: find x:****Choose the correct option.****A** $x = -1, 7$ (one value only)**B** $x = -1, 7$ **C** $x = +1, 7$ **D** $x = -17$

Classified Practice

Coordinate Geometry Basics

Q125 MCQ

A $(-1, 3)$, B $(x, 1)$, distance $\sqrt{5}$: find x:

Choose the correct option.

A $x = -20$

B $x = -2, 0$ (one value only)

C $x = -2, 0$

D $x = +2, 0$

Q126 MCQ**A** $(3, -1)$, **B** $(x, 2)$, **distance 5: find x:****Choose the correct option.****A** $x = -1, 7$ (one value only)**B** $x = -1, 7$ **C** $x = +1, 7$ **D** $x = -17$

Classified Practice

Coordinate Geometry Basics

Q127 MCQ

A (1, -3), B (-2, y), distance $\sqrt{13}$: find y:

Choose the correct option.

A $y = -5, -1$

B $y = +5, -1$

C $y = -5 - 1$

D $y = -5, -1$ (one value only)

Q128 MCQ**A** $(-4, -2)$, **B** $(x, 3)$, **distance** $5\sqrt{5}$: **find x:****Choose the correct option.****A** $x = +14, 6$ **B** $x = -146$ **C** $x = -14, 6$ *(one value only)***D** $x = -14, 6$

Classified Practice

Coordinate Geometry Basics

Q129 **Written****Distance from origin to A (2, -4)**

Q130 Written**Distance from origin to A ($-3, 4$)**

Classified Practice

Coordinate Geometry Basics

Q131 MCQ

Find distance from origin to A (2, -4):**Choose the correct option.**A $2(5)$

B 1

C $2\sqrt{5}$ D $22\sqrt{5}$

Q132 MCQ**Find distance from origin to A $(-3, 4)$:****Choose the correct option.**

A 4

B -5

C 5

D 6

Classified Practice

Coordinate Geometry Basics

Q133 **Challenge****Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x**

Classified Practice

Coordinate Geometry Basics

Q134 Challenge

Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x :

Choose the correct option.

A $x = -23$

B $x = -2, 3$ (one value only)

C $x = -2, 3$

D $x = +2, 3$

Classified Practice

Coordinate Geometry Basics

Q135 **Written****Point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$**

Q136 Written**Point on the y-axis equidistant from A (4, 2) and B (1, -1)**

Classified Practice

Coordinate Geometry Basics

Q137 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(4, 3)$**

Q138 Written**Point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$**

Classified Practice

Coordinate Geometry Basics

Q139 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$**

Q140 MCQ**Find point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$:****Choose the correct option.**A $(-2, 0)$ B $(-3, 0)$ C $(2, 0)$ D $(0, 2)$

Classified Practice

Coordinate Geometry Basics

Q141 MCQ

Find point on the y-axis equidistant from A (4, 2) and B (1, -1):**Choose the correct option.**

A (0, 3)

B (3, 0)

C (0, -3)

D (-1, 3)

Q142 MCQ**Find point on the x-axis equidistant from A $(-1, 2)$ and B $(4, 3)$:****Choose the correct option.**A $(-2, 0)$ B $(-3, 0)$ C $(2, 0)$ D $(0, 2)$

Classified Practice

Coordinate Geometry Basics

Q143 MCQ

Find point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$:**Choose the correct option.**A $(0, -5/6)$ B $(-1, 5/6)$ C $(0, 5/6)$ D $(5/6, 0)$

Q144 MCQ**Find point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$:****Choose the correct option.**A $(-19/2, 0)$ B $(17/2, 0)$ C $(0, 17/2)$ D $(-17/2, 0)$

Classified Practice

Coordinate Geometry Basics

Q145 **Written****Point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1)**

Q146 Written**Point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3)**

Classified Practice

Coordinate Geometry Basics

Q147 **Written****Point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2)**

Q148 Written**Point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$**

Classified Practice

Coordinate Geometry Basics

Q149 **Written****Point on $y = -x + 1$ equidistant from A (1, 2) and B (-3, 6)**

Q150 MCQ**Find point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1):****Choose the correct option.****A** $(-13/8, 9/4)$ **B** $(13/8, 9/4)$ **C** $(9/4, 13/8)$ **D** $(13/8, -9/4)$

Classified Practice

Coordinate Geometry Basics

Q151 MCQ

Find point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3):**Choose the correct option.**

A (-4, 11)

B (4, -11)

C (-4, -11)

D (-11, -4)

Q152 MCQ**Find point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2):****Choose the correct option.**

A (2, 1)

B (1, 2)

C (2, -1)

D (-2, 1)

Classified Practice

Coordinate Geometry Basics

Q153 MCQ

Find point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$:**Choose the correct option.**A $(3, -2)$ B $(-3, 2)$ C $(3, 2)$ D $(2, 3)$

Q154 MCQ**Find point on $y = -x + 1$ equidistant from A (1, 2) and B (−3, 6):****Choose the correct option.**

A (2, 3)

B (−2, 3)

C (3, −2)

D (−2, −3)

Classified Practice

Coordinate Geometry Basics

Q155 **Challenge**

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD

Classified Practice

Coordinate Geometry Basics

Q156 Challenge

A (2, 1), B (4, 5), C (3, y) with $CA=CB$ and D (x, 0) with $DA=DB$: find CD:

Choose the correct option.

A $3\sqrt{5}$

B $32\sqrt{5}$

C 3(5)

D 1

Classified Practice

Coordinate Geometry Basics

Q157 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Classified Practice

Coordinate Geometry Basics

Q158 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1.

Classified Practice

Coordinate Geometry Basics

Q159 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1.

Classified Practice

Coordinate Geometry Basics

Q160 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2.

Classified Practice

Coordinate Geometry Basics

Q161 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Classified Practice

Coordinate Geometry Basics

Q162 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Choose the correct option.

A $(4/3, -11/3)$

B $(-4/3, 11/3)$

C $(4/3, 11/3)$

D $(11/3, 4/3)$

Classified Practice

Coordinate Geometry Basics

Q163 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1:

Choose the correct option.

A $(-15/4, -13/4)$

B $(15/4, 13/4)$

C $(-15/4, 13/4)$

D $(13/4, -15/4)$

Classified Practice

Coordinate Geometry Basics

Q164 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1:

Choose the correct option.

A (2, 9/5)

B (-2, -9/5)

C (2, -9/5)

D (-9/5, 2)

Classified Practice

Coordinate Geometry Basics

Q165 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2:

Choose the correct option.

A $(-3, 3)$

B $(-3, -3)$

C $(3, 3)$

D $(3, -3)$

Classified Practice

Coordinate Geometry Basics

Q166 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Choose the correct option.

A $(-2/3, -5/3)$

B $(-5/3, -2/3)$

C $(-2/3, 5/3)$

D $(2/3, -5/3)$

Classified Practice

Coordinate Geometry Basics

Q167 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Classified Practice

Coordinate Geometry Basics

Q168 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Classified Practice

Coordinate Geometry Basics

Q169 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Classified Practice

Coordinate Geometry Basics

Q170 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2.

Classified Practice

Coordinate Geometry Basics

Q171 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Classified Practice

Coordinate Geometry Basics

Q172 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Choose the correct option.

A $(-2, 19/2)$

B $(2, 19/2)$

C $(19/2, 2)$

D $(2, -19/2)$

Classified Practice

Coordinate Geometry Basics

Q173 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Choose the correct option.

A $(11, 18)$

B $(-11, -18)$

C $(11, -18)$

D $(-18, 11)$

Classified Practice

Coordinate Geometry Basics

Q174 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Choose the correct option.

A (-10, -15)

B (10, -15)

C (-15, 10)

D (10, 15)

Classified Practice

Coordinate Geometry Basics

Q175 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2:

Choose the correct option.

A $(8, -11)$

B $(-8, -11)$

C $(-11, -8)$

D $(-8, 11)$

Classified Practice

Coordinate Geometry Basics

Q176 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Choose the correct option.

A $(0, 15/2)$

B $(-1, -15/2)$

C $(0, -15/2)$

D $(-15/2, 0)$

Classified Practice

Coordinate Geometry Basics

Q177 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA.

Classified Practice

Coordinate Geometry Basics

Q178 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q179 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB.

Classified Practice

Coordinate Geometry Basics

Q180 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q181 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA.

Classified Practice

Coordinate Geometry Basics

Q182 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q183 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB.

Classified Practice

Coordinate Geometry Basics

Q184 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q185 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC.

Classified Practice

Coordinate Geometry Basics

Q186 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q187 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA:

Choose the correct option.

A $(3/2, -1/2)$

B $(-1/2, 3/2)$

C $(3/2, 1/2)$

D $(-3/2, -1/2)$

Classified Practice

Coordinate Geometry Basics

Q188 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A $(-2, 2)$

B $(-2, -2)$

C $(2, 2)$

D $(2, -2)$

Classified Practice

Coordinate Geometry Basics

Q189 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB:

Choose the correct option.

A $(0, 0)$

B $(-1, 0)$

C $(0, -1)$

D $(1, 0)$

Classified Practice

Coordinate Geometry Basics

Q190 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A $(-2, 1)$

B $(1, -2)$

C $(-2, -1)$

D $(2, 1)$

Classified Practice

Coordinate Geometry Basics

Q191 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA:

Choose the correct option.

A (0, 3)

B (3, 0)

C (0, -3)

D (-1, 3)

Classified Practice

Coordinate Geometry Basics

Q192 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A (1, 2/3)

B (2/3, -1)

C (-2/3, 1)

D (2/3, 1)

Classified Practice

Coordinate Geometry Basics

Q193 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB:

Choose the correct option.

A $(-2, -1)$

B $(2, -1)$

C $(-1, 2)$

D $(2, 1)$

Classified Practice

Coordinate Geometry Basics

Q194 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A $(5/3, 5/3)$

B $(5/3, -5/3)$

C $(-5/3, 5/3)$

D $(-5/3, -5/3)$

Classified Practice

Coordinate Geometry Basics

Q195 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC:

Choose the correct option.

A $(-2, -5/2)$

B $(2, -5/2)$

C $(-5/2, 2)$

D $(2, 5/2)$

Classified Practice

Coordinate Geometry Basics

Q196 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A 1

B 2

C $(0, 0)$ D $(-1, 0)$

Classified Practice

Coordinate Geometry Basics

Q197 **Challenge****A $(-2, 3)$, B $(6, 1)$, C $(0, y)$, $CA=CB$: find the centroid G**

Classified Practice

Coordinate Geometry Basics

Q198 Challenge

A $(-2, 3)$, **B** $(6, 1)$, **C** $(0, y)$, **CA=CB**: find the centroid **G**:

Choose the correct option.

A $(4/3, 2/3)$

B $(-4/3, -2/3)$

C $(4/3, -2/3)$

D $(-2/3, 4/3)$

Classified Practice

Coordinate Geometry Basics

Q199 **Written****Point with A $(-3, 2)$ symmetric about centre $(2, 6)$**

Q200 Written**Point with A (1, -2) symmetric about centre (2, 3)**

Classified Practice

Coordinate Geometry Basics

Q201 **Written****Point with A (3, 2) symmetric about centre (-1, 5)**

Q202 Written**Point with A $(-3, 1)$ symmetric about centre $(2, 0)$**

Classified Practice

Coordinate Geometry Basics

Q203 **Written**

Point with A (7, 4) symmetric about centre (-3, 5)

Q204 Written**Point with A (4, -8) symmetric about centre (-3, 2)**

Classified Practice

Coordinate Geometry Basics

Q205 **Written****Point with A (0, -4) symmetric about centre (-4, 1)**

Q206 Written**Point with A $(-4, 2)$ symmetric about centre $(-1, -3)$**

Classified Practice

Coordinate Geometry Basics

Q207 MCQ

Find point with A $(-3, 2)$ symmetric about centre $(2, 6)$:**Choose the correct option.**A $(7, -10)$ B $(-7, 10)$ C $(7, 10)$ D $(10, 7)$

Q208 MCQ**Find point with A (1, -2) symmetric about centre (2, 3):****Choose the correct option.**

A (-3, 8)

B (3, 8)

C (8, 3)

D (3, -8)

Classified Practice

Coordinate Geometry Basics

Q209 MCQ

Find point with A (3, 2) symmetric about centre (-1, 5):**Choose the correct option.**

A (-5, 8)

B (8, -5)

C (-5, -8)

D (5, 8)

Q210 MCQ**Find point with A $(-3, 1)$ symmetric about centre $(2, 0)$:****Choose the correct option.**A $(7, 1)$ B $(-7, -1)$ C $(7, -1)$ D $(-1, 7)$

Classified Practice

Coordinate Geometry Basics

Q211 MCQ

Find point with A (7, 4) symmetric about centre (-3, 5):**Choose the correct option.**

A (-13, -6)

B (13, 6)

C (-13, 6)

D (6, -13)

Q212 MCQ**Find point with A (4, -8) symmetric about centre (-3, 2):****Choose the correct option.**

A (-10, -12)

B (10, 12)

C (-10, 12)

D (12, -10)

Classified Practice

Coordinate Geometry Basics

Q213 MCQ

Find point with A (0, -4) symmetric about centre (-4, 1):**Choose the correct option.**

A (8, 6)

B (-8, 6)

C (6, -8)

D (-8, -6)

Q214 MCQ**Find point with A $(-4, 2)$ symmetric about centre $(-1, -3)$:****Choose the correct option.**A $(2, -8)$ B $(-8, 2)$ C $(2, 8)$ D $(-2, -8)$

Classified Practice

Coordinate Geometry Basics

Q215 **Challenge****B (m , 5), C (4, n) symmetric about A (1, -1): find $m+n$**

Classified Practice

Coordinate Geometry Basics

Q216 Challenge

B (m , 5), **C** (4, n) symmetric about **A** (1, -1): find $m+n$:

Choose the correct option.

A -10

B 9

C -9

D -8

Classified Practice

Coordinate Geometry Basics

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Point A $(-3, 4)$ lies in which quadrant?:

Choose the correct option.

A *II*

B *I*

C *III*

D *IV*

Quiz MCQ 2 Concept Quiz · MCQ

Find the distance between A (1, 2) and B (4, 6):

Choose the correct option.

A -5

B 5

C 6

D 4

Classified Practice

Coordinate Geometry Basics

Quiz MCQ 3 [Concept Quiz](#) · MCQ

Point P divides the segment from A $(-2, 3)$ to B $(6, 1)$ internally in ratio 1:3. Find P:

Choose the correct option.

A $(0, -5/2)$

B $(-1, 5/2)$

C $(0, 5/2)$

D $(5/2, 0)$

Quiz MCQ 4 Concept Quiz · MCQ

Point B is symmetric to A (2, -1) about P (-3, 4). Find B:

Choose the correct option.

A (-8, -9)

B (8, 9)

C (-8, 9)

D (9, -8)

Classified Practice

Coordinate Geometry Basics

Quiz WRITTEN 5 Concept Quiz · Written

Find the point on $y = 2x + 1$ equidistant from A (0, 2) and B (4, 0).

Quiz WRITTEN 6 Concept Quiz · Written

For points A (-5) and B (7) , find the point dividing AB externally in ratio 2:5.

Classified Practice

Coordinate Geometry Basics

Quiz WRITTEN 7 Concept Quiz · Written

If the distance between A (3, -1) and B (x, 2) is 5, find x.

Quiz WRITTEN 8 Concept Quiz · Written

Find the centroid of A $(-2, 1)$, **B** $(6, -3)$, **C** $(1, 7)$.

Answer key

Use the question number shown on each page.

Q001 12	Q027 2	Q053 <i>II</i>	Q079 D
Q002 12	Q028 7	Q054 <i>IV</i>	Q080 C
Q003 8	Q029 C	Q055 <i>III</i>	Q081 D
Q004 6	Q030 A	Q056 <i>I</i>	Q082 C
Q005 C	Q031 D	Q057 C	Q083 (3, 2)
Q006 C	Q032 A	Q058 B	Q084 (1, 2)
Q007 C	Q033 5	Q059 A	Q085 (4, -2)
Q008 C	Q034 -7	Q060 C	Q086 (-1, -2)
Q009 4	Q035 -23	Q061 A	Q087 (3, 4)
Q010 3	Q036 -11/2	Q062 C	Q088 C
Q011 4	Q037 A	Q063 B	Q089 C
Q012 8	Q038 C	Q064 A	Q090 A
Q013 B	Q039 B	Q065 B	Q091 A
Q014 C	Q040 C	Q066 A	Q092 A
Q015 B	Q041 <i>I</i>	Q067 A	Q093 (3, -2)
Q016 B	Q042 <i>IV</i>	Q068 C	Q094 (1, -2)
Q017 -28	Q043 <i>II</i>	Q069 B	Q095 (4, 2)
Q018 -13	Q044 <i>III</i>	Q070 C	Q096 (-1, 2)
Q019 18	Q045 <i>II</i>	Q071 C	Q097 (3, -4)
Q020 1	Q046 <i>I</i>	Q072 D	Q098 B
Q021 A	Q047 <i>IV</i>	Q073 (-3, -2)	Q099 B
Q022 B	Q048 <i>III</i>	Q074 (-1, -2)	Q100 B
Q023 A	Q049 <i>II</i>	Q075 (-4, 2)	Q101 A
Q024 B	Q050 <i>IV</i>	Q076 (1, 2)	Q102 D
Q025 2	Q051 <i>I</i>	Q077 (-3, -4)	Q103 (-3, -4)
Q026 1	Q052 <i>III</i>	Q078 C	Q104 (3, 4)

Q105 (3, -4)

Q106 A

Q107 B

Q108 C

Q109 5

Q110 $\sqrt{10}$ Q111 $\sqrt{7}$ Q112 $5\sqrt{5}$ Q113 $4\sqrt{2}$

Q114 C

Q115 A

Q116 C

Q117 B

Q118 A

Q119 $x = -1, 7$ Q120 $x = -2, 0$ Q121 $x = -1, 7$ Q122 $y = -5, -1$ Q123 $x = -14, 6$

Q124 B

Q125 C

Q126 B

Q127 A

Q128 D

Q129 $2\sqrt{5}$

Q130 5

Q131 C

Q132 C

Q133 $x = -2, 3$

Q134 C

Q135 (2, 0)

Q136 (0, 3)

Q137 (2, 0)

Q138 (0, 5/6)

Q139 (17/2, 0)

Q140 C

Q141 A

Q142 C

Q143 C

Q144 B

Q145 (13/8, 9/4)

Q146 (-4, -11)

Q147 (2, 1)

Q148 (3, 2)

Q149 (-2, 3)

Q150 B

Q151 C

Q152 A

Q153 C

Q154 B

Q155 $3\sqrt{5}$

Q156 A

Q157 (4/3, 11/3)

Q158 (-15/4, 13/4)

Q159 (2, -9/5)

Q160 (3, 3)

Q161 (-2/3, -5/3)

Q162 C

Q163 C

Q164 C

Q165 C

Q166 A

Q167 (2, 19/2)

Q168 (11, -18)

Q169 (10, -15)

Q170 (-8, -11)

Q171 (0, -15/2)

Q172 B

Q173 C

Q174 B

Q175 B

Q176 C

Q177 (3/2, -1/2)

Q178 (2, 2)

Q179 (-1, 0)

Q180 (-2, 1)

Q181 (0, 3)

Q182 (2/3, 1)

Q183 (2, -1)

Q184 (5/3, 5/3)

Q185 (2, -5/2)

Q186 (0, 0)

Q187 A

Q188 C

Q189 B

Q190 A

Q191 A

Q192 D

Q193 B

Q194 A

Q195 B

Q196 C

Q197 (4/3, -2/3)

Q198 C

Q199 (7, 10)

Q200 (3, 8)

Q201 (-5, 8)

Q202 (7, -1)

Q203 (-13, 6)

Q204 (-10, 12)

Q205 (-8, 6)

Q206 (2, -8)

Q207 C

Q208 B

Q209 A

Q210 C

Q211 C

Q212 C

Q213 B

Q214 A

Q215 -9

Q216 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 C

Quiz WRITTEN 5 (1, 3)

Quiz WRITTEN 6 -13

Quiz WRITTEN 7 $x = -1, 7$

Quiz WRITTEN 8 (5/3, 5/3)

Concept 2 | Triangles and Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Line

$$y - y_1 = m(x - x_1)$$

Perpendicular

$$m_1 m_2 = -1$$

Area

$$A = \frac{1}{2}bh$$

Classified Practice

Triangles and Lines

Q001 **Written**

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$.

Classified Practice

Triangles and Lines

Q002 **Written**

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$.

Classified Practice

Triangles and Lines

Q003 MCQ

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$:

Choose the correct option.

A *isosceles (not right – angled)*

B *scalene (not right – angled)*

C *scalene right – angled*

D *equilateral*

Classified Practice

Triangles and Lines

Q004 MCQ

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$:

Choose the correct option.

A *scalene (not right – angled)*

B *scalene right – angled*

C *equilateral*

D *isosceles (not right – angled)*

Q005 Written**Determine the shape of triangle ABC with A $(-1, 2)$, B $(1, 1)$, C $(0, 4)$.**

Classified Practice

Triangles and Lines

Q006 Written

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1).

Q007 MCQ**Determine the shape of triangle ABC with A $(-1, 2)$, B $(1, 1)$, C $(0, 4)$:****Choose the correct option.****A** *scalene right – angled***B** *isosceles right – angled; right angle at A***C** *equilateral***D** *isosceles (not right – angled)*

Classified Practice

Triangles and Lines

Q008 MCQ

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1):**Choose the correct option.**A *equilateral*B *isosceles (not right – angled)*C *scalene right – angled*D *isosceles right – angled; right angle at A*

Q009 Written**Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4).**

Classified Practice

Triangles and Lines

Q010 MCQ

Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4):**Choose the correct option.**A *equilateral*B *isosceles (not right – angled)*C *scalene (not right – angled)*D *scalene right – angled; right angle at A*

Q011 Written**Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2).**

Classified Practice

Triangles and Lines

Q012 **Written****Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4).**

Q013 Written

Determine the shape of triangle ABC with A (−1, 1), B (4, 2), C (1, 4).

Classified Practice

Triangles and Lines

Q014 MCQ

Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2):**Choose the correct option.**A *scalene right – angled*B *isosceles right – angled; right angle at B*C *equilateral*D *isosceles (not right – angled)*

Q015 MCQ**Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4):****Choose the correct option.****A** *isosceles (not right – angled)***B** *scalene right – angled***C** *isosceles right – angled; right angle at B***D** *equilateral*

Classified Practice

Triangles and Lines

Q016 MCQ

Determine the shape of triangle ABC with A $(-1, 1)$, B $(4, 2)$, C $(1, 4)$:

Choose the correct option.

A *scalene right – angled*

B *isosceles right – angled; right angle at C*

C *equilateral*

D *isosceles (not right – angled)*

Classified Practice

Triangles and Lines

Q017 Challenge

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k.

Classified Practice

Triangles and Lines

Q018 Challenge

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k:

Choose the correct option.

A $k = 1$

B $k = -2$

C $k = 2$

D $k = 3$

Classified Practice

Triangles and Lines

Q019 Written

Find the equation of the line through (3,-4) parallel to the x-axis.

Q020 Written**Find the equation of the line through $(-3,8)$ parallel to the y -axis.**

Classified Practice

Triangles and Lines

Q021 Written

Find the equation of the line through (2,6) parallel to the x-axis.

Q022 Written**Find the equation of the line through (3,5) perpendicular to the x-axis.**

Classified Practice

Triangles and Lines

Q023 Written

Find the equation of the line through $(-4, 5)$ parallel to x-axis.

Q024 Written**Find the equation of the line through (2,1) parallel to x-axis.**

Classified Practice

Triangles and Lines

Q025 **Written****Find the equation of the line through $(-4, -7)$ parallel to x-axis.**

Q026 Written**Find the equation of the line through (1,-2) perpendicular to x-axis.**

Classified Practice

Triangles and Lines

Q027 Written

Find the equation of the line through $(-3, -4)$ perpendicular to x-axis.

Q028 Written**Find the equation of the line through (3,-5) perpendicular to x-axis.**

Classified Practice

Triangles and Lines

Q029 Written

Find the equation of the line through (2,-7) parallel to y-axis.

Q030 Written**Find the equation of the line through (4,3) parallel to y-axis.**

Classified Practice

Triangles and Lines

Q031 Written

Find the equation of the line through $(-6, -2)$ parallel to y-axis.

Q032 MCQ**Find the equation of the line through (3,-4) parallel to the x-axis:****Choose the correct option.**

A $y = 4$

B $y = -4$

C $y = -3$

D $y = -5$

Classified Practice

Triangles and Lines

Q033 MCQ

Find the equation of the line through $(-3,8)$ parallel to the y-axis:**Choose the correct option.**

A $x = -4$

B $x = 3$

C $x = -3$

D $x = -2$

Q034 MCQ**Find the equation of the line through (2,6) parallel to the x-axis:****Choose the correct option.**

A $y = 7$

B $y = 5$

C $y = -6$

D $y = 6$

Classified Practice

Triangles and Lines

Q035 MCQ

Find the equation of the line through (3,5) perpendicular to the x-axis:**Choose the correct option.**

A $x = 4$

B $x = 2$

C $x = -3$

D $x = 3$

Q036 MCQ**Find the equation of the line through $(-4,5)$ parallel to x-axis:****Choose the correct option.**

A $y = -5$

B $y = 5$

C $y = 6$

D $y = 4$

Classified Practice

Triangles and Lines

Q037 MCQ

Find the equation of the line through (2,1) parallel to x-axis:**Choose the correct option.**

A $y = 2$

B $y = 0$

C $y = -1$

D $y = 1$

Q038 MCQ**Find the equation of the line through $(-4, -7)$ parallel to x-axis:****Choose the correct option.**

A $y = 7$

B $y = -7$

C $y = -6$

D $y = -8$

Classified Practice

Triangles and Lines

Q039 MCQ

Find the equation of the line through (1,-2) perpendicular to x-axis:**Choose the correct option.**

A $x = 0$

B $x = -1$

C $x = 1$

D $x = 2$

Q040 MCQ**Find the equation of the line through (-3,-4) perpendicular to x-axis:****Choose the correct option.**

A $x = -2$

B $x = -4$

C $x = 3$

D $x = -3$

Classified Practice

Triangles and Lines

Q041 MCQ

Find the equation of the line through (3,-5) perpendicular to x-axis:**Choose the correct option.**

A $x = 4$

B $x = 2$

C $x = -3$

D $x = 3$

Q042 MCQ**Find the equation of the line through (2,-7) parallel to y-axis:****Choose the correct option.**

A $x = -2$

B $x = 2$

C $x = 3$

D $x = 1$

Classified Practice

Triangles and Lines

Q043 MCQ

Find the equation of the line through (4,3) parallel to y-axis:**Choose the correct option.**

A $x = 5$

B $x = 3$

C $x = -4$

D $x = 4$

Q044 MCQ**Find the equation of the line through $(-6, -2)$ parallel to y -axis:****Choose the correct option.**

A $x = -6$

B $x = -5$

C $x = -7$

D $x = 6$

Classified Practice

Triangles and Lines

Q045 **Written****Find the equation of the line through $(-3,1)$ and $(-3,2)$.**

Q046 Written**Find the equation of the line through $(-2,3)$ and $(-5,3)$.**

Classified Practice

Triangles and Lines

Q047 **Written****Find the equation of the line through (1,-1) and (4,0).**

Q048 Written**Find the equation of the line through $(-2,-2)$ and $(2,-2)$.**

Classified Practice

Triangles and Lines

Q049 **Written****Find the equation of the line through $(-4,2)$ and $(-4,8)$.**

Q050 Written**Find the equation of the line through $(-2,-6)$ and $(-4,1)$.**

Classified Practice

Triangles and Lines

Q051 Written

Find the equation of the line through (4,0) and (4,3).

Q052 Written**Find the equation of the line through $(-1,2)$ and $(-1,6)$.**

Classified Practice

Triangles and Lines

Q053 **Written****Find the equation of the line through $(-5,-3)$ and $(-5,2)$.**

Q054 Written**Find the equation of the line through $(-2,3)$ and $(-3,3)$.**

Classified Practice

Triangles and Lines

Q055 **Written****Find the equation of the line through (4,-2) and (1,-2).**

Q056 Written**Find the equation of the line through $(-3,6)$ and $(2,6)$.**

Classified Practice

Triangles and Lines

Q057 **Written****Find the equation of the line through (1,2) and (3,10).**

Q058 Written**Find the equation of the line through $(-1,-8)$ and $(4,2)$.**

Classified Practice

Triangles and Lines

Q059 MCQ

Find the equation of the line through (-3,1) and (-3,2):**Choose the correct option.**

A $x = 3$

B $x = -3$

C $x = -2$

D $x = -4$

Q060 MCQ**Find the equation of the line through $(-2,3)$ and $(-5,3)$:****Choose the correct option.**

A $y = 2$

B $y = -3$

C $y = 3$

D $y = 4$

Classified Practice

Triangles and Lines

Q061 MCQ

Find the equation of the line through (1,-1) and (4,0):**Choose the correct option.**

A $y = -x/3 - 4/3$

B $y = 4/3 - x/3$

C $y = x/3 - 4/3 + 1$

D $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$)

Q062 MCQ**Find the equation of the line through $(-2,-2)$ and $(2,-2)$:****Choose the correct option.**

A $y = -2$

B $y = -1$

C $y = -3$

D $y = 2$

Classified Practice

Triangles and Lines

Q063 MCQ

Find the equation of the line through (-4,2) and (-4,8):**Choose the correct option.**

A $x = 4$

B $x = -4$

C $x = -3$

D $x = -5$

Q064 MCQ**Find the equation of the line through (-2,-6) and (-4,1):****Choose the correct option.**

A $7x + 2y + 26 + 1 = 0$

B $7x - 2y + 26 = 0$

C $2(7x + 2y + 26) + 1 = 0$

D $7x + 2y + 26 = 0$

Classified Practice

Triangles and Lines

Q065 MCQ

Find the equation of the line through (4,0) and (4,3):**Choose the correct option.**

A $x = 3$

B $x = -4$

C $x = 4$

D $x = 5$

Q066 MCQ**Find the equation of the line through $(-1,2)$ and $(-1,6)$:****Choose the correct option.**

A $x = -1$

B $x = 0$

C $x = -2$

D $x = 1$

Classified Practice

Triangles and Lines

Q067 MCQ

Find the equation of the line through $(-5,-3)$ and $(-5,2)$:**Choose the correct option.**

A $x = -6$

B $x = 5$

C $x = -5$

D $x = -4$

Q068 MCQ**Find the equation of the line through $(-2,3)$ and $(-3,3)$:****Choose the correct option.**

A $y = 4$

B $y = 2$

C $y = -3$

D $y = 3$

Classified Practice

Triangles and Lines

Q069 MCQ

Find the equation of the line through (4,-2) and (1,-2):**Choose the correct option.**

A $y = -1$

B $y = -3$

C $y = 2$

D $y = -2$

Q070 MCQ**Find the equation of the line through (-3,6) and (2,6):****Choose the correct option.**

A $y = 5$

B $y = -6$

C $y = 6$

D $y = 7$

Classified Practice

Triangles and Lines

Q071 MCQ

Find the equation of the line through (1,2) and (3,10):**Choose the correct option.**

A $y = -4x - 2$

B $y = 4x - 2 + 1$

C 0

D $y = 4x - 2$

Q072 MCQ**Find the equation of the line through (-1,-8) and (4,2):****Choose the correct option.**

A $y = -2x - 6$

B $y = 2x - 6 + 1$

C 0

D $y = 2x - 6$

Classified Practice

Triangles and Lines

Q073 **Written****Find the equation of the line through (2,-1) with gradient 3.**

Q074 Written**Find the equation of the line through $(-3,4)$ with gradient 2.**

Classified Practice

Triangles and Lines

Q075 **Written****Find the equation of the line through $(-1, 5)$ with gradient -1 .**

Q076 Written**Find the equation of the line through $(-3, -8)$ with gradient 1.**

Classified Practice

Triangles and Lines

Q077 **Written****Find the equation of the line through (2,-1) with gradient -2.**

Q078 MCQ**Find the equation of the line through (2,-1) with gradient 3:****Choose the correct option.**

A $y = -3x - 7$

B $y = 3x - 7 + 1$

C 0

D $y = 3x - 7$

Classified Practice

Triangles and Lines

Q079 MCQ

Find the equation of the line through (-3,4) with gradient 2:**Choose the correct option.**

A $y = 2x + 10 + 1$

B 0

C $y = 2x + 10$

D $y = -2x + 10$

Q080 MCQ**Find the equation of the line through $(-1,5)$ with gradient -1 :****Choose the correct option.**

A $y = x + 4$

B $y = x - 4$

C $y = -x + 4 + 1$

D $y = -x + 4$

Classified Practice

Triangles and Lines

Q081 MCQ

Find the equation of the line through (-3,-8) with gradient 1:**Choose the correct option.**

A $y = x - 5 + 1$

B $y = x - 5$

C $y = -x - 5$

D $y = 5 - x$

Q082 MCQ**Find the equation of the line through (2,-1) with gradient -2:****Choose the correct option.**

A $y = 2x + 3$

B $y = -2x + 3 + 1$

C 0

D $y = -2x + 3$

Classified Practice

Triangles and Lines

Q083 **Challenge**

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B.

Classified Practice

Triangles and Lines

Q084 Challenge

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B:

Choose the correct option.

A $y = 3$

B $y = 1$

C $y = -2$

D $y = 2$

Classified Practice

Triangles and Lines

Q085 Written

A (2, 3), **B** (4, -1), and **C** ($a - 1$, $2 - a$) are collinear. Find **a**.

Q086 Written

A (2, 5), B (4, 9), and C (−1, a) are collinear. Find a.

Classified Practice

Triangles and Lines

Q087 **Written**

A (1, -1), **B** (-2, -3), and **C** (4, a) are collinear. Find a .

Q088 **Written**

A (1, 2), B (3, 1), and C (a, -1) are collinear. Find a.

Classified Practice

Triangles and Lines

Q089 Written

A $(-3, -9)$, **B** $(5, 7)$, and **C** $(a + 3, 3a - 2)$ are collinear. Find **a**.

Q090 MCQ**A** $(2, 3)$, **B** $(4, -1)$, and **C** $(a - 1, 2 - a)$ are collinear. Find a :**Choose the correct option.**

A $a = 8$

B $a = 6$

C $a = -7$

D $a = 7$

Classified Practice

Triangles and Lines

Q091 MCQ

A (2, 5), **B** (4, 9), and **C** (−1, a) are collinear. Find a :

Choose the correct option.

A $a = 0$

B $a = -2$

C $a = 1$

D $a = -1$

Q092 MCQ**A (1, -1), B (-2, -3), and C (4, a) are collinear. Find a:****Choose the correct option.****A** $a = 0$ **B** $a = -1$ **C** $a = 1$ **D** $a = 2$

Classified Practice

Triangles and Lines

Q093 MCQ

A (1, 2), **B** (3, 1), and **C** (a , -1) are collinear. Find a :

Choose the correct option.

A $a = 8$

B $a = 6$

C $a = -7$

D $a = 7$

Q094 MCQ**A** $(-3, -9)$, **B** $(5, 7)$, and **C** $(a + 3, 3a - 2)$ are collinear. Find **a**:**Choose the correct option.**

A $a = 6$

B $a = 4$

C $a = -5$

D $a = 5$

Classified Practice

Triangles and Lines

Q095 Written

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1, 3)$.

Q096 Written**Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6).**

Classified Practice

Triangles and Lines

Q097 Written

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$.

Q098 Written

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$.

Classified Practice

Triangles and Lines

Q099 MCQ

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1, 3)$:

Choose the correct option.

A $2(x + y - 2) + 1 = 0$

B $x + y - 2 = 0$

C $x + y - 2 + 1 = 0$

D $x - y - 2 = 0$

Q100 MCQ

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6):

Choose the correct option.

A $x + 3y - 23 = 0$

B $x + 3y - 23 + 1 = 0$

C $x - 3y - 23 = 0$

D $2(x + 3y - 23) + 1 = 0$

Classified Practice

Triangles and Lines

Q101 MCQ

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$:

Choose the correct option.

A $2(x + y - 1) + 1 = 0$

B $x + y - 1 = 0$

C $x + y - 1 + 1 = 0$

D $x - y - 1 = 0$

Q102 MCQ

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$:

Choose the correct option.

A $2x - 5y + 1 + 1 = 0$

B $2x - 5y - 1 = 0$

C $2(2x - 5y + 1) + 1 = 0$

D $2x - 5y + 1 = 0$

Classified Practice

Triangles and Lines

Q103 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory.

Classified Practice

Triangles and Lines

Q104 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory:

Choose the correct option.

A $3x - 13y = 0$

B $2(3x + 13y) + 1 = 0$

C $3x + 13y = 0$

D $3x + 13y + 1 = 0$

Classified Practice

Triangles and Lines

Q105 Written

Find the gradient of line $y = 2x + 3$.

Q106 Written**Find the gradient of line $y = -3x + 2$.**

Classified Practice

Triangles and Lines

Q107 Written

Find the gradient of line $x - 3y + 1 = 0$.

Q108 Written**Find the gradient of line $2x - y - 1 = 0$.**

Classified Practice

Triangles and Lines

Q109 **Written****Find the gradient of line $x + 3y + 3 = 0$.**

Q110 Written**Find the gradient of line $y = 3x + 1$.**

Classified Practice

Triangles and Lines

Q111 Written

Find the gradient of line $y = 4x - 1$.

Q112 Written**Find the gradient of line $y = -4x - 7$.**

Classified Practice

Triangles and Lines

Q113 Written

Find the gradient of line $6x - 2y = 5$.

Q114 Written**Find the gradient of line $8x + 2y - 3 = 0$.**

Classified Practice

Triangles and Lines

Q115 **Written****Find the gradient of line $x + 4y - 4 = 0$.**

Q116 Written**Find the gradient of line $y = 3x - 1$.**

Classified Practice

Triangles and Lines

Q117 **Written****Find the gradient of line $y = -3x + 4$.**

Q118 Written**Find the gradient of line $y = x - 3$.**

Classified Practice

Triangles and Lines

Q119 Written

Find the gradient of line $6x + 2y + 5 = 0$.

Q120 Written**Find the gradient of line $2x - 2y - 1 = 0$.**

Classified Practice

Triangles and Lines

Q121 Written

Find the gradient of line $x + 3y - 2 = 0$.

Q122 Written**Find the gradient of line $y = 2x - 1$.**

Classified Practice

Triangles and Lines

Q123 Written

Find the gradient of line $y = -x + 4$.

Q124 Written**Find the gradient of line $y = 3x/2 + 6$.**

Classified Practice

Triangles and Lines

Q125 Written

Find the gradient of line $x + y + 3 = 0$.

Q126 Written**Find the gradient of line $2x + 3y + 8 = 0$.**

Classified Practice

Triangles and Lines

Q127 **Written****Find the gradient of line $3x + 6y - 1 = 0$.**

Q128 MCQ

Find the gradient of line $y = 2x + 3$:**Choose the correct option.**

A $m = 3$

B $m = 1$

C $m = -2$

D $m = 2$

Classified Practice

Triangles and Lines

Q129 MCQ

Find the gradient of line $y = -3x + 2$:**Choose the correct option.**A $m = -4$ B $m = 3$ C $m = -3$ D $m = -2$

Q130 MCQ**Find the gradient of line $x - 3y + 1 = 0$:****Choose the correct option.**

A $m = 1/3$

B $m = 4/3$

C $m = -2/3$

D $m = -1/3$

Classified Practice

Triangles and Lines

Q131 MCQ

Find the gradient of line $2x - y - 1 = 0$:**Choose the correct option.**A $m = -2$ B $m = 2$ C $m = 3$ D $m = 1$

Q132 MCQ**Find the gradient of line $x + 3y + 3 = 0$:****Choose the correct option.**

A $m = -1/3$

B $m = 2/3$

C $m = -4/3$

D $m = 1/3$

Classified Practice

Triangles and Lines

Q133 MCQ

Find the gradient of line $y = 3x + 1$:**Choose the correct option.**

A $m = 3$

B $m = 4$

C $m = 2$

D $m = -3$

Q134 MCQ

Find the gradient of line $y = 4x - 1$:**Choose the correct option.**

A $m = -4$

B $m = 4$

C $m = 5$

D $m = 3$

Classified Practice

Triangles and Lines

Q135 MCQ

Find the gradient of line $y = -4x - 7$:**Choose the correct option.**

A $m = -3$

B $m = -5$

C $m = 4$

D $m = -4$

Q136 MCQ**Find the gradient of line $6x - 2y = 5$:****Choose the correct option.**

A $m = 3$

B $m = 4$

C $m = 2$

D $m = -3$

Classified Practice

Triangles and Lines

Q137 MCQ

Find the gradient of line $8x + 2y - 3 = 0$:**Choose the correct option.**

A $m = -5$

B $m = 4$

C $m = -4$

D $m = -3$

Q138 MCQ**Find the gradient of line $x + 4y - 4 = 0$:****Choose the correct option.**

A $m = -5/4$

B $m = 1/4$

C $m = -1/4$

D $m = 3/4$

Classified Practice

Triangles and Lines

Q139 MCQ

Find the gradient of line $y = 3x - 1$:**Choose the correct option.**

A $m = 4$

B $m = 2$

C $m = -3$

D $m = 3$

Q140 MCQ**Find the gradient of line $y = -3x + 4$:****Choose the correct option.**

A $m = -3$

B $m = -2$

C $m = -4$

D $m = 3$

Classified Practice

Triangles and Lines

Q141 MCQ

Find the gradient of line $y = x - 3$:**Choose the correct option.**

A $m = 2$

B $m = 0$

C $m = -1$

D $m = 1$

Q142 MCQ**Find the gradient of line $6x + 2y + 5 = 0$:****Choose the correct option.**

A $m = -4$

B $m = 3$

C $m = -3$

D $m = -2$

Classified Practice

Triangles and Lines

Q143 MCQ

Find the gradient of line $2x - 2y - 1 = 0$:**Choose the correct option.**

A $m = 0$

B $m = -1$

C $m = 1$

D $m = 2$

Q144 MCQ

Find the gradient of line $x + 3y - 2 = 0$:**Choose the correct option.**

A $m = -1/3$

B $m = 2/3$

C $m = -4/3$

D $m = 1/3$

Classified Practice

Triangles and Lines

Q145 MCQ

Find the gradient of line $y = 2x - 1$:**Choose the correct option.**

A $m = 3$

B $m = 1$

C $m = -2$

D $m = 2$

Q146 MCQ**Find the gradient of line $y = -x + 4$:****Choose the correct option.**

A $m = -1$

B $m = 0$

C $m = -2$

D $m = 1$

Classified Practice

Triangles and Lines

Q147 MCQ

Find the gradient of line $y = 3x/2 + 6$:**Choose the correct option.**

A $m = -3/2$

B $m = 3/2$

C $m = 5/2$

D $m = 1/2$

Q148 MCQ**Find the gradient of line $x + y + 3 = 0$:****Choose the correct option.**

A $m = 0$

B $m = -2$

C $m = 1$

D $m = -1$

Classified Practice

Triangles and Lines

Q149 MCQ

Find the gradient of line $2x + 3y + 8 = 0$:**Choose the correct option.**

A $m = 2/3$

B $m = -2/3$

C $m = 1/3$

D $m = -5/3$

Q150 MCQ**Find the gradient of line $3x + 6y - 1 = 0$:****Choose the correct option.****A** 0**B** $m = -1/2$ **C** $m = 1/2$ **D** $m = -3/2$

Classified Practice

Triangles and Lines

Q151 **Written****Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$.**

Q152 Written

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$.

Classified Practice

Triangles and Lines

Q153 **Written****Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$.**

Q154 Written

Find the equation of the line through $(-1,2)$ parallel to $2x + 3y = 0$.

Classified Practice

Triangles and Lines

Q155 Written

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$.

Q156 MCQ

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$:**Choose the correct option.**

- A $3x + 2y - 14 = 0$
- B $2(3x - 2y - 14) + 1 = 0$
- C $3x - 2y - 14 = 0$
- D $3x - 2y - 14 + 1 = 0$

Classified Practice

Triangles and Lines

Q157 MCQ

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$:**Choose the correct option.**

- A $2x + y + 1 + 1 = 0$
B $2x - y + 1 = 0$
C $2(2x + y + 1) + 1 = 0$
D $2x + y + 1 = 0$
-
-
-

Q158 MCQ

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$:**Choose the correct option.**

A $2x + y - 4 = 0$

B $2(2x - y - 4) + 1 = 0$

C $2x - y - 4 = 0$

D $2x - y - 4 + 1 = 0$

Classified Practice

Triangles and Lines

Q159 MCQ

Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$:**Choose the correct option.**

A $2x + 3y - 4 + 1 = 0$

B $2x - 3y - 4 = 0$

C $2(2x + 3y - 4) + 1 = 0$

D $2x + 3y - 4 = 0$

Q160 MCQ

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$:

Choose the correct option.

A $3x + 2y - 7 + 1 = 0$

B $3x - 2y - 7 = 0$

C $2(3x + 2y - 7) + 1 = 0$

D $3x + 2y - 7 = 0$

Classified Practice

Triangles and Lines

Q161 Written

Find the line through $(-2, 5)$ perpendicular to $3x + 5y + 1 = 0$.

Q162 Written

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$.

Classified Practice

Triangles and Lines

Q163 **Written**

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$.

Q164 Written

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$.

Classified Practice

Triangles and Lines

Q165 Written

Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$.

Q166 MCQ**Find the line through $(-2,5)$ perpendicular to $3x + 5y + 1 = 0$:****Choose the correct option.**

A $5x - 3y + 25 + 1 = 0$

B $5x - 3y - 25 = 0$

C $2(5x - 3y + 25) + 1 = 0$

D $5x - 3y + 25 = 0$

Classified Practice

Triangles and Lines

Q167 MCQ

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$:**Choose the correct option.**

A $3x - 2y - 4 + 1 = 0$

B $3x + 2y - 4 = 0$

C $2(3x - 2y - 4) + 1 = 0$

D $3x - 2y - 4 = 0$

Q168 MCQ**Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$:****Choose the correct option.**

A $2(x - 2y - 12) + 1 = 0$

B $x - 2y - 12 = 0$

C $x - 2y - 12 + 1 = 0$

D $x + 2y - 12 = 0$

Classified Practice

Triangles and Lines

Q169 MCQ

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$:

Choose the correct option.

A $2x + 5y - 13 = 0$

B $2x + 5y - 13 + 1 = 0$

C $2x - 5y - 13 = 0$

D $2(2x + 5y - 13) + 1 = 0$

Q170 MCQ

Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$:

Choose the correct option.

A $2x + 3y + 19 + 1 = 0$

B $2x - 3y + 19 = 0$

C $2(2x + 3y + 19) + 1 = 0$

D $2x + 3y + 19 = 0$

Classified Practice

Triangles and Lines

Q171 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k .

Classified Practice

Triangles and Lines

Q172 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k :

Choose the correct option.

- A $m/k = 1$ ($k = 2, m = 2$)
 - B $m/k = 3$ ($k = 2, m = 6$)
 - C $m/k = -2$ ($k = 1, m = -2$)
 - D $m/k = 2$ ($k = 3, m = 6$)
-
-
-

Classified Practice

Triangles and Lines

Q173 **Written**

For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB.

Q174 Written

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB.

Classified Practice

Triangles and Lines

Q175 **Written****Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$.**

Q176 Written**Find the perpendicular bisector of segment A $(-1, 2)$, B $(5, -2)$.**

Classified Practice

Triangles and Lines

Q177 **Written****Find the perpendicular bisector of segment A (0, 6), B (4, 4).**

Q178 MCQ**For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB:****Choose the correct option.**

A $x + 3y - 18 = 0$

B $x + 3y - 18 + 1 = 0$

C $x - 3y - 18 = 0$

D $2(x + 3y - 18) + 1 = 0$

Classified Practice

Triangles and Lines

Q179 MCQ

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB:

Choose the correct option.

A $2x + 3y - 5 = 0$

B $2(2x - 3y - 5) + 1 = 0$

C $2x - 3y - 5 = 0$

D $2x - 3y - 5 + 1 = 0$

Q180 MCQ**Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$:****Choose the correct option.**

A $2x - y - 2 = 0$

B $2(2x - y + 2) + 1 = 0$

C $2x - y + 2 = 0$

D $2x - y + 2 + 1 = 0$

Classified Practice

Triangles and Lines

Q181 MCQ

Find the perpendicular bisector of segment A (−1, 2), B (5, −2):**Choose the correct option.**

A $3x - 2y - 6 + 1 = 0$

B $3x + 2y - 6 = 0$

C $2(3x - 2y - 6) + 1 = 0$

D $3x - 2y - 6 = 0$

Q182 MCQ**Find the perpendicular bisector of segment A (0, 6), B (4, 4):****Choose the correct option.**

A $2x - y + 1 = 0$

B $2x - y + 1 + 1 = 0$

C $2x - y - 1 = 0$

D $2(2x - y + 1) + 1 = 0$

Classified Practice

Triangles and Lines

Q183 **Written**

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$.

Q184 Written**Find the point B symmetric to A (0, 4) with respect to $2x - y - 1 = 0$.**

Classified Practice

Triangles and Lines

Q185 **Written**

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$.

Q186 Written**Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$.**

Classified Practice

Triangles and Lines

Q187 MCQ

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$:**Choose the correct option.**A $B = (2, 1)$ B $B = (1, 2)$ C $B = (-2, 1)$ D $B = (2, -1)$

Q188 MCQ**Find the point B symmetric to A (0, 4) with respect to $2x - y - 1 = 0$:****Choose the correct option.**

A $B = (4, 2)$

B $B = (2, 4)$

C $B = (-4, 2)$

D $B = (4, -2)$

Classified Practice

Triangles and Lines

Q189 MCQ

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$:

Choose the correct option.

A $B = (-3, 6)$

B $B = (3, -6)$

C $B = (3, 6)$

D $B = (6, 3)$

Q190 MCQ**Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$:****Choose the correct option.**

A $B = (2, 2)$

B $B = (-2, -2)$

C $B = (-2, 2)$

D $B = (2, -2)$

Classified Practice

Triangles and Lines

Q191 **Challenge****Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B.**

Classified Practice

Triangles and Lines

Q192 Challenge

Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B:

Choose the correct option.

A $B = (3, 4)$

B $B = (-3, -4)$

C $B = (-3, 4)$

D $B = (4, -3)$

Classified Practice

Triangles and Lines

Q193 Written

Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$.

Q194 Written**Find the distance from (3,-2) to $x + 2y - 4 = 0$.**

Classified Practice

Triangles and Lines

Q195 **Written****Find the perpendicular distance from $(-3,2)$ to $2x - 3y + 6 = 0$.**

Q196 Written**Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$.**

Classified Practice

Triangles and Lines

Q197 **Written****Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$.**

Q198 Written**Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$.**

Classified Practice

Triangles and Lines

Q199 Written

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$.

Q200 Written**Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$.**

Classified Practice

Triangles and Lines

Q201 **Written****Find the perpendicular distance from (1,2) to $y = 3x + 1$.**

Q202 MCQ**Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$:****Choose the correct option.**

A $3/\sqrt{13} = 3\sqrt{13}/13$

B $3 \cdot \sqrt{13}/13 + 1$

C $-1 + 3 \cdot \sqrt{13}/13$

D $-3 \cdot \sqrt{13}/13$

Classified Practice

Triangles and Lines

Q203 MCQ

Find the distance from (3,-2) to $x + 2y - 4 = 0$:**Choose the correct option.**

A $1 + \sqrt{5}$

B $-1 + \sqrt{5}$

C $-\sqrt{5}$

D $\sqrt{5}$

Q204 MCQ

Find the perpendicular distance from $(-3,2)$ to $2x - 3y + 6 = 0$:**Choose the correct option.**

A $1 + 6 \cdot \sqrt{13}/13$

B $-1 + 6 \cdot \sqrt{13}/13$

C $-6 \cdot \sqrt{13}/13$

D $6/\sqrt{13} = 6\sqrt{13}/13$

Classified Practice

Triangles and Lines

Q205 MCQ

Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$:**Choose the correct option.**A -6 B 6 C 7 D 5

Q206 MCQ**Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$:****Choose the correct option.**

A $4\sqrt{5} + 1$

B $4\sqrt{5} - 1$

C 0

D $4\sqrt{5}$

Classified Practice

Triangles and Lines

Q207 MCQ

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$:**Choose the correct option.**

A 2

B 0

C -1

D 1

Q208 MCQ

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$:**Choose the correct option.**

- A $\sqrt{13}$
 - B $1 + \sqrt{13}$
 - C $-1 + \sqrt{13}$
 - D $-\sqrt{13}$
-
-
-

Classified Practice

Triangles and Lines

Q209 MCQ

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$:**Choose the correct option.**

A -10

B 10

C 11

D 9

Q210 MCQ**Find the perpendicular distance from (1,2) to $y = 3x + 1$:****Choose the correct option.**

- A $\sqrt{10}/5 + 1$
 - B $-1 + \sqrt{10}/5$
 - C $-\sqrt{10}/5$
 - D $\sqrt{10}/5$
-
-
-

Classified Practice

Triangles and Lines

Q211 Written

Find the perpendicular length from the origin to $3x - y = 5$.

Q212 Written**Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$.**

Classified Practice

Triangles and Lines

Q213 MCQ

Find the perpendicular length from the origin to $3x - y = 5$:**Choose the correct option.**

- A $1 + \sqrt{10}/2$
- B $-1 + \sqrt{10}/2$
- C $-\sqrt{10}/2$
- D $\sqrt{10}/2$

Q214 MCQ

Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$:**Choose the correct option.**

A $-\sqrt{10}$

B $\sqrt{10}$

C $1 + \sqrt{10}$

D $-1 + \sqrt{10}$

Classified Practice

Triangles and Lines

Q215 **Challenge****Find the shortest distance from P $(-1, 5)$ to the radiation line $y = x/3 + 2$.**

Classified Practice

Triangles and Lines

Q216 Challenge

Find the shortest distance from P $(-1, 5)$ to the radiation line $y = x/3 + 2$:

Choose the correct option.

- A $\sqrt{10}$
- B $1 + \sqrt{10}$
- C $-1 + \sqrt{10}$
- D $-\sqrt{10}$

Classified Practice

Triangles and Lines

Q217 **Written**

Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1).

Q218 Written

Find the area for A (1, 5), B (−3, −3), C (3, −6).

Classified Practice

Triangles and Lines

Q219 **Written****Find the area for A $(-2, 1)$, B $(2, -1)$, C $(0, 4)$.**

Q220 Written**Find the area for A (0, -6), B (2, -2), C (1, 5).**

Classified Practice

Triangles and Lines

Q221 **Written****Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$.**

Q222 Written**Find the area for A (2, 5), B (0, 1), C (4, 2).**

Classified Practice

Triangles and Lines

Q223 MCQ

Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1):**Choose the correct option.**

A 11

B 9

C −10

D 10

Q224 MCQ**Find the area for A (1, 5), B (−3, −3), C (3, −6):****Choose the correct option.****A** 29**B** −30**C** 30**D** 31

Classified Practice

Triangles and Lines

Q225 MCQ

Find the area for A $(-2, 1)$, B $(2, -1)$, C $(0, 4)$:**Choose the correct option.**

A 8

B 9

C 7

D -8

Q226 MCQ**Find the area for A (0, -6), B (2, -2), C (1, 5):****Choose the correct option.****A** -9**B** 9**C** 10**D** 8

Classified Practice

Triangles and Lines

Q227 MCQ

Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$:**Choose the correct option.**A -34 B 34 C 35 D 33

Q228 MCQ

Find the area for A (2, 5), B (0, 1), C (4, 2):**Choose the correct option.**

A 8

B 6

C -7

D 7

Classified Practice

Triangles and Lines

Q229 **Challenge****Find the area of the wound triangle A $(-2, 1)$, B $(1, 5)$, C $(-21, -21)$.**

Classified Practice

Triangles and Lines

Q230 Challenge

Find the area of the wound triangle A $(-2, 1)$, B $(1, 5)$, C $(-21, -21)$:

Choose the correct option.

A 6

B 4

C -5

D 5

Classified Practice

Triangles and Lines

Q231 **Written**

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it.

Q232 Written

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it.

Classified Practice

Triangles and Lines

Q233 Written

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it.

Q234 Written

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it.

Classified Practice

Triangles and Lines

Q235 Written

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it.

Q236 MCQ

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-3/2, 1/2)$

B $(-1/2, -3/2)$

C $(1/2, 3/2)$

D $(1/2, -3/2)$

Classified Practice

Triangles and Lines

Q237 MCQ

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-2, -3)$

B $(-2, 3)$

C $(3, -2)$

D $(2, 3)$

Q238 MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-3, 3)$

B $(-3, -3)$

C $(3, 3)$

D $(3, -3)$

Classified Practice

Triangles and Lines

Q239 MCQ

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-1, 1)$

B $(1, -1)$

C 0

D $(1, 1)$

Q240 MCQ

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-3, 1)$

B $(1, -3)$

C $(3, 1)$

D $(-3, -1)$

Classified Practice

Triangles and Lines

Q241 **Challenge**

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it.

Classified Practice

Triangles and Lines

Q242 Challenge

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it:

Choose the correct option.

A (4, 1)

B (1, 4)

C (-4, 1)

D (4, -1)

Classified Practice

Triangles and Lines

Q243 Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Q244 Written

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a .

Classified Practice

Triangles and Lines

Q245 **Written**

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a .

Q246 Written

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a .

Classified Practice

Triangles and Lines

Q247 MCQ

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a:

Choose the correct option.

A $a = -4, -3, \text{ or } 6$

B $a = -3, -3, \text{ or } 6$

C $a = -4, -2, \text{ or } 6$

D $a = -4, -3, \text{ or } 7$

Q248 MCQ

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a:

Choose the correct option.

A $a = -1, -1/2, \text{ or } 1$

B $a = 0, -1/2, \text{ or } 1$

C $a = -1, 1/2, \text{ or } 1$

D $a = -1, -1/2, \text{ or } 2$

Classified Practice

Triangles and Lines

Q249 MCQ

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a:

Choose the correct option.

A $a = 11/10$, $5/2$, or $-9/2$

B $a = 11/10$, $3/2$, or $-7/2$

C $a = 11/10$, $3/2$, or $-9/2$

D $a = 21/10$, $3/2$, or $-9/2$

Q250 MCQ

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a:

Choose the correct option.

A $a = 2, -3/2, \text{ or } 2$

B $a = 1, -1/2, \text{ or } 2$

C $a = 1, -3/2, \text{ or } 3$

D $a = 1, -3/2, \text{ or } 2$

Classified Practice

Triangles and Lines

Q251 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m .

Classified Practice

Triangles and Lines

Q252 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m :

Choose the correct option.

A 15

B 16

C 14

D -15

Classified Practice

Triangles and Lines

Quiz MCQ 1 **Concept Quiz · MCQ**

Determine the shape of triangle A (0, 0), B (3, 0), C (0, 4):

Choose the correct option.

A *isosceles (not right – angled)*

B *scalene (not right – angled)*

C *scalene right – angled*

D *equilateral*

Quiz MCQ 2 Concept Quiz · MCQ

Find the equation of the line through (2,-1) with gradient 3:

Choose the correct option.

A 0

B $y = 3x - 7$

C $y = -3x - 7$

D $y = 3x - 7 + 1$

Classified Practice

Triangles and Lines

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)**Find the perpendicular distance from (1,2) to $3x + 4y - 10 = 0$:****Choose the correct option.**A $6/5$ B $-4/5$ C $-1/5$ D $1/5$

Quiz MCQ 4 Concept Quiz · MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point. Find it:

Choose the correct option.

A $(-3, -3)$

B $(3, 3)$

C $(3, -3)$

D $(-3, 3)$

Classified Practice

Triangles and Lines

Quiz WRITTEN 5 Concept Quiz · Written

Find the area of triangle A (0, 0), B (4, 0), C (0, 3).

Quiz WRITTEN 6 Concept Quiz · Written

Find the perpendicular bisector of the segment joining A (1, 2) and B (5, 4).

Classified Practice

Triangles and Lines

Quiz WRITTEN 7 Concept Quiz · Written

Find the equation of the line through (1,2) perpendicular to $y = 3x - 1$.

Quiz WRITTEN 8 Concept Quiz · Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Answer key

Use the question number shown on each page.

Q001 <i>equilateral</i>	Q027 $x = -3$	Q053 $x = -5$	Q079 C
Q002 <i>equilateral</i>	Q028 $x = 3$	Q054 $y = 3$	Q080 D
Q003 D	Q029 $x = 2$	Q055 $y = -2$	Q081 B
Q004 C	Q030 $x = 4$	Q056 $y = 6$	Q082 D
Q005 <i>isosceles right – angled; right angle at A</i>	Q031 $x = -6$	Q057 $y = 4x - 2$	Q083 $y = 2$
Q006 <i>isosceles right – angled; right angle at A</i>	Q032 B	Q058 $y = 2x - 6$	Q084 D
Q007 B	Q033 C	Q059 B	Q085 $a = 7$
Q008 D	Q034 D	Q060 C	Q086 $a = -1$
Q009 <i>scalene right – angled; right angle at A</i>	Q035 D	Q061 D	Q087 $a = 1$
Q010 D	Q036 B	Q062 A	Q088 $a = 7$
Q011 <i>isosceles right – angled; right angle at B</i>	Q037 D	Q063 B	Q089 $a = 5$
Q012 <i>isosceles right – angled; right angle at B</i>	Q038 B	Q064 D	Q090 D
Q013 <i>isosceles right – angled; right angle at C</i>	Q039 C	Q065 C	Q091 D
Q014 B	Q040 D	Q066 A	Q092 C
Q015 C	Q041 D	Q067 C	Q093 D
Q016 B	Q042 B	Q068 D	Q094 D
Q017 $k = 2$	Q043 D	Q069 D	Q095 $x + y - 2 = 0$
Q018 C	Q044 A	Q070 C	Q096 $x + 3y - 23 = 0$
Q019 $y = -4$	Q045 $x = -3$	Q071 D	Q097 $x + y - 1 = 0$
Q020 $x = -3$	Q046 $y = 3$	Q072 D	Q098 $2x - 5y + 1 = 0$
Q021 $y = 6$	Q047 $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$)	Q073 $y = 3x - 7$	Q099 B
Q022 $x = 3$	Q048 $y = -2$	Q074 $y = 2x + 10$	Q100 A
Q023 $y = 5$	Q049 $x = -4$	Q075 $y = -x + 4$	Q101 B
Q024 $y = 1$	Q050 $7x + 2y + 26 = 0$	Q076 $y = x - 5$	Q102 D
Q025 $y = -7$	Q051 $x = 4$	Q077 $y = -2x + 3$	Q103 $3x + 13y = 0$
Q026 $x = 1$	Q052 $x = -1$	Q078 D	Q104 C

Q105 $m = 2$	Q137 C	Q169 A	Q201 $\sqrt{10}/5$
Q106 $m = -3$	Q138 C	Q170 D	Q202 A
Q107 $m = 1/3$	Q139 D	Q171 $m/k = 2 (k = 3, m = 6)$	Q203 D
Q108 $m = 2$	Q140 A	Q172 D	Q204 D
Q109 $m = -1/3$	Q141 D	Q173 $x + 3y - 18 = 0$	Q205 B
Q110 $m = 3$	Q142 C	Q174 $2x - 3y - 5 = 0$	Q206 D
Q111 $m = 4$	Q143 C	Q175 $2x - y + 2 = 0$	Q207 D
Q112 $m = -4$	Q144 A	Q176 $3x - 2y - 6 = 0$	Q208 A
Q113 $m = 3$	Q145 D	Q177 $2x - y + 1 = 0$	Q209 B
Q114 $m = -4$	Q146 A	Q178 A	Q210 D
Q115 $m = -1/4$	Q147 B	Q179 C	Q211 $\sqrt{10}/2$
Q116 $m = 3$	Q148 D	Q180 C	Q212 $\sqrt{10}$
Q117 $m = -3$	Q149 B	Q181 D	Q213 D
Q118 $m = 1$	Q150 B	Q182 A	Q214 B
Q119 $m = -3$	Q151 $3x - 2y - 14 = 0$	Q183 $B = (2, 1)$	Q215 $\sqrt{10}$
Q120 $m = 1$	Q152 $2x + y + 1 = 0$	Q184 $B = (4, 2)$	Q216 A
Q121 $m = -1/3$	Q153 $2x - y - 4 = 0$	Q185 $B = (3, 6)$	Q217 10
Q122 $m = 2$	Q154 $2x + 3y - 4 = 0$	Q186 $B = (-2, 2)$	Q218 30
Q123 $m = -1$	Q155 $3x + 2y - 7 = 0$	Q187 A	Q219 8
Q124 $m = 3/2$	Q156 C	Q188 A	Q220 9
Q125 $m = -1$	Q157 D	Q189 C	Q221 34
Q126 $m = -2/3$	Q158 C	Q190 C	Q222 7
Q127 $m = -1/2$	Q159 D	Q191 $B = (-3, 4)$	Q223 D
Q128 D	Q160 D	Q192 C	Q224 C
Q129 C	Q161 $5x - 3y + 25 = 0$	Q193 $3/\sqrt{13} = 3\sqrt{13}/13$	Q225 A
Q130 A	Q162 $3x - 2y - 4 = 0$	Q194 $\sqrt{5}$	Q226 B
Q131 B	Q163 $x - 2y - 12 = 0$	Q195 $6/\sqrt{13} = 6\sqrt{13}/13$	Q227 B
Q132 A	Q164 $2x + 5y - 13 = 0$	Q196 6	Q228 D
Q133 A	Q165 $2x + 3y + 19 = 0$	Q197 $4\sqrt{5}$	Q229 5
Q134 B	Q166 D	Q198 1	Q230 D
Q135 D	Q167 D	Q199 $\sqrt{13}$	Q231 $(1/2, -3/2)$
Q136 A	Q168 B	Q200 10	Q232 $(-2, 3)$

Q233 (3, -3)

Q234 (1, 1)

Q235 (-3, 1)

Q236 D

Q237 B

Q238 D

Q239 D

Q240 A

Q241 (4, 1)

Q242 A

Q243 $a = -4, -3, \text{ or } 6$ Q244 $a = -1, -1/2, \text{ or } 1$ Q245 $a = 11/10, 3/2, \text{ or } -9/2$ Q246 $a = 1, -3/2, \text{ or } 2$

Q247 A

Q248 A

Q249 C

Q250 D

Q251 15

Q252 A

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 D

Quiz MCQ 4 C

Quiz WRITTEN 5 6

Quiz WRITTEN 6 $y = -2x + 9$ Quiz WRITTEN 7 $y = -x/3 + 7/3$ Quiz WRITTEN 8 $a = -4, -3, \text{ or } 6$

Concept 3 | Circles

English Student Edition • Focused formulas and guided practice

Formula opener

Circle

$$(x - a)^2 + (y - b)^2 = r^2$$

General form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Tangent

$$xx_1 + yy_1 = r^2$$

Classified Practice

Circles

Q001 **Written****Find the equation of the circle with centre (3,-2) and radius 5.**

Q002 Written**Find the equation of the circle with centre $(-3,1)$ and radius 2.**

Classified Practice

Circles

Q003 **Written****Find the equation of the circle with centre the origin and radius 4.**

Q004 Written**Find the equation of the circle with centre (3,4), radius 6.**

Classified Practice

Circles

Q005 **Written****Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$.**

Q006 Written**Find the equation of the circle with centre the origin, radius 5.**

Classified Practice

Circles

Q007 MCQ

Find the equation of the circle with centre (3,-2) and radius 5:**Choose the correct option.**

A $(x - 3)^2 + (y + 2)^2 = 25$

B $(x + 3)^2 + (y + 2)^2 = 25$

C $(x - 3)^2 + (y + 2)^2 = 25 + 1$

D 4

Q008 MCQ**Find the equation of the circle with centre (-3,1) and radius 2:****Choose the correct option.**

A $(x + 3)^2 + (y - 1)^2 = 4$

B $(x + 3)^2 + (y + 1)^2 = 4$

C $(x + 3)^2 + (y - 1)^2 = 4 + 1$

D 6

Classified Practice

Circles

Q009 MCQ

Find the equation of the circle with centre the origin and radius 4:**Choose the correct option.**

A $x^2 + y^2 = 16 + 1$

B 6

C 7

D $x^2 + y^2 = 16$

Q010 MCQ**Find the equation of the circle with centre (3,4), radius 6:****Choose the correct option.****A** 9

B $(x - 3)^2 + (y - 4)^2 = 36$

C $(x + 3)^2 + (y - 4)^2 = 36$

D $(x - 3)^2 + (y - 4)^2 = 36 + 1$

Classified Practice

Circles

Q011 MCQ

Find the equation of the circle with centre (-2,6), radius $2\sqrt{3}$:**Choose the correct option.**

A 10

B $(x + 2)^2 + (y - 6)^2 = 12$ C $(x + 2)^2 + (y + 6)^2 = 12$ D $(x + 2)^2 + (y - 6)^2 = 12 + 1$

Q012 MCQ**Find the equation of the circle with centre the origin, radius 5:****Choose the correct option.**

A $x^2 + y^2 = 25$

B $x^2 + y^2 = 25 + 1$

C 10

D 11

Classified Practice

Circles

Q013 Written

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents.

Q014 Written**Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents.**

Classified Practice

Circles

Q015 **Written**

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents.

Q016 Written**Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents.**

Classified Practice

Circles

Q017 **Written**

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents.

Q018 Written**Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents.**

Classified Practice

Circles

Q019 MCQ

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents:**Choose the correct option.**A circle; centre $(3, -1)$, radius 2

B 3

C 4

D 5

Classified Practice

Circles

Q020 MCQ

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents:**Choose the correct option.**

A 7

B 8

C circle; centre (2, 3), radius 3

D 6

Classified Practice

Circles

Q021 MCQ

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents:**Choose the correct option.**

- A 10
- B 11
- C 12
- D circle; centre $(2, -4)$, radius 4

Classified Practice

Circles

Q022 MCQ

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents:**Choose the correct option.**

A 12

B 13

C circle; centre (2, 0), radius 2

D 11

Classified Practice

Circles

Q023 MCQ

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents:**Choose the correct option.**

- A 13
- B 14
- C circle; centre $(-1, 2)$, radius 5
- D circle; centre $(-1, 2)$, radius 10

Classified Practice

Circles

Q024 MCQ

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents:**Choose the correct option.**

A 14

B 15

C circle; centre $(-2, 3)$, radius $\sqrt{13}$

D 13

Classified Practice

Circles

Q025 **Challenge**

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius.

Classified Practice

Circles

Q026 Challenge

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius:

Choose the correct option.

A 0

B centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$

C centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13 + 1$

D centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13 - 1$

Classified Practice

Circles

Q027 **Written****Find the equation of the circle with centre (2,3) passing through (4,6).**

Q028 Written**Find the equation of the circle with centre $(-3,1)$ passing through $(4,-2)$.**

Classified Practice

Circles

Q029 Written

Find the equation of the circle with centre $(1,-3)$ through $(-3,0)$.

Q030 Written**Find the equation of the circle with centre $(-4,1)$ through $(3,-2)$.**

Classified Practice

Circles

Q031 Written

Find the equation of the circle with centre (2,-1) through the origin.

Q032 Written**Find the equation of the circle with centre the origin through (4,-3).**

Classified Practice

Circles

Q033 MCQ

Find the equation of the circle with centre (2,3) passing through (4,6):**Choose the correct option.**

A $(x - 2)^2 + (y - 3)^2 = 13 + 1$

B 4

C $(x - 2)^2 + (y - 3)^2 = 13$

D $(x + 2)^2 + (y - 3)^2 = 13$

Q034 MCQ

Find the equation of the circle with centre $(-3,1)$ passing through $(4,-2)$:**Choose the correct option.**

A $(x + 3)^2 + (y - 1)^2 = 58$

B $(x + 3)^2 + (y + 1)^2 = 58$

C $(x + 3)^2 + (y - 1)^2 = 58 + 1$

D 7

Classified Practice

Circles

Q035 MCQ

Find the equation of the circle with centre (1,-3) through (-3,0):**Choose the correct option.**

A $(x + 1)^2 + (y + 3)^2 = 25$

B $(x - 1)^2 + (y + 3)^2 = 25 + 1$

C 10

D $(x - 1)^2 + (y + 3)^2 = 25$

Q036 MCQ**Find the equation of the circle with centre $(-4,1)$ through $(3,-2)$:****Choose the correct option.**

A $(x + 4)^2 + (y - 1)^2 = 58$

B $(x + 4)^2 + (y + 1)^2 = 58$

C $(x + 4)^2 + (y - 1)^2 = 58 + 1$

D 11

Classified Practice

Circles

Q037 MCQ

Find the equation of the circle with centre (2,-1) through the origin:**Choose the correct option.**

A 12

B $(x - 2)^2 + (y + 1)^2 = 5$ C $(x + 2)^2 + (y + 1)^2 = 5$ D $(x - 2)^2 + (y + 1)^2 = 5 + 1$

Q038 MCQ**Find the equation of the circle with centre the origin through (4,-3):****Choose the correct option.**

A $x^2 + y^2 = 25$

B $x^2 + y^2 = 25 + 1$

C 12

D 13

Classified Practice

Circles

Q039 **Written****Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5).**

Q040 Written**Find the equation of the circle with diameter endpoints $(3,4)$, $(5,-2)$.**

Classified Practice

Circles

Q041 **Written****Find the equation of the circle with diameter endpoints $(4,7)$, $(-2,-1)$.**

Q042 Written**Find the equation of the circle with diameter endpoints $(4,-2)$, $(-6,-2)$.**

Classified Practice

Circles

Q043 **Written****Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$.**

Q044 Written**Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$.**

Classified Practice

Circles

Q045 MCQ

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5):**Choose the correct option.**

A $(x - 2)^2 + (y + 1)^2 = 17$

B $(x + 2)^2 + (y + 1)^2 = 17$

C $(x - 2)^2 + (y + 1)^2 = 17 + 1$

D 5

Q046 MCQ

Find the equation of the circle with diameter endpoints (3,4), (5,-2):**Choose the correct option.**

A $(x - 4)^2 + (y - 1)^2 = 10 + 1$

B 8

C $(x - 4)^2 + (y - 1)^2 = 10$

D $(x + 4)^2 + (y - 1)^2 = 10$

Classified Practice

Circles

Q047 MCQ

Find the equation of the circle with diameter endpoints (4,7), (-2,-1):**Choose the correct option.**

A $(x - 1)^2 + (y - 3)^2 = 25 + 1$

B 14

C $(x - 1)^2 + (y - 3)^2 = 25$

D $(x + 1)^2 + (y - 3)^2 = 25$

Q048 MCQ**Find the equation of the circle with diameter endpoints (4,-2), (-6,-2):****Choose the correct option.**

A $(x + 1)^2 + (y + 2)^2 = 25$

B $(x + 1)^2 + (y + 2)^2 = 25 + 1$

C 14**D** 15

Classified Practice

Circles

Q049 MCQ

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$:**Choose the correct option.**

A 16

B $(x + 2)^2 + (y - 1)^2 = 2$ C $(x + 2)^2 + (y + 1)^2 = 2$ D $(x + 2)^2 + (y - 1)^2 = 2 + 1$

Q050 MCQ**Find the equation of the circle with diameter endpoints (-3,6), (3,-2):****Choose the correct option.**

A $x^2 + (y - 2)^2 = 25 + 1$

B 17

C $x^2 + (y - 2)^2 = 25$

D $x^2 + (y + 2)^2 = 25$

Classified Practice

Circles

Q051 **Written****Find the equation of the circle passing through (1,1), (2,-1), and (3,2).**

Q052 Written**Find the equation of the circle passing through (0,1), (-1,2), and (3,1).**

Classified Practice

Circles

Q053 **Written****Find the equation of the circle with through $(0,0)$, $(-1,1)$, $(2,4)$.**

Q054 Written**Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$.**

Classified Practice

Circles

Q055 **Written****Find the equation of the circle with through $(1,1)$, $(5,-1)$, $(-3,-7)$.**

Q056 MCQ

Find the equation of the circle passing through (1,1), (2,-1), and (3,2):**Choose the correct option.**

A $x^2 + y^2 - 5x - y + 4 = 0$

B $x^2 + y^2 + 5x - y + 4 = 0$

C $x^2 + y^2 - 5x - y + 4 = 0 + 1$

D 6

Classified Practice

Circles

Q057 MCQ

Find the equation of the circle passing through (0,1), (-1,2), and (3,1):**Choose the correct option.**

A 9

B $x^2 + y^2 - 3x - 7y + 6 = 0$

C $x^2 + y^2 + 3x - 7y + 6 = 0$

D $x^2 + y^2 - 3x - 7y + 6 = 0 + 1$

Q058 MCQ**Find the equation of the circle with through (0,0), (-1,1), (2,4):****Choose the correct option.**

A $x^2 + y^2 + 2x - 4y = 0$

B $x^2 + y^2 - 2x - 4y = 0 + 1$

C 18

D $x^2 + y^2 - 2x - 4y = 0$

Classified Practice

Circles

Q059 MCQ

Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$:**Choose the correct option.**

A $x^2 + y^2 + 6y - 16 = 0$

B $x^2 + y^2 - 6y - 16 = 0 + 1$

C 19

D $x^2 + y^2 - 6y - 16 = 0$

Q060 MCQ**Find the equation of the circle with through (1,1), (5,-1), (-3,-7):****Choose the correct option.**

A $x^2 + y^2 + 2x + 8y - 8 = 0$

B $x^2 + y^2 - 2x + 8y - 8 = 0 + 1$

C 20

D $x^2 + y^2 - 2x + 8y - 8 = 0$

Classified Practice

Circles

Q061 **Challenge**

Find the circle through $(0,2)$, $(2,0)$, and $(4,6)$, giving its centre and radius.

Classified Practice

Circles

Q062 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius:

Choose the correct option.

A $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$

B $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10} + 1$

C $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10} - 1$

D 0

Classified Practice

Circles

Q063 **Written****Find the circle with centre $(4, -2)$ tangent to the x-axis.**

Q064 Written**Find the circle with centre $(-2,3)$ tangent to the y -axis.**

Classified Practice

Circles

Q065 **Written****Find the circle with centre $(-1,4)$ tangent to x-axis.**

Q066 Written**Find the circle with centre (3,-4) tangent to x-axis.**

Classified Practice

Circles

Q067 **Written****Find the circle with centre (5,-4) tangent to x-axis.**

Q068 Written**Find the circle with centre $(-1,6)$ tangent to y -axis.**

Classified Practice

Circles

Q069 **Written****Find the circle with centre $(2, -5)$ tangent to y -axis.**

Q070 Written**Find the circle with centre $(-3, -4)$ tangent to y-axis.**

Classified Practice

Circles

Q071 MCQ

Find the circle with centre (4,-2) tangent to the x-axis:**Choose the correct option.**

A radius zero

B $(x - 4)^2 + (y + 2)^2 = 4$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q072 MCQ

Find the circle with centre $(-2,3)$ tangent to the y -axis:**Choose the correct option.**

- A radius zero
- B $(x + 2)^2 + (y - 3)^2 = 4$
- C no real circle
- D centre on the wrong locus

Classified Practice

Circles

Q073 MCQ

Find the circle with centre $(-1,4)$ tangent to x-axis:**Choose the correct option.**

A radius zero

B $(x + 1)^2 + (y - 4)^2 = 16$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q074 MCQ

Find the circle with centre (3,-4) tangent to x-axis:**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 3)^2 + (y + 4)^2 = 16$

Classified Practice

Circles

Q075 MCQ

Find the circle with centre (5,-4) tangent to x-axis:**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 5)^2 + (y + 4)^2 = 16$

Classified Practice

Circles

Q076 MCQ

Find the circle with centre $(-1,6)$ tangent to y-axis:**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x + 1)^2 + (y - 6)^2 = 1$

Classified Practice

Circles

Q077 MCQ

Find the circle with centre (2,-5) tangent to y-axis:**Choose the correct option.**

A radius zero

B $(x - 2)^2 + (y + 5)^2 = 4$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q078 MCQ

Find the circle with centre $(-3,-4)$ tangent to y-axis:**Choose the correct option.**

- A centre on the wrong locus
- B radius zero
- C $(x + 3)^2 + (y + 4)^2 = 9$
- D no real circle

Classified Practice

Circles

Q079 **Written**

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis.

Q080 Written**Find the circles through (1,2) tangent to both axes.**

Classified Practice

Circles

Q081 **Written****Find the circle with through $(-1, -2)$ tangent to both axes.**

Q082 Written**Find the circle with through $(-2,-1)$ tangent to both axes.**

Classified Practice

Circles

Q083 Written

Find the circle with through $(8,-1)$ tangent to both axes.

Classified Practice

Circles

Q084 MCQ

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis:**Choose the correct option.**

A centre on the wrong locus

B radius zero

C $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$

D no real circle

Classified Practice

Circles

Q085 MCQ

Find the circles through (1,2) tangent to both axes:**Choose the correct option.**

A no real circle

B centre on the wrong locus

C radius zero

D $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$

Classified Practice

Circles

Q086 MCQ

Find the circle with through $(-1,-2)$ tangent to both axes:**Choose the correct option.**

A centre on the wrong locus

B radius zero

C $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$

D no real circle

Classified Practice

Circles

Q087 MCQ

Find the circle with through $(-2,-1)$ tangent to both axes:**Choose the correct option.**

A centre on the wrong locus

B radius zero

C $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$

D no real circle

Classified Practice

Circles

Q088 MCQ

Find the circle with through (8,-1) tangent to both axes:**Choose the correct option.**

A radius zero

B $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q089 **Written**

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3).

Q090 Written

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2).

Classified Practice

Circles

Q091 **Written****Find the circle with centre on $y = x + 1$ through (3,4) and (1,4).**

Q092 Written**Find the circle with centre on $y = 2x - 6$ through $(3,7)$ and $(-1,3)$.**

Classified Practice

Circles

Q093 MCQ

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3):**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 1)^2 + (y + 1)^2 = 25$

Classified Practice

Circles

Q094 MCQ

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2):**Choose the correct option.**

- A centre on the wrong locus
- B radius zero
- C $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$
- D no real circle

Classified Practice

Circles

Q095 MCQ

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4):**Choose the correct option.**

- A centre on the wrong locus
- B radius zero
- C $(x - 2)^2 + (y - 3)^2 = 2$
- D no real circle

Classified Practice

Circles

Q096 MCQ

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3):**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 4)^2 + (y - 2)^2 = 26$

Classified Practice

Circles

Q097 **Challenge**

A circle centre is constrained to the x-axis and passes through $P1(-2,2)$, $P2(1,1)$. Find its standard equation, centre and radius.

Classified Practice

Circles

Q098 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius:

Choose the correct option.

A $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5} - 1$

B 0

C $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$

D $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5} + 1$

Classified Practice

Circles

Q099 Written

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle.

Q100 Written

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle.

Classified Practice

Circles

Q101 **Written**

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle.

Q102 Written

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle.

Classified Practice

Circles

Q103 Written

Find the range of k for $x^2 + y^2 + 2kx - 2(k+3)y + 4k^2 + 1 = 0$ to represent a circle.

Classified Practice

Circles

Q104 MCQ

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle:

Choose the correct option.

- A all real parameter values
- B radius squared is negative
- C $k < -1$ or $k > 3$
- D radius squared is zero

Classified Practice

Circles

Q105 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle:

Choose the correct option.

- A radius squared is negative
- B $k < -\sqrt{6}$ or $k > \sqrt{6}$
- C radius squared is zero
- D all real parameter values

Classified Practice

Circles

Q106 MCQ

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle:**Choose the correct option.**

- A all real parameter values
- B radius squared is negative
- C $k < 5/4$
- D radius squared is zero

Classified Practice

Circles

Q107 MCQ

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle:

Choose the correct option.

- A radius squared is negative
- B $k < -1/5$ or $k > 3$
- C radius squared is zero
- D all real parameter values

Classified Practice

Circles

Q108 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 2(k + 3)y + 4k^2 + 1 = 0$ to represent a circle:

Choose the correct option.

- A radius squared is zero
- B all real parameter values
- C radius squared is negative
- D $-1 < k < 4$

Classified Practice

Circles

Q109 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k-range giving a positive radius.

Classified Practice

Circles

Q110 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius:

Choose the correct option.

A 0

B $0 < k < 4$

C $0 < k < 4 + 1$

D $0 < k < 4 - 1$

Classified Practice

Circles

Q111 Written

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$.

Q112 Written

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$.

Classified Practice

Circles

Q113 **Written**

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$.

Q114 Written

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$.

Classified Practice

Circles

Q115 Written

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$.

Classified Practice

Circles

Q116 MCQ

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$:

Choose the correct option.

- A no real intersection
- B two intersection points
- C $(1, -1)$ and $(-1, 3)$
- D tangent

Classified Practice

Circles

Q117 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$:**Choose the correct option.**

- A tangent
- B no real intersection
- C two intersection points
- D (0, 2) and (2, 0)

Classified Practice

Circles

Q118 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$:**Choose the correct option.**

- A no real intersection
- B two intersection points
- C $(3, -1)$
- D tangent

Classified Practice

Circles

Q119 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D $(-2, 1)$

Classified Practice

Circles

Q120 MCQ

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$:

Choose the correct option.

- A two intersection points
- B $(-2, 1)$ and $(4, -1)$
- C tangent
- D no real intersection

Classified Practice

Circles

Q121 Written

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$.

Q122 Written

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$.

Classified Practice

Circles

Q123 Written

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$.

Q124 Written**Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$.**

Classified Practice

Circles

Q125 Written

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$.

Classified Practice

Circles

Q126 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$:**Choose the correct option.**

- A tangent
- B no real intersection
- C two intersection points
- D 0 points

Classified Practice

Circles

Q127 MCQ

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$:**Choose the correct option.**

A 2points : (1, -3) and (3, 1)

B tangent

C no real intersection

D two intersection points

Classified Practice

Circles

Q128 MCQ

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$:**Choose the correct option.**

- A no real intersection
 - B two intersection points
 - C 2 points : $\left(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2\right)$ and $\left(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2\right)$
 - D tangent
-
-
-

Classified Practice

Circles

Q129 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$:**Choose the correct option.**

- A no real intersection
- B two intersection points
- C 0points
- D tangent

Classified Practice

Circles

Q130 MCQ

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D 1 point : $(-2, 0)$

Classified Practice

Circles

Q131 **Written**

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k .

Q132 Written

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r .

Classified Practice

Circles

Q133 Written

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m .

Q134 Written

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range.

Classified Practice

Circles

Q135 Written

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m .

Q136 Written

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n .

Classified Practice

Circles

Q137 **Written**

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m .

Q138 Written

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r .

Classified Practice

Circles

Q139 Written

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range.

Classified Practice

Circles

Q140 MCQ

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k :

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D $-5 < k < 5$

Classified Practice

Circles

Q141 MCQ

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r :

Choose the correct option.

A two intersection points

B $r = \sqrt{5}$

C tangent

D no real intersection

Classified Practice

Circles

Q142 MCQ

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m :

Choose the correct option.

A two intersection points

B $m = \pm\sqrt{3}$

C tangent

D no real intersection

Classified Practice

Circles

Q143 MCQ

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range:

Choose the correct option.

A two intersection points

B $0 < r < \sqrt{10}$

C tangent

D no real intersection

Classified Practice

Circles

Q144 MCQ

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m :

Choose the correct option.

- A no real intersection
- B two intersection points
- C $m = \pm 4$
- D tangent

Classified Practice

Circles

Q145 MCQ

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n :

Choose the correct option.

A $n < -10$ or $n > 10$

B tangent

C no real intersection

D two intersection points

Classified Practice

Circles

Q146 MCQ

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m :

Choose the correct option.

- A no real intersection
- B two intersection points
- C $-5 \leq m \leq 5$
- D tangent

Classified Practice

Circles

Q147 MCQ

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r :

Choose the correct option.

A two intersection points

B $r = \sqrt{2}$

C tangent

D no real intersection

Classified Practice

Circles

Q148 MCQ

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D $r > \sqrt{17}$

Classified Practice

Circles

Q149 **Challenge**

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r .

Classified Practice

Circles

Q150 Challenge

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r :

Choose the correct option.

A $0 < r < 2 + 1$

B $0 < r < 2 - 1$

C 0

D $0 < r < 2$

Classified Practice

Circles

Q151 **Written****Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$.**

Q152 Written**Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$.**

Classified Practice

Circles

Q153 Written

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$.

Q154 Written**Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$.**

Classified Practice

Circles

Q155 Written

Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$.

Q156 MCQ

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$:

Choose the correct option.

A $\sqrt{14}$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

Classified Practice

Circles

Q157 MCQ

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$:

Choose the correct option.

A $\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 - d^2} + 1$

C 4

D $2\sqrt{r^2 + d^2}$

Q158 MCQ

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$:

Choose the correct option.

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D $2\sqrt{5}$

Classified Practice

Circles

Q159 MCQ

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$:

Choose the correct option.

A $2\sqrt{r^2 - d^2} + 1$

B $3\sqrt{2}$

C $2\sqrt{r^2 + d^2}$

D $\sqrt{r^2 - d^2}$

Q160 MCQ**Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$:****Choose the correct option.**

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D 2

Classified Practice

Circles

Q161 **Written**

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2.

Q162 Written

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$.

Classified Practice

Circles

Q163 Written

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$.

Q164 Written

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$.

Classified Practice

Circles

Q165 MCQ

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2:

Choose the correct option.

A $\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 - d^2} + 1$

C $k = \pm 2$

D $2\sqrt{r^2 + d^2}$

Q166 MCQ

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$:

Choose the correct option.

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D $k = \pm\sqrt{31}$

Classified Practice

Circles

Q167 MCQ

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$:

Choose the correct option.

A $k = \pm 2$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

Classified Practice

Circles

Q168 MCQ

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$:

Choose the correct option.

A $2\sqrt{r^2 - d^2} + 1$

B $k = 0$ or $k = -2$

C $2\sqrt{r^2 + d^2}$

D $\sqrt{r^2 - d^2}$

Classified Practice

Circles

Q169 **Challenge**

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4.

Classified Practice

Circles

Q170 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4:

Choose the correct option.

A $k = -1$ or $k = 7$

B $k = -1$ or $k = 7 + 1$

C $k = -1$ or $k = 7 - 1$

D 0

Classified Practice

Circles

Q171 Written

Find the tangent to $x^2 + y^2 = 25$ at (4,3).

Q172 Written**Find the tangent to $x^2 + y^2 = 34$ at (5,3).**

Classified Practice

Circles

Q173 Written

Find the tangents $x^2 + y^2 = 10$ at (3,-1).

Q174 Written**Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$.**

Classified Practice

Circles

Q175 Written

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$.

Classified Practice

Circles

Q176 MCQ

Find the tangent to $x^2 + y^2 = 25$ at (4,3):**Choose the correct option.**

A $4x + 3y = 25$

B $4x - 3y = 25$

C $4x + 3y = 25 + 1$

D the radius line

Classified Practice

Circles

Q177 MCQ

Find the tangent to $x^2 + y^2 = 34$ at (5,3):**Choose the correct option.**

- A $5x + 3y = 34 + 1$
- B the radius line
- C $5x + 3y = 34$
- D $5x - 3y = 34$

Classified Practice

Circles

Q178 MCQ

Find the tangents $x^2 + y^2 = 10$ at (3,-1):**Choose the correct option.**

A $3x - y = 10$

B $3x - y = 10 + 1$

C the radius line

D 8

Classified Practice

Circles

Q179 MCQ

Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$:**Choose the correct option.**

A the radius line

B 9

C $-5x - 2y = 29$ D $-5x - 2y = 29 + 1$

Classified Practice

Circles

Q180 MCQ

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$:**Choose the correct option.**

A $-2x + y = 5$

B $-2x - y = 5$

C $-2x + y = 5 + 1$

D the radius line

Classified Practice

Circles

Q181 **Written****Find the tangents from (1,3) to $x^2 + y^2 = 5$.**

Q182 Written**Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$.**

Classified Practice

Circles

Q183 Written

Find the tangents from (-1,3) to $x^2 + y^2 = 5$.

Q184 Written**Find the tangents from (2,1) to $x^2 + y^2 = 1$.**

Classified Practice

Circles

Q185 **Written****Find the tangents from (2,6) to $x^2 + y^2 = 20$.**

Classified Practice

Circles

Q186 MCQ

Find the tangents from (1,3) to $x^2 + y^2 = 5$:**Choose the correct option.**

- A $2x - y = 5$ and $-x + 2y = 5$
- B $2x + y = 5$ and $-x + 2y = 5 + 1$
- C the radius line
- D $2x + y = 5$ and $-x + 2y = 5$

Classified Practice

Circles

Q187 MCQ

Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$:**Choose the correct option.**

- A $-3x + y = 10$ and $x + 3y = 10$
- B $-3x - y = 10$ and $x + 3y = 10$
- C $-3x + y = 10$ and $x + 3y = 10 + 1$
- D the radius line

Classified Practice

Circles

Q188 MCQ

Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$:**Choose the correct option.**

- A $-2x - y = 5$ and $x + 2y = 5$
- B $-2x + y = 5$ and $x + 2y = 5 + 1$
- C the radius line
- D $-2x + y = 5$ and $x + 2y = 5$

Classified Practice

Circles

Q189 MCQ

Find the tangents from (2,1) to $x^2 + y^2 = 1$:**Choose the correct option.**

A 12

B $y = 1$ and $4x - 3y = 5$ C $y = 1$ and $4x - 3y = 5 + 1$

D the radius line

Classified Practice

Circles

Q190 MCQ

Find the tangents from (2,6) to $x^2 + y^2 = 20$:**Choose the correct option.**

A $-x + 2y = 10$ and $2x + y = 10 + 1$

B the radius line

C $-x + 2y = 10$ and $2x + y = 10$

D $-x - 2y = 10$ and $2x + y = 10$

Classified Practice

Circles

Q191 **Challenge**

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points.

Classified Practice

Circles

Q192 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points:

Choose the correct option.

A 0

B $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

C $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2) + 1$

D $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2) - 1$

Classified Practice

Circles

Q193 **Written****Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$.**

Q194 Written**Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E.**

Classified Practice

Circles

Q195 **Written****Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E.**

Q196 Written**Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E.**

Classified Practice

Circles

Q197 **Written****Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E.**

Q198 Written**Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E.**

Classified Practice

Circles

Q199 Written

Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E.

Classified Practice

Circles

Q200 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$., the required value is:

Choose the correct option.

- A internally tangent
- B intersecting at two points
- C intersect at 2 points
- D separate

Classified Practice

Circles

Q201 MCQ

For the expression Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E., the required value is:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D touch externally

Classified Practice

Circles

Q202 MCQ

For the expression Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E., the required value is:

Choose the correct option.

- A intersecting at two points
- B E : intersect at 2 points
- C separate
- D internally tangent

Classified Practice

Circles

Q203 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E., the required value is:

Choose the correct option.

- A C : touch internally
- B separate
- C internally tangent
- D intersecting at two points

Classified Practice

Circles

Q204 MCQ

For the expression Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E., the required value is:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D touch internally

Classified Practice

Circles

Q205 MCQ

For the expression Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E., the required value is:

Choose the correct option.

- A intersecting at two points
- B E : intersect at 2 points
- C separate
- D internally tangent

Classified Practice

Circles

Q206 MCQ

For the expression Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E., the required value is:

Choose the correct option.

- A internally tangent
- B intersecting at two points
- C D : one is inside the other
- D separate

Classified Practice

Circles

Q207 Written

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally.

Q208 Written**Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally.**

Classified Practice

Circles

Q209 Written

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally.

Classified Practice

Circles

Q210 MCQ

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally:

Choose the correct option.

A $(x - 3)^2 + (y - 1)^2 = 45$

B separate

C internally tangent

D intersecting at two points

Classified Practice

Circles

Q211 MCQ

Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D $(x + 3)^2 + (y + 9)^2 = 16$

Classified Practice

Circles

Q212 MCQ

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally:**Choose the correct option.**

A $(x - 3)^2 + (y + 9)^2 = 144$

B separate

C internally tangent

D intersecting at two points

Classified Practice

Circles

Q213 **Challenge**

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally.

Classified Practice

Circles

Q214 Challenge

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally:

Choose the correct option.

A $(x + 1)^2 + (y + 2)^2 = 9 + 1$

B $(x + 1)^2 + (y + 2)^2 = 9 - 1$

C 0

D $(x + 1)^2 + (y + 2)^2 = 9$

Classified Practice

Circles

Q215 **Written**

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$.

Q216 Written

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$.

Classified Practice

Circles

Q217 Written

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$.

Q218 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$.

Classified Practice

Circles

Q219 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$:

Choose the correct option.

A $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)

B $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4 + 1$)

C the radical axis only

D 4

Classified Practice

Circles

Q220 MCQ

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$:

Choose the correct option.

- A $x + y - 1 = 0$
- B the radical axis only
- C $x + y - 1 = 0$
- D $x - y - 1 = 0$

Classified Practice

Circles

Q221 MCQ

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$:

Choose the correct option.

A 8

B $12x - 6y - 7 = 0$

C $12x - 6y - 7 = 0 + 1$

D the radical axis only

Classified Practice

Circles

Q222 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$:

Choose the correct option.

- A $x = 1/2 + 1$
- B the radical axis only
- C 9
- D $x = 1/2$

Classified Practice

Circles

Q223 Written

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$.

Q224 Written**Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$.**

Classified Practice

Circles

Q225 Written

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$.

Classified Practice

Circles

Q226 MCQ

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$:**Choose the correct option.**

A 5

B $(-3, -1)$ and $(-1, 3)$ C $(-3, -1)$ and $(-1, 3) + 1$

D the radical axis only

Classified Practice

Circles

Q227 MCQ

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$:

Choose the correct option.

A the radical axis only

B 7

C (0, 2) and $(8/5, -6/5)$

D (0, 2) and $(8/5, -6/5) + 1$

Classified Practice

Circles

Q228 MCQ

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$:**Choose the correct option.**

A 10

B (1, 2) and (2, 1)

C (1, 2) and (2, 1) + 1

D the radical axis only

Classified Practice

Circles

Q229 **Challenge**

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points.

Classified Practice

Circles

Q230 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points:

Choose the correct option.

A $(-4, 3)$ and $(3, -4) - 1$

B 0

C $(-4, 3)$ and $(3, -4)$

D $(-4, 3)$ and $(3, -4) + 1$

Classified Practice

Circles

Q231 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0).

Q232 Written

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin.

Classified Practice

Circles

Q233 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0).

Q234 Written

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3, 2)$.

Classified Practice

Circles

Q235 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0).

Classified Practice

Circles

Q236 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0):

Choose the correct option.

A $x^2 - y^2 + 2x + 6y - 15 = 0$

B $x^2 + y^2 + 2x + 6y - 15 = 0 + 1$

C the radical axis only

D $x^2 + y^2 + 2x + 6y - 15 = 0$

Classified Practice

Circles

Q237 MCQ

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin:

Choose the correct option.

A $x^2 + y^2 + 4x + 2y = 0$

B $x^2 - y^2 + 4x + 2y = 0$

C $x^2 + y^2 + 4x + 2y = 0 + 1$

D the radical axis only

Classified Practice

Circles

Q238 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0):

Choose the correct option.

- A $x^2 - y^2 + 10x + 20y - 39 = 0$
- B $x^2 + y^2 + 10x + 20y - 39 = 0 + 1$
- C the radical axis only
- D $x^2 + y^2 + 10x + 20y - 39 = 0$

Classified Practice

Circles

Q239 MCQ

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3,2)$:

Choose the correct option.

A $x^2 + y^2 + 24x + 32y - 5 = 0 + 1$

B the radical axis only

C $x^2 + y^2 + 24x + 32y - 5 = 0$

D $x^2 - y^2 + 24x + 32y - 5 = 0$

Classified Practice

Circles

Q240 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0):

Choose the correct option.

A the radical axis only

B $x^2 + y^2 - 10x - 6y + 24 = 0$

C $x^2 - y^2 - 10x - 6y + 24 = 0$

D $x^2 + y^2 - 10x - 6y + 24 = 0 + 1$

Classified Practice

Circles

Q241 **Challenge**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2).

Classified Practice

Circles

Q242 Challenge

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2):

Choose the correct option.

- A $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$
- B $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0 + 1$
- C $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0 - 1$
- D 0

Classified Practice

Circles

Quiz MCQ 1 Concept Quiz · MCQ

For the circle $(x - 2)^2 + (y + 3)^2 = 25$, determine its centre and radius:

Choose the correct option.

- A centre $(-2, -3)$, radius 5
- B centre $(2, -3)$, radius 5
- C centre $(2, 3)$, radius 5
- D centre $(2, -3)$, radius 10

Classified Practice

Circles

Quiz MCQ 2 **Concept Quiz · MCQ**

Determine the substitution discriminant when a line meets a circle at exactly one real point:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D 0

Classified Practice

Circles

Quiz MCQ 3 **Concept Quiz · MCQ**

Determine the chord length for radius r and centre-to-chord distance d :

Choose the correct option.

A $2\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

Classified Practice

Circles

Quiz MCQ 4 **Concept Quiz · MCQ**

Determine the relative position of two non-overlapping circles when $D = r_1 + r_2$:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D externally tangent

Classified Practice

Circles

Quiz WRITTEN 5 Concept Quiz · Written

Find the circle with centre $(1,-2)$ passing through $(4,2)$.

Quiz WRITTEN 6 Concept Quiz · Written

Find the intersections of $x^2 + y^2 = 25$ and $y = 3$.

Classified Practice

Circles

Quiz WRITTEN 7 Concept Quiz · Written

Find the tangent to $x^2 + y^2 = 25$ at (3,4).

Quiz WRITTEN 8 Concept Quiz · Written

The circles $S_1 = x^2 + y^2 - 5 = 0$ and $S_2 = x^2 + y^2 - 2x - 2y - 4 = 0$ meet. Find the circle $S_1 + \lambda S_2 = 0$ passing through (1,1).

Answer key

Use the question number shown on each page.

Q001 $(x - 3)^2 + (y + 2)^2 = 25$	Q027 $(x - 2)^2 + (y - 3)^2 = 13$	Q053 $x^2 + y^2 - 2x - 4y = 0$	Q079 $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$
Q002 $(x + 3)^2 + (y - 1)^2 = 4$	Q028 $(x + 3)^2 + (y - 1)^2 = 58$	Q054 $x^2 + y^2 - 6y - 16 = 0$	Q080 $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$
Q003 $x^2 + y^2 = 16$	Q029 $(x - 1)^2 + (y + 3)^2 = 25$	Q055 $x^2 + y^2 - 2x + 8y - 8 = 0$	Q081 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q004 $(x - 3)^2 + (y - 4)^2 = 36$	Q030 $(x + 4)^2 + (y - 1)^2 = 58$	Q056 A	Q082 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q005 $(x + 2)^2 + (y - 6)^2 = 12$	Q031 $(x - 2)^2 + (y + 1)^2 = 5$	Q057 B	Q083 $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$
Q006 $x^2 + y^2 = 25$	Q032 $x^2 + y^2 = 25$	Q058 D	Q084 C
Q007 A	Q033 C	Q059 D	Q085 D
Q008 A	Q034 A	Q060 D	Q086 C
Q009 D	Q035 D	Q061 $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$	Q087 C
Q010 B	Q036 A	Q062 A	Q088 B
Q011 B	Q037 B	Q063 $(x - 4)^2 + (y + 2)^2 = 4$	Q089 $(x - 1)^2 + (y + 1)^2 = 25$
Q012 A	Q038 A	Q064 $(x + 2)^2 + (y - 3)^2 = 4$	Q090 $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$
Q013 circle; centre (3, -1), radius 2	Q039 $(x - 2)^2 + (y + 1)^2 = 17$	Q065 $(x + 1)^2 + (y - 4)^2 = 16$	Q091 $(x - 2)^2 + (y - 3)^2 = 2$
Q014 circle; centre (2, 3), radius 3	Q040 $(x - 4)^2 + (y - 1)^2 = 10$	Q066 $(x - 3)^2 + (y + 4)^2 = 16$	Q092 $(x - 4)^2 + (y - 2)^2 = 26$
Q015 circle; centre (2, -4), radius 4	Q041 $(x - 1)^2 + (y - 3)^2 = 25$	Q067 $(x - 5)^2 + (y + 4)^2 = 16$	Q093 D
Q016 circle; centre (2, 0), radius 2	Q042 $(x + 1)^2 + (y + 2)^2 = 25$	Q068 $(x + 1)^2 + (y - 6)^2 = 1$	Q094 C
Q017 circle; centre (-1, 2), radius 5	Q043 $(x + 2)^2 + (y - 1)^2 = 2$	Q069 $(x - 2)^2 + (y + 5)^2 = 4$	Q095 C
Q018 circle; centre (-2, 3), radius $\sqrt{13}$	Q044 $x^2 + (y - 2)^2 = 25$	Q070 $(x + 3)^2 + (y + 4)^2 = 9$	Q096 D
Q019 A	Q045 A	Q071 B	Q097 $(x + 1)^2 + y^2 = 5$; centre (-1, 0), radius $\sqrt{5}$
Q020 C	Q046 C	Q072 B	Q098 C
Q021 D	Q047 C	Q073 B	Q099 $k < -1$ or $k > 3$
Q022 C	Q048 A	Q074 D	Q100 $k < -\sqrt{6}$ or $k > \sqrt{6}$
Q023 C	Q049 B	Q075 D	Q101 $k < 5/4$
Q024 C	Q050 C	Q076 D	Q102 $k < -1/5$ or $k > 3$
Q025 centre (5, 3), radius $^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$	Q051 $x^2 + y^2 - 5x - y + 4 = 0$	Q077 B	Q103 $-1 < k < 4$
Q026 B	Q052 $x^2 + y^2 - 3x - 7y + 6 = 0$	Q078 C	Q104 C

Q105 B	Q137 $-5 \leq m \leq 5$	Q169 $k = -1$ or $k = 7$	Q201 D
Q106 C	Q138 $r = \sqrt{2}$	Q170 A	Q202 B
Q107 B	Q139 $r > \sqrt{17}$	Q171 $4x + 3y = 25$	Q203 A
Q108 D	Q140 D	Q172 $5x + 3y = 34$	Q204 D
Q109 $0 < k < 4$	Q141 B	Q173 $3x - y = 10$	Q205 B
Q110 B	Q142 B	Q174 $-5x - 2y = 29$	Q206 C
Q111 (1, -1) and (-1, 3)	Q143 B	Q175 $-2x + y = 5$	Q207 $(x - 3)^2 + (y - 1)^2 = 45$
Q112 (0, 2) and (2, 0)	Q144 C	Q176 A	Q208 $(x + 3)^2 + (y + 9)^2 = 16$
Q113 (3, -1)	Q145 A	Q177 C	Q209 $(x - 3)^2 + (y + 9)^2 = 144$
Q114 (-2, 1)	Q146 C	Q178 A	Q210 A
Q115 (-2, 1) and (4, -1)	Q147 B	Q179 C	Q211 D
Q116 C	Q148 D	Q180 A	Q212 A
Q117 D	Q149 $0 < r < 2$	Q181 $2x + y = 5$ and $-x + 2y = 5$	Q213 $(x + 1)^2 + (y + 2)^2 = 9$
Q118 C	Q150 D	Q182 $-3x + y = 10$ and $x + 3y = 10$	Q214 D
Q119 D	Q151 $\sqrt{14}$	Q183 $-2x + y = 5$ and $x + 2y = 5$	Q215 $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)
Q120 B	Q152 4	Q184 $y = 1$ and $4x - 3y = 5$	Q216 $x + y - 1 = 0$
Q121 0 points	Q153 $2\sqrt{5}$	Q185 $-x + 2y = 10$ and $2x + y = 10$	Q217 $12x - 6y - 7 = 0$
Q122 2 points : (1, -3) and (3, 1)	Q154 $3\sqrt{2}$	Q186 D	Q218 $x = 1/2$
Q123 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$	Q155 2	Q187 A	Q219 A
Q124 0 points	Q156 A	Q188 D	Q220 C
Q125 1 point : (-2, 0)	Q157 C	Q189 B	Q221 B
Q126 D	Q158 D	Q190 C	Q222 D
Q127 A	Q159 B	Q191 $-x + 2y = 10$ at (-2, 4); $2x + y = 10$ at (4, 0)	Q223 (-3, -1) and (-1, 3)
Q128 C	Q160 D	Q192 B	Q224 (0, 2) and (8/5, -6/5)
Q129 C	Q161 $k = \pm 2$	Q193 intersect at 2 points	Q225 (1, 2) and (2, 1)
Q130 D	Q162 $k = \pm\sqrt{31}$	Q194 B : touch externally	Q226 B
Q131 $-5 < k < 5$	Q163 $k = \pm 2$	Q195 E : intersect at 2 points	Q227 C
Q132 $r = \sqrt{5}$	Q164 $k = 0$ or $k = -2$	Q196 C : touch internally	Q228 B
Q133 $m = \pm\sqrt{3}$	Q165 C	Q197 C : touch internally	Q229 (-4, 3) and (3, -4)
Q134 $0 < r < \sqrt{10}$	Q166 D	Q198 E : intersect at 2 points	Q230 C
Q135 $m = \pm 4$	Q167 A	Q199 D : one is inside the other	Q231 $x^2 + y^2 + 2x + 6y - 15 = 0$
Q136 $n < -10$ or $n > 10$	Q168 B	Q200 C	Q232 $x^2 + y^2 + 4x + 2y = 0$

Q233 $x^2 + y^2 + 10x + 20y - 39 = 0$

Q238 D**Quiz MCQ 1** B**Quiz WRITTEN 6** $(4, 3), (-4, 3)$

Q234 $x^2 + y^2 + 24x + 32y - 5 = 0$

Q239 C**Quiz MCQ 2** D**Quiz WRITTEN 7** $3x + 4y = 25$

Q235 $x^2 + y^2 - 10x - 6y + 24 = 0$

Q240 B**Quiz MCQ 3** A**Quiz WRITTEN 8** $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$ **Q236** D

Q241 $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Quiz MCQ 4 D**Q237** A**Q242** A

Quiz WRITTEN 5 $(x - 1)^2 + (y + 2)^2 = 25$

Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$S = r\theta$$

Trig

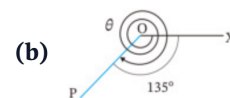
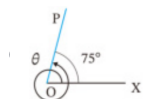
$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}$$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q001 Written

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75° ; (b) two full clockwise revolutions and a further 135° .



Classified Practice

Defining Angles and Basic Trigonometric Functions

Q002 **Written****Draw the terminal side for 300° and -450° .**

Q003 Written

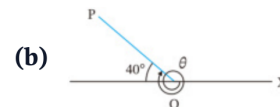
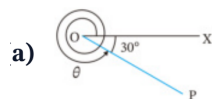
Express 420° and -210° as $\alpha + 360^\circ \times n$, where n is an integer and $0^\circ \leq \alpha < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x -axis; (b) a terminal side 40° above the negative x -axis, measured counter-clockwise from the positive x -axis).



2 Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q005 Written

Draw the terminal side for 500° and -120° .

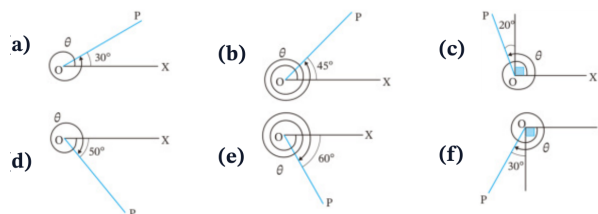
Q006 Written

Express 520° and -140° as $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q007 Written

Find the six diagram measures θ shown in the given figure (a)–(f).**2** Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for 470° , 780° , 225° , -380° , -990° , -200° .

Q009 Written

Express $610^\circ, 920^\circ, 840^\circ, -430^\circ, -960^\circ, -20^\circ$ **as** $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q010 Written

Express 129720° and -2592100° as $\alpha + 360^\circ \times n$, with $0^\circ \leq \alpha < 360^\circ$.

Q011 MCQ

The principal angle coterminal with 765° is:**Choose the correct option.**A 45° B 135° C 225° D 315°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q012 MCQ

The principal angle coterminal with -410° is:**Choose the correct option.**A -50° B 50° C 310° D 410°

Q013 MCQ**The terminal side of 540° lies on the:****Choose the correct option.**

- A positive x -axis
 - B negative x -axis
 - C positive y -axis
 - D negative y -axis
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q014 MCQ

A positive angle of 315° terminates in:**Choose the correct option.**

A QI

B QII

C QIII

D QIV

Q015 MCQ

Which is the general-angle form for -75° ?**Choose the correct option.**

- A $75^\circ + 360^\circ n$
- B $285^\circ + 360^\circ n$
- C $-285^\circ + 360^\circ n$
- D $360^\circ n - 75^\circ$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q016 MCQ

The direction of a negative angle is:**Choose the correct option.**

- A counter-clockwise
- B clockwise
- C radial only
- D undefined

Q017 MCQ**The terminal side of 1080° is the:****Choose the correct option.**

- A positive x -axis
- B negative x -axis
- C positive y -axis
- D negative y -axis

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q018 MCQ

For $\alpha = 0^\circ$, the general-angle family is:**Choose the correct option.**A $360^\circ n$ B $0^\circ n$ C $180^\circ + 360^\circ n$ D $90^\circ + 360^\circ n$

Q019 MCQ**The principal angle for -990° is:****Choose the correct option.**A 90° B 180° C 270° D -270°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q020 MCQ

A ray (P) is 25° above the negative x -axis. Its standard-position angle is:

Choose the correct option.

A 25°

B 155°

C 205°

D 335°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S .

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288^\circ, -400^\circ$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q023 Written

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ to radians.

Q024 Written

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ **to degrees.**

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q027 Written

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Q028 MCQ

Convert 150° to radians:**Choose the correct option.**

- A $\frac{5\pi}{6}$
B $\frac{3\pi}{5}$
C $\frac{5\pi}{3}$
D $\frac{3\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:**Choose the correct option.**A -120° B -150° C 150° D -300°

Q030 MCQ

For $r = 5$, $\theta = \frac{2\pi}{3}$, the arc length is:

Choose the correct option.

- A $\frac{5\pi}{3}$
- B $\frac{10\pi}{3}$
- C $\frac{25\pi}{3}$
- D 10π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q031 MCQ

For $r = 4$, $\theta = \frac{\pi}{2}$, the sector area is:

Choose the correct option.

- A π
- B 2π
- C 4π
- D 8π

Q032 MCQ

The radian measure of a full turn is:**Choose the correct option.**

- A π
 - B 2π
 - C 180
 - D 360
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Choose the correct option.

- A 210°
- B 270°
- C 315°
- D 360°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Choose the correct option.

- A $\frac{\pi}{6}$
- B $-\frac{\pi}{6}$
- C $\frac{5\pi}{6}$
- D $\frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q035 MCQ

For $r = 3, \theta = \pi$, the sector area is:**Choose the correct option.**

A $\frac{3\pi}{2}$

B 3π

C 9π

D $\frac{9\pi}{2}$

Q036 MCQ

The angle $\frac{5\pi}{3}$ lies in:

Choose the correct option.

A QI

B QII

C QIII

D QIV

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q037 Written

Convert 252° to radians.

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S .

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin \theta$, $\cos \theta$, $\tan \theta$.

Q040 Written

Find the exact values (write undefined where appropriate): $\sin \frac{7\pi}{4}$, $\cos \frac{\pi}{3}$, $\tan \frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q041 Written

Find the exact values of the eighteen trigonometric functions listed in the given exercise (a)–(r), including negative and undefined cases.

Find the values of the following trigonometric functions. If the value is undefined, write “undefined”.

(a) $\sin \frac{\pi}{3}$

(b) $\sin \frac{7}{6} \pi$

(c) $\sin \pi$

(d) $\cos \frac{5}{4} \pi$

(e) $\cos \frac{5}{3} \pi$

(f) $\cos \frac{7}{6} \pi$

(g) $\tan \frac{2}{3} \pi$

(h) $\tan \frac{5}{4} \pi$

(i) $\tan \pi$

(j) $\sin \left(-\frac{\pi}{6} \right)$

(k) $\sin \left(-\frac{5}{4} \pi \right)$

(l) $\sin \left(-\frac{5}{3} \pi \right)$

(m) $\cos \left(-\frac{\pi}{4} \right)$

(n) $\cos \left(-\frac{5}{6} \pi \right)$

(o) $\cos \left(-\frac{4}{3} \pi \right)$

(p) $\tan \left(-\frac{4}{3} \pi \right)$

(q) $\tan \left(-\frac{11}{6} \pi \right)$

(r) $\tan \left(-\frac{3}{2} \pi \right)$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q042 Written

The panel height is $h(t) = 3 \sin\left(\frac{2\pi}{3}t - \frac{\pi}{4}\right) + 1.5$. Find the exact height at $t = 0$.

Q043 MCQ

 $\sin \frac{\pi}{6}$ equals:

Choose the correct option.

A 0

B $\frac{1}{2}$ C $\frac{\sqrt{3}}{2}$

D 1

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q044 MCQ

 $\cos \frac{2\pi}{3}$ equals:

Choose the correct option.

A -1

B $-\frac{1}{2}$ C $\frac{1}{2}$ D $\frac{\sqrt{3}}{2}$

Q045 MCQ $\tan \frac{3\pi}{4}$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q046 MCQ

 $\sin\left(-\frac{\pi}{2}\right)$ equals:

Choose the correct option.

- A -1
- B 0
- C 1
- D undefined

Q047 MCQ $\tan \frac{\pi}{2}$ is:**Choose the correct option.**

A 0

B 1

C -1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q048 MCQ

 $\cos \frac{5\pi}{3}$ equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Q049 MCQ**If $P(x, y)$ lies on the unit circle, then $\sin \theta$ is:****Choose the correct option.**

- A** x
- B** y
- C** x/y
- D** $1/x$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q050 MCQ

 $\tan\left(-\frac{3\pi}{4}\right)$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Q051 MCQ $\sin \frac{11\pi}{6}$ equals:**Choose the correct option.**

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q052 Written

Find $\sin \frac{5\pi}{6}$, $\cos \frac{4\pi}{3}$, $\tan \frac{7\pi}{4}$.

Q053 Written

Find $\sin(-\frac{5\pi}{6})$, $\cos(-\frac{7\pi}{6})$, $\tan(-\frac{\pi}{3})$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q054 **Written**

For $P = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ on the unit circle, find the corresponding $\sin \theta$, $\cos \theta$, $\tan \theta$.

Q055 Written

Find the exact value of $3 \sin \frac{3\pi}{2} - 2 \cos \pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q056 Written

State the two standard angles where $\tan \theta$ is undefined in $[0, 2\pi)$.

Q057 Written

Solve for θ , $0 \leq \theta < 2\pi$: (a) $\sin \theta = -\frac{1}{2}$; (b) $\cos \theta = \frac{1}{\sqrt{2}}$; (c) $\tan \theta = -\sqrt{3}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q058 Written

Solve the six trigonometric equations in the given Try set for $0 \leq \theta < 2\pi$: $\sin \theta = 1/\sqrt{2}$, $\cos \theta = -\sqrt{3}/2$, $\tan \theta = 1/\sqrt{3}$, $\sin \theta = 0$, $\cos \theta = 1/2$, $\tan \theta = -1$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q059 **Written**

Solve the sixteen equations in the given exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

Q060 Written**Solve** $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta = \sqrt{3}$ for $0 \leq \theta < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = \frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{6}, \frac{5\pi}{6}$

B $\frac{\pi}{3}, \frac{2\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = -\frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{3}, \frac{5\pi}{3}$

B $\frac{2\pi}{3}, \frac{4\pi}{3}$

C $\frac{\pi}{6}, \frac{5\pi}{6}$

D $\frac{4\pi}{3}, \frac{5\pi}{3}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = 1$ gives:

Choose the correct option.

- A $\frac{\pi}{4}, \frac{5\pi}{4}$
B $\frac{3\pi}{4}, \frac{7\pi}{4}$
C $\frac{\pi}{4}, \frac{7\pi}{4}$
D $\frac{3\pi}{4}, \frac{5\pi}{4}$

Q064 MCQ

The solutions of $\sin \theta = 0$ in $[0, 2\pi)$ are:**Choose the correct option.**A $0, \pi$ B $\frac{\pi}{2}, \frac{3\pi}{2}$ C $0, 2\pi$ D π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q065 MCQ

If $\cos \theta = 1$, then in $[0, 2\pi)$, θ is:

Choose the correct option.

A 0

B $\frac{\pi}{2}$ C π D $\frac{3\pi}{2}$

Q066 MCQ**For $0 \leq \theta < 2\pi$, $\tan \theta = -\sqrt{3}$ gives:****Choose the correct option.**

- A $\frac{\pi}{3}, \frac{4\pi}{3}$
B $\frac{2\pi}{3}, \frac{5\pi}{3}$
C $\frac{\pi}{6}, \frac{7\pi}{6}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = -1$ gives:

Choose the correct option.

- A $\frac{\pi}{2}$
- B $\frac{3\pi}{2}$
- C $\frac{2}{\pi}$
- D 0

Q068 MCQ**For $0 \leq \theta < 2\pi$, $\cos \theta = 0$ gives:****Choose the correct option.**

- A $0, \pi$
- B $\frac{\pi}{2}, \frac{3\pi}{2}$
- C π
- D $\frac{\pi}{4}, \frac{5\pi}{4}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q069 MCQ

The equation $\tan \theta = 0$ has solutions in $[0, 2\pi)$:

Choose the correct option.

A $0, \pi$

B $\frac{\pi}{2}, \frac{3\pi}{2}$

C $\frac{\pi}{4}, \frac{5\pi}{4}$

D $\pi, 2\pi$

Q070 Written**Solve** $\cos \theta = \frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q071 Written

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Q072 Written**Solve** $\tan \theta = \frac{1}{\sqrt{3}}$ for $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q073 Written

Solve $2 \cos \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.

Q074 Written**Solve** $2 \sin \theta + \sqrt{2} = 0$ for $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 1 [Concept Quiz · MCQ](#)

The radian measure of 240° is:

Choose the correct option.

- A $\frac{2\pi}{3}$
- B $\frac{4\pi}{3}$
- C $\frac{3\pi}{4}$
- D $\frac{5\pi}{3}$

Quiz MCQ 2 Concept Quiz · MCQ

If $\sin \theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

Choose the correct option.

- A $\frac{\pi}{3}, \frac{2\pi}{3}$
B $\frac{4\pi}{3}, \frac{5\pi}{3}$
C $\frac{2\pi}{3}, \frac{4\pi}{3}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 3 Concept Quiz · MCQ

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Choose the correct option.

A 3π

B 6π

C 12π

D 18π

Quiz MCQ 4 **Concept Quiz · MCQ**

The general-angle family for -300° is:

Choose the correct option.

- A $60^\circ + 360^\circ n$
- B $300^\circ + 360^\circ n$
- C $-60^\circ + 360^\circ n$
- D $120^\circ + 360^\circ n$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\cos \theta = -\frac{\sqrt{2}}{2}$ **for** $0 \leq \theta < 2\pi$.

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Quiz WRITTEN 8 Concept Quiz · Written

Evaluate $3 \sin \frac{7\pi}{6} - 2 \cos \frac{4\pi}{3}$.

Use the question number shown on each page.

Eng Eslam Ahmed | 01120009622 85

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Tangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Secant

$$1 + \tan^2 \theta = \sec^2 \theta$$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin \theta = -\frac{1}{5}$ in QIII and (b) $\tan \theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q002 Written

Given one value and the quadrant, find the other two: (a) $\cos \theta = -1/\sqrt{3}$, QIII; (b) $\tan \theta = -2$, QIV; (c) $\sin \theta = 3/5$, QII; (d) $\tan \theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q003 **Written**

For each of the twelve numbered cases in the **given Exercise**, one of $\sin \theta$, $\cos \theta$, $\tan \theta$ and a quadrant/interval is given. Find the other two values.

Q004 Written

Using trigonometric identities, find the other two values given $\tan \theta + \cot \theta = 12$ and $\pi < \theta < 2\pi$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q005 **Written**

Given $\sin \theta = \frac{5}{13}$ **in QII, find** $\cos \theta$ **and** $\tan \theta$.

Q006 Written

Given $\cos \theta = -\frac{3}{5}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q007 **Written**

Given $\tan \theta = \frac{3}{4}$ **in QI, find** $\sin \theta$ **and** $\cos \theta$.

Q008 Written

Given $\tan \theta = -\frac{5}{12}$ in QIV, find $\sin \theta$ and $\cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q009 **Written**

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ **in QIII, find** $\cos \theta$ **and** $\tan \theta$.

Q010 MCQ

If $\sin \theta = -\frac{3}{5}$ in QIII, $\cos \theta$ is:

Choose the correct option.

A $\frac{4}{5}$

B $-\frac{4}{5}$

C $\frac{3}{5}$

D $-\frac{3}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q011 MCQ

If $\tan \theta = 2$ in QI, $\sin \theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $\frac{\sqrt{5}}{5}$

C $-\frac{2\sqrt{5}}{5}$

D $\frac{2}{5}$

Q012 MCQ

If $\cos \theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A 1

B -1

C $\sqrt{2}$

D $-\sqrt{2}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q013 MCQ

For $\tan \theta = -\sqrt{3}$ in QII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
- B $-\frac{1}{2}$
- C $\frac{\sqrt{3}}{2}$
- D $-\frac{\sqrt{3}}{2}$

Q014 MCQ

If $\sin \theta = \frac{7}{25}$ in QI, $\tan \theta$ is:

Choose the correct option.

- A $\frac{7}{24}$
- B $\frac{24}{7}$
- C $-\frac{7}{24}$
- D $\frac{25}{7}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin \theta + \cos \theta = \frac{1}{2}$, **with** θ **in** QII, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Q016 Written

Given $\sin \theta + \cos \theta = \frac{1}{3}$, **with** θ **in** QII, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q017 Written

Solve the three numbered Exercise problems: (1) given $\sin \theta - \cos \theta = \sqrt{3}/2$ in QI; (2) given $\sin \theta + \cos \theta = \sqrt{3}/2$ in QIV; (3) given $\sin \theta - \cos \theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q018 Written

Given $\sin \theta - \cos \theta = 0.5$ and θ in QIII, evaluate the exact value of $\sin^6 \theta - \cos^6 \theta$.

Q019 Written

Given $\sin \theta + \cos \theta = \frac{2}{3}$ **in QII, find** $\sin \theta \cos \theta$ **and** $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q020 Written

Given $\sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$ in QI, find $\sin \theta \cos \theta$ and $\sin^3 \theta - \cos^3 \theta$.

Q021 Written

Given $\sin \theta + \cos \theta = -\frac{1}{2}$ **in QIII, find** $\sin^3 \theta + \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q022 Written

Given $\sin \theta - \cos \theta = \frac{1}{3}$ **in QIV, find** $\sin \theta + \cos \theta$.

Q023 Written**Given** $\sin \theta - \cos \theta = 1$ **in QIII, find** $\sin^3 \theta - \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{5}{16}$
B $\frac{11}{16}$
C $-\frac{11}{16}$
D $\frac{3}{16}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Choose the correct option.

A $\frac{1}{8}$

B $-\frac{1}{8}$

C $\frac{3}{8}$

D $-\frac{3}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Choose the correct option.

A $-\frac{55\sqrt{7}}{256}$

B $\frac{55\sqrt{7}}{256}$

C $-\frac{11\sqrt{7}}{16}$

D $\frac{11}{16}$

Q029 Written

Prove $\frac{\sin \theta}{1 + \cos \theta} + \frac{1}{\tan \theta} = \frac{1}{\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan \theta}{1+\tan \theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$; (b) $\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \frac{1}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q031 Written

Prove the six identities in the given Exercise: (a) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$; (b) $\tan \theta + 1/\tan \theta = 1/(\sin \theta \cos \theta)$; (c) $\cos \theta/(1 - \sin \theta) - 1/\cos \theta = \tan \theta$; (d) $\tan \theta/\sin \theta - \sin \theta/\tan \theta = \sin \theta \tan \theta$; (e) $(1 + \sin \theta)/\cos \theta + \cos \theta/(1 + \sin \theta) = 2/\cos \theta$; (f) $(\tan^2 \theta - \sin^2 \theta)/\sin^2 \theta = \tan^2 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q032 Written

Prove that, for all defined values of x , $\frac{1+\sin x-\cos x}{1+\sin x+\cos x} + \frac{1+\sin x+\cos x}{1+\sin x-\cos x} = 2 \csc x$.

Q033 Written

Prove $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q034 Written

Prove $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 2\sec^2\theta$.

Q035 Written

Prove $\frac{\tan \theta}{1 + \tan \theta} = \frac{\sin \theta}{\sin \theta + \cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q036 Written

Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}$.

Q037 Written

Prove $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q038 MCQ

The first move in proving a trig identity is usually to:**Choose the correct option.**

- A Change both sides at once
- B Simplify one side
- C Substitute random values
- D Take a square root

Q039 MCQ

The identity used to replace $1 - \sin^2 \theta$ is:

Choose the correct option.

A $\cos^2 \theta$

B $\tan^2 \theta$

C $\sec^2 \theta$

D $\sin^2 \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q040 MCQ

$\tan \theta$ should be rewritten as:

Choose the correct option.

A $\sin \theta \cos \theta$

B $\sin \theta / \cos \theta$

C $\cos \theta / \sin \theta$

D $1 / \sin \theta$

Q041 MCQ**To add two trig fractions, first find a:****Choose the correct option.**

- A common numerator
- B common denominator
- C new quadrant
- D reference angle

Classified Practice

Trigonometric Identities and Expression Evaluation

Q042 MCQ

The result of a valid identity proof is:**Choose the correct option.**

- A LHS=0
- B LHS=RHS
- C RHS=1
- D undefined

Q043 MCQ

If $\cos \theta = -\frac{12}{13}$ in QII, $\sin \theta$ is:

Choose the correct option.

A $\frac{5}{13}$

B $-\frac{5}{13}$

C $\frac{12}{13}$

D $-\frac{12}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q044 MCQ

If $\tan \theta = -\frac{1}{2}$ in QII, $\cos \theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $-\frac{2\sqrt{5}}{5}$

C $\frac{\sqrt{5}}{5}$

D $-\frac{\sqrt{5}}{5}$

Q045 MCQ

If $\sin \theta = \frac{8}{17}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A $\frac{8}{15}$

B $-\frac{8}{15}$

C $\frac{15}{8}$

D $-\frac{15}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q046 MCQ

If $\tan \theta = \sqrt{3}$ and θ is in QIII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
- B $-\frac{1}{2}$
- C $\frac{\sqrt{3}}{2}$
- D $-\frac{\sqrt{3}}{2}$

Q047 MCQ**If $s + c = 1$, then sc equals:****Choose the correct option.****A** 0**B** $\frac{1}{2}$ **C** $-\frac{1}{2}$ **D** 1

Classified Practice

Trigonometric Identities and Expression Evaluation

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C -1

D 1

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{19}{27}$
B $\frac{5}{27}$
C $-\frac{19}{27}$
D $\frac{11}{27}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Choose the correct option.

A $\frac{11}{16}$

B $-\frac{11}{16}$

C $\frac{7}{16}$

D $-\frac{7}{16}$

Q051 MCQ

Which identity converts $\frac{1}{\tan \theta}$ into sine and cosine?

Choose the correct option.

- A $\frac{\sin \theta}{\cos \theta}$
- B $\frac{\cos \theta}{\sin \theta}$
- C $\sin \theta \cos \theta$
- D $\sec \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q052 MCQ

The identity $1 - \cos^2 \theta$ simplifies to:**Choose the correct option.**A $\sin^2 \theta$ B $\tan^2 \theta$ C $\cos^2 \theta$

D 1

Q053 MCQ**When proving an identity, which side should normally be transformed?****Choose the correct option.**

- A Both sides simultaneously
- B One side only
- C Neither side
- D Only the denominator

Classified Practice

Trigonometric Identities and Expression Evaluation

Q054 MCQ

After combining trig fractions, the next step is usually to use:

Choose the correct option.

- A A random angle
- B The Pythagorean identity
- C A logarithm
- D A derivative

Quiz MCQ 1 Concept Quiz · MCQ

If $\sin \theta + \cos \theta = \frac{1}{2}$, then $\sin \theta \cos \theta$ is:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\tan \theta = \frac{3}{4}$ in QI, then $\cos \theta$ is:

Choose the correct option.

- A $\frac{3}{5}$
- B $\frac{4}{5}$
- C $-\frac{4}{5}$
- D $\frac{3}{4}$

Quiz MCQ 3 Concept Quiz · MCQ

Which identity is always valid where defined?

Choose the correct option.

A $\tan \theta = \frac{\cos \theta}{\sin \theta}$

B $1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}$

C $\sin \theta + \cos \theta = 1$

D $\sin \theta \cos \theta = 1$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 Concept Quiz · MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Quiz WRITTEN 5 Concept Quiz · Written

Given $\cos \theta = -\frac{5}{13}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 6 Concept Quiz · Written**Given** $s + c = \frac{2}{3}$, **find** sc **and** $s^3 + c^3$.

Quiz WRITTEN 7 Concept Quiz · Written

Prove $\frac{1-\sin \theta}{\cos \theta} = \frac{\cos \theta}{1+\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 8 Concept Quiz · Written

Given $\tan \theta = -2$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$, **then verify** $1 + \tan^2 \theta = 1/\cos^2 \theta$.

Answer key

Use the question number shown on each page.

Q001 (a) $\cos \theta = -\frac{2\sqrt{6}}{5}$, $\tan \theta = \frac{\sqrt{6}}{12}$; (b) $\cos \theta = \frac{\sqrt{7}}{7}$, $\sin \theta = -\frac{\sqrt{42}}{7}$ **Q015** (a) $-\frac{3}{8}$ (b) $\frac{11}{16}$ (c) $\frac{\sqrt{7}}{2}$

Q002 (a) $\sin \theta = -\frac{\sqrt{6}}{3}$, $\tan \theta = \sqrt{2}$; (b) $\cos \theta = \frac{\sqrt{5}}{5}$, $\sin \theta = -\frac{2\sqrt{5}}{5}$; (c) $\cos \theta = -\frac{4}{5}$, $\tan \theta = -\frac{3}{4}$; (d) $\sin \theta = -\frac{\sqrt{21}}{7}$, $\cos \theta = -\frac{2\sqrt{7}}{7}$

Q003 $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (b) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (c) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (d) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$ **Q016** (a) $-\frac{4}{9}$ (b) $\frac{37}{27}$ (c) $\frac{\sqrt{17}}{3}$

Q004 $\sin \theta = -\frac{1}{2}$, $\cos \theta = -\frac{\sqrt{3}}{2}$ or the swapped pair **Q017** (1) $\sin \theta \cos \theta = \frac{1}{8}$, $\sin^2 \theta - \cos^2 \theta = \frac{7\sqrt{3}}{16}$, $\sin \theta + \cos \theta = \frac{\sqrt{5}}{2}$; (2) $-\frac{1}{8}$, $\frac{9\sqrt{3}}{16}$, $-\frac{\sqrt{5}}{2}$; (3) $\frac{3}{8}$, $\frac{11}{16}$ **Q031** LHS = RHS for (a)+(f)

Q005 $-\frac{12}{13}$, $-\frac{5}{12}$ **Q018** $-\frac{55\sqrt{7}}{256}$ **Q032** LHS = $2 \csc x$

Q006 $-\frac{4}{5}$, $\frac{4}{3}$ **Q019** $-\frac{5}{18}$, $\frac{\sqrt{14}}{3}$ **Q033** Identity proved

Q007 $\frac{3}{5}$, $\frac{4}{5}$ **Q020** $-\frac{1}{8}$, $\frac{7\sqrt{5}}{16}$ **Q034** Identity proved

Q008 $-\frac{5}{13}$, $\frac{12}{13}$ **Q021** $-\frac{11}{16}$ **Q035** Identity proved

Q009 $-\frac{\sqrt{2}}{2}$, 1 **Q022** $\frac{\sqrt{11}}{3}$ **Q036** Identity proved

Q010 B **Q023** $\frac{3}{2}$ **Q037** Identity proved

Q011 A **Q024** A **Q038** B

Q012 B **Q025** B **Q039** A

Q013 C **Q026** B **Q040** B

Q014 A **Q027** A **Q041** B

Q028 A

Q029 LHS = RHS

Q030 Both identities verified

Q031 LHS = RHS for (a)+(f)

Q032 LHS = $2 \csc x$

Q033 Identity proved

Q034 Identity proved

Q035 Identity proved

Q036 Identity proved

Q037 Identity proved

Q038 B

Q039 A

Q040 B

Q041 B

Q042 B

Q043 A

Q044 A

Q045 B

Q046 D

Q047 A

Q048 A

Q049 A

Q050 A

Q051 B

Q052 A

Q053 B

Q054 B

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $-\frac{12}{13}$, $\frac{12}{5}$

Quiz WRITTEN 6 $-\frac{5}{18}$, $\frac{19}{27}$

Quiz WRITTEN 7 Identity proved

Quiz WRITTEN 8 $\sin \theta = -\frac{2\sqrt{5}}{5}$, $\cos \theta = \frac{\sqrt{5}}{5}$; $6 = 6$

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a \sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \leq y \leq d + |a|$$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q001 Written

(1) Express $\sin \frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin \theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q002 Written

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin \frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2} - \theta) \sin(\pi - \theta) - \sin(\frac{3\pi}{2} + \theta) \cos(\pi - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three given demands: (1) express $\cos \frac{17\pi}{5}$, $\cos \frac{5\pi}{7}$, $\sin(-\frac{7\pi}{8})$, $\sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin \theta + \cos(\pi + \theta) - \cos(\pi - \theta) + \sin(\pi + \theta)$, (b) $\sin(\theta + \frac{\pi}{2}) \cos(\theta + \pi) + \sin(-\theta) \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q004 **Written**

Simplify $\sin\left(\frac{3\pi}{2} - \theta\right) \sec(\pi + \theta) - \tan\left(\theta - \frac{\pi}{2}\right) \cot\left(\frac{5\pi}{2} + \theta\right).$

Q005 Written

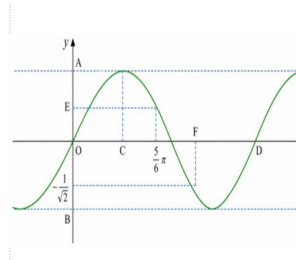
Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, and $y = \tan \theta$, and state each period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q006 **Written**

For the given graph of $y = \sin \theta$, determine the labelled scales A to F .

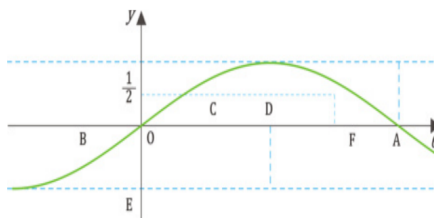


Classified Practice

Properties and Graphs of Trigonometric Functions

Q007 Written

(1) Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, $y = \tan \theta$, and state their periods. (2) For the given graph of $y = \cos \theta$, determine the labelled scales A to F .

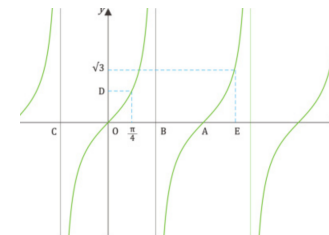
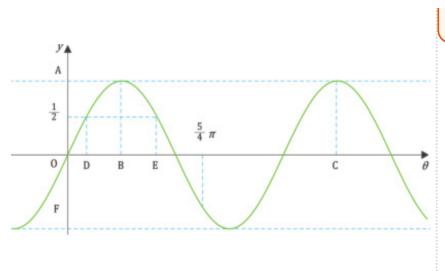
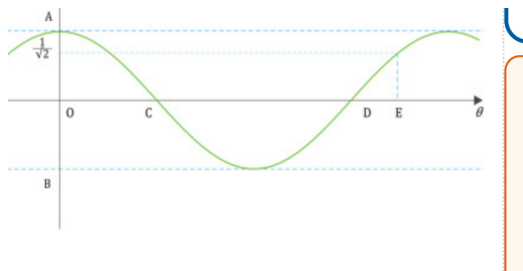


Classified Practice

Properties and Graphs of Trigonometric Functions

Q008 **Written**

Complete all four given graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the given cosine graph; read the labelled scales on the given sine graph; and read the labelled scales on the given tangent graph.

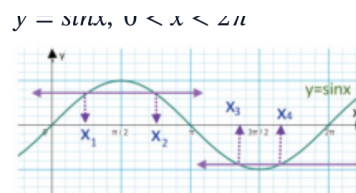


Classified Practice

Properties and Graphs of Trigonometric Functions

Q009 Written

The given graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



Then find: $x_1 + x_2 + x_3 + x_4$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q010 Written

Draw the graphs of $y = 2 \cos \theta$ and $y = \tan \frac{\theta}{2}$, and find their periods.

Q011 Written

Draw the graphs of $y = 2 \sin \theta$, $y = \tan 2\theta$, and $y = \cos \frac{\theta}{2}$, and find their periods.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q012 Written

Draw the nine given graphs and find each period: (a) $3 \sin \theta$, (b) $3 \cos \theta$, (c) $\frac{1}{2} \cos \theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin \frac{\theta}{2}$, (h) $\cos \frac{\theta}{3}$, (i) $\tan \frac{\theta}{3}$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q013 Written

Draw $y = -2|\sin(3x)|$ and find its period.

Q014 Written**Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q015 Written

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q016 Written

Draw all given graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q017 Written

Draw $y = 3 \cos \left| x + \frac{\pi}{3} \right|$ and find its period.

Q018 Written

Draw the graphs and find their periods: (1) $y = 2 \cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3 \sin \frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2 \cos(2\theta - \frac{\pi}{3})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q020 Written

Draw all nine given graphs and find their periods: (a) $2 \cos \frac{\theta}{2}$, (b) $3 \sin 2\theta$, (c) $2 \cos 2\theta$, (d) $2 \cos(\theta - \frac{\pi}{4})$, (e) $2 \sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3 \cos(2\theta - \frac{\pi}{2})$, (h) $3 \sin(2\theta - \frac{2\pi}{3})$, (i) $2 \cos(\frac{\theta}{2} - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q021 Written

Draw $y = 3 \sin |2x - \pi|$ and find its period.

Q022 MCQ $\sin(\theta + \pi)$ equals:**Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q023 MCQ

 $\cos(\theta + \pi)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q024 MCQ $\sin\left(\frac{\pi}{2} - \theta\right)$ equals:

Choose the correct option.

A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q025 MCQ

 $\cos(-\theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q026 MCQ **$\sin(-\theta)$ equals:****Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q027 MCQ

 $\cos(\pi - \theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q028 MCQ $\sin\left(\frac{3\pi}{2} + \theta\right)$ **equals:****Choose the correct option.**A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q029 MCQ

 $\tan(\theta - \frac{\pi}{2})$ equals:

Choose the correct option.

A $\tan \theta$ B $-\tan \theta$ C $\cot \theta$ D $-\cot \theta$

Q030 MCQ**The period of $y = \sin \theta$ is:****Choose the correct option.****A** $\pi/2$ **B** π **C** 2π **D** 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q031 MCQ

The range of $y = \cos \theta$ is:**Choose the correct option.**

A $y \geq 0$

B $-1 \leq y \leq 1$

C $y \in \mathbb{R}$

D $y > 1$

Q032 MCQ

The period of $y = \tan \theta$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan \theta$ occurs at:**Choose the correct option.**

A $\theta = k\pi$

B $\theta = \frac{\pi}{2} + k\pi$

C $\theta = 2k\pi$

D $\theta = \frac{\pi}{4} + k\pi$

Q034 MCQ

The maximum value of $\sin \theta$ is:**Choose the correct option.**

A -1

B 0

C 1

D π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q035 MCQ

The x-coordinate of the first positive sine maximum is:**Choose the correct option.**

- A $\pi/4$
- B $\pi/2$
- C π
- D $3\pi/2$

Q036 MCQ**At $\theta = 0$, $\cos \theta$ equals:****Choose the correct option.****A** -1**B** 0**C** 1**D** undefined

Classified Practice

Properties and Graphs of Trigonometric Functions

Q037 MCQ

At $\theta = 0$, $\tan \theta$ equals:**Choose the correct option.**

- A -1
- B 0
- C 1
- D undefined

Q038 MCQ

The range of $y = 3 \sin \theta$ is:**Choose the correct option.**A $[-1,1]$ B $[-3,3]$ C $[0,3]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q039 MCQ

The period of $y = \sin 4\theta$ is:**Choose the correct option.**A $\pi/4$ B $\pi/2$ C π D 2π

Q040 MCQ**The period of $y = \cos(\theta/3)$ is:****Choose the correct option.**A $2\pi/3$ B 2π C 3π D 6π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q041 MCQ

The period of $y = \tan 5\theta$ is:**Choose the correct option.**A $\pi/5$ B π C 5π D $2\pi/5$

Q042 MCQ

The range of $y = \frac{1}{2} \cos \theta$ is:

Choose the correct option.

A $[-2, 2]$

B $[-1, 1]$

C $[-1/2, 1/2]$

D $[0, 1/2]$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2 \sin \theta$ is the parent sine graph:**Choose the correct option.**

- A shifted only
- B reflected and stretched vertically
- C compressed horizontally only
- D undefined

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q045 MCQ

The period of $y = \sin(3\theta)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C 3π D 6π

Q046 MCQ**The graph of $y = \sin(\theta - \pi/4)$ is shifted:****Choose the correct option.**

- A left by $\pi/4$
 - B right by $\pi/4$
 - C up by $\pi/4$
 - D down by $\pi/4$
-
-
-

Classified Practice

Properties and Graphs of Trigonometric Functions

Q047 MCQ

The graph of $y = \cos(\theta + \pi/3)$ is shifted:**Choose the correct option.**A right by $\pi/3$ B left by $\pi/3$ C up by $\pi/3$ D down by $\pi/3$

Q048 MCQ**A horizontal shift changes the sine period from 2π to:****Choose the correct option.****A** π **B** 2π **C** 4π **D** 0

Classified Practice

Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Q050 MCQ**The graph of $y = \sin(\theta + \pi/2)$ equals:****Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:**Choose the correct option.**A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:

Choose the correct option.

A $\theta = 0$

B $\theta = \pi/2$

C $\theta = \pi$

D $\theta = 2\pi$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos |x|$ equals:**Choose the correct option.**A $\cos(x)$ B $-\cos(x)$ C $\sin(x)$ D $|\cos(x)|$

Q054 MCQ**In $y = a \sin(b\theta - q)$, the horizontal shift is:****Choose the correct option.**

- A q
- B q/b
- C b/q
- D a/b

Classified Practice

Properties and Graphs of Trigonometric Functions

Q055 MCQ

The period of $y = 2 \cos(3\theta - \pi)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C π D 2π

Q056 MCQ

The range of $y = 4 \sin(2\theta - 1)$ is:**Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[-4,4]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q057 MCQ

For $y = \tan(2\theta - \pi/3)$, the shift is:**Choose the correct option.**

- A left by $\pi/3$
- B right by $\pi/6$
- C right by $\pi/3$
- D left by $\pi/6$

Q058 MCQ

The period of $y = 3 \sin(\theta/2)$ is:**Choose the correct option.**

- A π
 - B 2π
 - C 3π
 - D 4π
-
-
-

Classified Practice

Properties and Graphs of Trigonometric Functions

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:

Choose the correct option.

- A vertical scale
- B horizontal shift
- C horizontal scale
- D reflection only

Q060 MCQ**The period of $y = \tan(4\theta + q)$ is:****Choose the correct option.****A** $\pi/4$ **B** $\pi/2$ **C** π **D** 2π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q061 MCQ

The range of $y = 3 \sin |2x - \pi|$ is:**Choose the correct option.**A $[-3,3]$ B $[0,3]$ C $[-1,1]$ D $\mathbb{R}$

Q062 Written

Express $\cos \frac{11\pi}{6}$ as a function of an acute angle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q063 Written

Simplify $\sin(\theta + \pi) - \cos\left(\frac{\pi}{2} - \theta\right)$.

Q064 Written**Simplify** $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q065 Written

Evaluate $\tan\left(-\frac{3\pi}{4}\right)$ **using an acute reference angle.**

Q066 Written

State the period and range of $y = \cos \theta$, then list its key points on one cycle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan \theta$ on $[-\pi, \pi]$.

Q068 Written

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q069 Written

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Q070 Written**Find the period and range of $y = 4 \cos \theta$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q071 Written

Find the period and range of $y = \sin(5\theta)$.

Q072 Written**Find the period and range of $y = \tan(\theta/4)$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q073 Written

Describe the transformations from $y = \cos \theta$ to $y = -\frac{1}{2} \cos(2\theta)$.

Q074 Written**Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.

Q076 Written**Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q077 Written

For $y = 2 \sin(4\theta - \pi)$, state the shift, period and range.

Q078 Written**For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q079 Written

For $y = 3 \tan(2\theta + \pi)$, state the shift and period.

Q080 Written

For $y = 2 \cos(\theta/2 - \pi/8)$, state the shift, period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 1 Concept Quiz · MCQ

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Quiz MCQ 2 Concept Quiz · MCQ

$\sin(\theta + \pi)$ equals:

Choose the correct option.

A $\sin \theta$

B $-\sin \theta$

C $\cos \theta$

D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)**The range of $y = 2 \cos \theta$ is:****Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $\mathbb{R}$

Quiz MCQ 4 Concept Quiz · MCQ

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Choose the correct option.

- A left by $\frac{\pi}{4}$
- B right by $\frac{\pi}{4}$
- C up by $\frac{\pi}{4}$
- D down by $\frac{\pi}{4}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

For $y = 3 \sin(2\theta - \pi)$, state the horizontal shift, period and range.

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written**Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.**

Quiz WRITTEN 8 Concept Quiz · Written

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Answer key

Use the question number shown on each page.

- Q001** (1) $-\sin \frac{\pi}{5}$ (2) 0
- Q002** (1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$
- Q003** (1) $-\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) $-\frac{1}{\sqrt{3}}$
- Q004** 0
- Q005** $\sin, \cos$: $T = 2\pi, -1 \leq y \leq 1$; $\tan$: $T = \pi, y \in \mathbb{R}$
- Q006** $A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$
- Q007** (1) $T_{\min} = 2\pi, T_{\max} = 2\pi, T_{\min} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph
- Q008** Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.
- Q009** Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.
- Q010** $y = 2 \cos \theta$: $T = 2\pi, -2 \leq y \leq 2$; $y = \tan \frac{\theta}{2}$: $T = 2\pi$
- Q011** $T = 2\pi, \frac{\pi}{2}, 4\pi$ respectively; ranges $[-2, 2], \mathbb{R}, [-1, 1]$
- Q012** (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) $\frac{\pi}{3}$ (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient
- Q013** $T = \frac{\pi}{3}, 0 \geq y \geq -2$
- Q014** shift right by $\frac{\pi}{6}, T = 2\pi$
- Q015** Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.
- Q016** All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.
- Q017** $y = 3 \cos(x + \frac{\pi}{3})$ since cosine is even; $T = 2\pi$
- Q018** (1) $y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2$; (2) $y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$
- Q019** (a) $T = 4\pi, -3 \leq y \leq 3$; (b) $T = \frac{\pi}{2}$; (c) $T = \pi, -2 \leq y \leq 2$
- Q020** Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.
- Q021** $y = 3|\sin(2x - \pi)|, T = \frac{\pi}{2}, 0 \leq y \leq 3$
- Q022** B
- Q023** B
- Q024** C
- Q025** A
- Q026** B
- Q027** B
- Q028** D
- Q029** D
- Q030** C
- Q031** B
- Q032** B
- Q033** B
- Q034** C
- Q035** B
- Q036** C
- Q037** B
- Q038** B
- Q039** B
- Q040** D
- Q041** A
- Q042** C
- Q043** B
- Q044** C
- Q045** B
- Q046** B
- Q047** B
- Q048** B
- Q049** B
- Q050** C
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Q055** B
- Q056** C
- Q057** B
- Q058** D
- Q059** B
- Q060** A
- Q061** B
- Q062** $\cos \frac{\pi}{6}$
- Q063** $-\sin \theta - \sin \theta = -2 \sin \theta$
- Q064** $-\cos \theta + \sin \theta$
- Q065** 1
- Q066** $T = 2\pi, -1 \leq y \leq 1$; (0, 1), $(\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$
- Q067** $T = \pi; \theta = \pm \frac{\pi}{2}$
- Q068** 0, $\pi, 2\pi$
- Q069** $\frac{\pi}{2}, \frac{3\pi}{2}$
- Q070** $T = 2\pi, -4 \leq y \leq 4$
- Q071** $T = \frac{2\pi}{5}, -1 \leq y \leq 1$
- Q072** $T = 4\pi, y \in \mathbb{R}$
- Q073** horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T = \pi$
- Q074** left by $\frac{\pi}{6}, T = 2\pi$
- Q075** right by $\frac{\pi}{2}, T = 2\pi$
- Q076** $\theta = -\frac{\pi}{4}; T = \pi$
- Q077** right by $\frac{\pi}{4}, T = \frac{\pi}{2}, -2 \leq y \leq 2$
- Q078** left by $\frac{\pi}{2}, T = 6\pi$
- Q079** left by $\frac{\pi}{2}, T = \frac{\pi}{2}$
- Q080** right by $\frac{\pi}{4}, T = 4\pi, -2 \leq y \leq 2$
- Quiz MCQ 1** B
- Quiz MCQ 2** B
- Quiz MCQ 3** B

Quiz MCQ 4 B**Quiz WRITTEN 6** $\cos \theta - \sin \theta$ **Quiz WRITTEN 7** $T = \pi; \theta = \frac{5\pi}{6} + k\pi$ **Quiz WRITTEN 8** $T = \frac{\pi}{3}, -2 \leq y \leq 0$; reflected below the**Quiz WRITTEN 5** right by $\frac{\pi}{2}$, $T = \pi, -3 \leq y \leq 3$

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Sine

$$\sin x = \sin \alpha$$

Cosine

$$\cos x = \cos \alpha$$

Tangent

$$\tan x = \tan \alpha$$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q001 Written

Solve (a) $-2 \cos \theta = \sqrt{2}$ and (b) $\tan \theta + \sqrt{3} = 0$, giving the general solution for (b).

Q002 Written

Solve (a) $2 \cos \theta - 1 = 0$ and (b) $\tan \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q003 Written

Solve all given equations on $0 \leq \theta \leq 2\pi$: (a) $2 \sin \theta + 1 = 0$, (b) $2 \sin \theta = \sqrt{3}$, (c) $3 \sin \theta + 3 = 0$, (d) $\sqrt{2} \cos \theta + 1 = 0$, (e) $2 \cos \theta = \sqrt{3}$, (f) $2 \cos \theta - \sqrt{2} = 0$, (g) $\tan \theta + 1 = 0$, (h) $\sqrt{3} \tan \theta = 1$, (i) $\sqrt{3} \tan \theta = 3$, **and give general solutions for** (j) $2 \sin \theta + \sqrt{3} = 0$, (k) $\cos \theta + 1 = 0$, (l) $\sqrt{3} \tan \theta + 3 = 0$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q004 **Written**

Solve $\tan^2 \theta - (2 + \sqrt{3}) \tan \theta + 2\sqrt{3} = 0$ and give the general solution.

Q005 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \cos \theta > -\sqrt{3}$, (b) $\tan \theta \leq \frac{1}{\sqrt{3}}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q006 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin \theta \leq -1$, (b) $\cos \theta > \frac{1}{2}$, (c) $\tan \theta > 1$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q007 Written

Solve all given inequalities for $0 \leq \theta < 2\pi$: (a) $\sin \theta > \frac{\sqrt{3}}{2}$, (b) $\sin \theta - \frac{\sqrt{2}}{2} > 0$, (c) $2 \sin \theta + \sqrt{3} \leq 0$, (d) $\cos \theta > 0$, (e) $\cos \theta - \frac{1}{2} \leq 0$, (f) $2 \cos \theta + \sqrt{2} \leq 0$, (g) $\tan \theta \leq -\frac{1}{\sqrt{3}}$, (h) $\tan \theta + \sqrt{3} > 0$, (i) $\tan \theta + 1 \geq 0$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q008 Written

Solve $2 \sin^2 \theta - \cos \theta - 1 > 0$ **for** $0 \leq \theta < 2\pi$.

Q009 Written**Solve for $0 \leq \theta < 2\pi$: (a) $2 \cos^2 \theta - 3 \sin \theta = 0$, (b) $2 \sin^2 \theta - 3 \cos \theta - 3 \leq 0$.**

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q010 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta - \sin \theta = 0$, (b) $2 \cos^2 \theta - \sin \theta = 2$, (c) $2 \sin^2 \theta - 4 < 5 \cos \theta$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q011 Written

Solve all given equations and inequalities for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta + \sin \theta - 1 = 0$, (b) $2 \sin^2 \theta + 3 \cos \theta = 0$, (c) $2 \sin^2 \theta - \cos \theta = 2$, (d) $2 \cos^2 \theta - 5 \cos \theta - 3 = 0$, (e) $2 \cos^2 \theta + \sin \theta - 1 = 0$, (f) $2 \cos^2 \theta + 3 \sin \theta = 3$, (g) $2 \cos^2 \theta + 3 \sin \theta \leq 0$, (h) $2 \sin^2 \theta + \cos \theta > 1$, (i) $2 \cos^2 \theta - 4 \leq 5 \sin \theta$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q012 Written

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

Q013 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, (b) $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q014 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q015 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, (b) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, (c) $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, (d) $\tan(\theta + \frac{\pi}{4}) = -1$, (e) $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, (f) $\tan(\theta + \frac{\pi}{6}) = 1$, (g) $\sin(\theta + \frac{\pi}{6}) \leq \frac{\sqrt{3}}{2}$, (h) $\sin(\theta + \frac{\pi}{4}) \geq \frac{\sqrt{2}}{2}$, (i) $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q016 **Written****Solve** $4 \sin^2\left(\theta + \frac{\pi}{6}\right) \geq 1$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q017 Written

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2 \cos x}$. Evaluate this strategy and propose a better substitution on $[0^\circ, 360^\circ[$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q018 MCQ

The period used for a general tangent solution is:**Choose the correct option.**A 2π B π C $\pi/2$ D 4π

Q019 MCQ**The solutions of $2 \cos \theta = 1$ on $[0, 2\pi]$ are:****Choose the correct option.****A $\pi/6, 11\pi/6$** **B $\pi/3, 5\pi/3$** **C $2\pi/3, 4\pi/3$** **D $\pi/4, 7\pi/4$**

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q020 MCQ

Which root is impossible when solving $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$?

Choose the correct option.

A $\sin \theta = 1/2$

B $\sin \theta = -2$

C $\sin \theta = 0$

D $\sin \theta = -1/2$

Q021 MCQ

The quadratic challenge factors into roots:**Choose the correct option.**

- A 1 and $\sqrt{3}$
- B 2 and $\sqrt{3}$
- C 2 and 3
- D $\pi/3$ and 2

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q022 MCQ

For $2 \cos \theta > -\sqrt{3}$, the excluded arc is:

Choose the correct option.

A $[0, 5\pi/6]$

B $[5\pi/6, 7\pi/6]$

C $[7\pi/6, 2\pi)$

D $(\pi/6, 5\pi/6)$

Q023 MCQ**The inequality $\tan \theta > 1$ on $[0, 2\pi)$ holds on:****Choose the correct option.****A** $(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$ **B** $[0, \pi/4]$ **C** $(\pi/2, \pi)$ **D** $[\pi, 2\pi)$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q024 MCQ

A closed endpoint in a trig inequality is used when the sign is:

Choose the correct option.

- A strictly greater
- B strictly less
- C greater than or equal to
- D never

Q025 MCQ**The 5-14 challenge reduces to which cosine condition?****Choose the correct option.**

A $\cos \theta > 1/2$

B $-1 < \cos \theta < 1/2$

C $\cos \theta \leq -1$

D $\cos \theta \geq 1$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q026 MCQ

When a quadratic gives $\sin \theta = -2$, we:**Choose the correct option.**

A keep it

B reject it

C square it

D divide by 2

Q027 MCQ

The first step for $2 \cos^2 \theta - 3 \sin \theta = 0$ is to use:**Choose the correct option.**

A $\sin^2 \theta + \cos^2 \theta = 1$

B $\tan \theta = \sin \theta$

C $\cos \theta = 1$

D the tangent period

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q028 MCQ

The solution of $2 \sin^2 \theta + \sin \theta - 1 = 0$ includes:

Choose the correct option.

A $\theta = \pi/6$

B $\theta = \pi/4$

C $\theta = \pi/3$

D $\theta = 2\pi/3$

Q029 MCQ**In the rational challenge, which angle must be excluded?****Choose the correct option.****A** 0**B** $\pi/4$ **C** $\pi/2$ **D** π

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q030 MCQ

After setting $t = \theta + \pi/3$ with $0 \leq \theta < 2\pi$, the t-range is:

Choose the correct option.

A $[0, 2\pi)$

B $[\pi/3, 7\pi/3)$

C $[-\pi/3, 5\pi/3)$

D $[\pi, 3\pi)$

Q031 MCQ**The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:****Choose the correct option.****A $\pi/12$** **B $\pi/6$** **C $\pi/3$** **D $\pi/2$**

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q032 MCQ

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ -solution is:

Choose the correct option.

A $[0, \pi/2]$

B $(0, \pi/2)$

C $[\pi/2, \pi]$

D $[0, 2\pi)$

Q033 MCQ**The 5-16 challenge is equivalent to:****Choose the correct option.**

A $|\sin(t)| \geq 1/2$

B $\sin(t) = 0$

C $\cos(t) > 1/2$

D $\tan(t) = 1$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q034 MCQ

In the Reflect problem, $\tan x \cos x$ simplifies to:**Choose the correct option.**A $\cos(x)$ B $\sin(x)$ C $\tan^2 x$

D 1

Quiz MCQ 1 **Concept Quiz · MCQ**

The general solution of $\tan \theta = -1$ is:

Choose the correct option.

A $-\pi/4 + 2n\pi$

B $-\pi/4 + n\pi$

C $\pi/4 + 2n\pi$

D $\pi/2 + n\pi$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 2 **Concept Quiz · MCQ**

The solution of $\cos \theta > 0$ on $[0, 2\pi)$ is:

Choose the correct option.

A $(\pi/2, 3\pi/2)$

B $[0, \pi/2) \cup (3\pi/2, 2\pi)$

C $[0, \pi)$

D $(\pi, 2\pi)$

Quiz MCQ 3 Concept Quiz · MCQ

The valid sine root of $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$ is:

Choose the correct option.

A $\sin \theta = -2$

B $\sin \theta = 1/2$

C $\sin \theta = 2$

D $\sin \theta = -1/2$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 4 [Concept Quiz · MCQ](#)

The first operation in a shifted-argument problem is to:

Choose the correct option.

- A divide by $\cos \theta$
- B set t equal to the whole argument and find its range
- C change radians to degrees
- D square both sides

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $2 \cos \theta = \sqrt{3}$ for $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 6 **Concept Quiz · Written****Solve** $\tan \theta \leq -1$ **for** $0 \leq \theta < 2\pi$.

Quiz WRITTEN 7 Concept Quiz · Written

Solve $2 \sin^2 \theta + \sin \theta - 1 = 0$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 8 Concept Quiz · Written**Solve** $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ **for** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

- Q001** (a) $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$ (b) $\theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$ **Q012** $0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$ **Q020** B
- Q002** (a) $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ **Q013** (a) $\theta = \frac{\pi}{2}, \frac{11\pi}{6}$ (b) $0 \leq \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$ **Q021** B
- Q003** (a) $\frac{3\pi}{4}, \frac{11\pi}{4}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q014** (a) $\theta = 0, \frac{2\pi}{3}$ (b) $\theta = \pi, \frac{13\pi}{12}$ (c) $\frac{\pi}{2} < \theta < \pi$ **Q022** B
- Q004** $\tan \theta = 2$ or $\tan \theta = \sqrt{3}$; $\theta = \arctan(2) + n\pi$ or $\theta = \frac{\pi}{3} + n\pi$ **Q015** (a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, \frac{5\pi}{12}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q023** A
- Q005** (a) $0 \leq \theta < \frac{5\pi}{6}$ or $\frac{7\pi}{6} < \theta < 2\pi$ (b) $0 \leq \theta \leq \frac{\pi}{6}$ or $\frac{\pi}{2} < \theta \leq \frac{7\pi}{6}$ or $\frac{3\pi}{2} < \theta < 2\pi$ **Q016** $0 \leq \theta \leq \frac{2\pi}{3}$ or $\pi \leq \theta \leq \frac{5\pi}{3}$ **Q024** C
- Q006** (a) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (b) $0 \leq \theta < \frac{\pi}{3}$ or $\frac{5\pi}{3} < \theta < 2\pi$ (c) $\frac{\pi}{4} < \theta < \frac{\pi}{2}$ or $\frac{5\pi}{4} < \theta < \frac{3\pi}{2}$ **Q017** Dividing by $\cos(x)$ separates the interacting factors and is not helpful, use $\tan(x) \cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30^\circ, 150^\circ$. **Q025** B
- Q007** (a) $\frac{\pi}{2}, \frac{3\pi}{2}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{\pi}{2}, \frac{3\pi}{2}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q026** B
- Q008** $\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$ **Q027** A
- Q009** (a) $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ (b) $0 \leq \theta \leq \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \leq \theta < 2\pi$ **Q028** A
- Q010** (a) $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$ (b) $\theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $0 \leq \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ **Q029** C
- Q011** (a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (b) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (c) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (d) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (e) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (g) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (h) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (i) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (j) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (k) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (l) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (m) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (n) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (o) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (p) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (q) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (r) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (s) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (t) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (u) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (v) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (w) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (x) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (y) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (z) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ **Q030** B
- Q018** B **Q019** B **Q031** A

Q032 A

Q033 A

Q034 B

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Quiz WRITTEN 6 $\frac{\pi}{2} < \theta \leq \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$

Quiz WRITTEN 7 $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Quiz WRITTEN 8 $\frac{4\pi}{3} < \theta < 2\pi$

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Classified Practice

Advanced Theorems and Applications

Q001 Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Q002 Written

Find the maximum and minimum of $y = \cos^2\theta - \sin\theta$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Classified Practice

Advanced Theorems and Applications

Q003 Written

Solve the four given maximum/minimum problems on $0 \leq \theta < 2\pi$, including $y = \sin^2 \theta + \sin \theta - 1$, $y = \cos^2 \theta - \cos \theta + 1$, $y = 2 \sin \theta + \cos^2 \theta - 1$, and $y = -2 \sin^2 \theta - 2 \cos \theta + 1$. Include the given challenge $y = \cos^2 \theta - \sin \theta$.

Classified Practice

Advanced Theorems and Applications

Q004 **Written****Use addition theorems to find $\sin 75^\circ$ and $\cos(\pi/12)$.**

Q005 Written

Given $\sin \alpha = \frac{\sqrt{3}}{2}$ **in QII** and $\cos \beta = -\frac{3}{5}$ **in QIII**, **find** $\sin(\alpha + \beta)$ **and** $\cos(\alpha - \beta)$.

Classified Practice

Advanced Theorems and Applications

Q006 Written

Complete the given addition-theorem exercises *exact values, quadrant – controlled α, β , and the three – angle challenge* ($\sin \alpha = 0.6, \cos \beta = 0.8, \tan \gamma = 0.75$).

Q007 Written**Find $\tan 105^\circ$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.**

Classified Practice

Advanced Theorems and Applications

Q008 **Written**

Find the angle between each given pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

Classified Practice

Advanced Theorems and Applications

Q009 Written

Find the exact angles for all given line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan \theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y -axis.

Classified Practice

Advanced Theorems and Applications

Q010 Written

Given $\sin \alpha = 1/3$ with α in QII, find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Q011 Written

Given $\sin \alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Classified Practice

Advanced Theorems and Applications

Q012 **Written**

Solve the given double-angle exercises for given sine/cosine values and the challenge $\cos \theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Q013 Written

Given $\cos \alpha = 3/5$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\cos(5\pi/8)$.

Classified Practice

Advanced Theorems and Applications

Q014 **Written**

Given $\cos \alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Q015 Written

Solve the given half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}, 4\pi < A < 5\pi,$
 $\cos B = -\frac{\sqrt{5}}{3}, 5\pi < B < 6\pi.$

Classified Practice

Advanced Theorems and Applications

Q016 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin \theta = 0$, $b \cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q017 Written**Solve the given equation** $\sin 2\theta + \cos \theta = 0$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q018 **Written**

Solve all given double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q019 Written

Rewrite $\sin \theta - \sqrt{3} \cos \theta$ as $r \sin(\theta + \alpha)$, $r > 0$, $-\pi < \alpha < \pi$.

Classified Practice

Advanced Theorems and Applications

Q020 Written

Rewrite the **given** expressions $\sin \theta - \cos \theta$, $\sin \theta + \cos \theta$, $3 \sin \theta - \sqrt{3} \cos \theta$, and $3 \sin \theta - \cos \theta$ in **harmonic-addition form**.

Q021 Written

Rewrite all given harmonic-addition exercises in the form $r \sin(\theta + \alpha)$, including $2 \sin \theta + 2 \cos \theta$ and the challenge $\sin \theta - \sqrt{3} \cos \theta$.

Classified Practice

Advanced Theorems and Applications

Q022 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin \theta - \cos \theta = 1$, $b \sin \theta + \cos \theta < 3/2$.

Q023 Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q024 **Written**

Solve all given harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin \theta + \cos \theta + 1 = 0$ and $\sin \theta - \cos \theta + 3/2 > 0$.

Q025 Written

Find the maximum and minimum of $y = 2 \sin(\theta - \pi/3)$, and of $y = 2 \cos 2\theta + 4 \cos \theta + 1$, $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q026 Written

Find the maximum and minimum of the given functions $y = -\frac{4}{3} \sin^2 \theta + \cos \theta$ and $y = \cos 2\theta - 2 \cos \theta$.

Q027 Written

Solve all given maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Classified Practice

Advanced Theorems and Applications

Q028 MCQ

The substitution for $\sin^2 \theta + \cos \theta + 1$ should be:

Choose the correct option.

A $t = \sin \theta$

B $t = \cos \theta$

C $t = \tan \theta$

D $t = \theta$

Q029 MCQ**The maximum of $\sin^2 \theta + \cos \theta + 1$ is:****Choose the correct option.****A** 0**B** 1**C** 9/4**D** 4

Classified Practice

Advanced Theorems and Applications

Q030 MCQ

An extremal vertex is valid only if t lies in:**Choose the correct option.**A $[-2,2]$ B $[-1,1]$ C $[0,1]$

D all real numbers

Q031 MCQ

The identity for $\sin(a+b)$ is:**Choose the correct option.**

- A $\sin \alpha \cos \beta + \cos \alpha \sin \beta$
- B $\sin \alpha \sin \beta$
- C $\cos \alpha \cos \beta$
- D $\tan \alpha + \tan \beta$

Classified Practice

Advanced Theorems and Applications

Q032 MCQ

sin 75 degrees equals:**Choose the correct option.**A $(\sqrt{6}-\sqrt{2})/4$ B $(\sqrt{6}+\sqrt{2})/4$ C $1/2$ D $\sqrt{3}/2$

Q033 MCQ**A missing trig value is selected using:****Choose the correct option.**

- A the quadrant
- B the denominator
- C the period only
- D a graph scale

Classified Practice

Advanced Theorems and Applications

Q034 MCQ

The slope of $y=mx+k$ is:**Choose the correct option.**

A k

B m

C $1/m$ D $m+k$

Q035 MCQ**The angle formula uses denominator:****Choose the correct option.****A** $1+m_1m_2$ **B** m_1-m_2 **C** m_1+m_2 **D** m_1m_2

Classified Practice

Advanced Theorems and Applications

Q036 MCQ

For an acute angle, if $\tan \theta_1$ is negative we use:

Choose the correct option.

A θ_1

B $\pi - \theta_1$

C $2\theta_1$

D $-\theta_1$ only

Q037 MCQ **$\sin(2a)$ equals:****Choose the correct option.**

- A $\sin \alpha$
- B $2 \sin \alpha \cos \alpha$
- C $\cos^2 \alpha$
- D $\tan \alpha$

Classified Practice

Advanced Theorems and Applications

Q038 MCQ

 $\cos(2\alpha)$ can be written as:**Choose the correct option.**A $\cos^2 \alpha - \sin^2 \alpha$ B $2\sin \alpha$ C $\tan \alpha$ D $1 + \sin \alpha$

Q039 MCQ**For theta in QIV, sin theta is:****Choose the correct option.**

- A positive
- B negative
- C zero always
- D undefined

Classified Practice

Advanced Theorems and Applications

Q040 MCQ

The half-angle formula for $\sin(a/2)$ uses:**Choose the correct option.**

- A $(1 - \cos \alpha)/2$
- B $(1 + \cos \alpha)/2$
- C $\tan \alpha$
- D $\cos 2\alpha$

Q041 MCQ**The sign of a half-angle root is chosen from:****Choose the correct option.**

- A the original quadrant interval
- B the coefficient only
- C the denominator
- D the period of $\tan$

Classified Practice

Advanced Theorems and Applications

Q042 MCQ

If $\cos a = -1/3$ and $0 < a < \pi$, $\cos(a/2)$ is:

Choose the correct option.

A negative

B positive

C zero

D undefined

Q043 MCQ**For double-angle equations, first rewrite in:****Choose the correct option.**

- A one trig function
- B two unrelated angles
- C degrees only
- D a logarithm

Classified Practice

Advanced Theorems and Applications

Q044 MCQ

The factorisation of $\sin(2\theta) + \sin\theta$ is:**Choose the correct option.**

- A $\sin\theta(2\cos\theta + 1)$
- B $\cos\theta(2\sin\theta - 1)$
- C $\sin\theta + \cos\theta$
- D $2\sin\theta$

Q045 MCQ**A tangent asymptote is always:****Choose the correct option.**

- A included
- B excluded
- C a maximum
- D an endpoint

Classified Practice

Advanced Theorems and Applications

Q046 MCQ

For $a \sin \theta + b \cos \theta$, r is:**Choose the correct option.**A $a+b$ B $\sqrt{a^2+b^2}$ C $a-b$ D ab

Q047 MCQ**The harmonic target form is:****Choose the correct option.****A** $r \sin(\theta + \alpha)$ **B** $r \cos \theta$ **C** $a \tan \theta$ **D** $\sin \theta$ only

Classified Practice

Advanced Theorems and Applications

Q048 MCQ

The angle a is located from point:**Choose the correct option.**

- A $P(a,b)$
 - B $P(r,\theta)$
 - C the origin only
 - D the x intercept
-
-
-

Q049 MCQ**After synthesis, t is usually:****Choose the correct option.****A** $\theta + \alpha$ **B** θ^2 **C** $\tan \theta$ **D** $\cos \theta$

Classified Practice

Advanced Theorems and Applications

Q050 MCQ

If $\max(\sin \theta + \cos \theta) = \sqrt{2}$, then $\sin \theta + \cos \theta < \frac{3}{2}$ is:

Choose the correct option.

- A never true
- B always true
- C true only at π
- D undefined

Q051 MCQ**In a strict inequality, equality endpoints are:****Choose the correct option.**

- A included
- B excluded
- C doubled
- D random

Classified Practice

Advanced Theorems and Applications

Q052 MCQ

The range of $2\sin(\theta - \pi/3)$ is:**Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $[-3,3]$

Q053 MCQ**For a double-angle maximum problem, use:****Choose the correct option.**

- A the double-angle formula then a bounded variable
- B a logarithm
- C only a graph
- D no substitution

Classified Practice

Advanced Theorems and Applications

Q054 MCQ

A vertex outside $[-1,1]$ is:**Choose the correct option.**

A an allowed extremum

B rejected

C always the maximum

D always the minimum

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2 \sin(\theta - \pi/3)$ is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice

Advanced Theorems and Applications

Quiz MCQ 2 Concept Quiz · MCQ

The addition-theorem value $\sin 75^\circ$ is:

Choose the correct option.

A $(\sqrt{6} + \sqrt{2})/4$

B $(\sqrt{6} - \sqrt{2})/4$

C $1/2$

D 1

Quiz MCQ 3 Concept Quiz · MCQ

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice

Advanced Theorems and Applications

Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Choose the correct option.

A $\pi/6$

B $\pi/4$

C $\pi/3$

D $\pi/2$

Quiz WRITTEN 5 Concept Quiz · Written

Find the maximum and minimum of $y = \sin^2 \theta + \cos \theta + 1$ on $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$ when $\sin \alpha = 1/3$ and α is in QII.

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3 \sin \theta - \sqrt{3} \cos \theta$ **as** $r \sin(\theta + \alpha)$.

Classified Practice

Advanced Theorems and Applications

Quiz WRITTEN 8 Concept Quiz · Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

- Q001** $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q002** $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q003** See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.
- Q004** $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$
- Q005** $\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$
- Q006** See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.
- Q007** $\tan 105^\circ = -2 + \sqrt{3}$; $\theta = \frac{\pi}{4}$
- Q008** The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.
- Q009** See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.
- Q010** $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$
- Q011** $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$
- Q012** For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.
- Q013** $\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan \frac{\alpha}{2} = \frac{1}{2}$, $\cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$
- Q014** $\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan \frac{\alpha}{2} = \sqrt{2}$, $\sin \frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$
- Q015** Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.
- Q016** $a\theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $b\frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$
- Q017** $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$
- Q018** See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.
- Q019** $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$
- Q020** $\sqrt{2} \sin \theta - \frac{\pi}{4}$, $\sqrt{2} \sin \theta + \frac{\pi}{4}$, $2\sqrt{3} \sin \theta - \frac{\pi}{6}$, $\sqrt{10} \sin \theta - \arctan \frac{1}{3}$
- Q021** See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.
- Q022** $a\theta = \frac{\pi}{2}, \pi$; $b\ 0 \leq \theta < 2\pi$
- Q023** $0 \leq \theta \leq \frac{\pi}{4}$ or $\frac{5\pi}{4} \leq \theta < 2\pi$
- Q024** See the numbered solution; synthesize first, then solve in the shifted variable over its full range.
- Q040** A
- Q041** A
- Q042** B
- Q043** A
- Q044** A
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** A
- Q050** B
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Quiz MCQ 1** D
- Quiz MCQ 2** A
- Quiz MCQ 3** A
- Quiz MCQ 4** B
- Quiz WRITTEN 5** $\max = \frac{9}{4}, \min = 0$
- Quiz WRITTEN 6** $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$
- Quiz WRITTEN 7** $2\sqrt{3} \sin \theta - \frac{\pi}{6}$
- Quiz WRITTEN 8** $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$

Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

Classified Practice

Laws of Exponents and Roots

Q001 Written

Find: (a) 10^0 , (b) 2^{-3} , (c) $(0.2)^{-2}$.

Q002 Written

Let $a \neq 0, b \neq 0$. Simplify: (a) $a^{-5}a^8$, (b) $a^5 \div a^{-3}$, (c) $(a^{-1}b)^2$, (d) $(2^2)^3 \div 2^{-2} \times 2^{-3}$.

Classified Practice

Laws of Exponents and Roots

Q003 Written

Find: (a) 5^0 , (b) 3^{-2} , (c) $(0.25)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q004 Written

Simplify the eight given expressions: a^5a^4 , $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, **and** $2^4 \div (2^{-1})^3 \times 2^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q005 Written

Find the nine values: 3^0 , 11^0 , $(2/3)^0$, 4^{-3} , 5^{-3} , 6^{-2} , $(1/3)^{-3}$, $(3/5)^{-2}$, $(0.04)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q006 Written

Simplify the twelve same-base expressions in given Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^53^{-3}, 4^44^{-3}, 5^55^{-5}$.

Classified Practice

Laws of Exponents and Roots

Q007 Written

Simplify given Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^23^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^42^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, **and** $(a/2)^3a^{-4} \div (2a)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q008 Written

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2} (4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

Q009 Written**Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.**

Classified Practice

Laws of Exponents and Roots

Q010 Written

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[3]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Q011 Written**Find** $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Classified Practice

Laws of Exponents and Roots

Q012 Written

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, and $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Q013 Written

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

Classified Practice

Laws of Exponents and Roots

Q014 **Written**

Perform the fifteen **given** calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

Classified Practice

Laws of Exponents and Roots

Q015 Written

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

Q016 Written

Express in the form a^r : $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

Classified Practice

Laws of Exponents and Roots

Q017 Written

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.

Q018 Written

Express in the form a^r : $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.

Classified Practice

Laws of Exponents and Roots

Q019 Written

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, and $[(27/125)^{1/2}]^{-4/3}$.

Q020 Written

Express the nine given expressions in the form a^r , including roots of a and numerical bases 2, 3, 5.

Classified Practice

Laws of Exponents and Roots

Q021 Written

Evaluate the twelve given expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/25)^{1/3}]^{1/2}$, $[(1/8)^{1/2}]^{1/3}$, $[(1/27)^{1/3}]^{1/2}$

Classified Practice

Laws of Exponents and Roots

Q022 Written

Simplify $\left[\left(\left(1000/27\right)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

Q023 Written

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[6]{32} \times \sqrt[3]{16}$.

Classified Practice

Laws of Exponents and Roots

Q024 Written

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

Q025 Written

For $a > 0, b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

Classified Practice

Laws of Exponents and Roots

Q026 **Written**

Perform the twelve numerical calculations in given Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Q027 Written

For $a > 0, b > 0$, simplify the six given expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[5]{a}$ and mixed a,b radicals.

Classified Practice

Laws of Exponents and Roots

Q028 Written

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}} \right)^{-3/2}$.

Q029 Written

Assess the statement $\sqrt{a^2} = a$ for all real a , using positive and negative examples.

Classified Practice

Laws of Exponents and Roots

Q030 MCQ

The value of 2^{-3} is:**Choose the correct option.**

A 8

B $1/8$ C -8 D $1/6$

Q031 MCQ**When multiplying equal bases, we:****Choose the correct option.**

- A subtract exponents
- B add exponents
- C multiply bases only
- D take reciprocals

Classified Practice

Laws of Exponents and Roots

Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:

Choose the correct option.

A a^2

B a^8

C a^{-8}

D a^{15}

Q033 MCQ**A negative exponent means:****Choose the correct option.**

- A multiply by -1
- B take the reciprocal power
- C set the base to zero
- D add one

Classified Practice

Laws of Exponents and Roots

Q034 MCQ

The value of $(0.04)^{-2}$ is:**Choose the correct option.**

- A 25
- B 400
- C 625
- D 1600

Q035 MCQ

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:**Choose the correct option.**

A -2

B -1

C 2

D 6

Classified Practice

Laws of Exponents and Roots

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:**Choose the correct option.**

A a^7b^6

B $1/(a^7b^6)$

C $a^{-7}b^{-6}$

D $1/(a^6b^7)$

Q037 MCQ**The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:****Choose the correct option.****A** $1/3$ **B** 1**C** 3**D** 9

Classified Practice

Laws of Exponents and Roots

Q038 MCQ

 $\sqrt[3]{64}$ equals:

Choose the correct option.

A 2

B 4

C 8

D 16

Q039 MCQ $\sqrt[3]{12}\sqrt[3]{18}$ equals:**Choose the correct option.**

A 4

B 5

C 6

D 9

Classified Practice

Laws of Exponents and Roots

Q040 MCQ

The fourth root of $1/16$ is:**Choose the correct option.**A $1/2$ B $1/4$

C 2

D 4

Q041 MCQ

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:

Choose the correct option.

- A $\sqrt[3]{2}$
 - B $2\sqrt[3]{2}$
 - C $3\sqrt[3]{2}$
 - D 0
-
-
-

Classified Practice

Laws of Exponents and Roots

Q042 MCQ

 $\sqrt[4]{625}$ equals:

Choose the correct option.

A 4

B 5

C 5^2

D 25

Q043 MCQ**When combining roots with the same index, multiply:****Choose the correct option.**

- A the indices
- B the radicands
- C the exponents only
- D the signs only

Classified Practice

Laws of Exponents and Roots

Q044 MCQ

The roots challenge evaluates to:**Choose the correct option.**

A 0

B 1

C 2

D 4

Q045 MCQ

$\sqrt[5]{2^4}$ in rational-index form is:

Choose the correct option.

A $2^{4/5}$

B $2^{5/4}$

C 2^9

D $4^{1/5}$

Classified Practice

Laws of Exponents and Roots

Q046 MCQ

The value of $27^{2/3}$ is:**Choose the correct option.**

A 3

B 6

C 9

D 18

Q047 MCQ**A reciprocal square root is written as:****Choose the correct option.**

A $a^{1/2}$

B $a^{-1/2}$

C a^{-2}

D a^2

Classified Practice

Laws of Exponents and Roots

Q048 MCQ

The value of $81^{-3/4}$ is:**Choose the correct option.**

A 27

B $1/9$ C $1/27$ D $1/81$

Q049 MCQ

The exponent of a in $\sqrt[4]{a^3}$ is:

Choose the correct option.

- A $3/4$
- B $4/3$
- C $-3/4$
- D 3

Classified Practice

Laws of Exponents and Roots

Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:

Choose the correct option.

A $4/9$

B $9/4$

C $16/9$

D $27/64$

Q051 MCQ**The 6^{-3} challenge simplifies to:****Choose the correct option.**A $3/10$ B $10/3$ C $100/9$ D $9/100$

Classified Practice

Laws of Exponents and Roots

Q052 MCQ

 $2^{2/3} 2^{1/2} \div 2^{7/6}$ equals:

Choose the correct option.

A $1/2$

B 1

C 2

D 4

Q053 MCQ $4^{5/6} 2^{4/3}$ equals:**Choose the correct option.****A** 4**B** 6**C** 8**D** 16

Classified Practice

Laws of Exponents and Roots

Q054 MCQ

 $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:

Choose the correct option.

A $a^{1/3}$ B $a^{2/3}$ C $a^{5/3}$

D a

Q055 MCQ**The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:****Choose the correct option.**

A 2

B 3

C 6

D 12

Classified Practice

Laws of Exponents and Roots

Q056 MCQ

The given expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Choose the correct option.

- A integer powers
- B combining rational exponents
- C logarithms
- D trigonometry

Q057 MCQ**The printed decimal in the 6-4 challenge is:****Choose the correct option.**

- A 0.04
- B 0.004
- C 0.4
- D 4

Classified Practice

Laws of Exponents and Roots

Q058 MCQ

For real a , $\sqrt{a^2}$ equals:**Choose the correct option.**

- A a always
- B $-a$ always
- C $|a|$
- D 0

Quiz MCQ 1 **Concept Quiz · MCQ**

Which law gives $a^m a^n = a^{m+n}$?

Choose the correct option.

- A same-base multiplication
- B power of a power
- C reciprocal law
- D root product

Classified Practice

Laws of Exponents and Roots

Quiz MCQ 2 **Concept Quiz · MCQ**

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Choose the correct option.

- A 2
- B 5
- C 10
- D 125

Quiz MCQ 3 Concept Quiz · MCQ

The value of $64^{-3/2}$ is:

Choose the correct option.

A 512

B $1/64$

C $1/512$

D 64

Classified Practice

Laws of Exponents and Roots

Quiz MCQ 4 Concept Quiz · MCQ

For a negative real a , $\sqrt{a^2}$ is:

Choose the correct option.

A a

B $-a$

C 0

D a^2

Quiz WRITTEN 5 Concept Quiz · Written**Simplify** $a^{-3}a^7 \div a^2$.

Classified Practice

Laws of Exponents and Roots

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Quiz WRITTEN 7 Concept Quiz · Written**Evaluate** $125^{-2/3}$.

Classified Practice

Laws of Exponents and Roots

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $\sqrt{a^2} = a$ is not true for every real a .

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3, a^8, \frac{b^2}{a^2}, 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2}, 2^{4/5}, 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 1, 6, 6, 2

Q024 5, 8, 7, 9

Q025 $a^{1/3}, b\sqrt{a}$

Q026 $1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$

Q027 $a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 $1/25$

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Concept 2 | Exponential Functions and Their Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Exponential

$$y = ab^x$$

Growth

$$A = A_0(1 + r)^t$$

Decay

$$A = A_0(1 - r)^t$$

Classified Practice

Exponential Functions and Their Applications

Q001 Written

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

Q002 Written**Draw $y = 2^x$ and $y = (1/2)^x$.**

Classified Practice

Exponential Functions and Their Applications

Q003 Written

Draw the six given graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

Q004 Written**Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.**

Classified Practice

Exponential Functions and Their Applications

Q005 **Written**

Order the given sets 0.7^{-3} , 0.7^{-1} , 0.7^2 and $5/8$, $5/64$, $6/16$ using inequality signs.

Q006 Written

Order the three displayed Try sets: $5^0, 5^{-4}, 5^{3/2}$; $(1/3)^7, (1/3)^2, (1/3)^{-5}$; **and** $3^0, 3^2, 3^{-2}$.

Classified Practice

Exponential Functions and Their Applications

Q007 **Written**

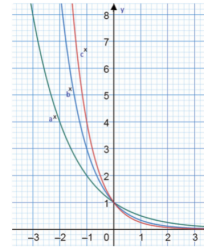
Order the given sets with bases 3, 2, $3/2$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

Classified Practice

Exponential Functions and Their Applications

Q008 Written

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a, b, c .



Classified Practice

Exponential Functions and Their Applications

Q009 Written

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

Q010 Written

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, and the reciprocal-base inequalities.

Classified Practice

Exponential Functions and Their Applications

Q011 **Written**

Solve all 24 given equations and inequalities in Exercise (a)–(x), including the 2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125 common-base forms.

Q012 Written

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

Classified Practice

Exponential Functions and Their Applications

Q013 Written

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

Q014 Written

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

Classified Practice

Exponential Functions and Their Applications

Q015 **Written**

Solve the twelve given items (a)–(l), each reducible to a quadratic in 2^x or 3^x .

Q016 Written

Solve completely for real x : $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0$.

Classified Practice

Exponential Functions and Their Applications

Q017 Written

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x -values.

Q018 Written

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x -values.

Classified Practice

Exponential Functions and Their Applications

Q019 Written

Solve the three given extrema models: $-3 \leq x \leq 2$, $y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3$, $y = -4 + 2^{2x} - 3^{x+2}$; and the third displayed interval/model.

Q020 Written**For $-1 \leq x \leq 2$, optimise $y = 2 \cdot 4^x - 2^{x+2} + 11$.**

Classified Practice

Exponential Functions and Their Applications

Q021 Written

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

Q022 MCQ

Which base gives a strictly decreasing exponential graph?**Choose the correct option.**

A 2

B 3

C $\frac{1}{2}$ D π

Classified Practice

Exponential Functions and Their Applications

Q023 MCQ

The horizontal asymptote of $y = a^x$ is:**Choose the correct option.**

A $x = 0$

B $y = 0$

C $y = 1$

D $x = 1$

Q024 MCQ

The range of $y = 5^x$ is:**Choose the correct option.**A $\mathbb{R}$ B $[0, \infty)$ C $(0, \infty)$ D $(-\infty, 0)$

Classified Practice

Exponential Functions and Their Applications

Q025 MCQ

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?

Choose the correct option.

A $(0, \pi)$

B $(1, \pi)$

C $(2, \pi)$

D $(-1, \pi)$

Q026 MCQ**For $0 < a < 1$ and $r < s$, which is true?****Choose the correct option.**

A $a^r < a^s$

B $a^r > a^s$

C $a^r = a^s$

D Cannot compare

Classified Practice

Exponential Functions and Their Applications

Q027 MCQ

The first step when bases differ is to:**Choose the correct option.**

- A add exponents
- B make the bases equal
- C take logarithms immediately
- D ignore the base

Q028 MCQ

Which is largest? 3^{-2} , 3^0 , 3^2 **Choose the correct option.**A 3^{-2} B 3^0 C 3^2

D They are equal

Classified Practice

Exponential Functions and Their Applications

Q029 MCQ

The challenge term $(\sqrt[3]{4})^3$ equals:

Choose the correct option.

A $2^{3/2}$

B 2^2

C 2^3

D 4^3

Q030 MCQ**Solving $4^{x-1} = 8^{x+3}$ gives:****Choose the correct option.**

A $x = -11$

B $x = -3$

C $x = 3$

D $x = 11$

Classified Practice

Exponential Functions and Their Applications

Q031 MCQ

For $0 < a < 1$, an exponential inequality requires:**Choose the correct option.**

- A the same exponent order
- B a reversed exponent order
- C no base conversion
- D squaring both sides

Q032 MCQ

The solution of $3^x = 81$ is:**Choose the correct option.**

A 2

B 3

C 4

D 5

Classified Practice

Exponential Functions and Their Applications

Q033 MCQ

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

Choose the correct option.

A $0 \leq x \leq 2$

B $x > 2$

C $x < -1$

D $x < 2$

Q034 MCQ

In substitution methods, $t = a^x$ must satisfy:

Choose the correct option.

A $t < 0$

B $t = 0$

C $t > 0$

D $t \in \mathbb{R}$

Classified Practice

Exponential Functions and Their Applications

Q035 MCQ

The positive root of $t^2 - 3t - 16 = 0$ is:**Choose the correct option.**

A $(3 - \sqrt{73})/2$

B $(3 + \sqrt{73})/2$

C 8

D 16

Q036 MCQ

The challenge quadratic in t is:**Choose the correct option.**

A $t^2 + 5t + 4$

B $t^2 - 5t + 4$

C $t^2 - 4t + 5$

D $t^2 + 4t - 5$

Classified Practice

Exponential Functions and Their Applications

Q037 MCQ

After solving an exponential quadratic, reject roots with:**Choose the correct option.**

A $t > 1$

B $t = 1$

C $t \leq 0$

D $t < 4$

Q038 MCQ**For extrema, the first required range is the range of:****Choose the correct option.****A** x **B** $t = a^x$ **C** a **D** y only

Classified Practice

Exponential Functions and Their Applications

Q039 MCQ

The minimum of $y = t^2 - 4t$ at $t = 2$ is:**Choose the correct option.**

A -8

B -4

C 0

D 4

Q040 MCQ**The C06-K02 challenge has minimum value:****Choose the correct option.****A** 9 at $x = 0$ **B** 9 at $x = 2$ **C** 27 at $x = 0$ **D** 27 at $x = 2$

Classified Practice

Exponential Functions and Their Applications

Q041 MCQ

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:**Choose the correct option.**

A 0

B 1

C 2

D It has no minimum

Q042 MCQ**At which value of x does $a^x + a^{-x}$ attain its minimum?****Choose the correct option.****A** $x = -1$ **B** $x = 0$ **C** $x = 1$ **D** It has no stationary point

Classified Practice

Exponential Functions and Their Applications

Quiz MCQ 1 Concept Quiz · MCQ

Which statement about $y = a^x$ is always true for $a > 0$, $a \neq 1$?

Choose the correct option.

- A range $(0, \infty)$
- B range all reals
- C vertical asymptote
- D decreasing for every a

Quiz MCQ 2 Concept Quiz · MCQ

For base 0.5, increasing the exponent makes the value:

Choose the correct option.

- A larger
- B smaller
- C unchanged
- D zero

Classified Practice

Exponential Functions and Their Applications

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

The positive-substitution condition is:

Choose the correct option.

A $t < 0$

B $t = 0$

C $t > 0$

D t can be any real

Quiz MCQ 4 Concept Quiz · MCQ

For $y = 2(t - 1)^2 + 9$, the minimum is:

Choose the correct option.

A 0

B 1

C 9

D 11

Classified Practice

Exponential Functions and Their Applications

Quiz WRITTEN 5 Concept Quiz · Written

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

Quiz WRITTEN 6 Concept Quiz · Written**Order** 2^{-3} , 2^0 , $2^{5/2}$.

Classified Practice

Exponential Functions and Their Applications

Quiz WRITTEN 7 Concept Quiz · Written

Solve $4^x - 5 \cdot 2^x + 4 = 0$.

Quiz WRITTEN 8 Concept Quiz · Written

Find the minimum of $a^x + a^{-x}$ for $a > 1$.

Answer key

Use the question number shown on each page.

- Q001** Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.
- Q002** 2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.
- Q003** Bases 4, 10, $3/2$: strictly increasing. Bases $2/3$, $1/4$, $1/10$: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.
- Q004** Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.
- Q005** $0.7^2 < 0.7^{-1} < 0.7^{-3}$; and $5/64 < 5/8 < 6/16$.
- Q006** $5^{-4} < 5^0 < 5^{3/2}$; $(1/3)^7 < (1/3)^2 < (1/3)^{-5}$; $3^{-2} < 3^0 < 3^2$.
- Q007** Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.
- Q008** Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.
- Q009** $x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.
- Q010** Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .
- Q011** Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.
- Q012** $x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.
- Q013** First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.
- Q014** Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.
- Q015** For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .
- Q016** Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.
- Q017** Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.
- Q018** Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.
- Q019** In each case set $t = a^x$, write the exact t -range, and compare the quadratic vertex and endpoints; report both extreme values and their x -locations.
- Q020** Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.
- Q021** h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.
- Q022** C
- Q023** B
- Q024** C
- Q025** B
- Q026** B
- Q027** B
- Q028** C
- Q029** B
- Q030** A
- Q031** B
- Q032** C
- Q033** B
- Q034** C
- Q035** B
- Q036** B
- Q037** C
- Q038** B
- Q039** B
- Q040** A
- Quiz MCQ 1** A
- Quiz MCQ 2** B
- Quiz MCQ 3** C
- Quiz MCQ 4** C
- Quiz WRITTEN 5** Decreasing; $D = \mathbb{R}, R = (0, \infty), y =$
- Quiz WRITTEN 6** $2^{-3} < 2^0 < 2^{5/2}$
- Quiz WRITTEN 7** $x = 0$ or $x = 2$
- Quiz WRITTEN 8** $2atx = 0$

Concept 3 | Introduction to Logarithms and Their Properties

English Student Edition • Focused formulas and guided practice

Formula opener

Definition

$$\log_b a = c \iff b^c = a$$

Product

$$\log_b(xy) = \log_b x + \log_b y$$

Power

$$\log_b(x^k) = k \log_b x$$

Classified Practice

Introduction to Logarithms and Their Properties

Q001 **Written**

Convert the given forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.

Q002 Written

Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.

Classified Practice

Introduction to Logarithms and Their Properties

Q003 **Written**

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

Q004 Written

If $\log_5 25 = x$, **evaluate** $\log_{1/2} x$ **without a calculator**.

Classified Practice

Introduction to Logarithms and Their Properties

Q005 Written

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1$, $\log_5 5$, $\log_2 8$, $\log_2 16$, $\log_5(1/5)$, $\log_7(49^{3/2})$.

Q006 Written

Find the six Try values $\log_5 1$, $\log_3 3$, $\log_4 4$, $\log_3 27$, $\log_2(1/4)$, $\log_5(1/125)$.

Classified Practice

Introduction to Logarithms and Their Properties

Q007 **Written****Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.**

Q008 Written

Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.

Classified Practice

Introduction to Logarithms and Their Properties

Q009 Written

Perform $\log_6 2 + \log_6 3$ **and** $\log_2 30 - \log_2 3 - \log_2 5$.

Q010 Written

Perform the six Try calculations using product, quotient, and power laws, including $4 \log_5 3 - 2 \log_5 15 - \log_5 45$.

Classified Practice

Introduction to Logarithms and Their Properties

Q011 Written

Perform all 18 given expressions (a)+(r), combining same-base logarithms exactly.

Q012 Written

Simplify the fractional expression $\log_5 75 + 3 \log_5 \sqrt{2} - 2 \log_5 \sqrt{6}$ to one reduced fraction.

Classified Practice

Introduction to Logarithms and Their Properties

Q013 Written

Given $\log_{10} 2 = a$ and $\log_{10} 3 = b$, express $\log_{10} 24$, $\log_{10}(27/4)$, $\log_{10} 45$, $\log_{10}(24/75)$ in terms of a, b .

Q014 Written

Given $\log_{10} 2 = a$, $\log_{10} 3 = b$, express the four Try arguments 72 , $3^8/10$, 15 , 12 in terms of a, b .

Classified Practice

Introduction to Logarithms and Their Properties

Q015 **Written**

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

Q016 Written

If $\log_c 3 = x$, $\log_c 2 = y$, **write** $\log_c 12$, $\log_c(3/8)$, $\log_c \sqrt{6}$, $\log_2 3$, $\log_{3^b}(2^b c)$ **in terms of** x, y .

Classified Practice

Introduction to Logarithms and Their Properties

Q017 Written

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

Q018 Written

Evaluate the five Try logarithms using a suitable common base.

Classified Practice**Introduction to Logarithms and Their Properties****Q019** Written**Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.**

Q020 Written**Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.**

Classified Practice

Introduction to Logarithms and Their Properties

Q021 Written

Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.

Q022 Written**Solve the four Try products of logarithms after making the bases the same.**

Classified Practice

Introduction to Logarithms and Their Properties

Q023 Written

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.

Q024 Written**Simplify** $\log_x(\sqrt{y}) \log_{y^2}(z^3) \log_{\sqrt{z}}(x^4)$ **for** $x > 1$.

Classified Practice

Introduction to Logarithms and Their Properties

Q025 Written

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.

Q026 Written

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.

Classified Practice

Introduction to Logarithms and Their Properties

Q027 Written

State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and $1/3$.

Q028 Written**Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.**

Classified Practice

Introduction to Logarithms and Their Properties

Q029 Written

For $f(x) = \log_5(x - 2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

Q030 Written

Order the displayed logarithmic sets, including $\log_2 3$, $\log_2 5$, $2\log_2(1/2)$, $\log_{1/2} 3$, $\log_{1/2} 5$.

Classified Practice

Introduction to Logarithms and Their Properties

Q031 **Written****Order the four Try sets of logarithms using monotonicity and change of base.**

Q032 Written**Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.**

Classified Practice

Introduction to Logarithms and Their Properties

Q033 Written

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

Q034 MCQ $\log_2 32$ equals:

Choose the correct option.

A 2

B 4

C 5

D 32

Classified Practice

Introduction to Logarithms and Their Properties

Q035 MCQ

 $\log_{1/2} 2$ equals:

Choose the correct option.

A -2

B -1

C 1

D 2

Q036 MCQ $\log_5(1/125)$ equals:**Choose the correct option.****A** -3**B** -2**C** 2**D** 3

Classified Practice

Introduction to Logarithms and Their Properties

Q037 MCQ

 $\log_7 49^{3/2}$ equals:

Choose the correct option.

A 1

B 2

C $3/2$ D $7/2$

Q038 MCQ

$\log_a M + \log_a N$ equals:

Choose the correct option.

A $\log_a(M + N)$

B $\log_a(MN)$

C $\log_a(M/N)$

D $\log_a(M - N)$

Classified Practice

Introduction to Logarithms and Their Properties

Q039 MCQ

The coefficient $3 \log_5 2$ becomes:**Choose the correct option.**

A $\log_5(2 + 3)$

B $\log_5 2^3$

C $\log_5(3^2)$

D $3^2 \log_5 2$

Q040 MCQ**If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:****Choose the correct option.**

- A $a + b$
 - B $2a + 3b$
 - C $3a + b$
 - D $3a + 2b$
-
-
-

Classified Practice

Introduction to Logarithms and Their Properties

Q041 MCQ

$\log_c \sqrt{6}$ in terms of $x = \log_c 3, y = \log_c 2$ is:

Choose the correct option.

A $x + y$

B $(x + y)/2$

C $2x + y$

D $x + 2y$

Q042 MCQ $\log_{16} 64$ equals:

Choose the correct option.

A $1/2$ B $3/2$

C 2

D 4

Classified Practice

Introduction to Logarithms and Their Properties

Q043 MCQ

Change of base gives $\log_a b =$:**Choose the correct option.**

A $\log_c a / \log_c b$

B $\log_c b / \log_c a$

C $\log_a b / \log_c b$

D $\log_c a \log_c b$

Q044 MCQ $\log_4 16 \cdot \log_9 3$ equals:

Choose the correct option.

A $1/2$

B 1

C 2

D 4

Classified Practice

Introduction to Logarithms and Their Properties

Q045 MCQ

The challenge product of three logarithms equals:**Choose the correct option.**

A 6

B 8

C 12

D 24

Q046 MCQ

The domain of $y = \log_a x$ is:**Choose the correct option.**

A all real x

B $x \geq 0$ C $x > 0$ D $x \neq 0$

Classified Practice

Introduction to Logarithms and Their Properties

Q047 MCQ

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

Choose the correct option.

A $x = 0$

B $y = 0$

C $y = x$

D $y = -x$

Q048 MCQ**For $a > 1$ and $M < N$, which is true?****Choose the correct option.**

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal**D** Cannot compare

Classified Practice

Introduction to Logarithms and Their Properties

Q049 MCQ

For $0 < a < 1$ and $M < N$, which is true?**Choose the correct option.**

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal

D Both are positive

Q050 MCQ

The vertical asymptote of $y = \log_a x$ is:**Choose the correct option.**

A $x = 0$

B $y = 0$

C $x = 1$

D $y = 1$

Classified Practice

Introduction to Logarithms and Their Properties

Q051 MCQ

A negative coefficient $-2 \log_a M$ can be written as:

Choose the correct option.

- A $\log_a M^{-2}$
- B $\log_a(-2M)$
- C $\log_a(M - 2)$
- D $2 \log_a M$

Q052 MCQ

The base of $\log_a M$ is:**Choose the correct option.**A M B a C p D $1/a$

Classified Practice

Introduction to Logarithms and Their Properties

Q053 MCQ

The argument of $\log_a M$ is:**Choose the correct option.**A a B M C p D 0

Q054 MCQ $\log_2(1/8)$ equals:

Choose the correct option.

A -3

B -2

C 2

D 3

Classified Practice

Introduction to Logarithms and Their Properties

Q055 MCQ

 $\log_4 4$ equals:

Choose the correct option.

A 0

B 1

C 2

D 4

Q056 MCQ $\log_3 1$ **equals:****Choose the correct option.****A** -1**B** 0**C** 1**D** 3

Classified Practice

Introduction to Logarithms and Their Properties

Q057 MCQ

 $\log_a M - \log_a N$ equals:

Choose the correct option.

A $\log_a(MN)$

B $\log_a(M/N)$

C $\log_a(M - N)$

D $\log_a(N/M)$

Q058 MCQ **$2 \log_3 5$ equals:****Choose the correct option.****A** $\log_3 10$ **B** $\log_3 25$ **C** $\log_3 7$ **D** $\log_3(5/2)$

Classified Practice

Introduction to Logarithms and Their Properties

Q059 MCQ

If $a = \log_{10} 2$, $b = \log_{10} 3$, $\log_{10} 12$ is:

Choose the correct option.

A $a + b$

B $2a + b$

C $a + 2b$

D $2a + 2b$

Q060 MCQ

$\log_{10} 5$ in terms of $a = \log_{10} 2$ is:

Choose the correct option.

A a

B $1 - a$

C $1 + a$

D $-a$

Classified Practice

Introduction to Logarithms and Their Properties

Q061 MCQ

 $\log_8 32$ equals:

Choose the correct option.

A $2/3$ B $3/2$ C $5/3$

D 3

Q062 MCQ

$\log_{25} 125$ equals:

Choose the correct option.

A $1/2$

B $3/2$

C 2

D $5/2$

Classified Practice

Introduction to Logarithms and Their Properties

Q063 MCQ

 $\log_a b \cdot \log_b a$ equals:

Choose the correct option.

A 0

B 1

C 2

D $\log_a b$

Q064 MCQ

$\log_2 8 \cdot \log_8 2$ equals:

Choose the correct option.

A $1/3$

B 1

C 3

D 8

Classified Practice

Introduction to Logarithms and Their Properties

Q065 MCQ

The best common base for 16 and 64 is:**Choose the correct option.**

A 2

B 4

C 8

D 10

Q066 MCQ

The range of $y = \log_a x$ is:**Choose the correct option.**A $x > 0$ B $\mathbb{R}$ C $[0, \infty)$ D $y > 0$

Classified Practice

Introduction to Logarithms and Their Properties

Quiz MCQ 1 **Concept Quiz · MCQ**

The logarithmic equivalent of $a^p = M$ is:

Choose the correct option.

A $\log_a M = p$

B $\log_p M = a$

C $\log_M a = p$

D $\log_a p = M$

Quiz MCQ 2 Concept Quiz · MCQ

$\log_2 8 + \log_2 4$ equals:

Choose the correct option.

A 2

B 3

C 5

D 12

Classified Practice

Introduction to Logarithms and Their Properties

Quiz MCQ 3 **Concept Quiz · MCQ**

$\log_{16} 64$ equals:

Choose the correct option.

A $1/2$

B $3/2$

C 2

D 4

Quiz MCQ 4 **Concept Quiz · MCQ**

For $a > 1$, $M < N$ implies:

Choose the correct option.

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C equality

D no relation

Classified Practice

Introduction to Logarithms and Their Properties

Quiz WRITTEN 5 Concept Quiz · Written

Find $\log_3(1/27)$.

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\log_5 25 + \log_5 5$.

Classified Practice

Introduction to Logarithms and Their Properties

Quiz WRITTEN 7 **Concept Quiz · Written**

Find $\log_{16} 64$ using change of base.

Quiz WRITTEN 8 Concept Quiz · Written

State the domain and range of $y = \log_a x$.

Answer key

Use the question number shown on each page.

- Q001** $\log_2 32 = 5$; $3^2 = 9$
- Q002** $\log_2 64 = 6$; $\log_4(1/16) = -2$, with the same swap rule for the remaining items.
- Q003** Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.
- Q004** $x = 2$, so $\log_{1/2} 2 = -1$.
- Q005** Values: 0, 1, 3, 4, -1, 3/2.
- Q006** 0, 1, 1, 3, -2, -3.
- Q007** Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.
- Q008** 2/3.
- Q009** 1 and 1.
- Q010** Each expression reduces to one logarithm and then an integer; the displayed worked example gives -3.
- Q011** Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.
- Q012** 1.
- Q013** $3a + b$, $2 - 2a + 3b$, $2a + 1 - a$, $b + 2 - 2a$.
- Q014** Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.
- Q015** Each result is a linear combination of $a, b, 1$ obtained from prime factors.
- Q016** $x + 2y$, $x - 3y$, $(x + y)/2$, x/y , $(1 + by)/(bx)$.
- Q017** 3/2 and -3.
- Q018** Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.
- Q019** All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.
- Q020** $\frac{\log_2 3 - 5/2}{2 \log_2 3}$.
- Q021** Use reciprocal and power identities; the worked values are exact constants.
- Q022** Each product collapses to a rational number after change of base and cancellation.
- Q023** Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.
- Q024** 12.
- Q025** They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.
- Q026** Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.
- Q027** Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.
- Q028** Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.
- Q029** They are inverse-reflected curves with symmetry line $y = x + 1$.
- Q030** Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.
- Q031** Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.
- Q032** Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.
- Q033** Use the same argument comparison; the relation reverses when the base crosses 1.
- Q040** C
- Q041** B
- Q042** B
- Q043** B
- Q044** A
- Q045** C
- Q046** C
- Q047** C
- Q048** B
- Q049** A
- Q050** A
- Q051** A
- Q052** B
- Q053** B
- Q054** A
- Q055** B
- Q056** B
- Q057** B
- Q058** B
- Q059** B
- Q060** B
- Q061** B
- Q062** B
- Q063** B
- Q064** B
- Q065** A

Q066 B**Quiz MCQ 1 A****Quiz MCQ 2 C****Quiz MCQ 3 B****Quiz MCQ 4 B****Quiz WRITTEN 5 -3** **Quiz WRITTEN 6 3****Quiz WRITTEN 7 $3/2$** **Quiz WRITTEN 8** Domain $x > 0$; range all real numbers

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

Classified Practice

Applications of Logarithms

Q001 **Written****Solve** $\log_2 x = 4$.

Q002 Written**Solve** $\log_2(2x - 1) = \log_2 3$.

Classified Practice

Applications of Logarithms

Q003 **Written**

Solve all 12 given logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

Q004 Written**Solve** $\log_3(x + 2) - \log_9((x - 2)^2) = 1$ **completely.**

Classified Practice

Applications of Logarithms

Q005 Written

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x + 2) + \log_{1/2}(x - 5) > -3$.

Q006 Written

Solve the six given logarithmic inequalities using base monotonicity and argument conditions.

Classified Practice

Applications of Logarithms

Q007 **Written**

Solve all 24 given logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Q008 Written

Solve the given challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

Classified Practice

Applications of Logarithms

Q009 Written

Solve $5(\log_2 x)^2 - 16 \log_2 x + 3 < 0$.

Q010 Written

Solve the three given substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.

Classified Practice

Applications of Logarithms

Q011 Written

Solve the **given** groups: three equations, three inequalities, and three exact logarithmic values.

Q012 Written**Solve** $(\log_3 x)^2 - \log_3 x - 6 < 0$ **for the positive wave-intensity multiplier** x .

Classified Practice

Applications of Logarithms

Q013 Written

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

Q014 Written

Find the maximum and minimum of the given logarithmic quadratic on its stated interval, with the corresponding values of x .

Classified Practice

Applications of Logarithms

Q015 **Written**

Solve the three given extrema problems, each with a different base and interval.

Q016 Written

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

Classified Practice

Applications of Logarithms

Q017 **Written**

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Q018 Written

Complete the given digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

Classified Practice

Applications of Logarithms

Q019 Written

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

Q020 Written**Solve** $\log_x(x + 2) \leq \log_x(4 - x)$ **for all real** x .

Classified Practice

Applications of Logarithms

Q021 MCQ

The argument condition for $\log_3(x - 5)$ is:**Choose the correct option.**A $x < 5$ B $x \geq 5$ C $x > 5$

D all real x

Q022 MCQ**If $\log_2(2x - 1) = \log_2 3$, then $x =$:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Applications of Logarithms

Q023 MCQ

The valid solution of the challenge equation is:**Choose the correct option.**

A 1

B 2

C 4

D 8

Q024 MCQ**For same-base logarithms, equality of logs gives:****Choose the correct option.**

- A arguments equal
- B bases equal
- C arguments add
- D no result

Classified Practice

Applications of Logarithms

Q025 MCQ

For base $a > 1$, $\log_a M < \log_a N$ means:

Choose the correct option.

A $M > N$

B $M < N$

C $M = N$

D $MN < 0$

Q026 MCQ**For base $0 < a < 1$, the inequality direction is:****Choose the correct option.**

- A preserved
- B reversed
- C removed
- D squared

Classified Practice

Applications of Logarithms

Q027 MCQ

The domain of $\log_{1/2}(x - 5)$ requires:**Choose the correct option.**A $x < 5$ B $x \leq 5$ C $x > 5$ D $x \neq 5$

Q028 MCQ

The warm-up intersection is:**Choose the correct option.**

A $0 < x < 5$

B $5 < x < 6$

C $x > 6$

D $-3 < x < 6$

Classified Practice

Applications of Logarithms

Q029 MCQ

The best substitution for a quadratic logarithmic equation is:**Choose the correct option.**

A $t = x^2$

B $t = \log_a x$

C $t = a^x$

D $t = 1/x$

Q030 MCQ**The inequality $5t^2 - 16t + 3 < 0$ gives:****Choose the correct option.****A** $t < 1/5$ **B** $1/5 < t < 3$ **C** $t > 3$ **D** $0 < t < 1$

Classified Practice

Applications of Logarithms

Q031 MCQ

The substitution variable must satisfy:**Choose the correct option.**A $t > 0$ B $t \in \mathbb{R}$ C $t < 0$ D $t = 1$

Q032 MCQ**The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:****Choose the correct option.****A** $x < 1/9$ **B** $1/9 < x < 27$ **C** $x > 27$ **D** $0 < x < 3$

Classified Practice

Applications of Logarithms

Q033 MCQ

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

Choose the correct option.

A $t = 0$

B $t = 1$

C $t = 2$

D $t = 3$

Q034 MCQ

The correct first step for extrema is:**Choose the correct option.**

- A differentiate in x
- B set $t = \log_a x$
- C ignore the interval
- D take $x = 0$

Classified Practice

Applications of Logarithms

Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:**Choose the correct option.**A $[-1,3]$ B $[0,3]$ C $[1,27]$

D all real

Q036 MCQ**The water-tank challenge maximum health rating is:****Choose the correct option.****A** 3**B** 4**C** 5**D** 8

Classified Practice

Applications of Logarithms

Q037 MCQ

The number of digits of N is determined by:**Choose the correct option.**

A $\log_{10} N$

B $\lfloor \log_{10} N \rfloor + 1$

C $N+1$

D 10^N

Q038 MCQ**If $-9 < \log_{10} N < -8$, the first non-zero digit is at:****Choose the correct option.**

- A 8th
- B 9th
- C 10th
- D 11th

Classified Practice

Applications of Logarithms

Q039 MCQ

The common logarithm base is:**Choose the correct option.**

A 2

B e

C 10

D any base

Q040 MCQ**The solution set of $\log_x(x + 2) \leq \log_x(4 - x)$ is:****Choose the correct option.****A** (0,1)**B** (1,4)**C** (0,4)**D** empty

Classified Practice

Applications of Logarithms

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $\log_5(x - 1) = 2$.

Choose the correct option.

A $x = 26$

B 0

C 1

D cannot be determined

Quiz MCQ 2 Concept Quiz · MCQ

For $0 < a < 1$ and $M < N$, compare $\log_a M$ and $\log_a N$.

Choose the correct option.

A $\log_a M > \log_a N$

B 0

C 1

D cannot be determined

Classified Practice

Applications of Logarithms

Quiz MCQ 3 **Concept Quiz · MCQ****Solve** $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.**Choose the correct option.**A $x = 2, \sqrt{2}$

B 0

C 1

D cannot be determined

Quiz MCQ 4 Concept Quiz · MCQ

Find the maximum of $y = -(\log_3 x)^2 + 2 \log_3 x + 4$ on $1 \leq x \leq 27$.

Choose the correct option.

A 5 at $x = 3$

B 0

C 1

D cannot be determined

Classified Practice

Applications of Logarithms

Quiz WRITTEN 5 Concept Quiz · Written

Solve $\log_3(x + 1) = 2$.

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\log_2 x \geq 3$.

Classified Practice

Applications of Logarithms

Quiz WRITTEN 7 Concept Quiz · Written

Solve $(\log_5 x)^2 - 4 < 0$.

Quiz WRITTEN 8 Concept Quiz · Written

How many digits does 6^{20} have using $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$?

Answer key

Use the question number shown on each page.

- Q001 16
- Q002 $x = 2$
- Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.
- Q004 $x = 4$
- Q005 $0 < x < 81$; $5 < x < 6$
- Q006 Each solution is the intersection of the argument domain with the monotonicity interval.
- Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.
- Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.
- Q009 $2^{1/5} < x < 8$
- Q010 Substitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.
- Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.
- Q012 $1/9 < x < 27$
- Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.
- Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.
- Q015 For every item, map the domain to t , locate the vertex, and check endpoints.
- Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.
- Q017 24 digits; 9th decimal place.
- Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.
- Q019 55 digits; leading digit 1.
- Q020 No real solution.
- Q021 C
- Q022 B
- Q023 C
- Q024 A
- Q025 B
- Q026 B
- Q027 C
- Q028 B
- Q029 B
- Q030 B
- Q031 B
- Q032 B
- Q033 C
- Q034 B
- Q035 B
- Q036 C
- Q037 B
- Q038 B
- Q039 C
- Q040 D
- Quiz MCQ 1 A
- Quiz MCQ 2 B
- Quiz MCQ 3 C
- Quiz MCQ 4 D
- Quiz WRITTEN 5 $x = 8$
- Quiz WRITTEN 6 $x \geq 8$
- Quiz WRITTEN 7 $1/25 < x < 25$
- Quiz WRITTEN 8 16 digits

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

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Formula opener

Derivative

$$\frac{d}{dx} x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q001 Written

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3 , and from 3 to $3 + h$.

Q002 Written**Find the average rate of change for the three given Try functions on their stated intervals.**

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q003 **Written**

Find the average rate of change for all nine given Exercise functions, including fixed and h-form intervals.

Q004 Written

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q005 Written

Find the **given** limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Q006 Written**Find the three given Try limits, including polynomial and h-quotient forms.**

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q007 **Written****Find all nine given Exercise limits, including x-limits and h-limits.**

Q008 Written

If $\lim_{x \rightarrow 3} (x^2 - 5x + k)/(x - 3) = L$ is finite, find Lk .

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q009 Written

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Q010 Written

Find the differential coefficients at the given values for the two given Try functions using the definition.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q011 Written

Find the differential coefficients for all six given Exercise functions at their stated values.

Q012 Written

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q013 Written

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.

Q014 Written**Differentiate the three given Try functions using the definition.**

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q015 **Written****Differentiate all nine given Exercise functions using the definition.**

Q016 Written

A regular hexagon has side length x . Write its area and differentiate it using the definition.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q017 Written

If $f(x) = 5$ for every x except that it is undefined at $x=2$, does $\lim_{x \rightarrow 2} f(x)$ exist?

Q018 MCQ**The average rate of change from a to b is:****Choose the correct option.**

A $[f(a)+f(b)]/(b-a)$

B $[f(b)-f(a)]/(b-a)$

C $f'(a)$

D $b-a$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q019 MCQ

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Choose the correct option.

A 2

B 3

C 4

D 8

Q020 MCQ**The average area-rate for diameter 10 to 18 is:****Choose the correct option.**A 7π B 14π C 28π D 56π

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q021 MCQ

The h-form average rate is found by:**Choose the correct option.**

- A setting $h = 1$
- B cancelling the common h
- C differentiating first
- D squaring h

Q022 MCQ**A direct-substitution limit is valid when:****Choose the correct option.**

- A the expression is defined
- B the denominator is zero
- C the numerator is infinite
- D h is negative

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q023 MCQ

The 0/0 form requires:**Choose the correct option.**

- A factor/cancel first
- B differentiate twice
- C ignore the denominator
- D change the base

Q024 MCQ

In the finite-limit challenge, k equals:**Choose the correct option.**

A 3

B 5

C 6

D 9

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q025 MCQ

For a polynomial, the limit equals:**Choose the correct option.**

- A its value at the point
- B zero
- C infinity
- D the slope

Q026 MCQ

The definition of the differential coefficient at a is:**Choose the correct option.**

A $\lim [f(a+h)-f(a)]/h$

B $f(a+h)+f(a)$

C $[f(a)-f(0)]/a$

D $h/f(a)$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:**Choose the correct option.**

A -12

B -6

C 6

D 12

Q028 MCQ**The stone's velocity at 10 s is:****Choose the correct option.**

- A 49
 - B 98
 - C 196
 - D 4.9
-
-
-

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q029 MCQ

After cancelling h , take the limit as:**Choose the correct option.**A $h \rightarrow 0$ B $h \rightarrow 1$ C $x \rightarrow 0$ D $a \rightarrow \infty$

Q030 MCQ**The derivative function is:****Choose the correct option.**

- A the coefficient at one point
- B the differential coefficient for every x
- C the average rate only
- D the original function

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q031 MCQ

The derivative of $x^2 + 3x$ from the definition is:**Choose the correct option.**A $x+3$ B $2x+3$ C $2x$ D x^2

Q032 MCQ

The area of a regular hexagon of side x is:**Choose the correct option.**

- A $3\sqrt{3}x$
 - B $(3\sqrt{3}/2)x^2$
 - C $6x^2$
 - D $\sqrt{3}x$
-
-
-

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q033 MCQ

A function need not be defined at a for its limit to:**Choose the correct option.**

A exist

B be infinite only

C equal zero only

D fail

Q034 MCQ

The derivative of a constant function is:**Choose the correct option.**

A 0

B 1

C x

D the constant

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Find the average rate of change of x^2 from 1 to 3.

Choose the correct option.

A 4

B 1

C 2

D 3

Quiz MCQ 2 Concept Quiz · MCQ**Find** $\lim_{x \rightarrow 2}(x^2 + 1)$.**Choose the correct option.****A** 0**B** 5**C** 2**D** 3

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)**Find $f'(x)$ for $f(x) = x^2$ using the definition.****Choose the correct option.**

A 0

B 1

C $2x$

D 3

Quiz MCQ 4 [Concept Quiz · MCQ](#)

If $s(t) = 4.9t^2$, find the velocity at $t=5$.

Choose the correct option.

A 0

B 1

C 2

D 49

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 5 Concept Quiz · Written

Find the average rate of change of $3x + 1$ from 2 to 7.

Quiz WRITTEN 6 Concept Quiz · Written**Evaluate** $\lim_{h \rightarrow 0} (6 + 2h)$.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 7 Concept Quiz · Written

Find the differential coefficient of $2x^2$ at $x=3$ from the definition.

Quiz WRITTEN 8 Concept Quiz · Written

State $\lim_{x \rightarrow 2} f(x)$ when $f(x)=5$ for all x except $x=2$ where it is undefined.

Answer key

Use the question number shown on each page.

Q001 4; $8 + h$	Q009 $f'(-2) = -12$	Q017 Yes; the limit is 5.	Q030 B
Q002 2; $-5/3$; $h - 6$	Q010 Substitute a into $[f(a + h) - f(a)]/h$	Q018 B	Q031 B
Q003 Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.	cancel h , and let h tend to zero.	Q019 C	Q032 B
Q004 $7\pi \text{ cm}^2/\text{cm}$	Q011 Use the definition at each specified point and simplify exactly.	Q020 A	Q033 A
Q005 -2 ; 6	Q012 98 m s^{-1}	Q021 B	Q034 A
Q006 Substitute after removing any common factor; polynomial limits are obtained directly.	Q013 $f'(x) = 2x + 3$	Q022 A	Quiz MCQ 1 A
Q007 Each limit is obtained by direct substitution or cancellation of the factor causing $0/0$.	Q014 Expand each $f(x + h) - f(x)$, cancel h and take the limit.	Q023 A	Quiz MCQ 2 B
Q008 6	Q015 Apply $f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)]/h$ to every printed polynomial.	Q024 C	Quiz MCQ 3 C
	Q016 $A = \frac{3\sqrt{3}}{2}x^2$, $A'(x) = 3\sqrt{3}x$	Q025 A	Quiz MCQ 4 D
		Q026 A	Quiz WRITTEN 5 3
		Q027 A	Quiz WRITTEN 6 6
		Q028 B	Quiz WRITTEN 7 12
		Q029 A	Quiz WRITTEN 8 5

Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

Classified Practice

Differentiation and Tangent Lines

Q001 Written

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r), and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

Q002 Written

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2)$, $f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Classified Practice

Differentiation and Tangent Lines

Q003 **Written**

Differentiate every given Exercise function 18 polynomial/product forms, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.

Classified Practice

Differentiation and Tangent Lines

Q004 Written

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$, find $f'(1) + f'(-1)$.

Q005 Written

Find the quadratic $f(x) = ax^2 + bx + c$ satisfying $f'(0) = -2$, $f'(2) = 2$, $f(1) = 2$.

Classified Practice

Differentiation and Tangent Lines

Q006 Written

Find the quadratic satisfying $f'(0) = 2$, $f'(1) = 4$, $f(2) = 6$.

Q007 Written

Find the four quadratics satisfying the four given triples of derivative and function conditions.

Classified Practice

Differentiation and Tangent Lines

Q008 Written

Find the quadratic $(f(x))$ satisfying $f'(2) + f'(4) = 0$, $f(3) = 0$, $f(1) = 12$.

Q009 Written**Find the tangent line to $y = x^2 - 4x + 1$ at $(3, -2)$.**

Classified Practice

Differentiation and Tangent Lines

Q010 Written

Find the tangent line for the three given functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at $((1, 0))$, and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

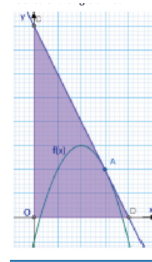
Q011 Written**Find the tangent line for all nine given Exercise functions at their given points.**

Classified Practice

Differentiation and Tangent Lines

Q012 **Written**

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Classified Practice

Differentiation and Tangent Lines

Q013 Written

Solve the two given Warm Up demands: tangents from $(C(0,2))$ to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Q014 Written

Solve the two given Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

Classified Practice

Differentiation and Tangent Lines

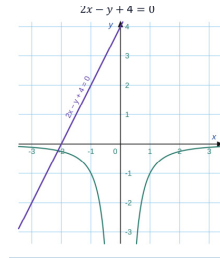
Q015 **Written****Solve all six given** Exercise demands involving external points or prescribed gradients.

Classified Practice

Differentiation and Tangent Lines

Q016 Written

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Classified Practice

Differentiation and Tangent Lines

Q017 MCQ

The derivative of x^n is:**Choose the correct option.**

A x^{n-1}

B nx^{n-1}

C $n + x$

D x^n

Q018 MCQ

The derivative of $V = \pi r^2 h$ with respect to r is:

Choose the correct option.

A πh

B $2\pi r h$

C $2\pi r$

D πr^2

Classified Practice

Differentiation and Tangent Lines

Q019 MCQ

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:**Choose the correct option.**

A -6

B -3

C 0

D 3

Q020 MCQ**The challenge function can be written as:****Choose the correct option.**

A $x^{5/3} + x^{-2}$

B $x^5 + x^2$

C $x^{3/5} + x^2$

D $x^{5/3} + x^2$

Classified Practice

Differentiation and Tangent Lines

Q021 MCQ

For $f(x) = ax^2 + bx + c$, $(f'(x))$ is:

Choose the correct option.

A $ax + b$

B $2ax + b$

C $2a + b + c$

D $ax^2 + b$

Q022 MCQ**The condition $f'(0) = -2$ gives:****Choose the correct option.**

A $a = -2$

B $b = -2$

C $c = -2$

D $2a = -2$

Classified Practice

Differentiation and Tangent Lines

Q023 MCQ

The warm-up quadratic is:**Choose the correct option.**

A $x^2 - 2x + 3$

B $x^2 + 2x + 3$

C $2x^2 - x + 3$

D $x^2 - 2x - 3$

Q024 MCQ**The challenge coefficient a is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 6

Classified Practice

Differentiation and Tangent Lines

Q025 MCQ

The tangent gradient at $x=a$ is:**Choose the correct option.**A $f(a)$ B a C $f'(a)$ D $1/f'(a)$

Q026 MCQ

The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:**Choose the correct option.**

A $y = x - 5$

B $y = 2x - 8$

C $y = -2x + 4$

D $y = 3x - 11$

Classified Practice

Differentiation and Tangent Lines

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:**Choose the correct option.**

A 2

B 3

C 4

D 8

Q028 MCQ**The area of triangle (ODC) in the challenge is:****Choose the correct option.****A** 8**B** 12**C** 16**D** 32

Classified Practice

Differentiation and Tangent Lines

Q029 MCQ

When the contact point is not given, set it to:**Choose the correct option.**A $((0,0))$ B $((a,f(a)))$ C $((f(a),a))$ D $((1,f'(a)))$

Q030 MCQ**The gradient of $2x - y + 4 = 0$ is:****Choose the correct option.****A** -2**B** -1**C** 1**D** 2

Classified Practice

Differentiation and Tangent Lines

Q031 MCQ

For $f(x) = -1/x^2$, $(f'(x))$ is:**Choose the correct option.**

A $-2/x^3$

B $2/x^3$

C $1/x^2$

D $-1/x^3$

Q032 MCQ

The parallel tangent in the challenge is:**Choose the correct option.**

A $y = 2x + 3$

B $y = 2x - 3$

C $y = -2x - 3$

D $y = x - 2$

Classified Practice

Differentiation and Tangent Lines

Quiz WRITTEN 1 Concept Quiz · Written

Differentiate $3x^3 - 2x$.

Quiz WRITTEN 2 Concept Quiz · Written

Find the quadratic with $f'(0) = 1, f'(1) = 3, f(1) = 2$.

Classified Practice

Differentiation and Tangent Lines

Quiz WRITTEN 3 Concept Quiz · Written

Find the tangent to $y = x^2$ at $((2,4))$.

Quiz WRITTEN 4 Concept Quiz · Written

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Classified Practice

Differentiation and Tangent Lines

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

The derivative of a constant is:

Choose the correct option.

A 0

B 1

C x

D the constant

Quiz MCQ 6 Concept Quiz · MCQ

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Choose the correct option.

- A three function values
- B two derivatives and one function value
- C one derivative only
- D the tangent gradient only

Classified Practice

Differentiation and Tangent Lines

Quiz MCQ 7 **Concept Quiz · MCQ****The tangent equation in point-slope form is:****Choose the correct option.**

A $y - y_1 = m(x - x_1)$

B $y = mx + c$

C $y + y_1 = m(x + x_1)$

D $x - x_1 = m(y - y_1)$

Quiz MCQ 8 Concept Quiz · MCQ

A tangent parallel to $y = 3x - 1$ has gradient:

Choose the correct option.

A -3

B $1/3$

C 3

D 1

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

Q002 $3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 20$

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Q004 $\frac{10}{3}$

Q005 $f(x) = x^2 - 2x + 3$

Q006 $f(x) = x^2 + 2x$

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x - \frac{3}{2}$

Q008 $f(x) = 3x^2 - 18x + 27$

Q009 $y = 2x - 8$

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = -4x - 1$

Q012 16 square units

Q013 $y = -7x + 2, y = x + 2; y = 3x + 15, y = 3x - \frac{85}{27}$

Q014 $y = 5x - 6, y = -3x + 2; y = 5x - 5, y = 5x - 1$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 4$

Q016 $y = 2x - 3$

Q017 B

Q018 B

Q019 C

Q020 A

Q021 B

Q022 B

Q023 A

Q024 C

Q025 C

Q026 B

Q027 C

Q028 C

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C

Concept 3 | Applications of Differentiation: Polynomial Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

Rate

$$\frac{dy}{dx}$$

Eng Eslam Ahmed | 01120009622

Classified Practice

Applications of Differentiation: Polynomial Functions

Q001 **Written**

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Q002 Written**Find the extreme values and sketch the three given Try cubics.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q003 **Written**

Find the extreme values and draw the graphs for all nine given cubic functions in Exercise.

Q004 Written

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q005 Written

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Q006 Written**Find the maximum and minimum values of the two given Try cubics on their stated closed intervals.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q007 **Written**

Find the maximum and minimum values of all six given cubics on their given closed intervals.

Q008 Written

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q009 Written

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a, b , and the local minimum.

Q010 Written

Solve the two given Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q011 Written

Solve all six given Exercise problems asking for cubic coefficients from extrema.

Q012 Written

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c .

Classified Practice

Applications of Differentiation: Polynomial Functions

Q013 Written

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k .

Q014 Written

Determine the parameter ranges in the given Try problems for a cubic to have no extreme values or to have both extremes.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q015 **Written**

Determine the parameter ranges for all six given Exercise cubics according to whether they have extreme values.

Q016 Written

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k -range that guarantees both a local maximum and minimum.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q017 Written

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k .

Q018 Written

Find the parameter ranges in the two given Try problems for the cubic to be monotonically increasing.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q019 **Written****Find the parameter ranges in all six given Exercise problems for monotonic increase.**

Q020 Written

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k + 6)x + 10$, find the k -range for monotonic increase on $x \in [1, \infty)$.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q021 Written

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.

Q022 Written**Find the number of distinct real roots for the three given Try equations.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q023 Written

Find the number of distinct real roots for all nine given cubic equations.

Q024 Written**Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q025 **Written**

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a .

Q026 Written**Solve the two given Try parameter problems for three or two distinct real roots.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q027 **Written****Solve all four given Exercise parameter problems for the required number of distinct roots.**

Q028 Written

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q029 Written

Prove the given inequalities using an auxiliary cubic and a derivative sign table.

Q030 Written**Prove all three given cubic inequalities on their stated domains.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q031 Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

Q032 Written**Find the extreme values and sketch the two given quartic functions.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q033 Written

Find the extreme values and graphs for all six given quartic functions.

Q034 Written

For $f(x) = x^4 - 4x^3 + k$, find the exact k-range for exactly two distinct real roots.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q035 MCQ

At a local maximum, the sign of f' changes:**Choose the correct option.**A $- \rightarrow +$ B $+ \rightarrow -$ C $+ \rightarrow +$ D $- \rightarrow -$

Q036 MCQ**For $P'(x) = 3[(x - a)^2 - 1]$, the local maximum occurs at:****Choose the correct option.****A** $a - 1$ **B** a **C** $a + 1$ **D** $1 - a$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q037 MCQ

Absolute extrema on a closed interval require comparing:**Choose the correct option.**

- A stationary points only
- B endpoints only
- C stationary points and endpoints
- D the y-intercept

Q038 MCQ

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:**Choose the correct option.**A $4\sqrt{2}$

B 9

C 6

D 0

Classified Practice

Applications of Differentiation: Polynomial Functions

Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Choose the correct option.

A 2

B 6

C 12

D -12

Q040 MCQ**The derivative roots in the challenge are:****Choose the correct option.**A $-4, 2$ B $-2, 4$ C $2, 4$ D $-3, 3$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q041 MCQ

No two distinct extrema for a cubic means the derivative discriminant is:**Choose the correct option.**A $D > 0$ B $D = 1$ C $D \leq 0$ D $D < 0$ only

Q042 MCQ

The challenge range for both extrema is:**Choose the correct option.**

A $0 < k < 4$

B $k < 0$ or $k > 4$

C $-4 < k < 0$

D $k \leq 4$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Choose the correct option.

- A negative everywhere
- B non-negative everywhere
- C zero everywhere
- D linear

Q044 MCQ

The challenge domain in 7-13 is:**Choose the correct option.**A $(-\infty, \infty)$ B $[1, \infty)$ C $[-1, 1]$ D $(0, 1)$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q045 MCQ

Three distinct roots occur when the x-axis lies:**Choose the correct option.**

- A above both extrema
- B between the local maximum and minimum
- C at an endpoint only
- D at the local maximum only

Q046 MCQ**The 7-14 challenge has how many distinct real roots?****Choose the correct option.****A** 0**B** 1**C** 2**D** 3

Classified Practice

Applications of Differentiation: Polynomial Functions

Q047 MCQ

Three intersections with $y = a$ require the level to lie:**Choose the correct option.**

- A strictly between the extrema
- B above the maximum
- C below the minimum
- D at an extremum

Q048 MCQ

The 7-15 warm-up range is:**Choose the correct option.**

A $-27 < a < 5$

B $-5 < a < 27$

C $a \leq -27$

D $a \geq 5$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q049 MCQ

To prove $LHS \geq RHS$, define:**Choose the correct option.**

A $f = LHS + RHS$

B $f = LHS - RHS$

C $f = RHS - 1$

D $f = f'$

Q050 MCQ

The largest k in the challenge is:**Choose the correct option.**

A $2 - 4\sqrt{2}$

B $4\sqrt{2} - 2$

C $2 + 4\sqrt{2}$

D $-2 - 4\sqrt{2}$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q051 MCQ

A repeated root of a derivative may be:**Choose the correct option.**

- A always a local maximum
- B always a local minimum
- C stationary but not an extreme
- D impossible

Q052 MCQ

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:**Choose the correct option.**A $k = 27$ B $k < 27$ C $k > 27$ D $k \leq 0$ only

Classified Practice

Applications of Differentiation: Polynomial Functions

Q053 MCQ

A local minimum is identified by the change:**Choose the correct option.**A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

Q054 MCQ

For $y = x^3 - 6x^2 + 16$, the critical x-values are:

Choose the correct option.

A 0, 4

B 1, 3

C -2, 2

D 2, 4

Classified Practice

Applications of Differentiation: Polynomial Functions

Q055 MCQ

The derivative sign table is used to determine:**Choose the correct option.**

- A only the y-intercept
- B increase/decrease and extrema
- C the constant term
- D the domain

Q056 MCQ**On a closed interval, an endpoint can be:****Choose the correct option.**

- A ignored
- B an absolute extreme
- C a derivative root only
- D never a minimum

Classified Practice

Applications of Differentiation: Polynomial Functions

Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Choose the correct option.

A $-2, 2$

B $-1, 1$

C $0, 4$

D $2, 4$

Q058 MCQ

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Choose the correct option.

A $x = -3$

B $x = -\sqrt{2}$

C $x = \sqrt{2}$

D $x = 3$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Choose the correct option.

A $f(a) = 0$

B $f'(a) = 0$

C $f''(a) = 0$

D $a = 0$

Q060 MCQ**In the warm-up** $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:**Choose the correct option.**

A $x = -2$

B $x = 0$

C $x = 2$

D $x = 7$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q061 MCQ

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Choose the correct option.

A $0 \leq k \leq 1/3$

B $k > 1/3$

C $k < 0$

D $k \geq 1/3$

Q062 MCQ

Two distinct stationary points occur when the derivative discriminant is:**Choose the correct option.**

A $D < 0$

B $D = 0$

C $D > 0$

D $D = 1$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Choose the correct option.

A non-negative for all x

B negative for all x

C linear

D undefined

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:**Choose the correct option.**A $[1, \infty)$ B $[-1, 1]$ C $(-\infty, 0)$ D $[0, 1]$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q065 MCQ

A cubic with both stationary values above the x-axis has:**Choose the correct option.**

- A three roots
- B one distinct real root
- C no real roots
- D two repeated roots

Q066 MCQ**To count cubic roots, compare the extrema with:****Choose the correct option.**

- A the y-axis
- B the x-axis
- C the derivative axis
- D the domain only

Classified Practice

Applications of Differentiation: Polynomial Functions

Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:**Choose the correct option.**

- A one distinct intersection
- B two distinct intersections
- C three distinct intersections
- D no intersection

Q068 MCQ**For $x^4 - 4x^3 + k$, the global minimum occurs at:****Choose the correct option.****A** $x = 0$ **B** $x = 1$ **C** $x = 3$ **D** $x = 4$

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 1 Concept Quiz · Written

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Quiz WRITTEN 2 Concept Quiz · Written

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at $x=1$.

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 3 Concept Quiz · Written**Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.**

Quiz WRITTEN 4 Concept Quiz · Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)**At a local minimum, f' changes:****Choose the correct option.**A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

Quiz MCQ 6 Concept Quiz · MCQ

A cubic with a positive-leading derivative and $D < 0$ is:

Choose the correct option.

- A always increasing
- B always decreasing
- C constant
- D undefined

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

The 7-15 challenge requires the level to satisfy:

Choose the correct option.

A $t < 0$

B $0 < t < 4$

C $t > 4$

D $t = 4$

Quiz MCQ 8 Concept Quiz · MCQ

The global minimum of $x^4 - 4x^3 + k$ is:

Choose the correct option.

A $k - 27$

B $k + 27$

C $k - 3$

D $27 - k$

Answer key

Use the question number shown on each page.

- Q001** Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.
- Q002** Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.
- Q003** Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.
- Q004** Local maximum at $x = a - 1$; local minimum at $x = a + 1$.
- Q005** Maximum 21 at $x = 2$; minimum -11 at $x = -2$.
- Q006** Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.
- Q007** For each function, compare the endpoint values with the values at all critical x-values lying in the interval.
- Q008** Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.
- Q009** $a = 12, b = -7$; local minimum -23 at $x = -2$.
- Q010** Use $f'(a) = 0, f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.
- Q011** Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.
- Q012** $a = -3, b = -24, c = 150$.
- Q013** $0 \leq k \leq \frac{1}{3}$.
- Q014** Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.
- Q015** Apply the derivative-discriminant criterion separately to each printed cubic.
- Q016** $k < 0$ or $k > 4$.
- Q017** $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$.
- Q018** Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.
- Q019** Differentiate each cubic and impose that its quadratic derivative never becomes negative.
- Q020** $k \leq 3$.
- Q021** Three distinct real roots.
- Q022** Use the cubic graph: count x-axis intersections from the stationary points and end behavior.
- Q023** Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.
- Q024** One distinct real root.
- Q025** $-27 < a < 5$.
- Q026** Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g .
- Q027** Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.
- Q028** $\frac{7-\sqrt{65}}{4} < a < \frac{7-\sqrt{33}}{4}$ or $\frac{7+\sqrt{33}}{4} < a < \frac{7+\sqrt{65}}{4}$.
- Q029** Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.
- Q030** Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.
- Q031** $k \leq 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.
- Q032** For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.
- Q033** Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.
- Q034** $k < 27$.
- Q035** B
- Q036** A
- Q037** C
- Q038** B
- Q039** C
- Q040** B
- Q041** C
- Q042** B
- Q043** B
- Q044** B
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** B
- Q050** A
- Q051** C
- Q052** B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at
 $x = 2$

Quiz WRITTEN 2 $a = 3, b = 2$

Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A

Concept 4 | Integration: Definite Integrals and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x) dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)| dx$$

Classified Practice

Integration: Definite Integrals and Applications

Q001 Written

Find the two indefinite integrals in the Warm Up.

Q002 Written

Find the six indefinite integrals in the given Try section, expanding a squared bracket first when needed.

Classified Practice

Integration: Definite Integrals and Applications

Q003 **Written****Solve the first ten given indefinite-integral exercises (constants, linear and quadratic polynomials).**

Q004 Written

Solve the remaining eight given integrals, including products and squared brackets.

Classified Practice

Integration: Definite Integrals and Applications

Q005 Written

Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.

Q006 Written**Find the two given** Try functions from their derivatives and one given value of F .

Classified Practice

Integration: Definite Integrals and Applications

Q007 **Written****Find $F(x)$ for all six given Exercise conditions.**

Q008 Written**Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.**

Classified Practice

Integration: Definite Integrals and Applications

Q009 Written

Solve the given Try tangent-gradient problem through $(1, 1)$.

Q010 Written

Solve the three given tangent-gradient exercises, finding each function from its point and gradient.

Classified Practice

Integration: Definite Integrals and Applications

Q011 Written

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Q012 Written

Evaluate the six given Try definite integrals, simplifying the integrand before applying the limits.

Classified Practice

Integration: Definite Integrals and Applications

Q013 **Written****Evaluate the first twelve given Exercise definite integrals, including reversed and equal limits.**

Q014 Written

Evaluate the remaining twelve given Exercise integrals, including sums, differences and products.

Classified Practice

Integration: Definite Integrals and Applications

Q015 Written

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Q016 Written**Solve the four given Try definite-integral transformations.**

Classified Practice

Integration: Definite Integrals and Applications

Q017 **Written****Solve the first six given Exercise questions using interval splitting and reversal.**

Q018 Written**Solve the final six given Exercise questions using the factorized definite-integral identity.**

Classified Practice

Integration: Definite Integrals and Applications

Q019 Written

Find $f(x)$ from the given equation $f(x) = x^2 + \int_0^1 f(t) dt$.

Q020 Written**Solve the two given Try integral equations for $f(x)$.**

Classified Practice

Integration: Definite Integrals and Applications

Q021 Written

Find $f(x)$ for all five given integral equations.

Q022 Written**Differentiate $\int_3^x (2t^2 + 3t - 5) dt$ with respect to x .**

Classified Practice

Integration: Definite Integrals and Applications

Q023 Written

Solve the two given Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Q024 Written**Differentiate the three given expressions with respect to x .**

Classified Practice

Integration: Definite Integrals and Applications

Q025 Written

Find $f(x)$ and a for the three given equations with a variable upper limit.

Q026 MCQ**The constant of integration is required because:****Choose the correct option.**

- A every primitive is unique
- B primitives differ by a constant
- C the integrand is constant
- D the limits are equal

Classified Practice

Integration: Definite Integrals and Applications

Q027 MCQ

The integral of x^n for $n \neq -1$ is:**Choose the correct option.**

A $nx^{n-1} + C$

B $x^{n+1}/(n+1) + C$

C $x^n/n + C$

D $x^{n-1}/(n-1) + C$

Q028 MCQ**Before integrating $(x - 1)^2$, first:****Choose the correct option.**

- A differentiate it
- B expand it
- C set $x = 0$
- D add limits

Classified Practice

Integration: Definite Integrals and Applications

Q029 MCQ

The integral of $3x^2 - 2x + 1$ is:**Choose the correct option.**

A $x^3 - x^2 + x + C$

B $3x^3 - 2x^2 + x + C$

C $x^3 - 2x + 1 + C$

D $6x - 2 + C$

Q030 MCQ**To find $F(x)$ from $F'(x)$, you should first:****Choose the correct option.**

- A differentiate again
- B integrate $F'(x)$
- C set $F = 0$
- D ignore the initial value

Classified Practice

Integration: Definite Integrals and Applications

Q031 MCQ

The condition $F(1) = 7$ determines:**Choose the correct option.**

- A the derivative
- B the domain
- C the constant C
- D the leading coefficient

Q032 MCQ**If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:****Choose the correct option.**

A $3x^2 - 2x + 1$

B $6x^2 - 2x + 1$

C $3x^2 - 2x$

D $6x - 2 + 1$

Classified Practice

Integration: Definite Integrals and Applications

Q033 MCQ

The gradient of the tangent at (x, y) is:**Choose the correct option.**A $f(x)$ B $f'(x)$ C $f''(x)$ D $x + y$

Q034 MCQ**A point (x_0, y_0) on $y = f(x)$ gives:****Choose the correct option.**

A $f'(x_0) = y_0$

B $f(x_0) = y_0$

C $f(0) = x_0$

D $C = 0$

Classified Practice

Integration: Definite Integrals and Applications

Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Choose the correct option.

A $3x^3 - 2x^2 + x$

B $x^3 - x^2 + x + C$

C $6x - 2$

D $x^3 - 2x + 1$

Q036 MCQ

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Choose the correct option.

A $F(a) - F(b)$

B $F(b) - F(a)$

C $F(a) + F(b)$

D 0 always

Classified Practice

Integration: Definite Integrals and Applications

Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:

Choose the correct option.

A 1

B a

C 0

D $f(a)$

Q038 MCQ

A reversed definite integral satisfies:**Choose the correct option.**

- A $\int_a^b f = \int_b^a f$
 - B $\int_a^b f = -\int_b^a f$
 - C both are zero
 - D only if $f = 0$
-
-
-

Classified Practice

Integration: Definite Integrals and Applications

Q039 MCQ

To evaluate a definite integral of a product, first:**Choose the correct option.**

- A integrate each factor separately
- B expand the product
- C differentiate the product
- D change the limits

Q040 MCQ**Adjacent integrals with the same integrand can be:****Choose the correct option.**

- A multiplied
- B combined over the joined interval
- C set equal to zero
- D reversed only

Classified Practice

Integration: Definite Integrals and Applications

Q041 MCQ

To use $\int_{\alpha}^{\beta} (x - \alpha)(x - \beta) dx$, the factors must have:

Choose the correct option.

- A different roots
- B the displayed roots (a,b)
- C zero coefficients
- D equal limits

Q042 MCQ

The identity for the factored integral is:**Choose the correct option.**

A $(\beta - \alpha)^2/6$

B $-(\beta - \alpha)^3/6$

C $(\beta + \alpha)^3/6$

D 0

Classified Practice

Integration: Definite Integrals and Applications

Q043 MCQ

A minus sign before an integral can be removed by:**Choose the correct option.**

- A squaring the integrand
- B reversing its limits
- C adding C
- D differentiating

Q044 MCQ

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Choose the correct option.

- A a function of x
- B a constant
- C always zero
- D a derivative

Classified Practice

Integration: Definite Integrals and Applications

Q045 MCQ

The best first step for an integral equation is to let:**Choose the correct option.**

A $A = f(x)$

B $A = \int_a^b f(t)dt$

C $A = x$

D $A = C$ only

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:

Choose the correct option.

A $1/3$

B $-1/3$

C 0

D 1

Classified Practice

Integration: Definite Integrals and Applications

Q047 MCQ

 $\frac{d}{dx} \int_a^x f(t)dt$ equals:

Choose the correct option.

A $f(a)$ B $f(x)$ C $F(x)$

D 0

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:

Choose the correct option.

- A the lower limit
- B the upper-limit variable
- C C
- D zero

Classified Practice

Integration: Definite Integrals and Applications

Q049 MCQ

If $\int_a^a f(t)dt = 0$, this can determine:

Choose the correct option.

- A the derivative
- B the unknown lower limit a
- C the integrand degree
- D the upper limit

Q050 MCQ**For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:****Choose the correct option.**

A $2x^2 + 3x - 5$

B $2x^3 + 3x^2 - 5x$

C $3x^2 + 3x - 5$

D 0

Classified Practice

Integration: Definite Integrals and Applications

Quiz WRITTEN 1 Concept Quiz · Written

Find $\int (4x^3 - 6x + 2)dx$.

Quiz WRITTEN 2 Concept Quiz · Written

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Classified Practice

Integration: Definite Integrals and Applications

Quiz WRITTEN 3 Concept Quiz · Written**Evaluate** $\int_{-1}^2 (3x^2 - 4x + 1)dx$.

Quiz WRITTEN 4 **Concept Quiz · Written**

Find $f(x)$ if $f(x) = x^2 + \int_0^1 f(t)dt$.

Classified Practice

Integration: Definite Integrals and Applications

Quiz MCQ 5 **Concept Quiz · MCQ****The derivative of $\int_0^x (3t^2 + 1)dt$ is:****Choose the correct option.**

A $3x^2 + 1$

B $x^3 + x$

C $6x$

D 0

Quiz MCQ 6 Concept Quiz · MCQ

A primitive of $5x^4$ is:

Choose the correct option.

A $x^5 + C$

B $5x^5 + C$

C $20x^3 + C$

D $x^4/5 + C$

Classified Practice

Integration: Definite Integrals and Applications

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Choose the correct option.

A $2x^2 + 2x + 3$

B $4x^2 + 2x + 3$

C $2x^2 + 2$

D $4x + 5$

Quiz MCQ 8 Concept Quiz · MCQ

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ equals:

Choose the correct option.

A $\int_1^5 f(x)dx$

B $\int_1^3 f(x)dx$

C $\int_3^5 f(x)dx$

D 0

Answer key

Use the question number shown on each page.

Q001 Integrate term by term and add C to each primitive.	Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.	Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.	Q036 B
Q002 Expand products or powers, collect like powers, then integrate every term.	Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.	Q021 Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$.	Q037 C
Q003 Apply linearity and the power rule; retain the constant of integration.	Q014 Use linearity, expand products, then apply the endpoint-substitution rule.	Q022 $2x^2 + 3x - 5$.	Q038 B
Q004 Expand each product or square, simplify, and integrate term by term.	Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.	Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a .	Q039 B
Q005 $F(x) = x^3 + 4x^2 - x + 3$.	Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.	Q024 Apply the variable-limit Fundamental Theorem directly to each integrand.	Q040 B
Q006 Integrate the derivative and use the given point to determine C .	Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits.	Q025 Differentiate to get $f'(x)$, then set the integral to zero at $x = a$ to solve for a .	Q041 B
Q007 For each item, integrate $F'(x)$, then impose the stated value of F .	Q018 Scale to $\int_a^b (x-a)(x-b) dx = -\frac{(b-a)^3}{6}$.	Q026 B	Q042 B
Q008 $f(x) = x^3 - x^2 + x - 19$.	Q019 Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$.	Q027 B	Q043 B
Q009 Integrate the given gradient and use the point on the curve to determine the constant.		Q028 B	Q044 B
Q010 Integrate the polynomial gradient and apply the given point condition in each case.		Q029 A	Q045 B
Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.		Q030 B	Q046 B
		Q031 C	Q047 B
		Q032 A	Q048 B
		Q033 B	Q049 B
		Q034 B	Q050 A
		Q035 B	Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$
			Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$
			Quiz WRITTEN 3 6
			Quiz WRITTEN 4 $f(x) = x^2 - 1/3$
			Quiz MCQ 5 A
			Quiz MCQ 6 A
			Quiz MCQ 7 A
			Quiz MCQ 8 A

Concept 5 | Applications of Integration to Geometry

English Student Edition • Focused formulas and guided practice

Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom}) dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \leq x \leq b$$

Classified Practice

Applications of Integration to Geometry

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x -axis, $x = -2$, and $x = 1$.

Q002 Written**Find the area bounded by $y = x^2 + x - 6$, the x -axis, $x = -2$, and $x = 1$.**

Classified Practice

Applications of Integration to Geometry

Q003 Written

Find the area bounded by $y = -x^2 + 4$ and the x -axis.

Q004 Written**Find the area under $y = -x^2 - 2x + 3$ and above the x -axis.**

Classified Practice

Applications of Integration to Geometry

Q005 Written

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.

Q006 Written**Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.**

Classified Practice

Applications of Integration to Geometry

Q007 Written

Find the area between $y = -x^2 + 4x$ and $y = x$.

Q008 Written**Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.**

Classified Practice

Applications of Integration to Geometry

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Q010 Written**Find the area bounded by $y = -x^3 + 4x$ and the x -axis.**

Classified Practice

Applications of Integration to Geometry

Q011 Written

Solve the given cubic area exercises by finding every x -intercept and adding the absolute signed pieces.

Q012 Written

Evaluate $\int_{-2}^1 |x + 1| dx$.

Classified Practice

Applications of Integration to Geometry

Q013 Written

Evaluate $\int_0^3 |x^2 - 4| dx$.

Q014 Written**Evaluate** $\int_0^2 |x^2 + 3x - 4| dx$.

Classified Practice

Applications of Integration to Geometry

Q015 Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Q016 Written**Find the area bounded by $y = x^3 + x^2 - 3x + 6$ and its tangent at $(1, 5)$.**

Classified Practice

Applications of Integration to Geometry

Q017 **Written**

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Q018 Written

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x -axis, find k .

Classified Practice

Applications of Integration to Geometry

Q019 **Written**

Solve the **given** bisection exercises for the given parabolas and lines, using the required area ratio.

Q020 MCQ**When the graph is below the x-axis, area is:****Choose the correct option.**

- A the signed integral
- B the negative of the integral
- C always zero
- D undefined

Classified Practice

Applications of Integration to Geometry

Q021 MCQ

A bounded area under a curve is found by first checking:**Choose the correct option.**

- A the graph and sign
- B only the y-intercept
- C the font size
- D the derivative only

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:

Choose the correct option.

A $8/3$

B $16/3$

C $32/3$

D $64/3$

Classified Practice

Applications of Integration to Geometry

Q023 MCQ

Between two curves, the integrand is:**Choose the correct option.**

- A lower-upper
- B upper-lower
- C product
- D derivative

Q024 MCQ**If curves cross inside the interval, area must be:****Choose the correct option.**

- A split at the crossing
- B ignored
- C doubled always
- D negative

Classified Practice

Applications of Integration to Geometry

Q025 MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A $16/3$

B $32/3$

C $64/3$

D $9/2$

Q026 MCQ**For cubic area, the sign changes at:****Choose the correct option.**

- A every turning point
- B each x-intercept
- C the y-intercept only
- D the midpoint

Classified Practice

Applications of Integration to Geometry

Q027 MCQ

The area for $y = -x^3 + 4x$ is:**Choose the correct option.**

A 4

B 6

C 8

D 12

Q028 MCQ**A negative lobe contributes:****Choose the correct option.**

- A negative area
- B positive absolute area
- C no area
- D a derivative

Classified Practice

Applications of Integration to Geometry

Q029 MCQ

For an absolute-value integral, first locate:**Choose the correct option.**

- A the roots
- B the vertex only
- C the y-axis
- D the tangent

Q030 MCQ

The value of $\int_0^2 |x^2 + 3x - 4| dx$ is:

Choose the correct option.

A 3

B 4

C 5

D 6

Classified Practice

Applications of Integration to Geometry

Q031 MCQ

If f changes sign at c , split the integral:**Choose the correct option.**A at c

B at 0 only

C never

D after doubling

Q032 MCQ**A tangent line at $x=a$ has slope:****Choose the correct option.**A $f(a)$ B $f'(a)$ C a^2

D the area

Classified Practice

Applications of Integration to Geometry

Q033 MCQ

For two tangent lines, split at their:**Choose the correct option.**

A intersection x-coordinate

B y-intercept only

C vertex only

D origin

Q034 MCQ

The area in the given parabola-tangent Warm Up is:**Choose the correct option.**A $\frac{1}{2}$

B 1

C $\frac{3}{2}$

D 3

Classified Practice

Applications of Integration to Geometry

Q035 MCQ

Bisection means each part has:**Choose the correct option.**

- A the same area
- B the same slope
- C the same root
- D zero area

Q036 MCQ**For $y=2x-x^2$, the total area above the x-axis is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Applications of Integration to Geometry

Q037 MCQ

The line $y=kx$ must satisfy the stated:**Choose the correct option.**

- A area ratio
- B font ratio
- C root count
- D period

Q038 MCQ**In a bisection problem, the cut-off area is:****Choose the correct option.**

- A the total area
- B half the total area
- C twice the total area
- D zero

Classified Practice

Applications of Integration to Geometry

Quiz MCQ 1 **Concept Quiz · MCQ**

The area under $y = -x^2 + 4$ and above the x -axis is:

Choose the correct option.

A $16/3$

B 8

C $32/3$

D $64/3$

Quiz MCQ 2 Concept Quiz · MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A $16/3$

B $32/3$

C $64/3$

D $9/2$

Classified Practice

Applications of Integration to Geometry

Quiz MCQ 3 Concept Quiz · MCQ

Evaluate $\int_{-2}^1 |x + 1| dx$:

Choose the correct option.

- A 2
- B $5/2$
- C 3
- D $7/2$

Quiz MCQ 4 Concept Quiz · MCQ

The area enclosed by $y = x^2$ and tangents through (3, 8) is:

Choose the correct option.

A $1/3$

B $1/2$

C $2/3$

D $4/3$

Classified Practice

Applications of Integration to Geometry

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Quiz WRITTEN 6 Concept Quiz · Written

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.

Classified Practice

Applications of Integration to Geometry

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Quiz WRITTEN 8 Concept Quiz · Written

If $y = kx$ bisects the area under $y = 2x - x^2$, find k .

Answer key

Use the question number shown on each page.

Q001 9

Q002 $\frac{33}{2}$ Q003 $\frac{32}{3}$ Q004 $\frac{32}{3}$ Q005 $\frac{32}{3}$ Q006 $\frac{64}{3}$ Q007 $\frac{9}{2}$

Q008 30

Q009 $\frac{37}{12}$

Q010 8

Q011 Use the numbered solution for each given cubic.

Q012 $\frac{5}{2}$ Q013 $\frac{23}{3}$

Q014 5

Q015 $\frac{3}{2}$ Q016 $\frac{64}{3}$ Q017 $\frac{2}{3}$ Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$

Q019 Use the numbered solution for each given ratio.

Q020 B

Q021 A

Q022 C

Q023 B

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 A

Q030 C

Q031 A

Q032 B

Q033 A

Q034 C

Q035 A

Q036 C

Q037 A

Q038 B

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $\frac{37}{12}$

Quiz WRITTEN 6 5

Quiz WRITTEN 7 $\frac{3}{2}$ Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$

Concept 1 | Fundamentals of Random Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Expectation

$$E(X) = \sum x p(x)$$

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

Classified Practice

Fundamentals of Random Variables

Q001 **Written**

For the sum of two dice, construct the probability distribution and find $P(5 \leq X \leq 8)$.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

Classified Practice

Fundamentals of Random Variables

Q003 **Written**

Construct both given probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Q004 Written

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

Classified Practice

Fundamentals of Random Variables

Q005 **Written**

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Q006 Written**Find EX for all six given random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).**

Classified Practice

Fundamentals of Random Variables

Q007 **Written**

For tails in four fair coins, find $E(3X+2)$, $E(-4X+1)$, and EX^2 .

Q008 Written

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

Classified Practice

Fundamentals of Random Variables

Q009 **Written****Solve both given Exercise groups of transformed expectations for coins and for balls.**

Q010 Written

For black balls drawn from the stated box, find EX and VX .

Classified Practice

Fundamentals of Random Variables

Q011 Written

Find the mean and variance for the two given Try random variables (a numbered card and winning tickets).

Q012 Written**Find EX and VX for all five given Exercise random variables.**

Classified Practice

Fundamentals of Random Variables

Q013 Written

Find the standard deviation of tails when two biased coins have $P(\text{head})=4/7$ and $P(\text{tail})=3/7$.

Q014 Written

Find the standard deviation from each given probability table and for white balls drawn from 4 red and 3 white.

Classified Practice

Fundamentals of Random Variables

Q015 Written

Find the standard deviation for both given probability-table exercises.

Q016 Written

For the number of winning tickets in the given draw, find VY and σY for $Y=2X+1$ and $Y=-3X-2$.

Classified Practice

Fundamentals of Random Variables

Q017 **Written**

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X.

Q018 Written

For a single die, find VY and σY for $Y=X+2$, $Y=3X+1$ and $Y=-2X+4$.

Classified Practice

Fundamentals of Random Variables

Q019 **Written**

Solve the two **given** transformation problems: a net prize after an entry fee, and finding a, b from $E(aX+b)=7/3$ and $\sigma(aX+b)=3$.

Q020 Written**Solve the two given Try problems involving net winnings and determining a,b for a die transformation.**

Classified Practice

Fundamentals of Random Variables

Q021 Written

Solve all four **given** transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Q022 MCQ**A probability distribution must have probabilities that sum to:****Choose the correct option.****A** 0**B** 1**C** the number of outcomes**D** the mean

Classified Practice

Fundamentals of Random Variables

Q023 MCQ

For two fair dice, $PX = 2$ is:**Choose the correct option.**A $1/6$ B $1/36$ C $2/36$

D 0

Q024 MCQ**The number of heads in three coins can be:****Choose the correct option.****A** 1,2,3**B** 0,1,2,3**C** 0,3 only**D** any integer

Classified Practice

Fundamentals of Random Variables

Q025 MCQ

The expected value is calculated as:**Choose the correct option.**

- A $\sum PX = x$
- B $\sum xPX = x$
- C largest x
- D variance squared

Q026 MCQ**The mean number of tails in four fair coins is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Fundamentals of Random Variables

Q027 MCQ

For a fair die, EX is:**Choose the correct option.**

A 3

B 3.5

C 4

D 6

Q028 MCQ **$E(aX+b)$ equals:****Choose the correct option.**A aEX B $aEX+b$ C $EX+b$ onlyD $a+b$

Classified Practice

Fundamentals of Random Variables

Q029 MCQ

In general, EX^2 is:**Choose the correct option.**A $E[X]^2$ B equal to EX

C a weighted square sum

D always zero

Q030 MCQ**If $EX=2$, then $E(3X+2)$ is:****Choose the correct option.****A** 6**B** 8**C** 10**D** 12

Classified Practice

Fundamentals of Random Variables

Q031 MCQ

The variance identity is:**Choose the correct option.**

A $E[X] - E[X^2]$

B $E[X^2] - E[X]^2$

C $E[X]^2$

D $\sqrt{V}$

Q032 MCQ**Variance can never be:****Choose the correct option.**

A positive

B zero

C negative

D fractional

Classified Practice

Fundamentals of Random Variables

Q033 MCQ

The first step in a variance problem is to:**Choose the correct option.**

- A draw the distribution
- B take a square root
- C ignore probabilities
- D add the values

Q034 MCQ

Standard deviation is:**Choose the correct option.**

- A VX
- B $\sqrt{V(X)}$
- C VX squared
- D EX

Classified Practice

Fundamentals of Random Variables

Q035 MCQ

Standard deviation has units:**Choose the correct option.**

A squared units

B the same units as X

C no units

D units cubed

Q036 MCQ **$V(aX+b)$ equals:****Choose the correct option.****A** $aVX+b$ **B** a squared times VX **C** $|a|VX$ **D** $VX+b$

Classified Practice

Fundamentals of Random Variables

Q037 MCQ

 $\sigma(aX+b)$ equals:**Choose the correct option.**A $a \sigma X$ B $|a| \sigma X$ C $a^2 \sigma X$ D $\sigma X+b$

Q038 MCQ**For $Y=X+3$, the variance is:****Choose the correct option.****A** $VX+3$ **B** VX **C** $3VX$ **D** $9VX$

Classified Practice

Fundamentals of Random Variables

Q039 MCQ

A net prize after a fixed fee is modeled as:**Choose the correct option.**

A $Y = X + b$

B $Y = aX + b$

C $Y = X^2$

D $Y = a + b$ only

Q040 MCQ**If $a > 0$ and $\sigma(aX+b)=3$, then a is:****Choose the correct option.****A** $3\sigma X$ **B** $3/\sigma X$ **C** $\sigma X/3$ **D** $3+b$

Classified Practice

Fundamentals of Random Variables

Q041 MCQ

Changing b in $aX+b$ changes:**Choose the correct option.**

- A variance only
- B standard deviation only
- C the mean only
- D nothing

Q042 MCQ**If $EX=2$ and $EX^2=7$, then VX is:****Choose the correct option.****A 3****B 5****C 7****D 9**

Classified Practice

Fundamentals of Random Variables

Quiz WRITTEN 1 **Concept Quiz · Written**

Find EX and VX for the number of heads in three fair coins.

Quiz WRITTEN 2 Concept Quiz · Written

For $Y=4X-1$ with $VX=2$, find VY and σY .

Classified Practice

Fundamentals of Random Variables

Quiz WRITTEN 3 Concept Quiz · Written

For a fair die, find $E(2X+5)$.

Quiz WRITTEN 4 Concept Quiz · Written

If $\sigma X=2$ and $EX=3$, find $\sigma(-5X+7)$.

Classified Practice

Fundamentals of Random Variables

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

If $VX=5$, $V(3X+1)$ is:

Choose the correct option.

A 5

B 8

C 15

D 45

Quiz MCQ 6 Concept Quiz · MCQ**For a fair die EX is:****Choose the correct option.****A** 2.5**B** 3**C** 3.5**D** 4

Classified Practice

Fundamentals of Random Variables

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

A probability table total must be:

Choose the correct option.

A 0

B 1

C 2

D the mean

Quiz MCQ 8 Concept Quiz · MCQ

If $\sigma X = 4$, $\sigma(X-9)$ is:

Choose the correct option.

A 0

B 4

C 13

D 16

Answer key

Use the question number shown on each page.

- Q001** Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$.
- Q002** Use the sample space or combination to list every possible X and sum the relevant probabilities.
- Q003** Use binomial or hypergeometric counts, then add probabilities for the requested values.
- Q004** For four fair coins $EX=2$; for the hypergeometric draw $EX=4/3$.
- Q005** Use $EX = \sum xPX = x$ after constructing each distribution.
- Q006** Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.
- Q007** Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table.
- Q008** Apply linearity for $aX+b$ and use the weighted square sum for X^2 .
- Q009** For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table.
- Q010** $EX = 6/7$; $VX = 20/49$.
- Q011** Construct each probability table, then use $VX = EX^2 - EX^2$.
- Q012** Use the distribution table for each experiment and the variance identity.
- Q013** The table gives $VX = 24/49$, so $\sigma X = 2 \sqrt{6/7}$.
- Q014** Use $\sigma X = \sqrt{V(X)}$ after evaluating the variance from the table.
- Q015** Compute EX , EX^2 , then $\sigma X = \sqrt{E(X^2) - E(X)^2}$.
- Q016** Use $V(aX + b) = a^2 VX$ and $\sigma(aX+b) = |a| \sigma X$.
- Q017** Apply the squared multiplier to VX and the absolute multiplier to σX .
- Q018** Use the variance and standard-deviation transformation formulas for each Y .
- Q019** Use $E(aX + b) = aEX + b$ and $\sigma(aX+b) = |a| \sigma X$, with $a > 0$.
- Q020** Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.
- Q021** Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.
- Q022** B
- Q023** B
- Q024** B
- Q025** B
- Q026** B
- Q027** B
- Q028** B
- Q029** C
- Q030** C
- Q031** B
- Q032** C
- Q033** A
- Q034** B
- Q035** B
- Q036** B
- Q037** B
- Q038** B
- Q039** B
- Q040** B
- Q041** C
- Q042** A
- Quiz WRITTEN 1** $EX=3/2$ and $VX=3/4$
- Quiz WRITTEN 2** $VY=32$ and $\sigma Y=4 \sqrt{2}$
- Quiz WRITTEN 3** 12
- Quiz WRITTEN 4** 10
- Quiz MCQ 5** D
- Quiz MCQ 6** C
- Quiz MCQ 7** B
- Quiz MCQ 8** B

Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Independent

$$\text{Var}(X + Y) = \text{Var } X + \text{Var } Y$$

Classified Practice

Expected Value and Variance of Multiple Variables

Q001 **Written**

Given the probability distributions of X and Y in the given tables, find $E(X+Y)$.

Q002 Written

Three coins are tossed once; let X , Y and Z be the head indicators. Find $E(X+Y+Z)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q003 Written

For the two given tables with $X=1,2,3$ and $Y=4,5,6$, find $E(X+Y)$.

Q004 Written

For the two bags in the given exercise, find the expected total number of red balls drawn.

Classified Practice

Expected Value and Variance of Multiple Variables

Q005 Written

Three independent subsystems each receive two signals with success probability 0.5. Find $E(X+Y+Z)$.

Q006 Written

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z .

Classified Practice

Expected Value and Variance of Multiple Variables

Q007 **Written****Two 500-point coins and two 100-point coins are tossed. Find EZ.**

Q008 Written

A die is thrown twice and the score is $Z=3X+5Y$. Find EZ .

Classified Practice

Expected Value and Variance of Multiple Variables

Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Q010 Written

In two independent raffles, buy two tickets from a $\frac{2}{10}$ winner raffle and two from a $\frac{1}{6}$ winner raffle. Find the expected prizes.

Classified Practice

Expected Value and Variance of Multiple Variables

Q011 Written

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find $E(XY)$.

Q012 Written

For independent X with probabilities $2/5, 2/5, 1/5$ at $1, 2, 3$ and Y uniform on $2, 4, 6$, find $E(XY)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q013 Written

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find $E(XY)$.

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find $E(XY)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q015 Written

For independent X and Y with $EX=1$ and $EY=2/7$, find $E(XY)$.

Q016 Written

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find $V(X+Y)$ and $\sigma(X+Y)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q017 **Written**

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Q018 Written

An independent coin head count X and die score Y are combined. Find $V(X+Y)$ and $\sigma(X+Y)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q019 Written

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Q020 Written**Independent X is uniform on 1 to 4 and Y on 1 to 6. Find $V(X+Y)$ and $\sigma(X+Y)$.**

Classified Practice

Expected Value and Variance of Multiple Variables

Q021 Written

Three independent special dice take values 2,4,6 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Q022 Written**Three independent dice take values 1,3,5 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.**

Classified Practice

Expected Value and Variance of Multiple Variables

Q023 Written

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Q024 Written

Three friends have 5, 4 and 3 independent rolls to score a six. Find $E(XYZ)$ and $V(X+Y+Z)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q025 Written

If X , Y and Z are independent head counts from 4, 3 and 2 fair coins, find $E(XYZ)$.

Q026 MCQ **$E(X+Y)$ equals:****Choose the correct option.****A** $EX-EY$ **B** $EX+EY$ **C** $EXEY$ **D** EX/EY

Classified Practice

Expected Value and Variance of Multiple Variables

Q027 MCQ

For three variables, $E(X+Y+Z)$ equals:**Choose the correct option.**

- A $EXEYEZ$
- B $EX+EY+EZ$
- C $EX-EY-EZ$
- D 0

Q028 MCQ**The expected heads in one fair coin toss is:****Choose the correct option.****A** 0**B** $1/2$ **C** 1**D** 2

Classified Practice

Expected Value and Variance of Multiple Variables

Q029 MCQ

For two signals with $p=0.5$, EX is:**Choose the correct option.**

A 0.5

B 1

C 2

D 4

Q030 MCQ**If $EX=2$ and $EY=5$, $E(X+Y)$ is:****Choose the correct option.****A** 3**B** 7**C** 10**D** 25

Classified Practice

Expected Value and Variance of Multiple Variables

Q031 MCQ

 $E(aX+bY)$ equals:**Choose the correct option.**A $aEX+bEY$ B $abE(XY)$ C $EX+EY$ D $a+b$

Q032 MCQ**The expected score of one fair 100-point coin is:****Choose the correct option.**

- A 25
- B 50
- C 100
- D 200

Classified Practice

Expected Value and Variance of Multiple Variables

Q033 MCQ

For a fair die, $E(3X+5Y)$ with two rolls is:**Choose the correct option.**

- A 14
- B 21
- C 28
- D 35

Q034 MCQ**A fixed entry fee changes:****Choose the correct option.**

- A variance only
- B the mean only
- C neither mean nor variance
- D standard deviation only

Classified Practice

Expected Value and Variance of Multiple Variables

Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:**Choose the correct option.**A $1/5$ B $2/5$ C $1/2$ D $4/5$

Q036 MCQ **$E(XY) = EXEY$ requires:****Choose the correct option.**

- A $X = Y$
- B independence
- C zero means
- D equal variances

Classified Practice

Expected Value and Variance of Multiple Variables

Q037 MCQ

For three fair coins, EX is:**Choose the correct option.**A $1/2$

B 1

C $3/2$

D 3

Q038 MCQ**For a fair die, EY is:****Choose the correct option.****A** 2.5**B** 3**C** 3.5**D** 6

Classified Practice

Expected Value and Variance of Multiple Variables

Q039 MCQ

With replacement, two card draws are:**Choose the correct option.**

- A dependent
- B independent
- C identical outcomes
- D impossible

Q040 MCQ**If $EX=2$ and $EY=3$ for independent variables, $E(XY)$ is:****Choose the correct option.****A** 5**B** 6**C** 1**D** 9

Classified Practice

Expected Value and Variance of Multiple Variables

Q041 MCQ

For independent X,Y, $V(X+Y)$ equals:**Choose the correct option.**A $VX-VY$ B $VX+VY$ C $VXVY$

D 0

Q042 MCQ

The variance of a fair die is:**Choose the correct option.**A $7/2$ B $35/12$ C $35/6$ D $1/4$

Classified Practice

Expected Value and Variance of Multiple Variables

Q043 MCQ

Sigma(X+Y) is:**Choose the correct option.**A $V(X+Y)$ B $\sqrt{V(X+Y)}$ C $VX+VY$ D $E(X+Y)$

Q044 MCQ**The variance of the sum of two independent fair dice is:****Choose the correct option.**A $35/12$ B $35/6$ C $70/6$ D $49/4$

Classified Practice

Expected Value and Variance of Multiple Variables

Q045 MCQ

For independent sums, a constant shift changes:**Choose the correct option.**

- A variance
- B standard deviation
- C neither spread measure
- D both spread measures

Q046 MCQ**For independent X, Y, Z , $E(XYZ)$ equals:****Choose the correct option.****A** $EX+EY+EZ$ **B** $EXEYEZ$ **C** 0**D** $E(XY)+EZ$

Classified Practice

Expected Value and Variance of Multiple Variables

Q047 MCQ

For independent X, Y, Z , $V(X+Y+Z)$ equals:**Choose the correct option.**A $VXVYVZ$ B $VX+VY+VZ$ C $VX-VY-VZ$

D 0

Q048 MCQ**For a binomial count with $n=5$ and $p=1/6$, EX is:****Choose the correct option.****A** $1/6$ **B** $5/6$ **C** $5/36$ **D** 6

Classified Practice

Expected Value and Variance of Multiple Variables

Q049 MCQ

For three values 2,4,6 equally likely, EX is:**Choose the correct option.**

A 2

B 3

C 4

D 6

Q050 MCQ**For three independent fair dice, $V(\text{sum})$ is:****Choose the correct option.**A $35/12$ B $35/6$ C $35/4$ D $343/8$

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 1 **Concept Quiz · Written**

If $EX=4$ and $EY=7$, find $E(3X-2Y)$.

Quiz WRITTEN 2 Concept Quiz · Written

For independent variables with $EX=2$ and $EY=5$, find $E(XY)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 3 **Concept Quiz · Written**

Two independent fair dice are thrown. Find $V(X+Y)$ and $\sigma(X+Y)$.

Quiz WRITTEN 4 Concept Quiz · Written

Three independent Binomial counts have $(n,p)=(2,1/2),(3,1/2),(4,1/2)$. Find $V(X+Y+Z)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

If $EX=3$ and $EY=4$, $E(2X+Y)$ is:

Choose the correct option.

A 7

B 10

C 14

D 24

Quiz MCQ 6 Concept Quiz · MCQ

For independent variables $EX=2$, $EY=5$, $E(XY)$ is:

Choose the correct option.

- A 7
- B 10
- C 25
- D $1/2$

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz MCQ 7 [Concept Quiz](#) · MCQ

For two independent die rolls, $V(X+Y)$ is:

Choose the correct option.

A $35/12$

B $35/6$

C $35/4$

D $7/2$

Quiz MCQ 8 Concept Quiz · MCQ

For three independent variables, $V(X+Y+Z)$ is:

Choose the correct option.

- A** the product of variances
- B** the sum of variances
- C** the sum of means
- D** always zero

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, \text{so } E(X + Y) = 97/6.$ **Q016** $VX=2/3, VY=3/4, \text{so } V(X + Y) = 17/12.$ **Q025** $E(XYZ) = 23/21=3.$

Q002 $E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$

and $\sigma = \sqrt{17/12}.$

Q026 B

Q003 $E(X + Y) = EX + EY = 2 + 5 = 7.$

Q017 $V(X + Y) = 35/12 + 35/12 = 35/6;$

Q027 B

Q004 Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7.$

$\sigma = \sqrt{35/6}.$

Q028 B

Q005 Each mean is $2(0.5) = 1, \text{so}$

Q018 $V = 1/4 + 35/12 = 19/6;$

Q029 B

$E(X + Y + Z) = 3.$

$\sigma = \sqrt{19/6}.$

Q030 B

Q006 $Z = 500X + 100Y, \text{so } EZ = 5001 + 1003/2 = 6500.5$

Q019 $V = 21/4 + 31/4 = 5/4.$

Q031 A

Q007 $EZ = 5001 + 1001 = 600.$

Q020 $V = 5/4 + 35/12 = 25/6;$

Q032 B

Q008 $EZ = 37/2 + 57/2 = 28.$

$\sigma = 5/\sqrt{6}.$

Q033 C

Q009 $EZ = 350 + 25 = 175.$

Q021 $EX=4, VX=8/3; E(XYZ) = 64 \text{ and}$

Q034 B

Q010 $EZ = 22/10 + 21/6 = 11/5.$

$V(\text{sum}) = 8.$

Q035 B

Q011 X and Y are independent, so

Q022 $EX=3, VX=8/3; E(XYZ) = 27 \text{ and}$

Q036 B

$E(XY) = EXEY = 3/27/2 = 21/4.$

$V(\text{sum}) = 8.$

Q037 C

Q012 $EX = 8/5, EY = 4, \text{so } E(XY) = 32/5.$

Q023 $E(\text{product}) = 7/2^3 = 343/8;$

Q038 C

Q013 $E(XY) = EXEY = 3/21 = 3/2.$

$V(\text{sum}) = 335/12 = 35/4.$

Q039 B

Q014 $EX = EY = 5/2, \text{so } E(XY) = 25/4.$

Q024 Means are $5/6, 2/3, 1/2;$

Q040 B

Q015 $E(XY) = EXEY = 2/7.$

$E(\text{product}) = 5/18 \text{ and } V(\text{sum}) = 5/3.$

Q041 B

Q042 B

Q043 B

Q044 B

Q045 C

Q046 B

Q047 B

Q048 B

Q049 C

Q050 C

Quiz WRITTEN 1 -2

Quiz WRITTEN 2 10

Quiz WRITTEN 3 $V = 35/6, \sigma = \sqrt{35/6}$

Quiz WRITTEN 4 $9/4$

Quiz MCQ 5 B

Quiz MCQ 6 B

Quiz MCQ 7 B

Quiz MCQ 8 B